# RECURRENCE COEFFICIENTS OF DISCRETE MEASURES WITH POWER-LAW ATOMS AND SLOWLY VARYING WEIGHTS

GORDON [SURNAME] (DRAFT)

ABSTRACT. Let $\sigma = \sum_{k\ge1} k^{-s}\ell(k)\,\delta_{k^{-p}}$ with $p>0$, $s>1$ and $\ell$ slowly varying; its Jacobi matrix is a compact operator with eigenvalues $k^{-p}$. We prove that the recurrence coefficients satisfy $n^p b_n \to c(p)$ and $n^p a_n \to 2c(p)$, where $c(p) = \frac14\big(\sqrt{\pi}\,\Gamma(1/p+1)/\Gamma(1/p+\frac12)\big)^p$ is the constant of the constrained equilibrium problem solved in a companion paper. This holds for every slowly varying $\ell$, including unbounded ones such as $\exp\sqrt{\log k}$. If $\ell$ extends to a slowly varying analytic function on some sector $|\arg t|\le\vartheta$ with half-angle $\vartheta>0$, however small, the error is $O(1/n)$; for pure powers this gives $n^p b_n = c(p)(1+O(1/n))$ and $a_n/(2b_n) = 1+O(1/n)$. The proof reduces, by a Christoffel transformation (which shifts $s$ by $p$ and preserves the law with its rate) and a smoothing step, to analytic factors and the window $1<s<1+2p$. There a Riemann–Hilbert analysis of discrete orthogonal polynomials whose nodes accumulate at a hard edge controls the non-analytic factor by a Conway–Deift comparison with a Bessel model, instead of a local parametrix. In a further paper we show that bounded but rough weights can destroy the pointwise law.

## 1. INTRODUCTION

1.1. **Setting.** A compact Jacobi operator is determined by its spectral measure. If its eigenvalues are $k^{-p}$ and its spectral weights decay like $k^{-s}$, how do its matrix entries behave? We show that the off-diagonal entries $b_n$ and the diagonal entries $a_n$ satisfy $n^p b_n \to c(p)$ and $n^p a_n \to 2c(p)$, with an explicit constant $c(p)$, for every $p>0$, every $s>1$ and every slowly varying perturbation of the weights, and that the convergence has rate $O(1/n)$ when the perturbation is analytic.

Concretely, for $p>0$, $s>1$ and a sequence $\ell\colon\mathbb{N}\to(0,\infty)$ we study the discrete measure

$$\sigma_{p,s,\ell} := \sum_{k\ge1} k^{-s}\ell(k)\,\delta_{k^{-p}}, \qquad \sigma_{p,s} := \sigma_{p,s,1}. \tag{1}$$

It lives on $(0,1]$, and its infinitely many atoms accumulate only at 0. Let $P_n$ be its monic orthogonal polynomials and $h_n := \int P_n^2\,\mathrm{d}\sigma$. They satisfy the three-term recurrence

$$xP_n = P_{n+1} + a_nP_n + b_n^2P_{n-1} \qquad (n\ge0,\ b_0:=0,\ b_n>0), \tag{2}$$

so that $b_n^2 = h_n/h_{n-1}$; the numbers $a_n$ and $b_n$ are the entries of the Jacobi matrix of $\sigma$. With

$$I(p) := \frac{\sqrt{\pi}}{2}\,\frac{\Gamma(\frac1p+\frac12)}{\Gamma(\frac1p+1)}, \qquad c(p) := \frac14\Big(\frac{\pi}{2I(p)}\Big)^p = \frac14\Big(\frac{\sqrt{\pi}\,\Gamma(\frac1p+1)}{\Gamma(\frac1p+\frac12)}\Big)^p, \tag{3}$$

for example $c(1)=\frac12$ and $c(2)=\pi^2/16$, the question is whether

$$n^p b_n \longrightarrow c(p) \qquad\text{and}\qquad n^p a_n \longrightarrow 2c(p) \qquad (n\to\infty). \tag{4}$$

We call (4) the *pointwise law*. The constant $c(p)$ comes from the companion paper [15], which proves (4) on average (see (7) below).

*Date*: Draft of 14 October 2026.

1.2. **Results.** We use the following form of slow variation, which needs no measurability or representation theorem.

**Definition 1.1** (slow variation)**.** A sequence $\ell\colon \mathbb{N} \to (0,\infty)$ is *slowly varying* if, for every $\lambda \geq 1$,

$$\operatorname{osc}_n(\lambda) := \max\bigl\{\, |\log \ell(k) - \log \ell(n)| : \ k \in \mathbb{N},\ n/\lambda \leq k \leq \lambda n \,\bigr\} \longrightarrow 0 \qquad (n \to \infty). \tag{5}$$

The case $\lambda = 2$ already implies the others (Remark 2.2). Examples are $(\log(k+1))^{\beta}$ with $\beta \in \mathbb{R}$, $\exp((\log k)^{\beta})$ with $0 < \beta < 1$, and $\exp(\sqrt{\log k}\,\sin\log\log k)$ for $k \geq 3$. Neither boundedness nor monotonicity is required.

For the sharp rate we need an analytic extension of $\ell$ to a sector, written in the logarithmic variable $\xi = \log t$. For $\vartheta > 0$ and $c_0 > 0$ put $S_\vartheta := \{\xi \in \mathbb{C} : \ \Re\xi \geq -c_0,\ |\Im\xi| \leq \vartheta\}$.

**Definition 1.2** (analytic slowly varying factors)**.** Let $\vartheta > 0$. A sequence $\ell$ satisfies $(\mathrm{A}_\vartheta)$ if $\ell(k) = \exp\Lambda(\log k)$ for all $k \geq 1$, where, for some $c_0 > 0$, the function $\Lambda$ is analytic on $S_\vartheta$, real on $S_\vartheta \cap \mathbb{R}$, bounded on $S_\vartheta \cap \{\Re\xi \leq 0\}$, and

$$\varepsilon_\Lambda(T) := \sup\bigl\{|\Lambda'(\xi)| : \ \xi \in S_\vartheta,\ \Re\xi \geq T\bigr\} \longrightarrow 0 \qquad (T \to \infty). \tag{6}$$

We then also say that $\Lambda$ satisfies $(\mathrm{A}_\vartheta)$. If $\vartheta' < \vartheta$, then $(\mathrm{A}_\vartheta)$ implies $(\mathrm{A}_{\vartheta'})$.

**Theorem 1** (main theorem)**.** *Let $p > 0$, $s > 1$, let $\ell\colon \mathbb{N} \to (0,\infty)$ be slowly varying, and let $\sigma = \sigma_{p,s,\ell}$. Then*

(i) *$n^p b_n \to c(p)$ and $n^p a_n \to 2c(p)$;*
(ii) *if $\ell$ satisfies $(\mathrm{A}_\vartheta)$ for some $\vartheta > 0$, then $n^p b_n = c(p)(1 + O(1/n))$ and $a_n/(2b_n) = 1 + O(1/n)$.*

In words, $(\mathrm{A}_\vartheta)$ asks that $\ell$ extend to an analytic function $\ell(t) = e^{\Lambda(\log t)}$ on the set $\{|t| \geq e^{-c_0},\ |\arg t| \leq \vartheta\}$, bounded near $|t| = e^{-c_0}$, with $t\ell'(t)/\ell(t) \to 0$. Such an $\ell$ is slowly varying, since $|\Lambda(\log k) - \Lambda(\log n)| \leq \varepsilon_\Lambda(\log n - \log\lambda)\log\lambda$ for $n/\lambda \leq k \leq \lambda n$. The half-angle $\vartheta$ may be arbitrarily small, and it does not have to depend on $p$: the Riemann–Hilbert analysis evaluates the weight only in a sector of the $z$-plane whose opening we are free to shrink (Lemma 5.7 and Remark 5.8). The sequences $\ell(k) = \exp\sqrt{\log(k+e)}$ and $\ell(k) = \exp((\log(k+e))^{\beta})$, $\beta < 1$, satisfy $(\mathrm{A}_\vartheta)$ for every $\vartheta < \pi$, because the singularities of $\Lambda(\xi) = (\log(e^{\xi}+e))^{\beta}$ lie on the lines $|\Im\xi| = \pi$. An example that is analytic only in a thin strip is $\Lambda(\xi) = \sqrt{\xi+1} + (\xi+2)^{-1/2}\log(1 - q\cos 2\xi)$ with $q = 1/\cosh(0.6)$: it has logarithmic branch points at $\xi = \pi m \pm 0.3i$ for every integer $m$, and it satisfies $(\mathrm{A}_\vartheta)$ for every $\vartheta < 0.3$ (§9).

*Why the pointwise law is subtle.* Corollary 1.2 of [15], specialised to the atoms $x_k = k^{-p}$, states: if the weights $w_k \geq 0$ satisfy $\sum_k w_k < \infty$ and $w_k \geq c_1 k^{-s'}$ for some constants $c_1 > 0$ and $s' \in \mathbb{R}$, then the recurrence coefficients of $\sum_k w_k \delta_{k^{-p}}$ satisfy

$$\frac{2}{n^2}\sum_{j=1}^{n-1}(n-j)\log\bigl(j^p b_j\bigr) \longrightarrow \log c(p). \tag{7}$$

The constant comes from an explicitly solvable constrained equilibrium problem [15, Theorem 1.3]. Its solution $\mu^*$ is an average of arcsine laws; it is the zero distribution that a Jacobi matrix with entries $a_n \approx 2b_n \approx 2c\,n^{-p}$ would produce, and the saturation condition of the problem fixes $c = c(p)$ [15, §1.2]. So $c(p)$ is the value predicted for the pointwise law. But (7) is a double Cesàro mean of $\log(j^p b_j)$, that is, a statement about the logarithm of the Hankel determinant $D_n$ of $\sigma$ at order $n^2$. Multiplying the weights by factors in $[e^{-\varepsilon}, e^{\varepsilon}]$ changes $\log D_n$ by at most $\varepsilon n$, so the averaged law cannot see such perturbations, and passing from it to (4)

is a Tauberian step that needs a regularity of $j^p b_j$ which the averaged argument does not provide. That regularity must come from the weights: the averaged law holds under the one-sided bound $w_k \geq c_1 k^{-s'}$ alone, while the pointwise law can fail for weights that oscillate within a fixed factor of $k^{-s}$ (§1.4). Theorem 1 shows that slow variation of the weights is enough.

**Definition 1.3** (smoothing)**.** Let $L_\ell\colon \mathbb{R} \to \mathbb{R}$ be the piecewise-linear interpolant of the points $(\log k, \log \ell(k))$, $k \geq 1$, extended by the constant $\log \ell(1)$ on $(-\infty, 0]$. Let $G(y) := e^{-y^2/2}/\sqrt{2\pi}$ and

$$\Lambda_\ell := L_\ell * G, \qquad \rho_\ell(m) := \sup_{k \geq m} \bigl|\log \ell(k) - \Lambda_\ell(\log k)\bigr| \qquad (m \geq 1). \tag{8}$$

*Quantitative form.* The proof of Theorem 1(i) gives a rate, measured by how well $\ell$ can be smoothed in the logarithmic variable. The smoothing error $\rho_\ell(m)$ of Definition 1.3 tends to 0 for every slowly varying $\ell$ (Lemma 4.1), and

$$|n^p b_n - c(p)| + |n^p a_n - 2c(p)| = O\bigl(1/n + \rho_\ell(\lceil n/M_0 \rceil)\bigr), \tag{9}$$

where the head margin $M_0 = M_0(p) \geq 4$ is an integer that depends only on $p$ (Lemma 4.3). The indices $k < n/M_0$ are those at which the spectral weights of $P_n$ are exponentially saturated.

*Remark* 1.4. No boundedness of $\ell$ is assumed. For example, the weights $k^{-s} \exp \sqrt{\log k}$ and $k^{-s} \exp(\sqrt{\log k}\, \sin \log \log k)$ $(k \geq 3)$ are covered; the second is not even eventually comparable with any fixed power of $\log k$. The measure is always finite, by the Potter bounds (§2.2), so its Jacobi matrix is a compact operator with eigenvalues $k^{-p}$.

*Remark* 1.5 (what is, and is not, proved at order $1/n$). Part (ii) bounds the deviation from $c(p)$ at order $1/n$ but does not identify the $1/n$ term. The proof compares the true problem with an exactly solvable comparison problem: relative to it, the non-analytic part of $\ell$ moves the first two coefficients at infinity by $O(\varepsilon'_n/n)$ with $\varepsilon'_n \to 0$ (Section 7), but the comparison problem itself depends on $\ell$ at order $1/n$. So the $O(1/n)$ is absolute, not relative: we do not claim that $b_n(\sigma_{p,s,\ell})/b_n(\sigma_{p,s}) - 1 = O(\varepsilon_\Lambda/n)$. A conjecture for the $1/n$ term, supported by numerics, is formulated in §10.1.

For pure powers, $\ell \equiv 1$ satisfies $(\mathrm{A}_\vartheta)$ for every $\vartheta$, and Theorem 1(ii) gives the following.

**Corollary 1.6** (pure powers)**.** *For every $p > 0$ and $s > 1$ the measure $\sigma_{p,s} = \sum_{k \geq 1} k^{-s} \delta_{k^{-p}}$ satisfies $n^p b_n = c(p)(1 + O(1/n))$ and $a_n/(2b_n) = 1 + O(1/n)$. For example, $nb_n = \frac{1}{2} + O(1/n)$ for $\sigma_{1,s}$ and $n^2 b_n = \pi^2/16 + O(1/n)$ for $\sigma_{2,s}$.*

*Remark* 1.7 (a classical instance). For $(p, s) = (2, 2)$ the coefficients are explicit. Van Assche [18, §3] records that the orthonormal Lommel polynomials satisfy $xp_n = \beta_{n+1} p_{n+1} + \beta_n p_{n-1}$ with $\beta_{n+1} = [2\sqrt{(n+\nu)(n+\nu+1)}]^{-1}$. For $\nu = \frac{3}{2}$ the orthogonality measure $\mu$ is identified as follows. By the Bessel recurrence [8, (10.6.1)], $u_n := \sqrt{n+\nu}\, J_{\nu+n}(1/x)$ solves $xu_n = \beta_{n+1} u_{n+1} + \beta_n u_{n-1}$ for $n \geq 0$ (with $\beta_0 := [2\sqrt{\nu(\nu-1)}]^{-1}$), and it is square summable by [8, (10.2.2)]. For $x \notin \mathbb{R}$ the resolvent vector $(J - x)^{-1} e_0$ of the Jacobi matrix $J$ solves the same recurrence for $n \geq 1$ and is square summable, and such solutions form a line, since the Wronskian $\beta_n(u_n v_{n-1} - u_{n-1} v_n)$ of two of them is constant and tends to 0. Hence $\int (t - x)^{-1}\, \mathrm{d}\mu(t)$, the first entry of the resolvent, equals $-u_0/(\beta_0 u_{-1}) = -2\nu J_\nu(1/x)/J_{\nu-1}(1/x)$. For $\nu = \frac{3}{2}$ this is $-3(z^{-1} - \cot z)$ with $z = 1/x$ [8, (10.16.1), (10.6.1)], and the partial fractions of cot [8, (4.22.3)] show that $\mu$ has mass $3/(\pi k)^2$ at $\pm 1/(\pi k)$, $k \geq 1$. Its image under $t \mapsto \pi^2 t^2$ is $(6/\pi^2)\sigma_{2,2}$, and the quadratic transformation (from $P_{2n}(t) = Q_n(t^2)$, where $Q_n$ are the monic

orthogonal polynomials of the image: $a_n = \pi^2(\beta_{2n}^2 + \beta_{2n+1}^2)$, $b_n = \pi^2\beta_{2n-1}\beta_{2n}$) gives

$$b_n = \frac{\pi^2}{(4n+1)\sqrt{(4n-1)(4n+3)}}, \qquad a_n = \frac{2\pi^2}{(4n+1)(4n+5)} \qquad (n \geq 1)$$

for $\sigma_{2,2}$. Thus $n^2 b_n/c(2) = 1 - \frac{1}{2n} + O(n^{-2})$ and $a_n/(2b_n) = 1 - \frac{1}{n} + O(n^{-2})$, in agreement with Corollary 1.6.

*Remark* 1.8 (Hankel determinants of zeta values). For $a > 0$ and $2a+b > 1$ the measure $\sigma_{a,2a+b}$ has moments $\int x^j \, \mathrm{d}\sigma_{a,2a+b} = \zeta(aj+2a+b)$, so its Hankel determinants are $\det_{1\leq i,j\leq n} \zeta(a(i+j)+b)$ [15, Corollary 1.4]. Corollary 1.6 gives $n^a b_n = c(a)(1+O(1/n))$ for the recurrence coefficients behind these determinants. For $a = 1$ Monien [13] reports a full asymptotic expansion of the determinants, obtained numerically by D. Zagier; it goes beyond Corollary 1.6 but is not proved.

*Remark* 1.9 (where $c(p)$ comes from). The equilibrium measure $\mu^*$ of [15] is supported on $(0, \infty)$ and coincides with the node density on $[4c(p), \infty)$ (it is *saturated* there); its unconstrained part is the band $(0, 4c(p))$ (Proposition 2.4). After rescaling by $n^p$, the Riemann–Hilbert analysis sees a band of length $4c(p)$, and a band of length $\beta$ produces off-diagonal entries close to $\beta/4$.

The second result is elementary, and it is the reason why the analytic part of the proof only has to cover a window of exponents $s$. A sequence $\eta_n > 0$ is a *regular rate* if $\eta_n$ is non-increasing, $n\eta_n$ is non-decreasing, and $\eta_n \to 0$; for example $\eta_n = 1/n$ (see §2.3).

**Theorem 2** (Christoffel shift: law and rate)**.** *Let $\sigma$ be a positive Borel measure on $[0, \infty)$ with infinite support and finite moments, let $p > 0$ and $c > 0$, and let $\eta_n$ be a regular rate. If*

$$|n^p a_n(\sigma) - 2c| + |n^p b_n(\sigma) - c| = O(\eta_n),$$

*then the measure $x\sigma$ ($\mathrm{d}(x\sigma) := x \, \mathrm{d}\sigma(x)$) satisfies the same estimate, with the same $c$ and rate $O(\eta_n)$. Without a rate: if $n^p a_n(\sigma) \to 2c$ and $n^p b_n(\sigma) \to c$, then the same holds for $x\sigma$. For $x_k = k^{-p}$ one has exactly $x \cdot \sigma_{p,s,\ell} = \sigma_{p,s+p,\ell}$.*

*Remark* 1.10. The Jacobi matrix of $x\sigma$ is obtained from that of $\sigma$ by factorising $J = LU$ and reversing the factors; these transfer formulas are classical [7, §1]. The factors are governed by a chain sequence, and the statement without rate follows from a proposition of Dai, Piao and Qiao [4, Proposition 2.1] together with those formulas. The part with a rate is the one that matters here: it lets the rate $O(1/n)$ survive the shift. The parameter sequence of the chain converges only at the rate $O(\sqrt{\eta_n})$ in general, and this is sharp; the coefficients of $x\sigma$ involve only its increments, which are $O(\eta_n)$ (Section 3). We have not found the rate statement in the literature.

*Remark* 1.11. Conversely, without rate: if $x\sigma$ satisfies (4) (with any $c > 0$ in place of $c(p)$), so does $\sigma$. In particular the pointwise law is insensitive to a point mass at 0 (Remark 3.6).

1.3. **Strategy of the proof.** The proof has a soft part and a hard part. The soft part (Sections 3 and 4) is elementary and self-contained; it reduces Theorem 1 to the following statement, which is the technical core of the paper.

**Theorem 3** (the Riemann–Hilbert engine: analytic factor, base window)**.** *Let $p > 0$ and $1 < s < 1 + 2p$, i.e. $\alpha := (s-1-p)/p \in (-1, 1)$. Let $\ell_1$ satisfy $(\mathrm{A}_\vartheta)$ for some $\vartheta > 0$. Then $\sigma_{p,s,\ell_1}$ satisfies*

$$n^p b_n = c(p)\big(1 + O(1/n)\big), \qquad a_n/(2b_n) = 1 + O(1/n).$$

*From Theorem 3 to Theorem 1.* Write $s = s_0 + mp$ with $s_0 \in (1, 1+2p)$ and an integer $m \geq 0$; this is possible because the window $(1, 1+2p)$ has length $2p > p$. Theorem 3 at $s_0$ and $m$ applications of Theorem 2 with $\eta_n = 1/n$ give the law with rate $O(1/n)$ for every analytic factor and every $s > 1$. This is Theorem 1(ii). For a general slowly varying $\ell$ we apply this to the analytic factor $\ell_1 := \exp \Lambda_\ell(\log \cdot)$ and compare: a multiplicative perturbation $\ell/\ell_1 \to 1$ of the weights changes $\log b_n^2$ and $n^p a_n$ by at most a constant times the supremum of $|\log(\ell/\ell_1)|$ over the indices $k \geq n/M_0$, where the spectral weight of $P_n$ lives, up to an exponentially small contribution from the head $k < n/M_0$ (Proposition 4.5). This gives (9), and Theorem 1(i) follows.

*Why the window* $|\alpha| < 1$. Near the accumulation point 0 the measure looks like a continuous weight: the nodes $x_k = k^{-p}$ have density $\frac{1}{p}x^{-1-1/p}$, so the mass per unit length near $x = k^{-p}$ is $k^{-s} \cdot \frac{1}{p}x^{-1-1/p} = \frac{1}{p}x^{\alpha}$. In the Riemann–Hilbert (RH) analysis 0 becomes a hard edge with Bessel order $\alpha = (s-1-p)/p$. Our comparison argument works for $|\alpha| < 1$ (Remark 7.8). Multiplication by $x$ raises $\alpha$ by 1, and Theorem 2 carries the law and its rate across. For $s > 1 + p/2$ one can even choose $s_0$ with $|\alpha| \leq \frac{1}{2}$; the case $\alpha = 0$ cannot be avoided in general (Remark 4.6).

*The frame.* After rescaling $z \mapsto n^p z$ the nodes $(n/k)^p$ fill $(0, n^p]$. The equilibrium measure $\mu^*$ of [15] governs the $n$-th root behaviour: on the band $(0, 4c(p))$ the polynomials oscillate, on the saturated region $(4c(p), n^p)$ the zero distribution coincides with the node density, and $4c(p)$ is a soft edge with an Airy parametrix. We set up the discrete interpolation problem and its first transformations as Wu, Lin, Xu and Zhao do [19], in the one-sided variable $x$ (Section 5). What depends on $p$ is the outer scalar function that removes the Gamma-function growth of the node masses: the index function $t(z) = nz^{-1/p}$ maps the slit plane onto the sector $|\arg t| < \pi/p$, which lies inside, exactly fills, or exceeds the sector $|\arg t| < \pi$ where Stirling's formula holds, according as $p > 1$, $p = 1$ or $p < 1$. Section 6 treats the three cases and ends with a Frame Proposition (Proposition 6.7), the only interface between the frame and the hard edge.

*The hard edge.* The slowly varying factor enters the band weight through the function $r_n(z) = \ell_1(nz^{-1/p})/\ell_1(n)$, which is not analytic at 0. Hence the standard Bessel parametrix, which needs the weight continued around 0, leaves a jump of size $O(n\varepsilon_\Lambda)$ at the Bessel scale: a quotient argument fails. Following Conway and Deift [2], we instead build the Bessel parametrix from the Szegő function of the whole band weight. The comparison problem is then exactly solvable, and it differs from the true problem only on the lips of the lens, by an amount that is $o(1)$ in the supremum norm but has no rate. A Neumann series closes once the comparison resolvent is bounded uniformly in $n$ (Proposition 7.4; this is where $|\alpha| < 1$ is used), and Bessel localisation turns the $o(1)$ mismatch into an $O(1/n)$ error in the coefficients (Section 7).

Figure 1 shows how the parts fit together.

1.4. **Context.** *Discrete orthogonal polynomials.* The RH approach to discrete orthogonal polynomials was developed by Baik, Kriecherbauer, McLaughlin and Miller [1] for measures on finite node sets in a bounded interval, with real-analytic node density and potential; their analysis includes saturated regions, which also appear here. Within the literature we have read, the only RH analysis of discrete orthogonal polynomials on infinitely many nodes accumulating at a finite point is that of Wu, Lin, Xu and Zhao [19], for the Tricomi–Carlitz polynomials; our frame follows theirs. Dai and Yao make the same observation: "Wu et al. [21] studied the case where the orthogonality lattice consists of infinite nodes with an accumulation point. Except [21], the orthogonality lattices are always equally spaced" [3, §1], [21] being [19].

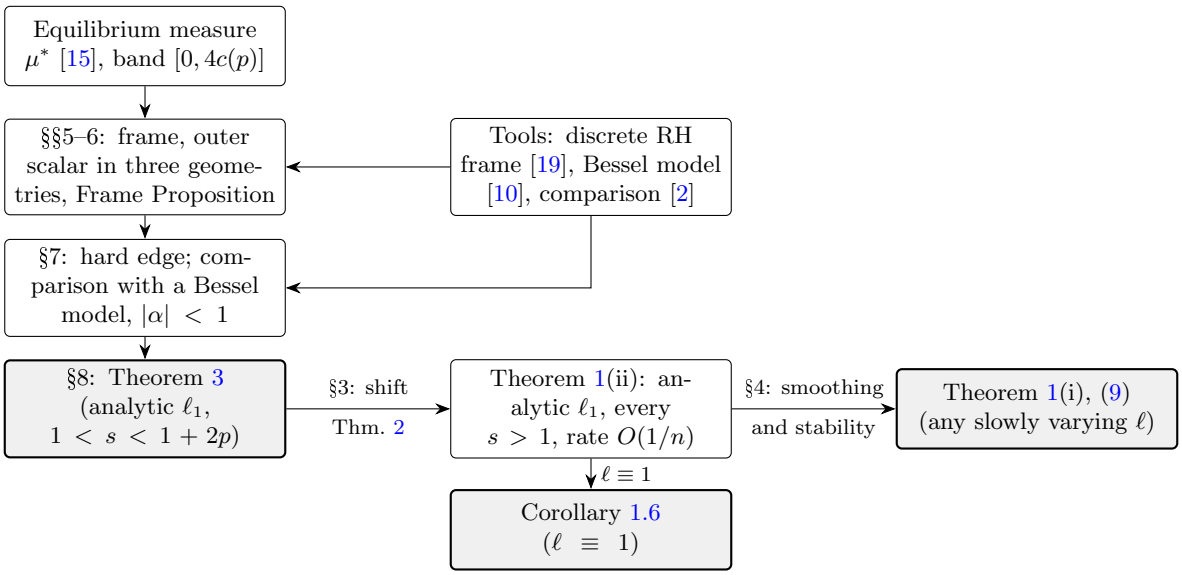



FIGURE 1. Structure of the proof. Sections 3 and 4 are elementary and reduce Theorem 1 to Theorem 3, which is proved in Sections 5–8.

*Non-analytic weights.* Our weights are not analytic at the hard edge, and the standard local parametrix does not exist there. Conway and Deift [2] met the same difficulty for the weight $\log \frac{2k}{1-x}$ on $(-1, 1)$, which carries a slowly varying factor at the endpoint $x = 1$; they write that "there is no known parametrix for the Riemann–Hilbert problem in a neighborhood of the logarithmic singularity at $x = 1$", and they compare the problem with the Legendre problem through the operator theory behind RH problems instead of through a quotient. The comparison argument in Section 7 is that of Conway and Deift [2, Theorem 4.7 and Proposition 4.9]. What we add is a comparison model built from the Szegő function of the full weight (an explicit parametrix rather than a second orthogonality problem), a uniform resolvent bound for a Bessel edge of any order $|\alpha| < 1$ (rather than the Legendre value $\alpha = 0$), and the treatment of an arbitrary analytic slowly varying factor (rather than one explicit logarithm), for a discrete problem whose nodes accumulate at the hard edge. Deift and Piorkowski [6] extended the analysis of [2] to the weight $\log \frac{2}{1-x}$ and write, about a comparison with a weight that has a simple zero at the endpoint: "the crucial Theorem 4.7 in [4] requires a different proof in this case: now one needs to control the behaviour of the Cauchy operator acting on spaces with Muckenhoupt weights" [6, §1], [4] being [2]. This is consistent with our restriction to Bessel orders $|\alpha| < 1$ (Remark 7.8). Other approaches to non-analytic weights are Piorkowski's parametrix-free method, which needs an a priori estimate for the exact solution [14]; the $\bar\partial$ methods of McLaughlin and Miller [12] and of Trogdon [17], which assume a finite number of Lipschitz derivatives of the weight (for [17], $C^{k,1}$ regularity up to the endpoints of the support); and the approximation of a smooth symbol by analytic ones in Deift, Its and Krasovsky [5, §6.2]. Our reduction from arbitrary to analytic slowly varying factors (Section 4) is of the last kind, but the approximation error is controlled outside the RH analysis.

*Recurrence asymptotics without analyticity.* For exponential weights $W = e^{-Q}$, Levin and Lubinsky [11] prove $o(1)$ asymptotics of the recurrence coefficients, with constants given by equilibrium quantities, under regularity conditions that do not involve analyticity, and without RH methods. Theorem 1 is an analogue for compact Jacobi operators. We have not found

Theorem 1 in the literature, even for bounded $\ell$. The closest results we know concern continuous weights on an interval [2, 6, 14, 17, 12] or exponential weights [11].

*Special cases.* Besides the Lommel case (Remark 1.7) and the Hankel determinants of zeta values studied by Monien [13] and by Haynes and Zudilin [9] (Remark 1.8), we mention the Tricomi–Carlitz polynomials. Their orthogonality measure is supported on $\{\pm 1/\sqrt{k+\alpha}\}$ [18, §3]; in the squared variable its atoms are $1/(k+\alpha)$, so it is an instance of $p=1$ with weights that are not pure powers. For it the recurrence coefficients are explicit [18, §3], and [19] obtain uniform asymptotics of the polynomials.

*Rough weights.* Slow variation cannot be replaced by boundedness. In a further paper [16] we show the following. For every $p>0$ and $s>1$ there is $\varepsilon_*>0$ such that, for every $\varepsilon\in(0,\varepsilon_*]$, there are weights $w_k=k^{-s}e^{\epsilon_k}$ with $|\epsilon_k|\le\varepsilon$ for which $n^p b_n$ does not converge, and weights of the same kind for which $n^p a_n$ does not converge. The averaged law (7) holds for all of them. Together with Theorem 1 this gives a dichotomy: the pointwise law is stable under slowly varying perturbations of the weights, of any size, and unstable under rough ones, however small.

1.5. **Organisation.** Section 2 fixes conventions, proves two facts about slowly varying sequences, quotes the input from [15] (only parts (i) and (ii) of its Theorem 1.3 are used), and states the external tools with their hypotheses. Section 3 proves Theorem 2. Section 4 reduces arbitrary slowly varying factors to analytic ones and proves Theorem 1 and Corollary 1.6, assuming Theorem 3. Sections 5–8 prove Theorem 3: the common frame (Section 5), the outer scalar function in the three geometries $p>1$, $p=1$ and $p<1$ (Section 6), the comparison at the hard edge (Section 7), and the extraction of $a_n$ and $b_n$ (Section 8). Section 9 illustrates the results numerically, and Section 10 collects remarks and open problems. The four appendices contain the longer estimates of the Riemann–Hilbert half: the construction of the outer scalar for $p\neq 1$ (Appendix A), the a priori bound at the hard edge (Appendix B), two $L^2$ bounds on a star of arcs (Appendix C), and the uniform invertibility of the comparison problem (Appendix D). A reader who is mainly interested in the Riemann–Hilbert analysis can start at Section 5, with the statement of Theorem 3 in hand.

Throughout, $p>0$ is fixed, $C, C'$ denote positive constants that may change from line to line (never $c$, which is reserved for $c(p)$; the only other constants written with $c$ are $c_0$ in Definition 1.2 and $c_1$ in (7)), and $n$ is the degree. We write "Section 5" for sections and "§5.3" for subsections.

## 2. Preliminaries

This section fixes conventions and collects the inputs of the proof. Two elementary facts about slowly varying sequences are used repeatedly: they are locally almost constant on dyadic scales, and they grow more slowly than any power. We prove both in §2.2. The only input from the companion paper [15] is the explicit equilibrium measure (§2.4). Every external result used later is stated in §2.5, with its hypotheses, under a label T1–T4 by which the later sections refer to it.

*Notation used throughout.* The data are $p>0$, $s>1$ and $\ell$ (§1.1); $\operatorname{osc}_n(\lambda)$, $\Lambda$, $\varepsilon_\Lambda$, $(\mathrm{A}_\vartheta)$, $L_\ell$, $G$, $\Lambda_\ell$, $\rho_\ell$ and the regular rates $\eta_n$ are defined in §1.2; the head margin $M_0$ in §4.3. In Sections 5–8 and the appendices the exponent $s$ lies in the window $(1,1+2p)$, the analytic factor is $\ell_1=e^{\Lambda(\log\cdot)}$, and $\alpha=(s-1-p)/p\in(-1,1)$ is the Bessel order of the hard edge; the Riemann–Hilbert notation is introduced in §§5.1–5.7 (the chain of matrices $Y\to U\to R\to T\to S\to V\to Q$ is displayed at the start of Section 5), and Table 1 lists the quantities that depend on whether $p>1$, $p=1$ or $p<1$. Symbols used only inside one proof are local to it.

2.1. **Orthogonal polynomials and Jacobi coefficients.** Let $\sigma$ be a positive Borel measure on $[0,\infty)$ with infinite support and finite moments. Its Hankel determinants are positive, so the monic orthogonal polynomials $P_n$ ($\deg P_n = n$) exist for every $n \ge 0$. Put $h_n := \int P_n^2 \,\mathrm{d}\sigma > 0$, $P_{-1} := 0$ and $b_0 := 0$. The polynomials satisfy the three-term recurrence

$$xP_n = P_{n+1} + a_n P_n + b_n^2 P_{n-1} \qquad (n \ge 0), \qquad b_n^2 = \frac{h_n}{h_{n-1}} \quad (n \ge 1), \tag{10}$$

with $a_n \in \mathbb{R}$ and $b_n > 0$ for $n \ge 1$. We call $a_n$ and $b_n$ the *Jacobi coefficients* of $\sigma$.[1] Multiplying $\sigma$ by a positive constant does not change them.

The orthonormal polynomials $p_n := P_n/\sqrt{h_n}$ satisfy

$$xp_n = b_{n+1}p_{n+1} + a_n p_n + b_n p_{n-1} \qquad (n \ge 0), \tag{11}$$

as one sees by dividing (10) by $\sqrt{h_n}$ and using $b_n^2\sqrt{h_{n-1}/h_n} = b_n$. Since $xp_j$ has degree $j+1$, orthonormality gives

$$a_n = \int x p_n^2 \,\mathrm{d}\sigma, \qquad b_n = \int x p_{n-1} p_n \,\mathrm{d}\sigma, \qquad \int x p_n p_j \,\mathrm{d}\sigma = 0 \quad (j \le n-2). \tag{12}$$

In matrix form, (11) says that the symmetric tridiagonal matrix $\mathcal{J}(\sigma)$ with diagonal $(a_0, a_1, \dots)$ and off-diagonal $(b_1, b_2, \dots)$ is the Jacobi matrix of $\sigma$. The monic form (10) corresponds to the non-symmetric matrix $J = J(\sigma)$ with $J_{n,n} = a_n$, $J_{n,n+1} = 1$ and $J_{n+1,n} = b_{n+1}^2$; this is the form used in Section 3.

*Christoffel's extremal property.* For $y \in \mathbb{R}$ and $m \ge 1$ let $\mathsf{K}_m(y) := \sum_{j<m} p_j(y)^2$. If $\deg P < m$, then $P = \sum_{j<m}\langle P, p_j\rangle p_j$ with $\langle P, p_j\rangle := \int P p_j \,\mathrm{d}\sigma$, and the Cauchy–Schwarz inequality gives $P(y)^2 \le \mathsf{K}_m(y)\sum_j \langle P, p_j\rangle^2 = \mathsf{K}_m(y)\int P^2 \,\mathrm{d}\sigma$, with equality for $\langle P, p_j\rangle = p_j(y)$. Hence

$$\frac{1}{\mathsf{K}_m(y)} = \min\Big\{ \int P^2 \,\mathrm{d}\sigma : \ \deg P < m, \ P(y) = 1 \Big\}. \tag{13}$$

If $\sigma$ has an atom of mass $w$ at $y$, then every competitor in (13) has $\int P^2 \,\mathrm{d}\sigma \ge w P(y)^2 = w$, and therefore

$$w\,\mathsf{K}_m(y) \le 1. \tag{14}$$

2.2. **Slow variation.** Let $\ell$ be slowly varying (Definition 1.1). Each $\operatorname{osc}_n(\lambda)$ is a maximum of finitely many finite numbers. Put

$$\operatorname{osc}_n^* := \sup_{m \ge n} \operatorname{osc}_m(2).$$

Since $\operatorname{osc}_m(2) \to 0$, every $\operatorname{osc}_n^*$ is finite, the sequence $(\operatorname{osc}_n^*)$ is non-increasing, and $\operatorname{osc}_n^* \to 0$.

**Lemma 2.1** (Dyadic chaining and Potter bounds)**.** *Let $\ell$ be slowly varying.*

(i) *For all $n \ge 1$ and all integers $k \ge n$, $|\log \ell(k) - \log \ell(n)| \le \operatorname{osc}_n^*\big(1 + \log_2(k/n)\big)$.*
(ii) *For every $\eta > 0$ there is $A_\eta < \infty$ such that $e^{-A_\eta} k^{-\eta} \le \ell(k) \le e^{A_\eta} k^{\eta}$ for all $k \ge 1$.*

*Proof.* (i) Let $j := \lfloor \log_2(k/n) \rfloor \ge 0$, so that $2^j n \le k < 2^{j+1} n$. For $0 \le i < j$ the integer $2^{i+1}n$ lies in $[2^i n/2,\, 2\cdot 2^i n]$, so $|\log \ell(2^{i+1}n) - \log \ell(2^i n)| \le \operatorname{osc}_{2^i n}(2) \le \operatorname{osc}_n^*$. Also $k \in [2^j n,\, 2\cdot 2^j n]$, so $|\log \ell(k) - \log \ell(2^j n)| \le \operatorname{osc}_n^*$. Summing these $j+1$ steps ($j$ dyadic steps and one final step) gives $|\log \ell(k) - \log \ell(n)| \le (j+1)\operatorname{osc}_n^* \le \operatorname{osc}_n^*(1 + \log_2(k/n))$.

[1] The companion paper [15] writes $xP_k = P_{k+1} + a_{k+1}P_k + b_k^2 P_{k-1}$; its $a_{n+1}$ is our $a_n$, and $b_n$ agrees.

(ii) Fix $\eta > 0$ and choose $m$ with $\operatorname{osc}_m^* \le \eta \log 2$. For $k \ge m$, part (i) with $n = m$ gives

$$|\log \ell(k)| \le |\log \ell(m)| + \operatorname{osc}_m^* + \frac{\operatorname{osc}_m^*}{\log 2} \log \frac{k}{m} \le |\log \ell(m)| + \operatorname{osc}_m^* + \eta \log k.$$

The finitely many $k < m$ are covered by taking

$$A_\eta := \max\bigl(|\log \ell(m)| + \operatorname{osc}_m^*,\ \max_{k<m} |\log \ell(k)|\bigr).$$

□

*Remark* 2.2 ($\lambda = 2$ suffices). If $\operatorname{osc}_n(2) \to 0$, then $\operatorname{osc}_n(\lambda) \to 0$ for every $\lambda \ge 1$. Indeed, Lemma 2.1(i) uses only osc(2). For $n \le k \le \lambda n$ it gives $|\log \ell(k) - \log \ell(n)| \le \operatorname{osc}_n^*(1 + \log_2 \lambda)$. For $n/\lambda \le k < n$, apply it with base point $k$: $|\log \ell(n) - \log \ell(k)| \le \operatorname{osc}_k^*(1 + \log_2(n/k)) \le \operatorname{osc}_{\lceil n/\lambda \rceil}^*(1 + \log_2 \lambda)$. Both bounds tend to 0.

For every $s > 1$, Lemma 2.1(ii) with $\eta < s - 1$ gives $\sum_k k^{-s}\ell(k) < \infty$. So $\sigma_{p,s,\ell} = \sum_k k^{-s}\ell(k)\delta_{k^{-p}}$ is a finite measure on $(0, 1]$ with infinitely many atoms, and §2.1 applies to it.

2.3. **Regular rates.** Regular rates were defined before Theorem 2. Every regular rate satisfies

$$\frac{1}{n} \le \frac{\eta_n}{\eta_1}, \qquad \eta_{n-1} \le 2\eta_n \quad (n \ge 2), \qquad \sum_n \eta_n = \infty. \tag{15}$$

The first holds because $n\eta_n \ge \eta_1$; the second because $(n-1)\eta_{n-1} \le n\eta_n$; the third follows from the first.

*Remark* 2.3 (Regular majorants). Every non-increasing sequence $\upsilon_n > 0$ with $\upsilon_n \to 0$ is dominated by a regular rate, namely $\eta_n := \max_{m \le n}(m\upsilon_m)/n$. Indeed $\eta_n \ge \upsilon_n$ and $n\eta_n$ is non-decreasing by construction; $\eta_{n+1} \le \max\bigl(\eta_n\, n/(n+1), \upsilon_{n+1}\bigr) \le \eta_n$ because $\upsilon_{n+1} \le \upsilon_n \le \eta_n$; and $\eta_n \to 0$, since $m\upsilon_m/n \le \upsilon_1/\sqrt{n}$ for $m \le \sqrt{n}$ and $m\upsilon_m/n \le \upsilon_{\lceil \sqrt{n} \rceil}$ for $\sqrt{n} < m \le n$. So Theorem 2 applies to any monotone rate. We use it only with $\eta_n = 1/n$.

2.4. **The equilibrium measure.** The Riemann–Hilbert analysis of Sections 5–8 is built on the equilibrium measure of the constrained problem solved in [15]. We use only its explicit form and the equality of its potential on the band; the optimality statement [15, Theorem 1.3(iii)] is not used anywhere in this paper.

Let $c = c(p)$ and $I(p)$ be as in (3), let $\gamma_p := \log c(p) + p$, and let $\lambda_p(x) := \frac{1}{p} x^{-1-1/p}$ be the density of the nodes $k^{-p}$ (the image of Lebesgue measure on $(0, \infty)$ under $u \mapsto u^{-p}$). For $V \in [0, 1]$ and $a, b > 0$ write $\mathcal{I}_V(a, b) := B(a, b)^{-1} \int_0^V t^{a-1}(1-t)^{b-1}\, \mathrm{d}t$ for the regularised incomplete beta function ($B$ is Euler's Beta function). For $\beta > 0$ let $\operatorname{arc}_\beta$ be the arcsine law on $[0, \beta]$, with density $1/(\pi\sqrt{y(\beta - y)})$, and put $\beta_\tau := 4c\tau^{-p}$ for $0 < \tau \le 1$.

**Proposition 2.4** ([15, Theorem 1.3(i)–(ii)]). *Let $p > 0$ and let $\mu^* := \int_0^1 \operatorname{arc}_{\beta_\tau}\, \mathrm{d}\tau$, a probability measure on $(0, \infty)$.*

(i) *$\mu^*$ has the density*

$$\psi^*(x) = \lambda_p(x)\, \mathcal{I}_{\min(1,\, x/4c)}\bigl(\tfrac{1}{p} + \tfrac{1}{2}, \tfrac{1}{2}\bigr), \qquad x > 0.$$

*In particular $\psi^* = \lambda_p$ on $[4c, \infty)$ (saturation) and $0 < \psi^* < \lambda_p$ on $(0, 4c)$.*

(ii) *The function $U^*(x) := \int \log|x - y|\, \mathrm{d}\mu^*(y)$ satisfies $U^*(x) = \gamma_p$ for $0 < x \le 4c$, and $U^*(x) \ge \gamma_p$ for all $x > 0$.*

(iii) *(Hard edge.) As $x \to 0^+$,*

$$\psi^*(x) = C_* x^{-1/2}\bigl(1 + O(x)\bigr), \qquad C_* := \frac{(4c)^{-1/p-1/2}}{p\, a_* B(a_*, \frac{1}{2})}, \quad a_* := \frac{1}{p} + \frac{1}{2}.$$

*Proof.* Parts (i) and (ii) are [15, Theorem 1.3(i)–(ii)], transported from the variable $u$ to $x = u^{-p}$: there, $\mu_u^*$ is the image of $\mu^*$ under $x \mapsto x^{-1/p}$ and has density $\rho^*(u) = \mathcal{I}_{\min(1,u^{-p}/4c)}(\frac{1}{p} + \frac{1}{2}, \frac{1}{2})$, with $\rho^* = 1$ exactly on $(0, (4c)^{-1/p}]$ and $0 < \rho^* < 1$ on $((4c)^{-1/p}, \infty)$. Since $|\,\mathrm{d}u/\,\mathrm{d}x| = \lambda_p(x)$, the density of $\mu^*$ in $x$ is $\lambda_p(x)\rho^*(x^{-1/p})$, which is (i). Part (ii) is quoted verbatim, with $\gamma_p = \log c + p$. For (iii), let $V = x/4c \le 1$. Then $\int_0^V t^{a_*-1}(1-t)^{-1/2}\,\mathrm{d}t = \frac{V^{a_*}}{a_*}(1 + O(V))$, so $\psi^*(x) = \frac{1}{p}x^{-1-1/p}(x/4c)^{1/p+1/2}\big(a_* B(a_*, \frac{1}{2})\big)^{-1}(1 + O(x))$, which is (iii). □

The point $x = 4c$ is the soft edge, $(0, 4c)$ is the band, and $x = 0$ is the hard edge, where $\mu^*$ has the inverse-square-root density of (iii).

2.5. **Tools.** This subsection states every external result used in Sections 5–8 and the appendices, with its hypotheses and the equation numbers of the source; the later sections refer to the items as T1–T4. It is needed only from Section 5 on, and a reader may skip it until then. Throughout, the Cauchy transform on an oriented contour $\Sigma$ is

$$\mathcal{C}f(z) := \frac{1}{2\pi i}\int_\Sigma \frac{f(s)\,\mathrm{d}s}{s - z}, \qquad z \in \mathbb{C} \setminus \Sigma,$$

and $\mathcal{C}_\pm f$ are its non-tangential boundary values from the left $(+)$ and the right $(-)$ of $\Sigma$, so that $\mathcal{C}_+ - \mathcal{C}_- = 1$. For a jump matrix $v$ on $\Sigma$ we write $\mathcal{C}_v f := \mathcal{C}_-\big(f(v - I)\big)$. Matrices $\sigma_1, \sigma_2, \sigma_3$ are the Pauli matrices, and $\sigma_3 = \mathrm{diag}(1, -1)$.

**T1. The discrete interpolation problem (Wu–Lin–Xu–Zhao [19]).** Wu, Lin, Xu and Zhao [19] develop the Riemann–Hilbert method for discrete orthogonal polynomials on infinitely many nodes with an accumulation point, for the Tricomi–Carlitz polynomials. We apply none of their theorems to our measures; we use the formulation of the problem and the order of the transformations, and our versions are written out in Sections 5 and 6.

*(a) The problem and its solution.* Let $\sigma = \sum_{k\ge1} w_k \delta_{x_k}$ with distinct nodes $x_k \in (0, 1]$, $w_k > 0$ and $\sum_k w_k < \infty$, and fix $n \ge 1$. Following [19, (2.14) and Theorem 2.3], put

$$Y(z) := \begin{pmatrix} P_n(z) & \displaystyle\sum_k \frac{w_k P_n(x_k)}{z - x_k} \\[2ex] h_{n-1}^{-1} P_{n-1}(z) & \displaystyle h_{n-1}^{-1}\sum_k \frac{w_k P_{n-1}(x_k)}{z - x_k} \end{pmatrix}. \tag{16}$$

Then $Y$ is analytic in $\mathbb{C} \setminus (\{x_k\}_k \cup \{0\})$; at each node it has a simple pole with $\mathrm{Res}_{z=x_k} Y(z) = \lim_{z\to x_k} Y(z)\left(\begin{smallmatrix} 0 & w_k \\ 0 & 0 \end{smallmatrix}\right)$ (the residue condition [19, (2.14)]); and $Y(z)z^{-n\sigma_3} = I + Y_1/z + Y_2/z^2 + O(z^{-3})$ as $z \to \infty$, with

$$(Y_1)_{12}(Y_1)_{21} = b_n^2, \qquad \frac{(Y_2)_{12}}{(Y_1)_{12}} - (Y_1)_{22} = a_n. \tag{17}$$

*Proof.* The sums converge locally uniformly off $\{x_k\} \cup \{0\}$, which gives analyticity, the simple poles and $\mathrm{Res}_{x_k} Y_{j2} = w_k Y_{j1}(x_k)$. Write $P_m(x) = x^m + \pi_m x^{m-1} + \dots$. For $|z| > 2$, expanding $(z - x)^{-1}$ gives $\sum_k w_k P_m(x_k)/(z - x_k) = \sum_{i\ge0} z^{-i-1}\int P_m x^i\,\mathrm{d}\sigma$, and $\int P_m x^i\,\mathrm{d}\sigma$ equals 0 for $i < m$, $h_m$ for $i = m$ and $-\pi_{m+1}h_m$ for $i = m + 1$. Hence $(Y_1)_{12} = h_n$, $(Y_1)_{21} = h_{n-1}^{-1}$, $(Y_2)_{12} = -\pi_{n+1}h_n$ and $(Y_1)_{22} = -\pi_n$; and $\pi_n = \pi_{n+1} + a_n$ by (10). □

*(b) Transformations.* The transformations of Section 5 follow [19, §§3–5]: removal of the poles by triangular factors built from $\Pi = \sin\theta/\theta'$ and $e^{\pm i\theta}$ [19, (3.9)–(3.15)], normalisation by the $g$-function [19, (4.1)], the lens and saturated-region factors $V_\pm$, $V_\pm^\triangle$ [19, (4.12)–(4.13)], and conjugation by an outer scalar [19, (5.1)]. In [19] these are written in a symmetric variable

with nodes $\pm x_k$; we use only the pattern, and every jump is recomputed in Section 5 for our one-sided problem.

*(c) The soft edge.* Our Airy parametrix (Lemma 6.6) follows the pattern of [19, §6.1, (6.4), (6.10)–(6.12)]; it is constructed from the Airy function directly (T4), and no statement of [19] is used.

**T2. The Bessel model problem (Kuijlaars, McLaughlin, Van Assche and Vanlessen [10]).** We use the explicit Bessel model of [10, §6] and its algebraic properties; we apply none of their theorems on orthogonal polynomials, so their standing hypothesis on the weight [10, (1.1)] plays no role.

*(a) The model.* Let $\alpha > -1$ and let $\Sigma_\Psi$ consist of the rays $\gamma_1 : \arg\zeta = 2\pi/3$, $\gamma_2 : \arg\zeta = \pi$ and $\gamma_3 : \arg\zeta = -2\pi/3$, all oriented towards the origin, with the $+$ side on the left [10, Figure 3]. The function

$$\Psi_\alpha(\zeta) = \begin{pmatrix} I_\alpha(2\zeta^{1/2}) & \frac{i}{\pi}K_\alpha(2\zeta^{1/2}) \\ 2\pi i\zeta^{1/2}I'_\alpha(2\zeta^{1/2}) & -2\zeta^{1/2}K'_\alpha(2\zeta^{1/2}) \end{pmatrix}, \qquad |\arg\zeta| < 2\pi/3$$

[10, (6.23)], continued to $2\pi/3 < \pm\arg\zeta < \pi$ by the Hankel-function formulas [10, (6.24)–(6.25)], solves the Bessel model problem [10, Theorem 6.3], with jumps

$$\begin{pmatrix} 1 & 0 \\ e^{\alpha\pi i} & 1 \end{pmatrix} \text{ on } \gamma_1, \qquad \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \text{ on } \gamma_2, \qquad \begin{pmatrix} 1 & 0 \\ e^{-\alpha\pi i} & 1 \end{pmatrix} \text{ on } \gamma_3$$

[10, (6.16)–(6.18)]. Uniformly as $\zeta \to \infty$ in $\mathbb{C} \setminus \Sigma_\Psi$,

$$\Psi_\alpha(\zeta) = (2\pi\zeta^{1/2})^{-\sigma_3/2}\frac{1}{\sqrt{2}}\begin{pmatrix} 1+O(\zeta^{-1/2}) & i+O(\zeta^{-1/2}) \\ i+O(\zeta^{-1/2}) & 1+O(\zeta^{-1/2}) \end{pmatrix} e^{2\zeta^{1/2}\sigma_3}$$

[10, (6.28)], which follows from the large-argument expansions (T4) of the Bessel and Hankel functions in (6.23)–(6.25); this also covers the rays moved by (b) (§7.2).

*(b) Angle freedom.* By the remark after [10, (6.21)], the angle $2\pi/3$ may be replaced by any angle in $(0, \pi)$; the lens contours in the disc are then the preimages of the rays under $\zeta = n^2 f(z)$, and they do not depend on $n$. We use rays at $\pm(\pi - \vartheta_{\mathrm{lip}})$.

*(c) The left endpoint.* Our hard edge has the band to its right, as the endpoint $-1$ of [10]. There the conformal map reverses orientation, and the reversal is absorbed by conjugation with $\sigma_3$: $\tilde\Psi(\zeta) = \sigma_3\Psi(\zeta; \alpha \to \beta)\sigma_3$ [10, (6.51)], with prefactor

$$\tilde E_n(z) = N(z)\tilde W(z)^{\sigma_3}\tfrac{1}{\sqrt{2}}\begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}\tilde f(z)^{\sigma_3/4}(2\pi n)^{\sigma_3/2}$$

[10, (6.53)], where $N$, $\tilde W$ and $\tilde f$ are the global parametrix, an analytic square root of the weight and the local conformal map of [10] (compare the right-endpoint formulas [10, (6.27), (6.31)] and $\det E_n \equiv 1$ [10, (6.33)]). The prefactor is analytic by removability: its entries have at most a square-root singularity at the endpoint and it is analytic in a punctured neighbourhood [10, proof of Proposition 6.5]. In Section 7 this argument is applied only to a prefactor that does not involve the weight.

**T3. The comparison argument (Conway–Deift [2]).** The following are from [2, §4], whose Cauchy operator is normalised as ours. Lebesgue exponents are written $r, r'$ ([2] writes $p, q$).

*(a) Boundedness and the class $\partial\mathcal{C}$* [2, §4.1]. $\mathcal{C}_\pm$ are bounded on $L^2(\Sigma)$ (and on $L^r(\Sigma)$, $1 < r < \infty$) if and only if $\Sigma$ is Carleson. A pair $A_\pm \in L^r(\Sigma)$ belongs to $\partial\mathcal{C}(L^r(\Sigma))$ if $A_\pm = \mathcal{C}_\pm f$ for some (unique) $f \in L^r(\Sigma)$; we write $A_\pm \in I + \partial\mathcal{C}(L^r)$ if $A_\pm - I \in \partial\mathcal{C}(L^r)$.

*(b) Theorem 4.4 of* [2]. *Let $v_A, v_B \in L^\infty(\Sigma)$ with $v_B^{-1} \in L^\infty(\Sigma)$. Let $A_\pm \in I + \partial\mathcal{C}(L^r(\Sigma))$ satisfy $A_+ = A_- v_A$ on $\Sigma$, and let $B_\pm \in I + \partial\mathcal{C}(L^r(\Sigma))$ satisfy $B_+ = B_- v_B$ on $\Sigma$, with $B_\pm^{-1} \in I + \partial\mathcal{C}(L^{r'}(\Sigma))$, $1/r + 1/r' = 1$. If $A, B$ are the analytic extensions of $A_\pm, B_\pm$, then*

$$AB^{-1} = I + \mathcal{C}\big(A_-(v_A v_B^{-1} - I)B_-^{-1}\big).$$

*Consequence (moments at infinity).* If in addition $\Sigma$ is bounded and $A = I + A_1/z + A_2/z^2 + O(z^{-3})$, $B = I + B_1/z + B_2/z^2 + O(z^{-3})$ as $z \to \infty$, then with $\mathcal{E} := A_-(v_A - v_B)v_B^{-1}B_-^{-1} \in L^1(\Sigma)$ (take $r = r' = 2$),

$$A_1 - B_1 = -\frac{1}{2\pi i}\int_\Sigma \mathcal{E}(s)\,\mathrm{d}s, \qquad A_2 - B_2 = -\frac{1}{2\pi i}\int_\Sigma s\mathcal{E}(s)\,\mathrm{d}s + (A_1 - B_1)B_1. \tag{18}$$

Indeed $v_A v_B^{-1} - I = (v_A - v_B)v_B^{-1}$; for $|z| > \max_\Sigma |s|$ expand $1/(s - z) = -\sum_{j\ge 0} s^j z^{-j-1}$ in $\mathcal{C}\mathcal{E}(z)$ and compare with $AB^{-1} = I + (A_1 - B_1)z^{-1} + \big(A_2 - B_2 - (A_1 - B_1)B_1\big)z^{-2} + O(z^{-3})$. This is the mechanism of [2, Proposition 4.9], whose proof identifies $\mu = Q_-$ and $\tilde\mu = \tilde Q_-$. Formula (4.14) of [2] is printed without the factor $v_B^{-1}$; it agrees with (18) when $(v_A - v_B)v_B^{-1} = v_A - v_B$, as in their setting. We use the form derived from Theorem 4.4.

*(c) Solution formula* [2, §4.2]. If $1 - \mathcal{C}_v$ is invertible on $L^2(\Sigma)$, then $\mu := (1 - \mathcal{C}_v)^{-1}I$ gives the unique solution $X(z) = I + \mathcal{C}(\mu(v - I))(z)$ of the RH problem normalised at infinity with jump $v$.

*(d) The Neumann-series step* [2, proof of Theorem 4.7, (4.12)–(4.13)]. If $\|(1 - \mathcal{C}_{\tilde v})^{-1}\|_{L^2\to L^2} \le K$ and $\|\mathcal{C}_v - \mathcal{C}_{\tilde v}\|_{L^2\to L^2} \le 1/(2K)$, then

$$(1 - \mathcal{C}_v)^{-1} = (1 - \mathcal{C}_{\tilde v})^{-1}\big(1 - (\mathcal{C}_v - \mathcal{C}_{\tilde v})(1 - \mathcal{C}_{\tilde v})^{-1}\big)^{-1}$$

exists and has norm at most $2K$, since the inner factor is 1 minus an operator of norm at most $\tfrac12$. In [2] the smallness comes from $\|v - \tilde v\|_{L^\infty(\Sigma)} \to 0$.

*(e) Proposition 4.6 of* [2] (attributed there to Deift and Zhou). The operators $(1 - \mathcal{C}_{h_n})^{-1}$ are bounded uniformly in $n$ if and only if the inhomogeneous problem $m_+ = m_- h_n + g$ on $\Sigma$, $m_\pm \in \partial\mathcal{C}(L^2(\Sigma))$, is uniquely solvable for every $g \in L^2(\Sigma)$ with $\|m_\pm\|_{L^2} \le C\|g\|_{L^2}$, $C$ independent of $n$ and $g$. We use only the direction "solvable with bounds $\Rightarrow$ bounded inverse", and we re-prove it in Appendix D.

We do *not* use [2, Theorem 4.5], whose proof is specific to the Legendre comparison problem.

**T4. Special functions (NIST DLMF [8]).** Principal branches throughout. "Uniformly in closed subsectors" of a sector means uniformly in every closed sector contained in it.

- *Gamma function.* (5.5.1) $\Gamma(z+1) = z\Gamma(z)$; (5.5.3) $\Gamma(z)\Gamma(1-z) = \pi/\sin(\pi z)$, $z \notin \mathbb{Z}$; (5.11.1) $\operatorname{Ln}\Gamma(z) \sim (z - \tfrac12)\ln z - z + \tfrac12\ln(2\pi) + \sum_{k\ge1} \frac{B_{2k}}{2k(2k-1)z^{2k-1}}$ as $z \to \infty$, uniformly in closed subsectors of $|\operatorname{ph} z| < \pi$, where by §5.11(ii) the remainder after the term $k = m - 1$ is at most $\sec^{2m}(\tfrac12 \operatorname{ph} z)$ times the first neglected term; (5.12.3), (5.12.1) $\int_0^\infty t^{a-1}(1+t)^{-a-b}\,\mathrm{d}t = B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$ for $\Re a, \Re b > 0$.
- *Airy functions* (§§9.2, 9.7). Ai is entire, and $\mathrm{Ai}(z)$, $\mathrm{Ai}(ze^{\mp 2\pi i/3})$ solve $w'' = zw$; (9.2.8) $\mathcal{W}\{\mathrm{Ai}(z), \mathrm{Ai}(ze^{\mp 2\pi i/3})\} = e^{\pm\pi i/6}/(2\pi)$; (9.2.12) $\mathrm{Ai}(z) + e^{-2\pi i/3}\mathrm{Ai}(ze^{-2\pi i/3}) + e^{2\pi i/3}\mathrm{Ai}(ze^{2\pi i/3}) = 0$. By (9.7.5)–(9.7.6), with $u_0 = v_0 = 1$ (9.7.2), as $z \to \infty$ uniformly in closed subsectors of $|\operatorname{ph} z| < \pi$,

$$\mathrm{Ai}(z) = \frac{e^{-\frac23 z^{3/2}}}{2\sqrt{\pi}\,z^{1/4}}\big(1 + O(|z|^{-3/2})\big), \qquad \mathrm{Ai}'(z) = -\frac{z^{1/4}e^{-\frac23 z^{3/2}}}{2\sqrt{\pi}}\big(1 + O(|z|^{-3/2})\big).$$

- *Modified Bessel functions.* (10.25.2) $I_\nu(z) = (\tfrac12 z)^\nu \sum_{k\ge0} (\tfrac14 z^2)^k/(k!\,\Gamma(\nu + k + 1))$, analytic in $\mathbb{C} \setminus (-\infty, 0]$ (§10.25(ii)); (10.27.4) $K_\nu = \tfrac12\pi(I_{-\nu} - I_\nu)/\sin(\nu\pi)$; (10.31.2)

$K_0(z) = -(\ln(\frac{1}{2}z) + \gamma_{\mathrm{E}})I_0(z) + \frac{1}{4}z^2 + \cdots$, $\gamma_{\mathrm{E}}$ being Euler's constant and the dots an entire function of $z^2$; (10.28.2) $\mathcal{W}\{K_\nu, I_\nu\}(z) = 1/z$, which with (6.23) of T2 gives $\det \Psi_\alpha = 1$.

- *Large argument* (§10.40(i), $\nu$ fixed). Let $\mathsf{s}_\pm(z) := \sum_k (\pm 1)^k a_k(\nu) z^{-k}$, where $a_0(\nu) = 1$ by (10.17.1). The expansions (10.40.1), (10.40.2) and (10.40.5) read

$$\begin{aligned} I_\nu(z) &\sim e^z (2\pi z)^{-1/2} \mathsf{s}_-(z), && |\mathrm{ph}\, z| < \tfrac{1}{2}\pi, \\ K_\nu(z) &\sim (\pi/2z)^{1/2} e^{-z} \mathsf{s}_+(z), && |\mathrm{ph}\, z| < \tfrac{3}{2}\pi, \\ I_\nu(z) &\sim (2\pi z)^{-1/2} \big(e^z \mathsf{s}_-(z) \pm i e^{\pm \nu\pi i} e^{-z} \mathsf{s}_+(z)\big), && -\tfrac{1}{2}\pi < \pm \mathrm{ph}\, z < \tfrac{3}{2}\pi, \end{aligned}$$

uniformly in closed subsectors; and (10.40.3)–(10.40.4) are the expansions of $I'_\nu, K'_\nu$ in the sectors of (10.40.1)–(10.40.2), with $b_k(\nu)$ in place of $a_k(\nu)$ and an overall minus sign for $K'_\nu$. For the Hankel functions, with $\omega = z - \frac{1}{2}\nu\pi - \frac{1}{4}\pi$ (10.17.2) and $\delta > 0$ arbitrarily small, (10.17.5) $H_\nu^{(1)}(z) \sim (2/(\pi z))^{1/2} e^{i\omega} \sum_k i^k a_k(\nu) z^{-k}$ for $-\pi + \delta \le \mathrm{ph}\, z \le 2\pi - \delta$, and (10.17.6) $H_\nu^{(2)}(z) \sim (2/(\pi z))^{1/2} e^{-i\omega} \sum_k (-i)^k a_k(\nu) z^{-k}$ for $-2\pi + \delta \le \mathrm{ph}\, z \le \pi - \delta$; (10.17.11)–(10.17.12) give $H_\nu^{(1,2)\prime}(z) \sim \pm i (2/(\pi z))^{1/2} e^{\pm i\omega} \sum_k (\pm i)^k b_k(\nu) z^{-k}$ in the same two sectors, $b_0(\nu) = 1$ (10.17.8).

## 3. The Christoffel shift

Multiplying the measure by $x$ shifts the exponent $s$ by exactly $p$: for $x_k = k^{-p}$,

$$x\, \sigma_{p,s,\ell} = \sum_{k \ge 1} k^{-p}\, k^{-s} \ell(k)\, \delta_{k^{-p}} = \sigma_{p,s+p,\ell}, \tag{19}$$

with the same $\ell$. This section proves Theorem 2: the pointwise law, with its rate, passes from $\sigma$ to $x\sigma$. The Jacobi matrix of $x\sigma$ is obtained by factoring the Jacobi matrix of $\sigma$ as a product $LU$ of bidiagonal matrices and multiplying the factors in the opposite order. The entries of the factors are not known in advance, since they involve the values $P_n(0)$. The point of the section is that no information about $P_n(0)$ is needed beyond its sign: the factors satisfy a recursion whose limiting map has a double (parabolic) fixed point, and positivity alone forces them to converge to it.

Throughout, $a_n, b_n$ are the Jacobi coefficients of $\sigma$ (§2.1), and $\tilde{a}_n, \tilde{b}_n$ are those of $x\sigma$, where $\mathrm{d}(x\sigma) := x\, \mathrm{d}\sigma(x)$.

### 3.1. The transfer formulas.

**Lemma 3.1.** *Let $\sigma$ be a positive Borel measure on $[0, \infty)$ with infinite support and finite moments. Then for $n \ge 1$ the polynomial $P_n$ has $n$ simple zeros, all in $(0, \infty)$. In particular $(-1)^n P_n(0) > 0$ for all $n \ge 0$.*

*Proof.* Let $y_1 < \cdots < y_m$ be the zeros of $P_n$ in $(0, \infty)$ of odd multiplicity, and put $\pi(x) := \prod_j (x - y_j)$ ($\pi := 1$ if $m = 0$). Suppose $m < n$. Then $\int P_n \pi\, \mathrm{d}\sigma = 0$ by orthogonality. On the other hand, every remaining factor of $P_n$ is of the form $(x - y)^{2i}$ with $y$ real, or $(x - y)$ with $y \le 0$, or $(x - z)(x - \bar{z}) = |x - z|^2$ with $z \notin \mathbb{R}$; each is non-negative on $[0, \infty)$. Hence $P_n \pi \ge 0$ on $[0, \infty)$. Since $P_n \pi$ is a non-zero polynomial and $\sigma$ has infinite support, $\int P_n \pi\, \mathrm{d}\sigma > 0$, a contradiction. So $m = n$, the $n$ zeros are simple and positive, and $P_n(0) = \prod_j (-y_j)$ has the sign $(-1)^n$. □

By Lemma 3.1 we may define

$$u_n := -\frac{P_{n+1}(0)}{P_n(0)} > 0 \quad (n \geq 0), \qquad l_0 := 0, \qquad l_n := \frac{b_n^2}{u_{n-1}} > 0 \quad (n \geq 1).$$

**Lemma 3.2.** *Let $\sigma$ be as in Lemma 3.1. Then:*

(i) *$a_n = u_n + l_n$ for $n \geq 0$, and $b_n^2 = u_{n-1} l_n$ for $n \geq 1$. Equivalently, $u_0 = a_0$ and $u_n = a_n - b_n^2/u_{n-1}$ for $n \geq 1$.*

(ii) *The measure $x\sigma$ has infinite support, its monic orthogonal polynomials are $\widetilde{P}_n(x) := \big(P_{n+1}(x) + u_n P_n(x)\big)/x$, and its Jacobi coefficients are*

$$\tilde{a}_n = u_n + l_{n+1} \quad (n \geq 0), \qquad \tilde{b}_n^2 = u_n l_n \quad (n \geq 1). \tag{20}$$

In matrix form: $J(\sigma) = LU$, with $L$ unit lower bidiagonal with subdiagonal $(l_1, l_2, \dots)$ and $U$ upper bidiagonal with diagonal $(u_0, u_1, \dots)$ and superdiagonal 1; and $J(x\sigma) = UL$. This is the classical Christoffel (or Darboux) transformation; see Derevyagin–Derkach [7, §1], where it is written $J = LU \mapsto UL$. We include the short proof.

*Proof.* (i) Evaluate (10) at $x = 0$: $0 = P_{n+1}(0) + a_n P_n(0) + b_n^2 P_{n-1}(0)$. Dividing by $P_n(0) \neq 0$ gives $a_n = u_n - b_n^2 P_{n-1}(0)/P_n(0) = u_n + b_n^2/u_{n-1} = u_n + l_n$ for $n \geq 1$, and $a_0 = u_0$ since $P_{-1} = 0$. The identity $b_n^2 = u_{n-1} l_n$ is the definition of $l_n$.

(ii) The support of $x\sigma$ is $\operatorname{supp}\sigma \setminus \{0\}$, which is infinite, and $x\sigma$ has finite moments. The polynomial $P_{n+1} + u_n P_n$ vanishes at 0 by the definition of $u_n$, so $\widetilde{P}_n$ is a monic polynomial of degree $n$. For $j \leq n-1$, $\int \widetilde{P}_n x^j \,\mathrm{d}(x\sigma) = \int (P_{n+1} + u_n P_n) x^j \,\mathrm{d}\sigma = 0$. Hence the $\widetilde{P}_n$ are the monic orthogonal polynomials of $x\sigma$.

Next, for $n \geq 1$ the polynomial $P_n - \widetilde{P}_n$ has degree at most $n-1$, and for $j \leq n-2$

$$\int (P_n - \widetilde{P}_n) x^j \,\mathrm{d}(x\sigma) = \int P_n x^{j+1} \,\mathrm{d}\sigma - \int \widetilde{P}_n x^j \,\mathrm{d}(x\sigma) = 0.$$

So $P_n - \widetilde{P}_n = l_n' \widetilde{P}_{n-1}$ for some constant $l_n'$; put $l_0' := 0$, consistent with $P_0 = \widetilde{P}_0 = 1$. Then

$$x P_n = x\widetilde{P}_n + l_n' \, x\widetilde{P}_{n-1} = (P_{n+1} + u_n P_n) + l_n'(P_n + u_{n-1} P_{n-1}).$$

Comparing with (10) (the $P_j$ are linearly independent) gives $a_n = u_n + l_n'$ and $b_n^2 = l_n' u_{n-1}$, so $l_n' = l_n$ by (i). Finally,

$$x\widetilde{P}_n = P_{n+1} + u_n P_n = (\widetilde{P}_{n+1} + l_{n+1}\widetilde{P}_n) + u_n(\widetilde{P}_n + l_n \widetilde{P}_{n-1}),$$

which is the recurrence of $x\sigma$ with $\tilde{a}_n = u_n + l_{n+1}$ and $\tilde{b}_n^2 = u_n l_n$. □

3.2. **The chain sequence converges.** Assume now that $\sigma$ satisfies the pointwise law at scale $n^p$ with constant $c > 0$. We work with the rescaled quantities

$$\hat{a}_n := n^p a_n, \quad \hat{b}_n := n^p b_n, \quad \hat{u}_n := n^p u_n, \quad \hat{l}_n := n^p l_n, \quad \kappa_n := \Big(\frac{n-1}{n}\Big)^p.$$

Multiplying the identities of Lemma 3.2(i) by $n^p$, and using $n^p \cdot (n-1)^p = n^{2p}\kappa_n$, we get

$$\hat{u}_n + \hat{l}_n = \hat{a}_n, \qquad \hat{u}_{n-1}\hat{l}_n = \kappa_n \hat{b}_n^2, \qquad \text{so} \qquad \hat{u}_n = \hat{a}_n - \frac{\kappa_n \hat{b}_n^2}{\hat{u}_{n-1}} \qquad (n \geq 1). \tag{21}$$

If $\hat{a}_n \to 2c$, $\hat{b}_n \to c$ and $\hat{u}_{n-1} \to c$, the last identity tends to

$$\hat{u}_n \approx F(\hat{u}_{n-1}), \qquad F(t) := 2c - \frac{c^2}{t}, \qquad F(t) - t = -\frac{(t-c)^2}{t}, \qquad F(c+y) = c + \frac{cy}{c+y}. \tag{22}$$

The map $F$ is increasing on $(0,\infty)$, has the single fixed point $c$, and this fixed point is double: $F(t) \le t$ for every $t > 0$. An orbit that settles below $c$ is therefore pushed down towards 0, and one above $c$ is pulled down to $c$. Positivity of $u_n$ rules out the first possibility.

**Lemma 3.3.** *Let $\sigma$ be as in Lemma 3.1, $p > 0$ and $c > 0$. If $\hat{a}_n \to 2c$ and $\hat{b}_n \to c$, then $\hat{u}_n \to c$ and $\hat{l}_n \to c$.*

Lemma 3.3 is a special case of Dai–Piao–Qiao [4, Proposition 2.1]: in the variables $\mathsf{r}_n := b_n^2/(a_{n-1}a_n)$ and $\mathsf{g}_n := l_n/a_n$ one has $\mathsf{r}_n = (1-\mathsf{g}_{n-1})\mathsf{g}_n$ with $\mathsf{g}_0 = 0$ and $0 < \mathsf{g}_n < 1$, so $(\mathsf{r}_n)$ is a chain sequence with parameter sequence $(\mathsf{g}_n)$; the hypothesis gives $\mathsf{r}_n = \kappa_n\hat{b}_n^2/(\hat{a}_{n-1}\hat{a}_n) \to \frac{1}{4}$, and [4, Proposition 2.1] states that then every parameter sequence tends to $\frac{1}{2}$. We include the short proof, because the rate version, Lemma 3.4, follows the same lines.

*Proof. A priori bounds.* By Lemma 3.1, $u_n, l_n > 0$, so $\hat{u}_n < \hat{a}_n$ and $\hat{l}_n < \hat{a}_n$. Hence $\hat{u}_{n-1} = \kappa_n\hat{b}_n^2/\hat{l}_n > \kappa_n\hat{b}_n^2/\hat{a}_n \to c/2$. So there is $n_1$ with $\hat{u}_n \in [c/3, 3c]$ for $n \ge n_1$.

*Recursion.* For $n > n_1$, (21) reads $\hat{u}_n = F(\hat{u}_{n-1}) + e_n$ with

$$e_n := (\hat{a}_n - 2c) - \frac{\kappa_n\hat{b}_n^2 - c^2}{\hat{u}_{n-1}}, \qquad |e_n| \le |\hat{a}_n - 2c| + \frac{3}{c}\,|\kappa_n\hat{b}_n^2 - c^2| \longrightarrow 0.$$

Fix $\delta \in (0, c/3)$ and put $d := \delta^2/(3c)$. By (22), $F(t) \le t - d$ for every $t \in [c/3, 3c]$ with $|t - c| \ge \delta$. Choose $n_2 > n_1$ with $|e_n| < \min(d/2, \delta)$ for $n \ge n_2$.

*Lower bound.* Suppose $\hat{u}_m < c - \delta$ for some $m \ge n_2$. Then $\hat{u}_{m+1} \le \hat{u}_m - d + |e_{m+1}| < \hat{u}_m - d/2 < c - \delta$, and by induction $\hat{u}_{m+j} < \hat{u}_m - jd/2$ for all $j \ge 1$. This contradicts $\hat{u}_n \ge c/3$. So $\hat{u}_n \ge c - \delta$ for all $n \ge n_2$.

*Upper bound.* Let $n \ge n_2$. If $\hat{u}_n \ge c + \delta$, then $\hat{u}_{n+1} < \hat{u}_n - d/2$. If $\hat{u}_n < c + \delta$, then, $F$ being increasing, $\hat{u}_{n+1} < F(c+\delta) + \delta < c + 2\delta$, because $F(c+\delta) < c + \delta$. The first alternative cannot hold for all $n \ge n_2$, since $\hat{u}_n$ would then decrease by $d/2$ at every step while staying above $c/3$. So $\hat{u}_m < c + \delta$ for some $m \ge n_2$. From then on $\hat{u}_n < c + 2\delta$: if $\hat{u}_n < c + \delta$ this was just shown, and if $c + \delta \le \hat{u}_n < c + 2\delta$ then $\hat{u}_{n+1} < \hat{u}_n < c + 2\delta$.

Hence $c - \delta \le \liminf \hat{u}_n \le \limsup \hat{u}_n \le c + 2\delta$ for every small $\delta > 0$, so $\hat{u}_n \to c$, and $\hat{l}_n = \hat{a}_n - \hat{u}_n \to c$. □

3.3. **Rates.** With a rate, the same argument gives a rate for $\hat{u}_n$, but only the square root of the input rate. What saves the transfer is that the coefficients of $x\sigma$ depend on $\hat{u}_n$ only through its increments, which converge at the full rate.

**Lemma 3.4.** *Let $\sigma$ be as in Lemma 3.1, $p > 0$, $c > 0$, and let $\eta_n$ be a regular rate with $|\hat{a}_n - 2c| + |\hat{b}_n - c| \le M\eta_n$ for $n \ge 1$. Then*

$$|\hat{u}_n - c| = O(\sqrt{\eta_n}) \qquad \text{and} \qquad |\hat{u}_n - \hat{u}_{n-1}| = O(\eta_n).$$

*Proof.* We keep the notation of the previous proof; in particular $\hat{u}_n \in [c/3, 3c]$ for $n \ge n_1$, and $\hat{u}_n \to c$ by Lemma 3.3. We first bound the perturbation $e_n$. For every $p > 0$ and $n \ge 2$ we have $0 \le 1 - \kappa_n \le \max(p, 1)/n$: for $p \ge 1$ by Bernoulli's inequality $(1 - 1/n)^p \ge 1 - p/n$, and for $p < 1$ because $(1 - 1/n)^p \ge 1 - 1/n$. By (15), $1/n \le \eta_n/\eta_1$. Also $\hat{b}_n$ is bounded, so $|\hat{b}_n^2 - c^2| = O(\eta_n)$. Hence $|\kappa_n\hat{b}_n^2 - c^2| \le (1 - \kappa_n)\hat{b}_n^2 + |\hat{b}_n^2 - c^2| = O(\eta_n)$, and there is $M_1$ with $|e_n| \le M_1\eta_n$ for $n > n_1$.

Put $y_n := \hat{u}_n - c$. By (22),

$$y_{n+1} = \frac{cy_n}{c + y_n} + e_{n+1} = y_n - \frac{y_n^2}{\hat{u}_n} + e_{n+1} \qquad (n \ge n_1).$$

*Lower bound.* Let $\lambda_- := \sqrt{6cM_1}$. Suppose $y_m \le -\lambda_-\sqrt{\eta_m}$ for some $m \ge n_1$. Then $y_m^2/\hat{u}_m \ge \lambda_-^2\eta_m/(3c) = 2M_1\eta_m$, so $y_{m+1} \le y_m - 2M_1\eta_m + M_1\eta_{m+1} \le y_m - M_1\eta_m \le -\lambda_-\sqrt{\eta_m} \le -\lambda_-\sqrt{\eta_{m+1}}$. By induction $y_{m+j} \le y_m - M_1\sum_{i<j}\eta_{m+i}$, which tends to $-\infty$ because $\sum\eta_n = \infty$. This contradicts $\hat{u}_n > 0$. Hence $y_n > -\lambda_-\sqrt{\eta_n}$ for all $n \ge n_1$.

*Upper bound.* Choose $n_3 \ge n_1$ with $|y_n| \le c/2$ for $n \ge n_3$; then $\hat{u}_n \le 2c$ there. Choose $\lambda_+ > 0$ with $\lambda_+^2/(2c) \ge 2M_1 + \lambda_+/(2\sqrt{\eta_1})$ (possible, since the left side is quadratic in $\lambda_+$), and $n_4 \ge n_3$ with $\lambda_+\sqrt{\eta_n} \le c$ for $n \ge n_4$.

*Step.* Let $n \ge n_4$ and $y_n \le t := \lambda_+\sqrt{\eta_n}$. The map $y \mapsto cy/(c+y)$ is increasing on $(-c,\infty)$ and $y_n \ge -c/2$, so

$$y_{n+1} \le \frac{ct}{c+t} + M_1\eta_{n+1} \le t - \frac{t^2}{2c} + M_1\eta_n = t - \Big(\frac{\lambda_+^2}{2c} - M_1\Big)\eta_n \le t - \frac{\lambda_+\eta_n}{2\sqrt{\eta_1}},$$

using $t \le c$ and the choice of $\lambda_+$. On the other hand, $\eta_{n+1} \ge \eta_n n/(n+1)$ gives $t - \lambda_+\sqrt{\eta_{n+1}} \le \lambda_+\sqrt{\eta_n}\big(1 - \sqrt{n/(n+1)}\big) \le \lambda_+\sqrt{\eta_n}/(2n)$, where the last step is $\sqrt{n(n+1)} \ge n - 1$. Since $1/n \le \eta_n/\eta_1 \le \sqrt{\eta_n/\eta_1}$, this is at most $\lambda_+\eta_n/(2\sqrt{\eta_1})$. Together, $y_{n+1} \le \lambda_+\sqrt{\eta_{n+1}}$.

*Entry.* If $y_n > \lambda_+\sqrt{\eta_n}$ for some $n \ge n_4$, then $y_n^2/\hat{u}_n \ge y_n^2/(2c) \ge 2M_1\eta_n$, so $y_{n+1} \le y_n - M_1\eta_n$. While $y$ stays above $\lambda_+\sqrt{\eta} \ge 0$ it decreases by the non-summable amounts $M_1\eta_n$, which is impossible for ever. So $y_m \le \lambda_+\sqrt{\eta_m}$ for some $m \ge n$, and by the step $y_k \le \lambda_+\sqrt{\eta_k}$ for all $k \ge m$.

Hence $|y_n| = O(\sqrt{\eta_n})$ (the finitely many earlier terms only change the constant). Finally, $|\hat{u}_n - \hat{u}_{n-1}| = |y_n - y_{n-1}| \le y_{n-1}^2/\hat{u}_{n-1} + |e_n| = O(\eta_{n-1}) + O(\eta_n) = O(\eta_n)$ by (15). □

3.4. **Proof of Theorem 2.** By Lemma 3.2(ii), $x\sigma$ is again a positive measure on $[0,\infty)$ with infinite support and finite moments, and by (20) and (21),

$$n^{2p}\tilde{b}_n^2 = \hat{u}_n\hat{l}_n = \frac{\kappa_n\hat{b}_n^2\,\hat{u}_n}{\hat{u}_{n-1}},$$
$$n^p\tilde{a}_n = \hat{u}_n + \Big(\frac{n}{n+1}\Big)^p\hat{l}_{n+1} = \hat{a}_{n+1} - (\hat{u}_{n+1} - \hat{u}_n) + \Big(\Big(\frac{n}{n+1}\Big)^p - 1\Big)\hat{l}_{n+1}.$$

*Without a rate.* By Lemma 3.3, $\hat{u}_n \to c$ and $\hat{l}_n \to c$, so $n^p\tilde{b}_n \to c$ and $n^p\tilde{a}_n \to 2c$.

*With a regular rate $\eta_n$.* By Lemma 3.4, $\hat{u}_n/\hat{u}_{n-1} = 1 + O(\eta_n)$, since $\hat{u}_{n-1} \ge c/3$ for large $n$. Also $\kappa_n = 1 + O(1/n) = 1 + O(\eta_n)$ and $\hat{b}_n = c + O(\eta_n)$. Hence $n^p\tilde{b}_n = \hat{b}_n\big(\kappa_n\hat{u}_n/\hat{u}_{n-1}\big)^{1/2} = c + O(\eta_n)$. For the diagonal, $\hat{a}_{n+1} = 2c + O(\eta_{n+1})$, $\hat{u}_{n+1} - \hat{u}_n = O(\eta_{n+1})$, $\hat{l}_{n+1}$ is bounded and $(n/(n+1))^p - 1 = O(1/n)$. Since $\eta_{n+1} \le \eta_n$ and $1/n \le \eta_n/\eta_1$, this gives $n^p\tilde{a}_n = 2c + O(\eta_n)$. □

*Remark* 3.5 (Only increments enter). The bound $|\hat{u}_n - c| = O(\sqrt{\eta_n})$ of Lemma 3.4 cannot be improved under the hypotheses of the lemma. Take $\eta_n = 1/n$, any $\theta_0 > 0$, and define $u_n := n^{-p}(c + \theta_0 n^{-1/2})$ for $n \ge 1$, $u_0 := 1$, $b_n := cn^{-p}$, $l_n := b_n^2/u_{n-1}$ $(n \ge 1)$, $l_0 := 0$ and $a_n := u_n + l_n$. All $u_n, l_n$ are positive. The symmetric tridiagonal matrix with diagonal $a_n$ and off-diagonal $b_n$ equals $\mathcal{L}\mathcal{L}^{\mathsf{T}}$, where $\mathcal{L}$ is lower bidiagonal with diagonal $\sqrt{u_n}$ and subdiagonal $\sqrt{l_{n+1}}$. It is therefore a bounded positive semi-definite operator, so it is the Jacobi matrix of a unique positive measure with infinite compact support in $[0,\infty)$. By Lemma 3.2(i), the $u_n$ of that measure are the ones just defined, because $u_0 = a_0$ and $u_n = a_n - b_n^2/u_{n-1}$ determine them. Here $\hat{b}_n = c$. Since $\hat{l}_n = n^p b_n^2/u_{n-1} = \kappa_n c^2/(c + \theta_0(n-1)^{-1/2})$, $\kappa_n = 1 - p/n + O(n^{-2})$ and $(n-1)^{-1/2} = n^{-1/2} + O(n^{-3/2})$, we get $\hat{l}_n = c - \theta_0 n^{-1/2} + (\theta_0^2/c - pc)n^{-1} + O(n^{-3/2})$; the terms $\pm\theta_0 n^{-1/2}$ cancel in $\hat{a}_n = \hat{u}_n + \hat{l}_n$, so $\hat{a}_n - 2c = (\theta_0^2/c - pc)/n + O(n^{-3/2})$. So the law holds with rate $1/n$, while $\hat{u}_n - c = \theta_0\, n^{-1/2}$: the parameters $\hat{u}_n$ themselves converge only at

the rate $\sqrt{\eta_n}$, and the transfer keeps the full rate only because $\tilde{a}_n$ and $\tilde{b}_n$ involve $\hat{u}_n$ through the increments $\hat{u}_n - \hat{u}_{n-1}$ and the ratio $\hat{u}_n/\hat{u}_{n-1}$.

*Remark* 3.6 (The converse; a point mass at 0). Without a rate, the implication of Theorem 2 can be reversed: *if $x\sigma$ satisfies $n^p\tilde{a}_n \to 2c$ and $n^p\tilde{b}_n \to c$, then so does $\sigma$.* This direction is also a consequence of [4, Proposition 2.1], since that proposition covers every parameter sequence of the chain sequence, not only the minimal one. Since $\sigma$ and $\sigma + m\delta_0$ ($m \geq 0$) have the same image $x\sigma$, it follows that the pointwise law is insensitive to a point mass at 0: by the converse and by Theorem 2, the laws for $\sigma$ and for $\sigma + m\delta_0$ are both equivalent to the law for their common image $x\sigma$.

For completeness we give the argument. By (20), $l_{n+1} = \tilde{a}_n - u_n > 0$ and $u_{n+1} = \tilde{b}_{n+1}^2/l_{n+1}$, so $z_n := n^p u_n/c$ satisfies

$$z_{n+1} = \frac{B_n}{A_n - z_n}, \qquad 0 < z_n < A_n, \qquad A_n := \frac{n^p\tilde{a}_n}{c} \to 2, \qquad B_n := \frac{(n+1)^p n^p \tilde{b}_{n+1}^2}{c^2} \to 1.$$

The limiting map $z \mapsto 1/(2-z)$ has the double fixed point 1, now with $1/(2-z) \geq z$ for $z < 2$. Fix $\delta \in (0, \frac{1}{5}]$, put $\epsilon := \delta^3/16$, and take $n_5$ with $|A_n - 2|, |B_n - 1| \leq \epsilon$ for $n \geq n_5$. For $n \geq n_5$, $z_{n+1} < A_{n+1}$ gives $A_n - z_n > B_n/A_{n+1} \geq (1-\epsilon)/(2+\epsilon) > 0.49$; hence $z_n < 1.52$, $2 - z_n > 0.48$, and

$$\left| z_{n+1} - \frac{1}{2 - z_n} \right| \leq \frac{|B_n - 1|}{A_n - z_n} + \frac{|A_n - 2|}{(A_n - z_n)(2 - z_n)} \leq 7\epsilon.$$

If $z_m \geq 1 + \delta$ for some $m \geq n_5$, then $z_{m+1} \geq z_m + (z_m - 1)^2/(2 - z_m) - 7\epsilon \geq z_m + \delta^2 - 7\epsilon \geq z_m + \delta^2/2$, and inductively $z$ would exceed 1.52; so $z_n < 1 + \delta$ for $n \geq n_5$. If $z_n \leq 1 - \delta$, then $z_{n+1} \geq z_n + \delta^2/2 - 7\epsilon \geq z_n + \delta^2/4$, so $z$ enters $(1 - \delta, 1 + \delta)$ after finitely many steps; and if $z_n > 1 - \delta$, then $z_{n+1} \geq 1/(1+\delta) - 7\epsilon > 1 - \delta$, because $1/(1+\delta) - (1-\delta) = \delta^2/(1+\delta) > 7\epsilon$. Hence $z_n \to 1$. Then $a_n = u_n + l_n = u_n + (\tilde{a}_{n-1} - u_{n-1})$ and $b_n^2 = u_{n-1} l_n$ give $n^p a_n \to c + c = 2c$ and $n^{2p} b_n^2 \to c^2$.

## 4. From analytic to arbitrary slowly varying factors

A slowly varying sequence is "almost analytic". Gaussian smoothing in the logarithmic variable turns $\log \ell$ into an entire function $\Lambda_\ell$ whose derivatives tend to 0 in every horizontal strip, and which differs from $\log \ell$ by $o(1)$ at the integers (§4.1). The factor $\ell_1 := e^{\Lambda_\ell(\log \cdot)}$ is therefore covered by the Riemann–Hilbert analysis, and it remains to compare the measures with factors $\ell$ and $\ell_1$. Their ratio $\psi := \ell/\ell_1$ tends to 1. A multiplicative perturbation that tends to 1 changes $\log b_n^2$ by at most the supremum of $|\log \psi|$ over the bulk of the spectral weights, which sit at indices $k \gtrsim n$ (§4.2). The only danger is the head $k < n/M_0$, where $|\log \psi|$ need not be small; there the spectral weights are exponentially small, because those atoms are saturated (§4.3).

One cannot instead compare directly with the pure power $k^{-s}$: $\ell$ may be unbounded and need not be close to 1 anywhere, and controlling $\log \ell(k) - \log \ell(n)$ against the spectral weights far beyond $k \asymp n$ would require information about the hard edge at the level of $1/n$ corrections. Comparing $\ell$ with its own smoothing avoids this. §4.4 then proves Theorem 1 and Corollary 1.6, given Theorem 3.

Throughout this section $p > 0$ and $s > 1$ are fixed, $x_k := k^{-p}$, and for a positive sequence $w = (w_k)$ with $\sum_k w_k < \infty$ we write $a_n(w)$, $b_n(w)$ for the Jacobi coefficients of $\sum_k w_k \delta_{x_k}$.

With $p_n$ the orthonormal polynomials (§2.1), the spectral weights are

$$q_n(k) := w_k^{1/2} p_n(x_k), \qquad m_n(k) := q_n(k)^2, \qquad K_n(k) := \sum_{j<n} m_j(k) = w_k \mathsf{K}_n(x_k).$$

Each $m_n$ is a probability vector on $\mathbb{N}$ (orthonormality of $p_n$), $0 \le K_n(k) \le 1$ by (14), and $m_n = K_{n+1} - K_n$. For sequences $h_1, h_2$ on $\mathbb{N}$ we write $\langle h_1, h_2 \rangle := \sum_k h_1(k) h_2(k)$ whenever the sum converges absolutely, and $x$ also denotes the sequence $(x_k)$.

4.1. **Gaussian smoothing.** Let $L_\ell$, $G$, $\Lambda_\ell = L_\ell * G$ and $\rho_\ell$ be as in Definition 1.3; $G$ extends to an entire function, and $(\mathrm{A}_\vartheta)$ is the hypothesis of Definition 1.2.

Smoothing statements of this kind are familiar from the theory of regular variation; we include the short proof, which uses only the sequence form of slow variation.

**Lemma 4.1** (Gaussian smoothing)**.** *Let $\ell$ be slowly varying and write $L = L_\ell$, $\Lambda = \Lambda_\ell$.*

- (o) *For every $\eta > 0$, $|L(y)| \le A_\eta + \eta(|y| + \log 2)$ for all $y \in \mathbb{R}$, with $A_\eta$ from Lemma 2.1(ii). The function $\Lambda$ is entire and real on $\mathbb{R}$.*
- (i) *$\Lambda(\log k) - \log \ell(k) \to 0$ as $k \to \infty$; that is, $\rho_\ell(m) \to 0$ as $m \to \infty$.*
- (ii) *For every $\vartheta \ge 0$, as $\Re\xi \to +\infty$ we have $\Lambda(\xi) - \Lambda(\Re\xi) \to 0$ and $\Lambda^{(j)}(\xi) \to 0$ for every $j \ge 1$, uniformly in $|\Im\xi| \le \vartheta$; and $\Lambda$ is bounded on $\{\Re\xi \le 0,\ |\Im\xi| \le \vartheta\}$.*
- (iii) *Consequently $\Lambda$ satisfies $(\mathrm{A}_\vartheta)$ for every $\vartheta > 0$ (with any $c_0 > 0$). The sequence $\ell_1(k) := e^{\Lambda(\log k)}$ is positive, and $\psi := \ell/\ell_1$ satisfies $\sup_{k \ge m} |\log \psi(k)| = \rho_\ell(m) \to 0$.*

*Proof.* We prove (o), then (ii), from which (i) and (iii) follow.

*(o).* Let $y \ge 0$ and choose $k \ge 1$ with $y \in [\log k, \log(k+1)]$. Then $L(y)$ is a convex combination of $\log \ell(k)$ and $\log \ell(k+1)$, each of absolute value at most $A_\eta + \eta \log(k+1) \le A_\eta + \eta(y + \log 2)$ by Lemma 2.1(ii). For $y < 0$, $|L(y)| = |\log \ell(1)| \le A_\eta$. For $\xi = T + i\upsilon$ $(T, \upsilon \in \mathbb{R})$ and $y \in \mathbb{R}$,

$$|G(\xi - y)| = G(T - y)\, e^{\upsilon^2/2}. \tag{23}$$

Since $L$ grows at most linearly, the integrand $L(y)G(\xi - y)$ is dominated, for $\xi$ in any bounded set, by an integrable function of $y$; as it is entire in $\xi$ for each $y$, $\Lambda$ is entire, and its derivatives may be taken under the integral sign. $\Lambda$ is real on $\mathbb{R}$ because $L$ and $G$ are.

*Local oscillation of $L$.* For $T \ge T_0 := 1 + \log 2$ put

$$\operatorname{osc}_L(T) := \sup\bigl\{|L(y) - L(y')| :\ y, y' \in [T-1, T+1]\bigr\}, \qquad \operatorname{osc}_L^*(T) := \sup_{T' \ge T} \operatorname{osc}_L(T').$$

Let $y \in [T-1, T+1]$ and $y \in [\log k, \log(k+1)]$. Then $k \ge e^{T-1} - 1 \ge e^{T-1}/2$ and $k+1 \le e^{T+1} + 1$. With $m := \lceil e^T \rceil$ and $\lambda := 2e^2$ we have $m/\lambda \le e^{T-2} \le e^{T-1}/2$ and $\lambda m \ge 2e^{T+2} \ge e^{T+1} + 1$, so $k, k+1 \in [m/\lambda, \lambda m]$. Hence $L(y)$ and $L(y')$ are convex combinations of values $\log \ell(k')$ with $k' \in [m/\lambda, \lambda m]$, and $\operatorname{osc}_L(T) \le 2 \operatorname{osc}_m(2e^2)$. Slow variation gives $\operatorname{osc}_L(T) \to 0$, so $\operatorname{osc}_L^*(T)$ is finite and $\operatorname{osc}_L^*(T) \to 0$ as $T \to \infty$.

*Chaining.* Let $T \ge 2T_0$. If $y \ge T$, join $T$ to $y$ by $\lfloor y - T \rfloor$ steps of length 1 and one shorter step; every step lies in an interval $[T'-1, T'+1]$ with $T' \ge T$, so $|L(y) - L(T)| \le \operatorname{osc}_L^*(T)(1 + |y - T|)$. If $T/2 \le y < T$, the same argument started at $y \ge T_0$ gives $|L(y) - L(T)| \le \operatorname{osc}_L^*(T/2)(1 + |y - T|)$. If $y < T/2$, then $|y| + T \le |y - T| + 2T \le 5|y - T|$, and (o) with $\eta = 1$ gives $|L(y) - L(T)| \le 2A_1 + 2\log 2 + |y| + T \le C_1(1 + |y - T|)$ with $C_1 := \max(2A_1 + 2, 5)$. In all cases

$$|L(y) - L(T)| \le \bigl(\operatorname{osc}_L^*(T/2)\, \mathbf{1}_{\{y \ge T/2\}} + C_1 \mathbf{1}_{\{y < T/2\}}\bigr)(1 + |y - T|). \tag{24}$$

*(ii).* The derivatives of $G$ are $G^{(j)}(w) = \mathsf{h}_j(w) G(w)$ with polynomials $\mathsf{h}_j$ of degree $j$. Let $\bar{\mathsf{h}}_j$ be the polynomial whose coefficients are the absolute values of those of $\mathsf{h}_j$; then $|\mathsf{h}_j(t + i\upsilon)| \le$

$\bar{\mathsf{h}}_j(|t|+\vartheta)$ for $t\in\mathbb{R}$, $|\upsilon|\le\vartheta$. The function $\xi\mapsto\int_{\mathbb{R}}G(\xi-y)\,\mathrm{d}y$ is entire and equals 1 on $\mathbb{R}$, hence everywhere; therefore $\int_{\mathbb{R}}G^{(j)}(\xi-y)\,\mathrm{d}y=\delta_{j0}$ for all $\xi\in\mathbb{C}$. For $\xi=T+i\upsilon$ with $|\upsilon|\le\vartheta$ and $T\ge 2T_0$, this, (23), (24) and the substitution $t=T-y$ give

$$\begin{aligned}\big|\Lambda^{(j)}(\xi)-\delta_{j0}L(T)\big| &= \Big|\int_{\mathbb{R}}\big(L(y)-L(T)\big)G^{(j)}(\xi-y)\,\mathrm{d}y\Big|\\ &\le e^{\vartheta^2/2}\Big[\operatorname{osc}^*_L(T/2)\int_{\mathbb{R}}\mathsf{g}_j(t)\,\mathrm{d}t+C_1\int_{|t|>T/2}\mathsf{g}_j(t)\,\mathrm{d}t\Big],\end{aligned}$$

where $\mathsf{g}_j(t):=\bar{\mathsf{h}}_j(|t|+\vartheta)(1+|t|)G(t)$ is integrable. The right side tends to 0 as $T\to\infty$, uniformly in $|\upsilon|\le\vartheta$. For $j=0$, applying this at $\xi$ and at $T=\Re\xi$ and subtracting gives $\sup_{|\Im\xi|\le\vartheta}|\Lambda(\xi)-\Lambda(\Re\xi)|\to 0$. For $j\ge 1$ it gives $\Lambda^{(j)}\to 0$ uniformly in the strip.

For the bound on $\Re\xi\le 0$, note that $L(y)-L(0)=0$ for $y\le 0$ and $|L(y)-L(0)|\le 2A_1+2\log 2+y$ for $y>0$. If $T=\Re\xi\le 0$ and $y>0$, then $|T-y|\ge y$, so $G(T-y)\le G(y)$. Hence $|\Lambda(\xi)-L(0)|\le e^{\vartheta^2/2}\int_0^\infty(2A_1+2+y)G(y)\,\mathrm{d}y$ for $|\Im\xi|\le\vartheta$.

*(i).* Since $L(\log k)=\log\ell(k)$, the case $j=0$, $\Im\xi=0$ of the estimate in (ii) gives $\Lambda(\log k)-\log\ell(k)\to 0$.

*(iii).* $\Lambda$ is entire, real on $\mathbb{R}$, bounded on $\{-c_0\le\Re\xi\le 0,|\Im\xi|\le\vartheta\}$ by (ii), and $\varepsilon_\Lambda(T)\to 0$ by (ii) with $j=1$. So $(\mathrm{A}_\vartheta)$ holds. Finally $\log\psi(k)=\log\ell(k)-\Lambda(\log k)$, so $\sup_{k\ge m}|\log\psi(k)|=\rho_\ell(m)$, which tends to 0 by (i). □

4.2. **First variation.** The comparison of $\ell$ and $\ell_1$ runs along the path $w^{(\tau)}:=w^{(0)}\psi^\tau$, $\tau\in[0,1]$. The next lemma computes the derivatives of the Jacobi coefficients along such a path. It is classical in substance (for $h=x$ it is the Toda flow); we give the short proof.

**Lemma 4.2** (First variation)**.** *Let $x_k\in(0,1]$ be distinct, $v_k>0$ with $\sum_k v_k<\infty$, and let $h:\mathbb{N}\to\mathbb{R}$ be bounded. For $\tau\in\mathbb{R}$ put $w_k^{(\tau)}:=v_ke^{\tau h(k)}$, and let $a_n,b_n,q_n,m_n$ denote the quantities of $w^{(\tau)}$. Then $\tau\mapsto a_n$ $(n\ge 0)$ and $\tau\mapsto\log b_n^2$ $(n\ge 1)$ are $C^1$ on $\mathbb{R}$, and*

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\log b_n^2=\langle h,m_n-m_{n-1}\rangle, \tag{25}$$

$$\frac{\mathrm{d}}{\mathrm{d}\tau}a_n=\langle(x-a_n)h,m_n\rangle-2b_n\langle h,q_{n-1}q_n\rangle. \tag{26}$$

*Proof.* Write $\sigma_\tau:=\sum_k w_k^{(\tau)}\delta_{x_k}$ and $\|h\|_\infty:=\sup_k|h(k)|$.

*Smoothness.* The moments $\int x^i\,\mathrm{d}\sigma_\tau=\sum_k v_ke^{\tau h(k)}x_k^i$ are $C^\infty$ in $\tau$, with derivatives computed term by term: the $r$-th termwise derivative is bounded by $v_k\|h\|_\infty^re^{|\tau|\|h\|_\infty}$, which is summable uniformly for $\tau$ in compact sets. The coefficients of $P_n$ solve the linear system $\int P_nx^i\,\mathrm{d}\sigma_\tau=0$ $(i<n)$, whose determinant is the Hankel determinant $D_n(\sigma_\tau)>0$; by Cramer's rule they are $C^\infty$ in $\tau$. So are $h_n=\int P_n^2\,\mathrm{d}\sigma_\tau>0$, $p_n=P_n/\sqrt{h_n}$, $a_n$ and $b_n^2=h_n/h_{n-1}$. Moreover, if $g(x;\tau)$ is a polynomial in $x$ whose coefficients are $C^1$ in $\tau$, then

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\int g\,\mathrm{d}\sigma_\tau=\int\partial_\tau g\,\mathrm{d}\sigma_\tau+\sum_k h(k)w_k^{(\tau)}g(x_k;\tau), \tag{27}$$

since $\int g\,\mathrm{d}\sigma_\tau$ is a finite linear combination of moments.

*The b-formula.* Apply (27) to $g=P_n^2$. Since $P_n$ is monic, $\partial_\tau P_n$ has degree less than $n$ and is orthogonal to $P_n$. Hence $\frac{\mathrm{d}}{\mathrm{d}\tau}h_n=\sum_k h(k)w_k^{(\tau)}P_n(x_k)^2=h_n\langle h,m_n\rangle$, i.e. $\frac{\mathrm{d}}{\mathrm{d}\tau}\log h_n=\langle h,m_n\rangle$. Since $\log b_n^2=\log h_n-\log h_{n-1}$, this is (25).

*The a-formula.* Write $\dot p_n:=\partial_\tau p_n$, a polynomial of degree at most $n$, and expand $\dot p_n=\sum_{j\le n}\mathsf{c}_jp_j$ with $\mathsf{c}_j=\int\dot p_np_j\,\mathrm{d}\sigma_\tau$. Differentiating $\int p_np_j\,\mathrm{d}\sigma_\tau=\delta_{nj}$ with (27) gives $\int\dot p_np_j\,\mathrm{d}\sigma_\tau+$

$\int p_n \dot{p}_j \,\mathrm{d}\sigma_\tau + \langle h, q_n q_j\rangle = 0$. For $j < n$ the middle term vanishes ($\deg \dot{p}_j \le j < n$), so $\mathsf{c}_j = -\langle h, q_n q_j\rangle$; for $j = n$ we get $\mathsf{c}_n = -\frac{1}{2}\langle h, m_n\rangle$. Now $a_n = \int x p_n^2 \,\mathrm{d}\sigma_\tau$ by (12), and by (27) and (12),

$$\begin{aligned}\frac{\mathrm{d}a_n}{\mathrm{d}\tau} &= 2\int x p_n \dot{p}_n \,\mathrm{d}\sigma_\tau + \langle xh, m_n\rangle = 2\mathsf{c}_n a_n + 2\mathsf{c}_{n-1} b_n + \langle xh, m_n\rangle \\ &= -a_n\langle h, m_n\rangle - 2b_n\langle h, q_{n-1}q_n\rangle + \langle xh, m_n\rangle,\end{aligned}$$

which is (26). (For $n = 0$ the term with $\mathsf{c}_{n-1}$ is absent, consistent with $b_0 = 0$.) All the sums converge absolutely, since $h$ and $x$ are bounded, $m_n$ is a probability vector and $|q_{n-1}q_n| \le \frac{1}{2}(m_{n-1} + m_n)$. □

4.3. **The head is saturated; stability under $\ell/\ell_1 \to 1$.** For $p > 0$ and $0 < y < 1$ let

$$g_p(y) := (p-y)\log\frac{1}{y} + (1-y)\log(1-y) - p + (p-1)y, \qquad g_p^+ := \begin{cases} g_p, & p \ge 1, \\ g_p - \log\frac{1}{p}, & 0 < p < 1. \end{cases} \tag{28}$$

For $y < p/2$ the first term is at least $\frac{p}{2}\log\frac{1}{y}$, and the other terms are bounded; so $g_p^+(y) \to +\infty$ as $y \to 0^+$. Hence there is $y_1 > 0$ with $g_p^+ \ge 1$ on $(0, y_1]$. We fix the *head margin* $M_0 = M_0(p)$ used in the rate (9): it is the least integer $M_0 \ge 4$ such that $g_p^+ \ge 1$ on $(0, 2/M_0]$.

The next lemma says that the atoms $x_k$ with $k$ much smaller than $n$ are almost fully occupied by the first $n$ orthonormal polynomials. Its proof combines Christoffel's extremal property with a Lagrange test polynomial.

**Lemma 4.3** (Saturation of the head)**.** *Let $w_k > 0$ with $\|w\|_1 := \sum_k w_k < \infty$, and consider $\sum_k w_k \delta_{k^{-p}}$. For integers $1 \le k < n$ and $y := k/n$,*

$$1 - K_n(k) \le \frac{\|w\|_1}{w_k}\Big(\frac{n+1}{k}\Big)^{2(1-p)_+} e^{-2n g_p^+(y)}.$$

*Moreover $m_n(k) \le 1 - K_n(k)$ for all $k$ and $n$.*

*Proof.* The last statement holds because $m_n(k) = K_{n+1}(k) - K_n(k)$ and $K_{n+1}(k) \le 1$. For the main bound we use the test polynomial

$$P(x) := \prod_{j \le n,\, j \ne k} \frac{x - x_j}{x_k - x_j},$$

which has degree $n-1$, satisfies $P(x_k) = 1$, and vanishes at $x_j$ for $j \le n$, $j \ne k$. For $i > n$ and $j \le n$ we have $0 < x_i < x_j$, so $|x_i - x_j| \le x_j$ and $|P(x_i)| \le \mathcal{P} := \prod_{j \le n,\, j \ne k} x_j/|x_k - x_j|$. Therefore $\int P^2 \,\mathrm{d}\sigma \le w_k + \|w\|_1 \mathcal{P}^2$, and (13) gives $1/\mathsf{K}_n(x_k) \le w_k + \|w\|_1\mathcal{P}^2$. Hence $K_n(k) = w_k \mathsf{K}_n(x_k) \ge w_k/(w_k + \|w\|_1\mathcal{P}^2)$ and

$$1 - K_n(k) \le \frac{\|w\|_1 \mathcal{P}^2}{w_k + \|w\|_1\mathcal{P}^2} \le \frac{\|w\|_1}{w_k}\mathcal{P}^2.$$

It remains to bound $\mathcal{P}$. Here $x_j/|x_k - x_j| = k^p/|k^p - j^p|$. Put $\varsigma_p := \min(1, p)$. We claim $|k^p - j^p| \ge \varsigma_p \max(j,k)^{p-1}|k - j|$. For $p \ge 1$ and $j < k$: $k^p - j^p = k^{p-1}k - j^{p-1}j \ge k^{p-1}(k - j)$; for $p \ge 1$ and $j > k$ the same computation gives $j^p - k^p \ge j^{p-1}(j - k)$. For $p < 1$ the mean value theorem gives $|k^p - j^p| = p\xi^{p-1}|k - j|$ with $\xi$ between $j$ and $k$, and $\xi^{p-1} \ge \max(j,k)^{p-1}$. Hence

$$\mathcal{P} \le \varsigma_p^{-(n-1)} \prod_{j<k} \frac{k}{k-j} \prod_{j=k+1}^{n} \frac{k^p}{j^{p-1}(j-k)} = \varsigma_p^{-(n-1)} \frac{k^{k-1}}{(k-1)!} \cdot \frac{k^{p(n-k)}\,(n!/k!)^{1-p}}{(n-k)!}.$$

We bound the three factors. First, $k^{k-1}/(k-1)! = k^k/k! \le e^k$. Second, $\log(n-k)! \ge (n-k)\log(n-k) - (n-k)$. Third, $S := \log(n!/k!) = \sum_{j=k+1}^{n} \log j$ satisfies

$$I_{k,n} := \int_k^n \log t \, \mathrm{d}t \le S \le I_{k,n} + \log \frac{n+1}{k},$$

because log is increasing ($\log j \ge \int_{j-1}^{j} \log$ and $\log j \le \int_j^{j+1} \log$). We use the lower bound if $p \ge 1$ (then $1-p \le 0$) and the upper bound if $p < 1$; in both cases $(1-p)S \le (1-p)I_{k,n} + (1-p)_+ \log \frac{n+1}{k}$. With $I_{k,n} = n\log n - n - k\log k + k$ and $k = yn$, the terms in $\log n$ cancel and

$$k + p(n-k)\log k + (1-p)I_{k,n} - (n-k)\log(n-k) + (n-k) = -n\, g_p(y),$$

as a direct expansion shows. (The coefficient of $n\log n$ is $p(1-y)+(1-p)(1-y)-(1-y) = 0$.) Hence

$$\log \mathcal{P} \le -n\, g_p(y) + (n-1)\log\frac{1}{\varsigma_p} + (1-p)_+ \log\frac{n+1}{k}.$$

For $p \ge 1$, $\varsigma_p = 1$ and squaring gives the claim. For $p < 1$, $\varsigma_p = p$ and $(n-1)\log\frac{1}{p} \le n\log\frac{1}{p}$, so $\mathcal{P}^2 \le e^{-2n(g_p(y)-\log(1/p))}((n+1)/k)^{2(1-p)}$, which is the claim with $g_p^+$. □

We now fix the setting of the comparison. Let $\ell$ be slowly varying, let $\Lambda_\ell$, $\ell_1$, $\psi = \ell/\ell_1$ be as in Lemma 4.1, and put

$$\bar\rho := \sup_k |\log \psi(k)| = \rho_\ell(1) < \infty, \qquad w_k^{(\tau)} := k^{-s}\ell_1(k)\psi(k)^\tau \quad (\tau \in [0,1]).$$

So $w^{(0)}$ is the weight of $\sigma_{p,s,\ell_1}$ and $w^{(1)}$ that of $\sigma_{p,s,\ell}$. Since $\ell_1\psi^\tau = \ell\psi^{\tau-1}$, we have $e^{-\bar\rho}\ell(k) \le \ell_1(k)\psi(k)^\tau \le e^{\bar\rho}\ell(k)$, so every $w^{(\tau)}$ has finite total mass. We call the indices $k < n/M_0$ the *head*, and let

$$\mathfrak{h}_n := \sup_{\tau\in[0,1]} \max_{m\in\{n-1,n\}} \sum_{k<n/M_0} m_m^{(\tau)}(k)$$

be the largest head mass of the spectral weights $m_{n-1}^{(\tau)}$, $m_n^{(\tau)}$ of $w^{(\tau)}$.

**Lemma 4.4** (A priori size)**.** *In this setting there are constants $C$ and $n_0$, depending on $p, s, \ell$, such that for all $n \ge 2$*

$$\mathfrak{h}_n \le C n^{s+5} e^{-2n}, \tag{29}$$

*and for all $n \ge n_0$ and $\tau \in [0,1]$,*

$$a_n(w^{(\tau)}) \le 2M_0^p n^{-p}, \qquad b_n(w^{(\tau)}) \le 2M_0^p n^{-p}.$$

*Proof. Head.* By Lemma 2.1(ii) with $\eta = 1$, we have $\|w^{(\tau)}\|_1 \le e^{\bar\rho}\sigma_{p,s,\ell}(\mathbb{R}) < \infty$ and $w_k^{(\tau)} \ge e^{-\bar\rho}k^{-s}\ell(k) \ge e^{-\bar\rho-A_1}k^{-s-1}$, so $\|w^{(\tau)}\|_1/w_k^{(\tau)} \le C'k^{s+1}$ with $C' := e^{2\bar\rho+A_1}\sigma_{p,s,\ell}(\mathbb{R})$. Let $n \ge 2$, $m \in \{n-1, n\}$ and $k < n/M_0$. Then $k < n/4 < m$, and $k/m \le n/(M_0(n-1)) \le 2/M_0$, so $g_p^+(k/m) \ge 1$. Lemma 4.3, with $(m+1)/k \le n+1$ and $2(1-p)_+ \le 2$, gives

$$m_m^{(\tau)}(k) \le 1 - K_m^{(\tau)}(k) \le C'k^{s+1}(n+1)^2 e^{-2(n-1)}.$$

Summing over the fewer than $n$ indices $k < n/M_0$ gives (29).

*Size.* By (12), $a_m = \langle x, m_m\rangle$ and $b_n = \langle xq_{n-1}, q_n\rangle \le \frac12\langle x, m_{n-1} + m_n\rangle$, since $x \ge 0$ and $|q_{n-1}q_n| \le \frac12(m_{n-1}+m_n)$. For $k \ge n/M_0$ we have $x_k = k^{-p} \le (M_0/n)^p$, and $x_k \le 1$ always. Hence, for $m \in \{n-1, n\}$, $\langle x, m_m^{(\tau)}\rangle \le (M_0/n)^p + \mathfrak{h}_n$, and both $a_n$ and $b_n$ are at most $(M_0/n)^p + \mathfrak{h}_n$. By (29), $\mathfrak{h}_n \le (M_0/n)^p$ for $n \ge n_0$. □

**Proposition 4.5** (Stability under $\ell/\ell_1 \to 1$)**.** *Let $p > 0$, $s > 1$, let $\ell$ be slowly varying, and let $\ell_1 := e^{\Lambda_\ell(\log\cdot)}$. There are constants $C$ and $n_0$, depending on $p, s, \ell$, such that for $n \ge n_0$*

$$\big|\log b_n^2(\sigma_{p,s,\ell}) - \log b_n^2(\sigma_{p,s,\ell_1})\big| \le 2\rho_\ell(\lceil n/M_0\rceil) + Cn^{s+5}e^{-2n},$$
$$n^p\big|a_n(\sigma_{p,s,\ell}) - a_n(\sigma_{p,s,\ell_1})\big| \le 6M_0^p\,\rho_\ell(\lceil n/M_0\rceil) + Cn^{p+s+5}e^{-2n}.$$

*Proof.* Apply Lemma 4.2 with $v_k = k^{-s}\ell_1(k)$ and $h = \log\psi$, which is bounded by $\bar\rho$; then $w^{(\tau)}$ is the path defined above. Write $\rho := \rho_\ell(\lceil n/M_0\rceil)$. For $k \ge n/M_0$ we have $k \ge \lceil n/M_0\rceil$, so $|h(k)| \le \rho$; for $k < n/M_0$, $|h(k)| \le \bar\rho$.

*The b-part.* By (25) and the fundamental theorem of calculus,

$$\big|\log b_n^2(w^{(1)}) - \log b_n^2(w^{(0)})\big| \le \sup_{\tau\in[0,1]}\sum_k |h(k)|\big(m_n^{(\tau)}(k) + m_{n-1}^{(\tau)}(k)\big) \le 2\rho + 2\bar\rho\mathfrak{h}_n,$$

because $m_n^{(\tau)}$ and $m_{n-1}^{(\tau)}$ are probability vectors. Lemma 4.4 bounds $\mathfrak{h}_n$.

*The a-part.* By (26) and $2|q_{n-1}q_n| \le m_{n-1} + m_n$,

$$\Big|\frac{\mathrm{d}a_n}{\mathrm{d}\tau}\Big| \le \sum_k |h(k)|\Big(|x_k - a_n|\,m_n(k) + b_n\big(m_{n-1}(k) + m_n(k)\big)\Big).$$

Let $n \ge n_0$, where $n_0$ is as in Lemma 4.4, enlarged so that $2M_0^p n_0^{-p} \le 1$. For $k \ge n/M_0$, $0 \le x_k \le (M_0/n)^p$ and $0 \le a_n \le 2M_0^p n^{-p}$, so $|x_k - a_n| \le 2M_0^p n^{-p}$; also $b_n \le 2M_0^p n^{-p}$. These $k$ contribute at most $\rho\big(2M_0^p n^{-p} + 2\cdot 2M_0^p n^{-p}\big) = 6M_0^p\rho\, n^{-p}$. For $k < n/M_0$ use $|x_k - a_n| \le 1$ and $b_n \le 1$; these contribute at most $3\bar\rho\mathfrak{h}_n$. Integrating over $\tau \in [0,1]$ and multiplying by $n^p$ gives the second bound, again with Lemma 4.4. □

4.4. **Proof of Theorem 1, assuming Theorem 3.** For a sequence of measures with Jacobi coefficients $a_n, b_n$ and a constant $c > 0$, the two forms of the rate statement are equivalent:

$$\Big\{n^p b_n = c\big(1 + O(1/n)\big),\quad \frac{a_n}{2b_n} = 1 + O(1/n)\Big\} \iff |n^p a_n - 2c| + |n^p b_n - c| = O(1/n). \tag{30}$$

Indeed, from the left side, $n^p a_n = 2n^p b_n \cdot a_n/(2b_n) = 2c(1 + O(1/n))^2$; from the right side, $n^p b_n = c + O(1/n)$ and $a_n/(2b_n) = (2c + O(1/n))/(2c + O(1/n))$. We write $c = c(p)$.

*Step 1: analytic factors, every $s > 1$.* Let $\ell_1$ satisfy $(\mathrm{A}_\vartheta)$ for some $\vartheta > 0$ and let $s > 1$. We claim

$$|n^p a_n(\sigma_{p,s,\ell_1}) - 2c| + |n^p b_n(\sigma_{p,s,\ell_1}) - c| = O(1/n). \tag{31}$$

Let $m \ge 0$ be the least integer with $s_0 := s - mp < 1 + 2p$. If $m = 0$ then $s_0 = s > 1$; if $m \ge 1$ then $s - (m-1)p \ge 1 + 2p$, so $s_0 \ge 1 + p$. In both cases $s_0 \in (1, 1 + 2p)$. Theorem 3 at $s_0$ and (30) give (31) for $\sigma_{p,s_0,\ell_1}$. By (19), $\sigma_{p,s,\ell_1} = x^m\sigma_{p,s_0,\ell_1}$, and each $x^j\sigma_{p,s_0,\ell_1} = \sigma_{p,s_0+jp,\ell_1}$ is a positive measure on $(0,1]$ with infinite support and finite moments. Theorem 2, applied $m$ times with the regular rate $\eta_n = 1/n$, gives (31) for $\sigma_{p,s,\ell_1}$.

*Step 2: Theorem 1(ii).* If $\ell$ itself satisfies $(\mathrm{A}_\vartheta)$ for some $\vartheta > 0$, Step 1 with $\ell_1 = \ell$ and (30) give $n^p b_n = c(1 + O(1/n))$ and $a_n/(2b_n) = 1 + O(1/n)$.

*Step 3: Corollary 1.6.* The constant sequence $\ell \equiv 1$ is $e^{\Lambda(\log k)}$ with $\Lambda \equiv 0$, which satisfies $(\mathrm{A}_\vartheta)$ for every $\vartheta$ (with $\varepsilon_\Lambda \equiv 0$). Step 2 applies.

*Step 4: the rate (9).* Let $\ell$ be slowly varying and $\ell_1 := e^{\Lambda_\ell(\log\cdot)}$. By Lemma 4.1(iii), $\Lambda_\ell$ satisfies $(\mathrm{A}_\vartheta)$ for every $\vartheta > 0$, so Step 1 gives (31) for $\sigma_{p,s,\ell_1}$. Write $\rho := \rho_\ell(\lceil n/M_0\rceil)$ and $\Delta_n := \log b_n^2(\sigma_{p,s,\ell}) - \log b_n^2(\sigma_{p,s,\ell_1})$. By Proposition 4.5, $|\Delta_n| \le 2\rho + Cn^{s+5}e^{-2n}$ for $n \ge n_0$; in particular $|\Delta_n|$ is bounded, since $\rho \le \rho_\ell(1)$. Hence $|e^{\Delta_n/2} - 1| \le C'|\Delta_n|$ and

$$|n^p b_n(\sigma_{p,s,\ell}) - c| \le |n^p b_n(\sigma_{p,s,\ell_1}) - c| + n^p b_n(\sigma_{p,s,\ell_1})\,|e^{\Delta_n/2} - 1| = O(1/n) + O(\rho) + O(n^{s+5}e^{-2n}).$$

Similarly, by the second bound of Proposition 4.5, $|n^p a_n(\sigma_{p,s,\ell}) - 2c| \le |n^p a_n(\sigma_{p,s,\ell_1}) - 2c| + 6M_0^p\rho + Cn^{p+s+5}e^{-2n}$. The exponentially small terms are $O(1/n)$, and finitely many $n$ only affect the constant. This is

$$|n^p b_n(\sigma_{p,s,\ell}) - c| + |n^p a_n(\sigma_{p,s,\ell}) - 2c| = O\big(1/n + \rho_\ell(\lceil n/M_0\rceil)\big).$$

*Step 5: Theorem 1(i)* follows from (9), because $\rho_\ell(m) \to 0$ by Lemma 4.1(i). □

*On the order of the steps.* The smoothing is applied last, at the final exponent $s$, after the shift. This is possible because Proposition 4.5 holds for every $p > 0$ and $s > 1$, while Theorem 2 is applied only to the analytic factor $\ell_1$, with the rate $1/n$ of Theorem 3. Smoothing first and shifting afterwards would also work, but the shift would then have to carry the rate $1/n + \rho_\ell(\lceil n/M_0\rceil)$, which in general is not regular and must be replaced by a regular majorant (Remark 2.3).

*Remark* 4.6 (Which part of Theorem 3 is used). Any half-open $s$-window of length $p$ inside $(1, 1+2p)$ can serve as the base in Step 1. For $s > 1 + p/2$ one may take $s_0$ to be the unique point of $s - p\mathbb{N}_0$ in $(1 + p/2, 1 + 3p/2]$, so that $\alpha = (s_0 - 1 - p)/p \in (-\frac{1}{2}, \frac{1}{2}]$. Thus for every $s > 1 + p/2$ only the case $|\alpha| \le \frac{1}{2}$ of Theorem 3 is used, and the range $\frac{1}{2} < |\alpha| < 1$ is needed only for $1 < s \le 1 + p/2$, where no shift takes place. The case $\alpha = 0$ cannot be avoided: for $s \in 1 + p + p\mathbb{N}_0$ every admissible base point has $s_0 \in 1 + p\mathbb{Z}$, i.e. $\alpha \in \mathbb{Z} \cap (-1, 1) = \{0\}$.

## 5. The Riemann–Hilbert problem and the common frame

This section and the next prove the part of Theorem 3 that does not concern the hard edge. We encode the discrete measure by a meromorphic interpolation problem of Fokas–Its–Kitaev type, rescale it by $n^p$, and normalise it with the $g$-function of the equilibrium measure $\mu^*$ of the companion paper [15]. After rescaling, the nodes fill $(0, n^p]$. On the band $(0, 4c)$ the orthogonal polynomials oscillate; on the saturated region $(4c, n^p)$ the nodes are as dense as the equilibrium measure allows, and the problem is close to a pure interpolation problem; near 0 the band meets infinitely many nodes, and 0 is a hard edge. The frame is that of Wu, Lin, Xu and Zhao for the Tricomi–Carlitz polynomials (T1), written in the one-sided variable $x$.

*Roadmap of the proof of Theorem 3.* The matrices are

$$Y \xrightarrow{z\mapsto n^p z} U \xrightarrow{\text{poles}} R \xrightarrow{g} T \xrightarrow{\text{lens}} S \xrightarrow{F} V \xrightarrow{D_W} Q.$$

§§5.2–5.7 carry out the first four steps for any outer scalar $F$ with the properties (O1)–(O5) of §5.9. Section 6 constructs $F$ (the only geometry-dependent step) and collects everything the hard edge needs in the Frame Proposition 6.7, items (F1)–(F8). Section 7 compares $Q$ with an explicit model $\tilde{Q}$ built from the global, Airy and Bessel parametrices and shows that their first two moments at infinity agree up to $O(1/n)$. Section 8 reads off $a_n$ and $b_n$. A reader who trusts the frame can go from the Frame Proposition directly to Section 7.

### 5.1. Standing assumptions.

Throughout Sections 5–8 and the appendices we fix $p > 0$ and $1 < s < 1 + 2p$, and we put

$$\alpha := \frac{s - 1 - p}{p} \in (-1, 1). \tag{32}$$

We fix $\vartheta > 0$ and $\ell_1(t) = e^{\Lambda(\log t)}$ satisfying $(A_\vartheta)$ (Definition 1.2), with the constant $c_0 > 0$ of that hypothesis. We write $\sigma := \sigma_{p,s,\ell_1} = \sum_{k\ge1} w_k\delta_{x_k}$, with $x_k = k^{-p}$ and $w_k = k^{-s}\ell_1(k)$, and $c = c(p)$, $\gamma_p = \log c + p$. The degree $n$ is large, $n \ge n_0$, where $n_0$ may increase from line to

line. Constants $C, C', \dots$ depend only on $p$, $s$, $\ell_1$ (including $\vartheta$) and the contour parameters fixed in §5.6; they do not depend on $n$ unless they carry the subscript $n$. We put

$$\vartheta_{\mathrm{w}} := \mathsf{m}\,\vartheta^0_{\mathrm{w}}, \qquad \vartheta_{\mathrm{lip}} := \mathsf{m}\,\vartheta^0_{\mathrm{lip}}, \qquad \mathsf{m} := \min\Big(1, \frac{p\vartheta}{\vartheta^0_{\mathrm{w}}}\Big),$$
$$(\vartheta^0_{\mathrm{w}}, \vartheta^0_{\mathrm{lip}}) := \begin{cases} (7\pi/16,\ \pi/3), & p \ge 1, \\ (3\pi p/8,\ \pi p/4), & 0 < p < 1, \end{cases} \tag{33}$$

Then $\mathsf{m} \in (0, 1]$, and

$$0 < \vartheta_{\mathrm{lip}} < \tfrac{5}{4}\vartheta_{\mathrm{lip}} < \vartheta_{\mathrm{w}} < \tfrac{\pi}{2}, \qquad \vartheta_{\mathrm{w}} < \tfrac{\pi p}{2} \ \text{ if } p < 1, \qquad \vartheta_{\mathrm{w}} \le p\vartheta, \tag{34}$$

since the first two chains hold for the default angles $\vartheta^0_{\mathrm{w}}, \vartheta^0_{\mathrm{lip}}$ and are preserved by the common factor $\mathsf{m}$, and $\mathsf{m}\vartheta^0_{\mathrm{w}} \le p\vartheta$ by the choice of $\mathsf{m}$. The angle $\vartheta_{\mathrm{w}}$ bounds the arguments at which the weight is ever evaluated (Lemma 5.7); $\vartheta_{\mathrm{lip}}$ is the angle at which the lens leaves the hard edge. Only the relations (34) are used below, never the particular values; see Remark 5.8.

5.2. **The interpolation problem.** Let $P_n$ be the monic orthogonal polynomials of $\sigma$ and $h_n = \int P_n^2 \,\mathrm{d}\sigma$. For a $2\times 2$ matrix function $X$ with an expansion $X(z) = I + X_1/z + X_2/z^2 + O(z^{-3})$ at $\infty$ we write

$$\mathsf{b}(X) := (X_1)_{12}(X_1)_{21}, \qquad \mathsf{a}(X) := \frac{(X_2)_{12}}{(X_1)_{12}} - (X_1)_{22} \quad \text{(the latter when } (X_1)_{12} \neq 0). \tag{35}$$

**Lemma 5.1** (The interpolation problem)**.** *Let $Y$ be the matrix (16) of T1(a), built from the monic orthogonal polynomials $P_n$, $P_{n-1}$ of $\sigma$ and $h_{n-1} = \int P_{n-1}^2 \,\mathrm{d}\sigma$. Then $\det Y \equiv 1$. The rescaled matrix*

$$U(z) := n^{pn\sigma_3}\, Y(zn^{-p}) \tag{36}$$

*is analytic off $\{X_k\}_{k\ge 1} \cup \{0\}$, where*

$$X_k := n^p x_k = (n/k)^p \quad (k \ge 1), \qquad X_1 = n^p \ \textit{(the top node)}; \tag{37}$$

*it has simple poles at the $X_k$ in its second column only, with $\operatorname{Res}_{z=X_k} U_{j2} = n^p w_k\, U_{j1}(X_k)$ $(j = 1, 2)$; and $U(z)z^{-n\sigma_3} = I + U_1/z + U_2/z^2 + \dots$ for $|z| > n^p$, with*

$$\mathsf{b}(Uz^{-n\sigma_3}) = n^{2p} b_n^2, \qquad \mathsf{a}(Uz^{-n\sigma_3}) = n^p a_n. \tag{38}$$

*Proof.* For the determinant, write $h_{n-1} \det Y$ as

$$\int \frac{P_n(z)P_{n-1}(x) - P_{n-1}(z)P_n(x)}{z - x}\,\mathrm{d}\sigma(x) = \int \frac{P_n(z) - P_n(x)}{z - x} P_{n-1}(x)\,\mathrm{d}\sigma(x) - \int \frac{P_{n-1}(z) - P_{n-1}(x)}{z - x} P_n(x)\,\mathrm{d}\sigma(x).$$

The first difference quotient is a polynomial in $x$ of degree $n-1$ with leading coefficient 1, so the first integral is $h_{n-1}$; the second has degree $n-2$ in $x$, so the second integral vanishes. Hence $\det Y \equiv 1$. The statements for $U$ follow from those of T1(a) by the substitution $z = n^p\zeta$: the residue picks up the factor $n^p$ from $\mathrm{d}z = n^p\,\mathrm{d}\zeta$, and $U_j = n^{jp}\, n^{pn\sigma_3} Y_j n^{-pn\sigma_3}$ $(j = 1, 2)$. The constant conjugation by $n^{pn\sigma_3}$ does not change $\mathsf{a}$ and $\mathsf{b}$ (Lemma 5.2 below with $k$ constant), and the factors $n^{jp}$ give $\mathsf{b}(Uz^{-n\sigma_3}) = n^{2p}\mathsf{b}(Yz^{-n\sigma_3})$ and $\mathsf{a}(Uz^{-n\sigma_3}) = n^p\mathsf{a}(Yz^{-n\sigma_3})$. Now (17) gives (38). □

Every transformation below acts, near infinity, by a constant diagonal matrix on the left and a scalar diagonal factor on the right. The extraction functionals do not see such factors.

**Lemma 5.2** (Invariance of the extraction functionals)**.** *Let $X = I + X_1/z + X_2/z^2 + O(z^{-3})$, and let $k$ be a scalar function, analytic and non-zero near infinity, with $k(z) = k_\infty\big(1 + k_1/z + k_2/z^2 + O(z^{-3})\big)$, $k_\infty \neq 0$. Then $\hat X := k_\infty^{-\sigma_3} X k^{\sigma_3}$ satisfies $\hat X \to I$, $\mathsf{b}(\hat X) = \mathsf{b}(X)$ and $\mathsf{a}(\hat X) = \mathsf{a}(X)$. Conversely, if $C$ is a constant matrix and $CXk^{\sigma_3} \to I$, then $C = k_\infty^{-\sigma_3}$.*

*Proof.* Put $Z_j := k_\infty^{-\sigma_3} X_j k_\infty^{\sigma_3}$; then $(Z_j)_{12} = k_\infty^{-2}(X_j)_{12}$, $(Z_j)_{21} = k_\infty^{2}(X_j)_{21}$, and the diagonal entries are unchanged. Since $k^{\sigma_3} = k_\infty^{\sigma_3}\operatorname{diag}\big(1 + k_1 z^{-1} + k_2 z^{-2},\, 1 - k_1 z^{-1} + (k_1^2 - k_2) z^{-2}\big) + O(z^{-3})$,

$$\hat X_1 = Z_1 + k_1\sigma_3, \qquad \hat X_2 = Z_2 + Z_1\operatorname{diag}(k_1, -k_1) + \operatorname{diag}(k_2, k_1^2 - k_2).$$

Hence $(\hat X_1)_{12}(\hat X_1)_{21} = (Z_1)_{12}(Z_1)_{21} = \mathsf{b}(X)$, and with $(\hat X_2)_{12} = (Z_2)_{12} - k_1(Z_1)_{12}$ and $(\hat X_1)_{22} = (X_1)_{22} - k_1$,

$$\mathsf{a}(\hat X) = \frac{(Z_2)_{12}}{(Z_1)_{12}} - k_1 - (X_1)_{22} + k_1 = \frac{(X_2)_{12}}{(X_1)_{12}} - (X_1)_{22} = \mathsf{a}(X).$$

If $CXk^{\sigma_3} \to I$ then $Ck_\infty^{\sigma_3} = I$. □

We never use that $Y$ is the *unique* solution of an interpolation problem, and no existence theorem for a Riemann–Hilbert problem is needed: every object below is built from $Y$ explicitly.

5.3. **The index function and the band weight.** The node set $\{X_k\}$ is the set of points where the *index function*

$$t(z) := n z^{-1/p} \qquad (z \in \mathbb{C} \setminus (-\infty, 0],\ \text{principal power}) \tag{39}$$

takes positive integer values: $t(X_k) = k$. We put

$$\theta(z) := \pi t(z), \qquad \Pi(z) := \frac{\sin\theta(z)}{\theta'(z)}, \qquad \theta'(z) = -\frac{\pi}{p}\,\frac{t(z)}{z}, \tag{40}$$

so that $\Pi$ has simple zeros at the nodes. On $\mathbb{C} \setminus (-\infty, 0]$ we have $\arg t = -\arg z/p$, hence

$$\pm\operatorname{Im}\theta(z) < 0 \quad \text{for } 0 < \pm\arg z < \min(\pi, \pi p), \qquad |e^{\mp 2i\theta(z)}| = \exp\big(-2\pi n |z|^{-1/p}\sin(|\arg z|/p)\big). \tag{41}$$

The weight enters through the log variable $\xi := \log t = \log n - (\operatorname{Log} z)/p$. Let

$$\mathcal{Z}_n := \{z :\ 0 < |z| < e^{pc_0} n^p,\ |\arg z| < \vartheta_{\mathrm{w}}\}. \tag{42}$$

For $z \in \mathcal{Z}_n$ we have $|\operatorname{Im}\xi| < \vartheta_{\mathrm{w}}/p \le \vartheta$ and $\operatorname{Re}\xi > -c_0$, so $\xi \in S_\vartheta$ and the *residue weight* and the *slowly varying factor*

$$w(z) := n^p\, t(z)^{-s}\, \ell_1(t(z)), \qquad r_n(z) := \frac{\ell_1(t(z))}{\ell_1(n)} = e^{\Lambda(\xi) - \Lambda(\log n)} \tag{43}$$

are analytic on $\mathcal{Z}_n$. By Lemma 5.1, $\operatorname{Res}_{X_k} U_{j2} = w(X_k) U_{j1}(X_k)$. The combination that appears in all jumps is

$$W(z) := 2i\theta'(z) w(z) \qquad (z \in \mathcal{Z}_n). \tag{44}$$

**Lemma 5.3** (Band weight)**.** *On $\mathcal{Z}_n$,*

$$W(z) = \mathsf{w}_n\, z^\alpha\, r_n(z), \qquad \mathsf{w}_n := -\frac{2\pi i}{p}\, n^{1+p-s}\ell_1(n), \tag{45}$$

*with the principal power. In particular $|W(x)| = \frac{2\pi}{p} n^{1+p-s}\ell_1(n)\, x^\alpha r_n(x)$ and $\arg W(x) = -\pi/2$ for $x \in (0, e^{pc_0}n^p)$, where $r_n(x) > 0$. The weight has a hard edge at 0 with Bessel order $\alpha$, and $\alpha > -1$ if and only if $s > 1$.*

*Proof.* By (40), $2i\theta' w = -\frac{2\pi i}{p}\frac{t}{z} n^p t^{-s}\ell_1(t)$, and $t^{1-s} = n^{1-s}z^{(s-1)/p}$. The exponent of $z$ is $(s-1)/p - 1 = \alpha$. Finally $\ell_1(t) = \ell_1(n)r_n$, and $r_n > 0$ on $(0,\infty)$ because $\Lambda$ is real on the real axis. □

5.4. **The $g$-function.** We use the equilibrium measure $\mu^* = \int_0^1 \operatorname{arc}_{\beta_\tau}\,\mathrm{d}\tau$, its density $\psi^*$, the node density $\lambda_p$ and the potential $U^*$ exactly as in Proposition 2.4. Note $\int_x^\infty \lambda_p = x^{-1/p} = t(x)/n$. Put

$$g(z) := \int \log(z-y)\,\mathrm{d}\mu^*(y) \quad (z\in\mathbb{C}\setminus\mathbb{R}), \qquad \varphi := \gamma_p - g, \qquad \tilde\varphi := \varphi \pm \frac{i\theta}{n} \text{ in } \mathbb{C}_\pm, \tag{46}$$

with principal logarithms. Nothing here depends on $\ell_1$: the external field of the weight is $\frac{1}{n}\log w = O(\frac{\log n}{n})$ on the contour (Lemma 5.9) and is not put into $g$. Since $\theta/n = \pi z^{-1/p}$, the function $\tilde\varphi$ does not depend on $n$ either.

**Lemma 5.4** ($g$-function)**.** (i) *$g$ has continuous boundary values $g_\pm$ on $\mathbb{R}\setminus\{0\}$, $g_+ - g_- = 2\pi i$ on $(-\infty,0)$ and $g_+ - g_- = 2\pi i\mu^*((x,\infty))$ on $(0,\infty)$. In particular $\varphi_+ + \varphi_- = 0$ on $(0,4c)$, and $\tilde\varphi$ has no jump on $(4c,\infty)$.*

(ii) *$g(z) - \operatorname{Log} z \to 0$ as $z\to\infty$, uniformly in $\mathbb{C}\setminus\mathbb{R}$.*

(iii) *$U^*$ is strictly increasing on $[4c,\infty)$, and $U^*(x) = \log x + o(1)$ as $x\to\infty$. Hence $\operatorname{Re}\tilde\varphi(x) = \gamma_p - U^*(x) < 0$ for $x > 4c$, and $\operatorname{Re}\tilde\varphi(x) \le -\frac12\log x$ for $x\ge x_0$.*

*Proof.* (i) For $y>0$, $\log(z-y)$ jumps by $2\pi i$ across $(-\infty,y)$. Integrating against $\mu^*$ gives the jumps; continuity of the boundary values follows from the continuity of $\psi^*$ on $(0,\infty)$. On $(0,4c)$, $g_+ + g_- = 2U^* = 2\gamma_p$, so $\varphi_+ + \varphi_- = 0$. On $(4c,\infty)$, $\mu^*((x,\infty)) = \int_x^\infty \lambda_p = t(x)/n$, so $g_+ - g_- = 2i\theta/n$, and $\tilde\varphi_+ - \tilde\varphi_- = -(g_+ - g_-) + 2i\theta/n = 0$.

(ii) For $z\in\mathbb{C}_+$ and $y>0$ we have $\operatorname{Log}(z-y) = \operatorname{Log} z + \operatorname{Log}(1-y/z)$, because $\arg(z-y) - \arg z \in (0,\pi)$; the case $z\in\mathbb{C}_-$ is the mirror image. So $g(z) - \operatorname{Log} z = \int \operatorname{Log}(1-y/z)\,\mathrm{d}\mu^*(y)$. On $\{y\le |z|/2\}$ we use $|\operatorname{Log}(1-y/z)| \le 2y/|z|$ and $\int_{y\le R} y\,\mathrm{d}\mu^* \le 1 + \int_1^R y\lambda_p(y)\,\mathrm{d}y = o(R)$ (as $\psi^*\le\lambda_p$). On $\{y > |z|/2\}$ we use $\psi^*\le\lambda_p\le C|z|^{-1-1/p}$ there and $|\operatorname{Log}(1-y/z)| \le \pi + \log(1+u) + |\log|1-u||$ with $u = y/|z|$, which follows from $|1-u|\le|1-y/z|\le 1+u$. The resulting integral is $O(|z|^{-1/p})$. Both parts tend to 0 uniformly.

(iii) For $\beta>0$ put $\mathfrak{g}_\beta(z) := 2\log\big((\sqrt{z}+\sqrt{z-\beta})/2\big)$. It is analytic off $(-\infty,\beta]$, equals $\log z + o(1)$ at $\infty$, and $\mathfrak{g}_\beta'(z) = 1/\sqrt{z(z-\beta)} = \int \mathrm{d}\operatorname{arc}_\beta(y)/(z-y)$, the Stieltjes transform of the arcsine law. Hence $\mathfrak{g}_\beta(z) = \int \log(z-y)\,\mathrm{d}\operatorname{arc}_\beta(y)$. Taking real parts, $\mathfrak{u}_\beta(x) := \int \log|x-y|\,\mathrm{d}\operatorname{arc}_\beta(y)$ equals $\log(\beta/4)$ on $[0,\beta]$ (there $|\sqrt{x}+i\sqrt{\beta-x}|^2 = \beta$), and it is strictly increasing on $[\beta,\infty)$. So $U^* = \int_0^1 \mathfrak{u}_{\beta_\tau}\,\mathrm{d}\tau$ is non-decreasing on $(0,\infty)$. For $x > 4c$ the set of $\tau$ with $\beta_\tau < x$ is the interval $((4c/x)^{1/p},1]$, of positive length, on which $\mathfrak{u}_{\beta_\tau}$ is strictly increasing at $x$. So $U^*$ is strictly increasing on $[4c,\infty)$. For the asymptotics, $U^*(x) - \log x = \int \log|1-y/x|\,\mathrm{d}\mu^*(y)$, and the argument of (ii) on the real axis shows that this tends to 0. Finally, $\operatorname{Re}\tilde\varphi = \operatorname{Re}\varphi = \gamma_p - U^*$ on $(4c,\infty)$, because $\theta$ is real there; $U^*(4c) = \gamma_p$, which gives the sign, and $U^*(x)\ge\frac12\log x + \gamma_p$ for large $x$. □

5.5. **Signs and the local variables at the two edges.** The lens opening needs $\operatorname{Re}\varphi < 0$ off the band, and the triangular factors that appear later need $\operatorname{Re}\tilde\varphi > 0$ just above and below the band. Near the edges we need the local variables of the Bessel and Airy models.

**Lemma 5.5** (Signs and local variables)**.** (i) *$\operatorname{Re}\varphi(z) < 0$ for every $z\in\mathbb{C}\setminus[0,4c]$ (for $z$ on $(4c,\infty)$ and $(-\infty,0)$ this refers to the common real part of the two boundary values), and $\operatorname{Re}\varphi_\pm = 0$ on $(0,4c)$.*

(ii) *For every compact $K \subset (0,4c)$ there is $y_K > 0$ such that $\operatorname{Re}\tilde\varphi(x \pm iy) > 0$ for $x \in K$ and $0 < y \le y_K$. Moreover $\partial_y \operatorname{Re}\tilde\varphi(x+iy)|_{y=0^+} = \pi(\lambda_p - \psi^*)(x) > 0$ on $(0,4c)$.*

(iii) *(Hard edge.) There are $\delta_0 > 0$ and a function $f$, analytic and univalent on $U(0,2\delta_0)$, real on the real axis, with $f(0) = 0$ and $f'(0) < 0$, such that $(\varphi \pm i\pi)^2 = 4f$ on $U(0,2\delta_0) \cap \mathbb{C}_\pm$. More precisely $\varphi = \mp i\pi - 2f^{1/2}$ in $\mathbb{C}_\pm$, with the principal root. In particular $f$ does not depend on $n$, $|\operatorname{Re}\varphi(z)| \le C|z|^{1/2}$ on $U(0,\delta_0)$, and*

$$\operatorname{Re}\tilde\varphi(z) \ge |\arg z|\Big(\frac{2}{p}|z|^{-1/p} - C|z|^{1/2}\Big) \qquad (z \in U(0,\delta_0),\ 0 < |\arg z| \le \vartheta_{\mathrm{w}}). \tag{47}$$

(iv) *(Soft edge.) There are $\varepsilon_0 > 0$ and a function $f_r$, analytic and univalent on $U(4c,2\varepsilon_0)$, real on the real axis, with $f_r(4c) = 0$ and $f_r'(4c) > 0$, such that $\tilde\varphi^2 = \frac{4}{9}f_r^3$ on $U(4c,2\varepsilon_0)$, and $\tilde\varphi = -\frac{2}{3}f_r^{3/2}$ (principal branch, cut along $f_r^{-1}((-\infty,0])$). Hence, for $z \in U(4c,2\varepsilon_0)\setminus\mathbb{R}$,*

$$\operatorname{Re}\tilde\varphi(z) > 0 \iff \tfrac{\pi}{3} < |\arg f_r(z)| < \pi, \qquad \operatorname{Re}\tilde\varphi(z) < 0 \iff |\arg f_r(z)| < \tfrac{\pi}{3},$$

*and $\operatorname{Re}\tilde\varphi < 0$ on $(4c, 4c+2\varepsilon_0)$. Shrinking $\varepsilon_0$, we may assume $|\arg f_r(z) - \arg(z-4c)| \le \pi/12$ on $U(4c,2\varepsilon_0)\setminus\mathbb{R}$; in particular $\operatorname{Re}\tilde\varphi > 0$ on the vertical segments $\{4c \pm iy : 0 < y < 2\varepsilon_0\}$.*

*Proof.* (i) We have $\operatorname{Re}\varphi(z) = \gamma_p - \int \log|z-y|\,\mathrm{d}\mu^*(y)$. If $\operatorname{Re} z > 0$ and $\operatorname{Im} z \ne 0$, then $|z-y| > |\operatorname{Re} z - y|$ for all $y$, so $\operatorname{Re} g(z) > U^*(\operatorname{Re} z) \ge \gamma_p$ (Proposition 2.4 and Lemma 5.4(iii)). On $(4c,\infty)$, $\operatorname{Re} g_\pm = U^* > \gamma_p$ by Lemma 5.4(iii). If $\operatorname{Re} z \le 0$ and $z \ne 0$, choose $0 < x_0 \le \min(|z|/2, 4c)$. Then $|z-y| \ge y > y - x_0$ for $y \ge x_0$, and $|z-y| \ge |z| > x_0 \ge |x_0 - y|$ for $0 < y < x_0$; so $|z-y| > |x_0-y|$ for every $y > 0$, and integrating against $\mu^*$ gives $\operatorname{Re} g(z) > U^*(x_0) = \gamma_p$. On $(0,4c)$, $\operatorname{Re} g_\pm = U^* = \gamma_p$.

(ii) In $\mathbb{C}_+$, $\tilde\varphi = \varphi + i\theta/n = \gamma_p - g + i\pi z^{-1/p}$. Since $\psi^*$ is real-analytic on $(0,4c)$ (it is $\lambda_p$ times a regularised incomplete beta function of $x/4c$), $g|_{\mathbb{C}_+}$ continues analytically across $(0,4c)$, and so does $\tilde\varphi$. By the Cauchy–Riemann equations, $\partial_y \operatorname{Re} g(x+i0) = -\operatorname{Im} g'(x+i0) = \pi\psi^*(x)$ and $\partial_y \operatorname{Re}(i\pi z^{-1/p})|_{z=x} = -\pi\,\frac{\mathrm{d}}{\mathrm{d}x}x^{-1/p} = \pi\lambda_p(x)$. Since $\operatorname{Re}\tilde\varphi = 0$ on $(0,4c)$, a Taylor expansion with a uniform remainder on $K$ gives the claim. $\mathbb{C}_-$ is the mirror image.

(iii) On $(0,4c)$, $\varphi_+(x) = \gamma_p - U^*(x) - i\pi\mu^*((x,\infty)) = -i\pi + i\pi\mu^*((0,x])$. By Proposition 2.4, $\psi^*(x) = \lambda_p(x)\,\mathcal{I}_{x/4c}(\frac{1}{p}+\frac12,\frac12)$ near 0, and $\mathcal{I}_y(a,\frac12) = y^a\mathfrak{a}_0(y)$ with $\mathfrak{a}_0$ analytic near 0 and $\mathfrak{a}_0(0) > 0$. Since $\lambda_p(x)(x/4c)^{1/p+1/2} = Cx^{-1/2}$, we get $\mu^*((0,x]) = x^{1/2}\mathfrak{a}(x)$ with $\mathfrak{a}$ analytic near 0 and $\mathfrak{a}(0) > 0$ (indeed $\mathfrak{a}(0) = 2C_*$ by Proposition 2.4(iii)). Put $f(z) := -\frac{\pi^2}{4}z\mathfrak{a}(z)^2$. Then $f$ is analytic and real on the real axis, $f(0) = 0$, $f'(0) = -\pi^2\mathfrak{a}(0)^2/4 < 0$, so $f$ is univalent on a disc $U(0,2\delta_0)$. The functions $(\varphi+i\pi)^2$ (on $\mathbb{C}_+$) and $4f$ are analytic in $U(0,2\delta_0)\cap\mathbb{C}_+$ and have the same continuous boundary values $-\pi^2 x\mathfrak{a}(x)^2$ on $(0,2\delta_0)$; by the reflection principle they coincide. The same holds in $\mathbb{C}_-$ with $\varphi - i\pi$. In $U(0,2\delta_0)\cap\mathbb{C}_+$ the two analytic functions $f^{1/2}$ (principal) and $-\frac{i\pi}{2}z^{1/2}\mathfrak{a}(z)$ have the same square, so they differ by a constant sign; on the band both equal $-\frac{i\pi}{2}x^{1/2}\mathfrak{a}(x)$ (there $f(x+i0)$ lies on the negative axis approached from below), so they are equal, and $\varphi_+ + i\pi = -2f^{1/2}$ on the band gives $\varphi = -i\pi - 2f^{1/2}$ in $U(0,2\delta_0)\cap\mathbb{C}_+$. The bound on $\operatorname{Re}\varphi$ follows from $|f(z)| \le C|z|$. For (47) let $z = re^{i\phi}$ with $0 < \phi \le \vartheta_{\mathrm{w}}$ (the lower half-disc is the mirror image). Then

$$\operatorname{Re}\tilde\varphi(z) = \operatorname{Re}\big(-2f^{1/2} + i\pi z^{-1/p}\big) = \pi r^{-1/p}\sin(\phi/p) - \pi\operatorname{Im}\big(z^{1/2}\mathfrak{a}(z)\big).$$

Since $z^{1/2}\mathfrak{a}(z)$ is real for $\phi = 0$ and $|\partial_\phi(z^{1/2}\mathfrak{a}(z))| = |z(z^{1/2}\mathfrak{a})'(z)| \le Cr^{1/2}$, its imaginary part is at most $Cr^{1/2}\phi$; and $\sin(\phi/p) \ge \frac{2}{\pi}\frac{\phi}{p}$ because $\phi/p \le \vartheta_{\mathrm{w}}/p < \pi/2$.

(iv) On $(4c-2\varepsilon_0, 4c)$ we have $\tilde\varphi_+ + \tilde\varphi_- = \varphi_+ + \varphi_- = 0$ and, using Lemma 5.4(i), $\tilde\varphi_+ - \tilde\varphi_- = 2\pi i \int_x^\infty (\lambda_p - \psi^*) = 2\pi i \int_x^{4c} (\lambda_p - \psi^*)$. Near $y = 1$, $1 - \mathcal{I}_y(a, \frac12) = (1-y)^{1/2}\mathfrak{a}_1(y)$ with $\mathfrak{a}_1$ analytic and $\mathfrak{a}_1(1) > 0$, so $\tilde\varphi_+(x) = i\pi(4c-x)^{3/2}\mathfrak{a}_2(x)$ with $\mathfrak{a}_2$ analytic near $4c$ and $\mathfrak{a}_2(4c) > 0$. As in (iii), $\tilde\varphi^2$ and $\mathfrak{h}(z) := \pi^2(z-4c)^3\mathfrak{a}_2(z)^2$ are analytic in $U(4c, 2\varepsilon_0) \setminus [4c - 2\varepsilon_0, 4c + 2\varepsilon_0]$ with equal boundary values on $(4c - 2\varepsilon_0, 4c)$, and $\tilde\varphi$ has no jump on $(4c, 4c + 2\varepsilon_0)$ (Lemma 5.4(i)). So $\tilde\varphi^2 = \mathfrak{h}$. Put $f_r := (9\mathfrak{h}/4)^{1/3} = (\frac{3\pi}{2})^{2/3}(z-4c)\mathfrak{a}_2(z)^{2/3}$. On $(4c, \infty)$, $\tilde\varphi$ is real and negative (Lemma 5.4(iii)), which fixes the branch $\tilde\varphi = -\frac23 f_r^{3/2}$. Then $\operatorname{Re}\tilde\varphi = -\frac23 |f_r|^{3/2}\cos(\frac32 \arg f_r)$, which is positive exactly when $\frac32|\arg f_r| \in (\frac\pi2, \frac{3\pi}{2})$. Finally $f_r(z) = f_r'(4c)(z-4c)(1+O(z-4c))$ with $f_r'(4c) > 0$ gives the statement about $\arg f_r$, and on the vertical segments $|\arg f_r| \in [\frac{5\pi}{12}, \frac{7\pi}{12}]$. □

5.6. **The contour.** The lens is bounded by two arcs, the lips, on which the triangular factors of the lens are exponentially small. We first show that such arcs exist, with all the properties we need.

**Lemma 5.6** (The lips)**.** *There is $\delta_1 > 0$ such that for every $\delta \in (0, \delta_1]$ there is $h_1 = h_1(\delta) > 0$ with the following property. For every $h_0 \in (0, h_1]$ there are $C^2$ arcs $\Sigma^\pm_{\mathrm{lip}} \subset \mathbb{C}_\pm$ from $0$ to $4c \pm ih_0$, mirror images of each other and independent of $n$, such that:*

(a) *$\Sigma^\pm_{\mathrm{lip}} \cap U(0,\delta)$ is the preimage under $f$ (equivalently, under $\zeta = n^2 f(z)$) of a segment of the ray $\arg\zeta = \mp(\pi - \vartheta_{\mathrm{lip}})$; on it $\frac34\vartheta_{\mathrm{lip}} \le |\arg z| \le \frac54\vartheta_{\mathrm{lip}}$;*
(b) *on $\Sigma^\pm_{\mathrm{lip}} \setminus \{0, 4c \pm ih_0\}$: $|\arg z| < \vartheta_{\mathrm{w}}$, $\operatorname{Re}\varphi < 0$ and $\operatorname{Re}\tilde\varphi > 0$;*
(c) *$\Sigma^\pm_{\mathrm{lip}} \subset \{\operatorname{Re} z \le 4c\} \setminus U(4c, h_0)$, and $\Sigma^\pm_{\mathrm{lip}}$ meets the line $\operatorname{Re} z = 4c$ only at $4c \pm ih_0$.*

*Proof.* We construct the upper lip from four pieces and round the three corners inside the open set $G_+ := \{z \in \mathbb{C}_+ : \ |\arg z| < \vartheta_{\mathrm{w}},\ \operatorname{Re}\tilde\varphi(z) > 0\}$, which contains every piece except its two end points, and outside $\overline{U(0,\delta)}$; the lower lip is the mirror image. By Lemma 5.5(i), $\operatorname{Re}\varphi < 0$ on all of $\mathbb{C}_+$, so only $\operatorname{Re}\tilde\varphi > 0$ and $|\arg z| < \vartheta_{\mathrm{w}}$ have to be checked.

Choose $r_* \in (0, \delta_0)$ with $\frac2p r^{-1/p} > Cr^{1/2}$ for $0 < r \le r_*$; then (47) gives $S_0 := \{0 < |z| \le r_*,\ 0 < \arg z < \vartheta_{\mathrm{w}}\} \subset G_+$. Since $f(z) = f'(0)z(1+O(z))$ with $f'(0) < 0$ and $f$ is univalent, the preimage of the ray $\arg\zeta = -(\pi - \vartheta_{\mathrm{lip}})$ near 0 is an analytic arc leaving 0 at the angle $\vartheta_{\mathrm{lip}}$, and for $\delta_1 \le r_*/4$ small its part in $U(0, 2\delta)$ satisfies $\frac34\vartheta_{\mathrm{lip}} \le \arg z \le \frac54\vartheta_{\mathrm{lip}}$ (as $\frac54\vartheta_{\mathrm{lip}} < \vartheta_{\mathrm{w}}$, it lies in $S_0$). This is the first piece; let $z_1$ be its end point on $|z| = 2\delta$, so that the corner at $z_1$ can be rounded in $\{\delta < |z| < 2\delta\}$ and (a) is not affected. The second piece runs in $S_0$: along the ray $\arg z = \arg z_1$ to $|z| = \frac34 r_*$, and then along the circle $|z| = \frac34 r_*$ down to height $y_1 := h_0 \sin\frac{\pi}{12}$.

Let $K := [\frac12 r_*, 4c - \frac12\varepsilon_0]$ and let $y_K$ be as in Lemma 5.5(ii); choose $h_1 \le \min(y_K, \frac12\varepsilon_0)$ so small that the angle $\phi$ with $\frac34 r_* \sin\phi = y_1$ satisfies $\phi < \arg z_1$ and $\frac34 r_* \cos\phi \ge \frac12 r_*$, that $\arctan(2y_1/r_*) < \vartheta_{\mathrm{w}}$, and that $\arcsin(h_0/4c) < \vartheta_{\mathrm{w}}$ (the largest argument on the fourth piece). The third piece is the horizontal segment from the end of the second piece to the point $4c + h_0 e^{i11\pi/12}$, at height $y_1$. For $\operatorname{Re} z \in K$ it lies in $G_+$ by Lemma 5.5(ii), as $y_1 < y_K$. For $4c - \frac12\varepsilon_0 \le \operatorname{Re} z \le 4c - h_0\cos\frac{\pi}{12}$ it lies in $U(4c, \varepsilon_0)$ with $\arg(z-4c) \ge \frac{11\pi}{12}$, hence $\frac{5\pi}{6} \le \arg f_r(z) < \pi$ and $\operatorname{Re}\tilde\varphi > 0$ by Lemma 5.5(iv). The fourth piece is the arc $\{4c + h_0 e^{i\phi} : \ \frac\pi2 \le \phi \le \frac{11\pi}{12}\}$, on which $\frac{5\pi}{12} \le \arg f_r < \pi$, so again $\operatorname{Re}\tilde\varphi > 0$ (except at the end point $4c + ih_0$ itself, where $\arg f_r \in [\frac{5\pi}{12}, \frac{7\pi}{12}]$ also gives $\operatorname{Re}\tilde\varphi > 0$). All four pieces have arguments below $\vartheta_{\mathrm{w}}$ by the choice of $h_1$, and none depends on $n$, since $f$, $f_r$ and $\tilde\varphi$ do not. Property (c) holds by construction. □

We fix, once and for all and independently of $n$, numbers $\delta, h_0, \varepsilon, \delta'$ with

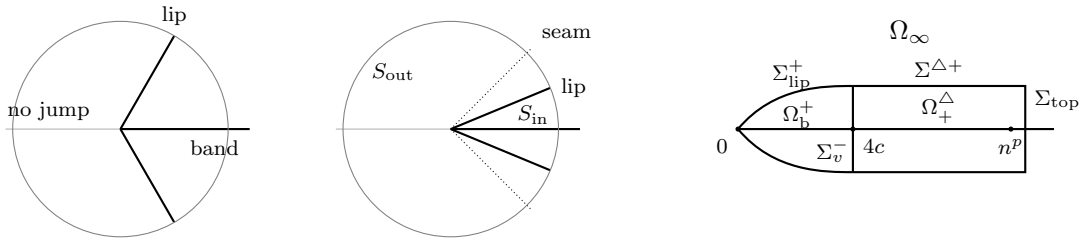
$$0<\delta\le\min\bigl(\delta_1,\tfrac14,\tfrac{c}{2}\bigr),\qquad 0<h_0\le\min\bigl(h_1(\delta),\tfrac12\delta\tan\vartheta_{\mathrm w}\bigr),$$
$$0<\varepsilon<\min(h_0,\varepsilon_0,c\sin\vartheta_{\mathrm w}),\qquad 0<\delta'<1,$$

(the condition $\varepsilon<c\sin\vartheta_{\mathrm w}$ gives $U(4c,4\varepsilon)\subset\mathcal Z_n$, which Lemma 6.3(vi)(b) uses), and the following arcs (Figure 2).

- The *band* $[0,4c]$, the *saturated segment* $(4c,n^p+\delta')$ and the *top segment* $\Sigma_{\mathrm{top}}:=\{n^p+\delta'+iy:\ 0<|y|\le h_0\}$. The top node $X_1=n^p$ lies inside, and no node lies on $[n^p+\delta',\infty)$.
- The *lips* $\Sigma^{\pm}_{\mathrm{lip}}$ of Lemma 5.6.
- The *vertical segments* $\Sigma^{\pm}_v:=\{4c\pm iy:\ 0<y\le h_0\}$.
- The *saturated lips* $\Sigma^{\triangle\pm}:=\{x\pm ih_0:\ 4c\le x\le n^p+\delta'\}$.

These arcs bound the *lens* $\Omega^{\pm}_{\mathrm b}$ (between the band and $\Sigma^{\pm}_{\mathrm{lip}}$, left of $\Sigma^{\pm}_v$) and the *saturated regions* $\Omega^{\triangle}_{\pm}:=\{4c<\operatorname{Re}z<n^p+\delta',\ 0<\pm\operatorname{Im}z<h_0\}$. The rest of $\mathbb C$ is $\Omega_\infty$. We write $\Sigma$ for the union of all these arcs. Only finitely many arcs meet at each vertex, and for fixed $n$, $\Sigma$ is bounded. By Lemma 5.6(c) the lips meet the saturated lips only at $4c\pm ih_0$. In $U(4c,\varepsilon)$ the only arcs are the band, the saturated segment and $\Sigma^{\pm}_v$; at 0 only the band and the two lips meet. Nothing lies on $(-\infty,0)$ or on $(n^p+\delta',\infty)$.

The closures of $\Omega^{\pm}_{\mathrm b}$ and $\Omega^{\triangle}_{\pm}$, minus the point 0, lie in $\mathcal Z_n$. Indeed, the lens lies in $|\arg z|<\vartheta_{\mathrm w}$ by Lemma 5.6(b); every point of $\Omega^{\triangle}$ has $\operatorname{Re}z\ge4c$ and $|\operatorname{Im}z|\le h_0<4c\tan\vartheta_{\mathrm w}$, so $|\arg z|<\vartheta_{\mathrm w}$; and $|z|\le n^p+\delta'+h_0<e^{pc_0}n^p$ for $n\ge n_0$.




$p\ge1$: lips at $\pm\vartheta_{\mathrm{lip}}$ $p<1$: lips at $\pm\vartheta_{\mathrm{lip}}$, seams at $\pm\pi p/2$ all $p$ (not to scale)

FIGURE 2. The contour near the hard edge for $p\ge1$ (the cases $p>1$ and $p=1$ have the same local picture) and for $p<1$, and the whole contour. The lips are drawn at the default angles $\vartheta_{\mathrm{lip}}=\pi/3$ and $\pi p/4$ (with $p=\frac12$); for a thin strip in $(\mathrm A_\vartheta)$ they are rotated towards the band. The negative axis and, for $p<1$, the seams carry no jump of $V$ or $Q$. The seams are used only to construct the outer scalar (Appendix A).

5.7. **The transformations.** We now remove the poles, normalise at infinity and open the lens. The pattern is that of T1(b); since we work in a one-sided variable, we write every factor out. Put, in $\mathbb C_\pm$,

$$\kappa_\pm:=\frac{1}{1-e^{\mp2i\theta}},\qquad\text{so that}\qquad \kappa_\pm-1=e^{\mp2i\theta}\kappa_\pm, \tag{48}$$

and note $|e^{\mp2i\theta}|<1$ in $\mathbb C_\pm\cap\mathcal Z_n$ by (41), so $\kappa_\pm$ is analytic there. Define, on $\mathcal Z_n\cap\mathbb C_\pm$,

$$L_\pm:=\begin{pmatrix}1 & \mp We^{\mp2i\theta}\kappa_\pm\\ 0 & 1\end{pmatrix},\qquad A_\pm:=\begin{pmatrix}1 & 0\\ \mp e^{2n\varphi}/W & 1\end{pmatrix},\qquad A^{\triangle}_\pm:=\begin{pmatrix}1 & 0\\ \mp1/(\kappa_\pm W) & 1\end{pmatrix}. \tag{49}$$

*Step 1: removal of the poles.* Let

$$R := U \text{ in } \Omega_\infty, \qquad R := UL_\pm \text{ in } \Omega_{\mathrm{b}}^{\pm}, \qquad R := UA_\pm^{\triangle}\,\kappa_\pm^{\sigma_3} \text{ in } \Omega_\pm^{\triangle}. \tag{50}$$

The restriction of $R$ to each of the four regions $\Omega_{\mathrm{b}}^{\pm}$, $\Omega_\pm^{\triangle}$ extends continuously to the real axis, including the nodes. Indeed, near $X_k$ all factors are meromorphic, and $1-e^{\mp 2i\theta} = \pm 2i\theta'(X_k)(z-X_k)(1+O(z-X_k))$ because $e^{2i\theta(X_k)}=1$. In $\Omega_{\mathrm{b}}^{\pm}$ the entry $\mp We^{\mp 2i\theta}\kappa_\pm$ then has the simple pole $W(X_k)/(2i\theta'(X_k)(z-X_k)) = w(X_k)/(z-X_k)$, and $R_{j2} = U_{j2} - U_{j1}w(X_k)/(z-X_k)+O(1)$ is regular by Lemma 5.1. In $\Omega_\pm^{\triangle}$, $\kappa_\pm^{-1}$ vanishes simply at $X_k$, which removes the pole of the second column; the first column is $\kappa_\pm(U_{j1} \mp U_{j2}/(\kappa_\pm W))$, and the bracket vanishes at $X_k$ since $U_{j2}/(\kappa_\pm W) \to \pm U_{j1}(X_k)$, so the first column is regular too.

*Step 2: normalisation.* $T := e^{-n\gamma_p\sigma_3} R e^{n\varphi\sigma_3}$.

*Step 3: the lens.* $S := TA_\pm$ in $\Omega_{\mathrm{b}}^{\pm}$, and $S := T$ elsewhere.

*Step 4: the outer scalar.* Given a function $F$ with the properties (O1)–(O2) of §5.9 below and its lens version $\tilde F$, let

$$V := C_\infty\, S\, \tilde F^{\sigma_3} \text{ in } \Omega_{\mathrm{b}} \cup \Omega^{\triangle}, \qquad V := C_\infty\, S\, F^{\sigma_3} \text{ in } \Omega_\infty, \tag{51}$$

with a constant diagonal matrix $C_\infty$ fixed in Lemma 6.4.

Composing the four steps, and using $\kappa_\pm^{\sigma_3}\tilde F^{\sigma_3} = F^{\sigma_3}$ in $\Omega_\pm^{\triangle}$ (definition (53) below),

$$V = C_\infty e^{-n\gamma_p\sigma_3}\, U\, M, \qquad M := \begin{cases} e^{n\varphi\sigma_3}F^{\sigma_3} & \text{in } \Omega_\infty, \\ L_\pm e^{n\varphi\sigma_3}A_\pm \tilde F^{\sigma_3} & \text{in } \Omega_{\mathrm{b}}^{\pm}, \\ A_\pm^{\triangle} e^{n\varphi\sigma_3}F^{\sigma_3} & \text{in } \Omega_\pm^{\triangle}. \end{cases} \tag{52}$$

Since $U$ is analytic off the nodes, the jump of $V$ on any arc of $\Sigma$ away from the nodes is $M_-^{-1}M_+$.

In [19] the saturated region is treated in two steps, by the factors $\Pi^{-\sigma_3}(\cdot)$ of [19, (3.10)–(3.11)] and $V_\pm^{\triangle}$ of [19, (4.12)–(4.13)] (T1(b)). Our factor $A_\pm^{\triangle}\kappa_\pm^{\sigma_3}$ combines them into one step. In $\Omega_-^{\triangle}$ it carries a minus sign relative to the literal product of the two factors; this is the sign of the factor $-V_-^{\triangle}$ that [19, (4.12)] uses in the regions $\Omega_{l,+}^{\triangle} \cup \Omega_{r,-}^{\triangle}$, which lie in the second and fourth quadrants and are therefore mapped into $\mathbb{C}_-$ by $z \mapsto z^2$. Without this sign, $V$ would have the constant jump $-I$ on the saturated segment.

5.8. **Where the weight is evaluated.** The following bookkeeping explains why a strip of any width $\vartheta > 0$ suffices.

**Lemma 5.7** (Where the weight is evaluated)**.** *The functions $w$, $W$, $r_n$ and $W_0$ (defined in (55)) are evaluated only at points of $\mathcal{Z}_n$. Specifically, they are evaluated (a) in $\overline{\Omega_{\mathrm{b}} \cup \Omega^{\triangle}} \setminus \{0\}$, which contains every arc of $\Sigma$ except the top segment and the parts of the lips outside the lens; (b) on the lips and on the top segment; and (c) in the sectors $\{|\arg z| \le \frac{5}{4}\vartheta_{\mathrm{lip}}\} \cap U(0,\delta)$ used in Section 7 and Appendix B. The region $\Omega_\infty$ never uses the weight: there $M = e^{n\varphi\sigma_3}F^{\sigma_3}$ contains neither $w$ nor $W$. Consequently $\xi(z) = \log t(z)$ always lies in $\{\operatorname{Re}\xi \ge -c_0,\ |\operatorname{Im}\xi| \le \vartheta_{\mathrm{w}}/p\} \subset S_\vartheta$.*

*Proof.* By (52), $w$ and $W$ enter $V$ only in $\Omega_{\mathrm{b}} \cup \Omega^{\triangle}$, whose closure minus $\{0\}$ lies in $\mathcal{Z}_n$ (§5.6); the top segment has $|z| \le n^p + \delta' + h_0 < e^{pc_0}n^p$ and $|\arg z| < \vartheta_{\mathrm{w}}$. The sectors used near 0 lie in $\mathcal{Z}_n$ because $\frac{5}{4}\vartheta_{\mathrm{lip}} < \vartheta_{\mathrm{w}}$. For $z \in \mathcal{Z}_n$, $\operatorname{Re}\xi = \log n - \frac{1}{p}\log|z| > -c_0$ and $|\operatorname{Im}\xi| = |\arg z|/p < \vartheta_{\mathrm{w}}/p$. □

*Remark* 5.8 (the angles). The width $\vartheta$ of the strip in $(\mathrm{A}_\vartheta)$ enters the proof only through $\vartheta_{\mathrm{w}} \le p\vartheta$, and $\vartheta_{\mathrm{w}}$ and $\vartheta_{\mathrm{lip}}$ enter only through (34). The angle $\vartheta_{\mathrm{lip}}$ can be taken as small as

we wish because the Bessel model admits rays at $\pm(\pi-\vartheta_{\mathrm{lip}})$ for every $\vartheta_{\mathrm{lip}}\in(0,\pi)$ (T2(b)), and the lens near 0 is then the preimage of these rays. Every step that uses the angles needs only a fixed positive lower bound, which (34) provides: the decay factors $\sin(\vartheta_{\mathrm{lip}}/2)$ in (84) and $\sin(\frac{3}{4}\vartheta_{\mathrm{lip}}/p)$ in Lemma 6.5; the bound $\sin(\phi/p)\ge\frac{2}{\pi}\frac{\phi}{p}$ in (47), which needs $\vartheta_{\mathrm{w}}/p<\pi/2$; the separation $|x-z|\ge(x+|z|)/C$ for $|\arg z|\ge\frac{1}{2}\vartheta_{\mathrm{lip}}$ in Lemma 6.3(vi) and Appendix B; and the large-argument expansions of the Bessel functions, which are used in the closed sector $|\arg 2\zeta^{1/2}|\le\frac{1}{2}(\pi-\vartheta_{\mathrm{lip}})$ (§7.2). The order of choices is: $\vartheta$, then $\vartheta_{\mathrm{w}},\vartheta_{\mathrm{lip}}$ by (33), then $\delta_1$, then $\delta\le\delta_1$, which is also taken so small that the angle-dependent conditions in Lemmas 6.5(b), 7.3 and 7.2(i) hold (they depend only on $p$, $s$, $\ell_1$, $\vartheta_{\mathrm{w}}$, $\vartheta_{\mathrm{lip}}$, never on $n$), then $h_0$, $\varepsilon$, $\delta'$, and $n_0$ last. Smaller angles make the radii $\delta$, $h_0$, $\varepsilon$ smaller and the threshold $n_0$ and the constants larger, but they do not change the statements. The upper bounds in (34) are what the construction needs: $\vartheta_{\mathrm{w}}<\pi/2$ (convexity of the sector $Z(r)$ in Lemma 6.3, and $|y-z|\ge(|y|+|z|)/\sqrt{2}$ for $y<0$ in Lemma A.1), $\vartheta_{\mathrm{w}}<\pi p/2$ for $p<1$ (the lens must lie in the inner sector of the seam construction), and $\frac{5}{4}\vartheta_{\mathrm{lip}}<\vartheta_{\mathrm{w}}$ (the sectors near 0 in which the weight is used lie in $\mathcal{Z}_n$). With the default angles ($\mathsf{m}=1$) the strip needed is $|\Im\xi|\le 7\pi/(16p)$ for $p\ge1$ and $|\Im\xi|\le 3\pi/8$ for $p<1$; it is never necessary to continue $\Lambda$ to the strip $|\Im\xi|\le\pi/p$ that corresponds to the whole slit plane in $z$.

**Lemma 5.9** (Potter bounds in the sector)**.** *For every $\eta>0$ there is $C_\eta$ such that:*

(i) $|\Lambda(\xi)|\le C_\eta+\eta\operatorname{Re}\xi$ *for* $\xi\in S_\vartheta$, $\operatorname{Re}\xi\ge0$, *and* $|\Lambda|\le C$ *on* $S_\vartheta\cap\{\operatorname{Re}\xi\le0\}$. *Hence* $|\ell_1(t)|^{\pm1}\le C_\eta|t|^\eta$ *for* $|\arg t|\le\vartheta$, $|t|\ge1$.

(ii) $|r_n(z)|^{\pm1}\le C_\eta n^{2\eta}(1+|z|^{-\eta/p})$ *for* $z\in\mathcal{Z}_n$.

(iii) *For* $R\ge1$, $z\in\mathcal{Z}_n$ *with* $|z|\le R$ *and* $x\in(0,R]$,

$$|\log r_n(z)-\log r_n(x)|\le\frac{\varepsilon_\Lambda(T)}{p}\big(|\arg z|+|\log(x/|z|)|\big),$$
$$|\log r_n(z)|\le\frac{\varepsilon_\Lambda(T)}{p}\big(|\arg z|+|\log|z||\big),$$

*with* $T:=\log n-\frac{1}{p}\log R$. *Here* $\log r_n=\Lambda(\xi)-\Lambda(\log n)$.

*Proof.* (i) For $\operatorname{Re}\xi\ge T_0$, integrate $\Lambda'$ along the horizontal segment from $T_0+i\operatorname{Im}\xi$ to $\xi$: $|\Lambda(\xi)-\Lambda(T_0+i\operatorname{Im}\xi)|\le\varepsilon_\Lambda(T_0)\operatorname{Re}\xi$. Choose $T_0$ with $\varepsilon_\Lambda(T_0)\le\eta$; $\Lambda$ is bounded on the compact set $S_\vartheta\cap\{0\le\operatorname{Re}\xi\le T_0\}$ and, by hypothesis, on $\operatorname{Re}\xi\le0$. Then $|\ell_1(t)|^{\pm1}=e^{\pm\operatorname{Re}\Lambda(\log t)}$. (ii) $|r_n|^{\pm1}=|\ell_1(t)|^{\pm1}\ell_1(n)^{\mp1}$ with $|t|=n|z|^{-1/p}$; apply (i) to both factors (for $|t|\le1$ use the bound on $\operatorname{Re}\xi\le0$). (iii) Join $\xi(x)$ to $\xi(z)$ by the horizontal segment from $\log n-\frac{1}{p}\log x$ to $\log n-\frac{1}{p}\log|z|$ followed by the vertical segment to $\xi(z)$. Along this path $\operatorname{Re}\xi\ge T$ and $|\operatorname{Im}\xi|\le|\arg z|/p$, and its length is $(|\log(x/|z|)|+|\arg z|)/p$. The second inequality is the case $x=1$, since $r_n(1)=1$. □

5.9. **The outer scalar and the jumps of $V$.** Section 6 constructs, for each geometry, a function $F$ with the following five properties. The constants in (O3) and (O4) do not depend on $n$; $\nu_p:=\min(1/p,\frac{1}{2})$.

(O1) $F$ is analytic and zero-free in $\mathbb{C}\setminus[0,\infty)$ (in particular, it has no jump on $(-\infty,0)$), it has continuous boundary values on $(0,\infty)\setminus\{X_k\}$, and $F_+=F_-e^{2i\theta}$ on $(4c,\infty)\setminus\{X_k\}$.

(O2) The *lens scalar*

$$\tilde F:=(1-e^{\mp2i\theta})F=F/\kappa_\pm\qquad\text{in }\mathbb{C}_\pm\cap\mathcal{Z}_n \tag{53}$$

extends to an analytic, zero-free function on $\mathcal{Z}_n\backslash[4c,\infty)$, with $\tilde F_+ = -\tilde F_-$ on $(4c, e^{pc_0}n^p)$. Thus $\tilde F^2$ is analytic and zero-free on $\mathcal{Z}_n$, and so is the *band factor*

$$H_n(z) := \frac{1}{(4c-z)\tilde F(z)^2} \qquad (z\in\mathcal{Z}_n). \tag{54}$$

We require $H_n > 0$ on $(0, e^{pc_0}n^p)$ and that $H_n(0^+) := \lim_{z\to 0,\, z\in\mathcal{Z}_n} H_n(z)$ exists in $(0,\infty)$.

(O3) For $z\in\mathcal{Z}_n$ with $|z|\le 1$: $|\log H_n(z) - \log H_n(0^+)| \le C|z|^{\nu_p}$ and $|(\log H_n)'(z)| \le C|z|^{\nu_p-1}$.

(O4) For $z\in\mathcal{Z}_n$ with $|z|\ge 1$: $|\log H_n(z) - \log H_n(0^+)| \le C\big(1+\log(1+|z|)\big)^2$.

(O5) (For each fixed $n$.) $|F|^{\pm 1}$ is bounded on $U(0,\delta)\setminus(\overline{\Omega_{\mathrm b}}\cup[0,\infty))$, and $\tilde F^{\pm 1}$ is bounded on $U(0,\delta)\cap\Omega_{\mathrm b}$. There is $N_n$ with $|F(z)|^{\pm 1}\le C_n|z|^{N_n}$ for $|z|\ge n^p+\delta'$, $z\notin\mathbb{R}$, and $F(iy)$ is bounded and bounded away from 0 as $y\to\pm\infty$.

The *band weight* is

$$W_0 := \frac{W}{\tilde F^2} = \mathsf{w}_n\, z^\alpha (4c-z)H_n(z)r_n(z) \quad \text{on } \mathcal{Z}_n \tag{55}$$

(Lemma 5.3). On $(0,4c)$ it equals the constant $\mathsf{w}_n$ times a positive function.

**Lemma 5.10** (Jumps of $V$)**.** *Assume (O1) and (O2). Then $V$ is analytic in $\mathbb{C}\setminus\Sigma$, has continuous boundary values on $\Sigma$ minus its vertices (including at the nodes), and $V_+ = V_- v_V$ with*

(a) *on the band $(0,4c)$: $v_V = \begin{pmatrix} 0 & W_0 \\ -1/W_0 & 0\end{pmatrix}$;*

(b) *on $\Sigma^\pm_{\mathrm{lip}}$ (oriented away from 0), with $\iota_+ = 1$ and $\iota_- = 0$:*

$$v_V = \begin{pmatrix} 1 & 0 \\ e^{2n\varphi}/W_0 & 1\end{pmatrix} + \begin{pmatrix} \iota_\pm(\kappa_\pm - 1) & W_0 e^{-2n\tilde\varphi} \\ (\kappa_\pm - 1)e^{2n\varphi}/W_0 & \iota_\mp(\kappa_\pm - 1)\end{pmatrix};$$

(c) *on $\Sigma_v^\pm$ (oriented away from the real axis): $v_V = \begin{pmatrix} 1 & -W_0 e^{-2n\tilde\varphi} \\ 0 & 1\end{pmatrix}$;*

(d) *on the saturated segment $(4c, n^p+\delta')$: $v_V = \begin{pmatrix} 1 & 0 \\ e^{2n\tilde\varphi}/W_0 & 1\end{pmatrix}$;*

(e) *on $\Sigma^{\triangle\pm}$ (oriented from left to right): $v_V = \begin{pmatrix} 1 & 0 \\ \kappa_\pm e^{2n\varphi}/W_0 & 1\end{pmatrix}$;*

(f) *on $\Sigma_{\mathrm{top}}\cap\mathbb{C}_\pm$ (oriented upwards): $v_V = \begin{pmatrix} 1 & 0 \\ \mp\kappa_\pm e^{2n\varphi}/W_0 & 1\end{pmatrix}$;*

(g) *there is no jump on $(-\infty,0)$ or on $(n^p+\delta',\infty)$.*

*Proof.* We use $M_-^{-1}M_+$ from (52) and the identities

$$e^{\mp 2i\theta}e^{-2n\varphi} = e^{-2n\tilde\varphi}, \qquad F = \kappa_\pm\tilde F, \qquad W/\tilde F^2 = W_0 \qquad \text{in } \mathbb{C}_\pm. \tag{56}$$

For a diagonal matrix $D=\operatorname{diag}(d_1,d_2)$, conjugation $D^{-1}XD$ multiplies the $(1,2)$ entry by $d_2/d_1$ and the $(2,1)$ entry by $d_1/d_2$.

(a) $L_-^{-1}L_+ = \left(\begin{smallmatrix} 1 & W \\ 0 & 1\end{smallmatrix}\right)$: its $(1,2)$ entry is $-W(e^{-2i\theta}\kappa_+ + e^{2i\theta}\kappa_-) = -W\big(\frac{e^{-2i\theta}}{1-e^{-2i\theta}} + \frac{1}{e^{-2i\theta}-1}\big) = W$. Conjugating by $e^{n\varphi_\pm\sigma_3}$ and using $\varphi_- = -\varphi_+$ (Lemma 5.4(i)) gives $\left(\begin{smallmatrix} e^{2n\varphi_+} & W \\ 0 & e^{2n\varphi_-}\end{smallmatrix}\right)$, and

$$A_-^{-1}\begin{pmatrix} e^{2n\varphi_+} & W \\ 0 & e^{2n\varphi_-}\end{pmatrix}A_+ = \begin{pmatrix} 0 & W \\ -1/W & 0\end{pmatrix}.$$

Since $\tilde F$ has no jump on the band (O2), the final conjugation gives (a) by (56).

(b) Upper lip: the + side is $\Omega_\infty$, so $v_V = M_{\Omega_\mathrm{b}}^{-1} M_{\Omega_\infty} = \tilde F^{-\sigma_3} A_+^{-1}\big(e^{-n\varphi\sigma_3} L_+^{-1} e^{n\varphi\sigma_3}\big) F^{\sigma_3}$. The bracket is $\left(\begin{smallmatrix} 1 & W\kappa_+ e^{-2n\tilde\varphi} \\ 0 & 1 \end{smallmatrix}\right)$, and

$$A_+^{-1}\begin{pmatrix} 1 & W\kappa_+ e^{-2n\tilde\varphi} \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & W\kappa_+ e^{-2n\tilde\varphi} \\ e^{2n\varphi}/W & \kappa_+ \end{pmatrix},$$

because $1 + \kappa_+ e^{2n\varphi} e^{-2n\tilde\varphi} = 1 + \kappa_+ e^{-2i\theta} = \kappa_+$. Conjugating with $\tilde F^{-\sigma_3}(\cdot)F^{\sigma_3}$, $F = \kappa_+ \tilde F$, gives $\left(\begin{smallmatrix} \kappa_+ & W_0 e^{-2n\tilde\varphi} \\ \kappa_+ e^{2n\varphi}/W_0 & 1 \end{smallmatrix}\right)$, which is (b). The lower lip (where the + side is the lens) is the mirror computation and gives $\left(\begin{smallmatrix} 1 & W_0 e^{-2n\tilde\varphi} \\ \kappa_- e^{2n\varphi}/W_0 & \kappa_- \end{smallmatrix}\right)$.

(c) Upper segment, + side $= \Omega_\mathrm{b}^+$: $v_V = M_{\Omega^\triangle}^{-1} M_{\Omega_\mathrm{b}} = F^{-\sigma_3} e^{-n\varphi\sigma_3} (A_+^\triangle)^{-1} L_+ e^{n\varphi\sigma_3} A_+ \tilde F^{\sigma_3}$. One checks $(A_+^\triangle)^{-1} L_+ = \left(\begin{smallmatrix} 1 & -We^{-2i\theta}\kappa_+ \\ 1/(\kappa_+ W) & 1-e^{-2i\theta} \end{smallmatrix}\right)$, and after the conjugation by $e^{n\varphi\sigma_3}$ and right multiplication by $A_+$ the $(2,1)$ entry vanishes and the diagonal becomes $(\kappa_+, \kappa_+^{-1})$; the final conjugation with $F = \kappa_+ \tilde F$ removes the diagonal and gives (c). The lower segment, whose + side is $\Omega_-^\triangle$, is the mirror image.

(d) On $(4c, n^p + \delta')$ both sides lie in $\Omega^\triangle$. Since $\kappa_\pm^{-1} = 1 - e^{\mp 2i\theta}$,

$$(A_-^\triangle)^{-1} A_+^\triangle = \begin{pmatrix} 1 & 0 \\ -(\kappa_+^{-1} + \kappa_-^{-1})/W & 1 \end{pmatrix}, \qquad \kappa_+^{-1} + \kappa_-^{-1} = 2 - 2\cos 2\theta = \kappa_+^{-1}\kappa_-^{-1}.$$

Conjugating by $e^{n\varphi\sigma_3} F^{\sigma_3}$ with boundary values, the diagonal becomes

$$(e^{n(\varphi_+ - \varphi_-)} F_+/F_-)^{\sigma_3} = (e^{-2i\theta} e^{2i\theta})^{\sigma_3} = I$$

by Lemma 5.4(i) and (O1), and the $(2,1)$ entry becomes

$$-e^{n(\varphi_+ + \varphi_-)} F_+ F_- / (\kappa_+ \kappa_- W) = -e^{2n\tilde\varphi} \tilde F_+ \tilde F_- / W.$$

By (O2), $-\tilde F_+ \tilde F_- = \tilde F^2$, which gives (d).

(e), (f) Here $v_V = (F^{-\sigma_3} e^{-n\varphi\sigma_3})(A_\pm^\triangle)^{\pm 1}(e^{n\varphi\sigma_3} F^{\sigma_3})$ with the exponent fixed by the orientation, and the $(2,1)$ entry $\mp 1/(\kappa_\pm W)$ of $A_\pm^\triangle$ becomes $\mp F^2 e^{2n\varphi}/(\kappa_\pm W) = \mp \kappa_\pm e^{2n\varphi}/W_0$.

(g) On $(-\infty, 0)$: $U$ has no jump, $e^{n\varphi}$ jumps by $e^{-2\pi i n} = 1$ (Lemma 5.4(i)) and $F$ has no jump (O1). On $(n^p + \delta', \infty)$ there are no nodes, and $M_-^{-1} M_+ = (e^{n(\varphi_+ - \varphi_-)} F_+/F_-)^{\sigma_3} = I$ as in (d).

Continuity at the nodes: in (a) and (d) all entries are analytic across the real axis; together with Step 1 this shows that $V$ has continuous boundary values there. □

## 6. The outer scalar, the parametrices away from the hard edge, and the Frame Proposition

The transformations of Section 5 reduce everything away from the hard edge to one scalar problem: find an outer scalar $F$ with the properties (O1)–(O5) of §5.9. The natural candidate comes from Stirling's formula. The node masses carry the factor $1/\Gamma(t)$ through $\Pi = \sin\theta/\theta'$, and

$$\tilde D(z) := e^{-j(t(z))}, \qquad j(t) := \operatorname{Ln}\Gamma(t) - \big(t - \tfrac12\big)\ln t + t - \tfrac12 \ln(2\pi),$$

removes this growth. Here $j$ is Binet's function, $\operatorname{Ln}\Gamma$ is the branch analytic on $\mathbb{C} \setminus (-\infty, 0]$ and real on $(0,\infty)$, and $\ln t$ is principal; so $\tilde D = \sqrt{2\pi}\, t^{t-1/2} e^{-t}/\Gamma(t)$, and this sign convention is used throughout. Then

$$D := \kappa_\pm \tilde D = \frac{\tilde D}{1 - e^{\mp 2i\theta}} \quad \text{in } \mathbb{C}_\pm \tag{57}$$

satisfies $D_+/D_- = (1-e^{2i\theta})/(1-e^{-2i\theta}) = -e^{2i\theta}$ on $(0,\infty)$ off the nodes, and $K(z) := (4c-z)^{-1/2}$ (principal branch, cut $[4c,\infty)$) supplies the missing factor $-1$, so that $DK$ has the jump required in (O1).

The difficulty is the cut of $t(z)$ on $(-\infty,0)$, and how it is handled depends on how far the sector $|\arg t| < \pi/p$ reaches (Table 1): for $p>1$ the jump of $DK$ on $(-\infty,0)$ is removed by a Cauchy integral, the Binet-cut scalar $B$; at $p=1$ the reflection formula shows that $D$ has no jump there at all; for $p<1$ the poles of $\Gamma(t)$ enter, and $D$ is used only in the inner sector $|\arg z| < \pi p/2$ and glued to an elementary function outside along two seams. Lemma 6.1 states the result (the constructions for $p\neq 1$ are in Appendix A). The rest of the section (§6.2) is common to all geometries: Szegő functions, the normalisation at infinity, the parametrices away from 0, and the Frame Proposition, the only input of Sections 7–8 from this section.

Two facts about $j$ are used in all three geometries. By T4 (DLMF 5.11.1 with the remainder bound of §5.11(ii), first term $1/(12t)$), for every $\phi<\pi$,

$$|j(t)| \le \frac{\sec^2(\phi/2)}{12|t|}, \qquad |j'(t)| \le \frac{C_\phi}{|t|^2} \qquad (|\arg t|\le\phi,\ |t|\ge 1), \tag{58}$$

the second by Cauchy's estimate on a disc of radius $|t|/C_\phi$ in a slightly larger sector. And for $|t|\le 1$, $\operatorname{Re} t\ge 0$, $t\neq 0$, DLMF 5.5.1 gives $\operatorname{Ln}\Gamma(t) = \operatorname{Ln}\Gamma(1+t) - \ln t$, hence

$$j(t) = -\tfrac12 \ln t - \tfrac12 \ln(2\pi) + O(|t|\log(2/|t|)) \qquad (t\to 0,\ \operatorname{Re} t\ge 0). \tag{59}$$

### 6.1. **The outer scalar.**

**Lemma 6.1** (The outer scalar)**.** *Define*

$$F := \begin{cases} DBK, & p>1,\ \textit{with the Binet-cut scalar } B \textit{ of } (96),\\ DK, & p=1,\\ DB_{\mathrm{in}}Ke^{\chi} \textit{ in } S_{\mathrm{in}},\quad B_{\mathrm{out}}Ke^{\chi} \textit{ in } S_{\mathrm{out}}, & 0<p<1,\ \textit{with } B_{\mathrm{in}}, B_{\mathrm{out}}, \chi \textit{ of } §A.2. \end{cases}$$

*Then $F$ satisfies (O1)–(O5). The band factor is $H_n = (\tilde D B)^{-2}$, $\tilde D^{-2}$, $(\tilde D B_{\mathrm{in}} e^{\chi})^{-2}$ respectively, with $H_n(0^+) = B(0)^{-2}$, $1$, $e^{-2\chi(0)}$. At $p=1$,*

$$D(z) = \frac{e^{-j(t)}}{1-e^{\mp 2\pi i t}} = e^{j(-n/z)} \qquad (z\in\mathbb{C}_\pm), \tag{60}$$

*so $D$ extends analytically to $\mathbb{C}\setminus[0,\infty)$, it is real and positive on $(-\infty,0)$, and $H_n = e^{2j(n/z)}$.*

*Proof.* For $p>1$ and $p<1$ this is Lemma A.1(iv), respectively Lemma A.2(iv). Let $p=1$, so $t = n/z$.

*The identity* (60). Let $z\in\mathbb{C}_+$, so $\arg t\in(-\pi,0)$ and $t\notin\mathbb{R}$. By DLMF 5.5.3 (T4), $1/\Gamma(t) = \Gamma(1-t)\sin(\pi t)/\pi$, and $\sin(\pi t)/(1-e^{-2\pi i t}) = e^{i\pi t}/(2i)$. Hence $D = \sqrt{2\pi}\, t^{t-1/2}e^{-t}\Gamma(1-t)e^{i\pi t}/(2\pi i)$. Put $u := -t = -n/z$; then $\arg u = \arg t + \pi \in (0,\pi)$, so $\ln u = \ln t + i\pi$ for the principal branches, and $t^{t-1/2}e^{i\pi t} = \exp((t-\frac12)(\ln u - i\pi) + i\pi t) = e^{i\pi/2}u^{-u-1/2}$. So $D = u^{-u-1/2}e^{u}\Gamma(1+u)/\sqrt{2\pi}$, and DLMF 5.5.1 ($\Gamma(1+u) = u\Gamma(u)$) together with the definition of $j$ gives $D = e^{j(u)}$. For $z\in\mathbb{C}_-$ the same computation with $\sin(\pi t)/(1-e^{2\pi i t}) = -e^{-i\pi t}/(2i)$ and $\ln u = \ln t - i\pi$ gives the same result. Since $j$ is analytic on $\mathbb{C}\setminus(-\infty,0]$ and $u\in(-\infty,0]$ exactly when $z\in(0,\infty)$, $e^{j(-n/z)}$ is analytic on $\mathbb{C}\setminus[0,\infty)$; on $(-\infty,0)$, $u>0$ and $j(u)$ is real.

*(O1).* $F = DK$ is analytic and zero-free off $[0,\infty)$, and $F_+/F_- = (-e^{2i\theta})(-1)$ on $(4c,\infty)$ off the nodes.

*(O2).* $\tilde F = F/\kappa_\pm = \tilde D K$ with $\tilde D = e^{-j(t)}$ analytic and zero-free on $\mathbb{C}\setminus(-\infty,0]$. So $H_n = \tilde D^{-2} = e^{2j(n/z)} > 0$ on $(0,\infty)$, and $H_n(0^+) = 1$.

| | $p>1$ | $p=1$ | $0<p<1$ |
|---|---|---|---|
| $\arg t$ on $\mathbb{C}\setminus(-\infty,0]$ | $(-\pi/p,\pi/p)\subset(-\pi,\pi)$ | $(-\pi,\pi)$, open | wider than $(-\pi,\pi)$; poles of $\Gamma(t)$ on $\arg z=\pm\pi p$ |
| where $\tilde D=e^{-j(t)}$ is used | slit plane | $\lvert\arg z\rvert<\vartheta_{\mathrm w}$ | inner sector $S_{\mathrm{in}}=\{\lvert\arg z\rvert<\pi p/2\}$ |
| outer scalar $F$ (Lemma 6.1) | $DBK$, Binet-cut scalar $B$ | $DK$, $D=e^{j(-n/z)}$, no cut | $DB_{\mathrm{in}}Ke^{\chi}$ in $S_{\mathrm{in}}$, $B_{\mathrm{out}}Ke^{\chi}$ in $S_{\mathrm{out}}$ |
| $\lvert\log\frac{H_n(z)}{H_n(0^+)}\rvert$ near 0 | $\le C\lvert z\rvert^{1/p}$ | $\le C\lvert z\rvert$ | $\le C\lvert z\rvert(1+\log\frac{1}{\lvert z\rvert})$ |
| lip angle $\vartheta_{\mathrm{lip}}$ (default); Bessel rays $\pm(\pi-\vartheta_{\mathrm{lip}})$ | $\pi/3$ | $\pi/3$ | $\pi p/4$ |
| angle $\vartheta_{\mathrm w}$ (default; weight evaluated) | $7\pi/16$ | $7\pi/16$ | $3\pi p/8$ |
| super-small entries $\lvert e^{\mp2i\theta}\rvert$ on the rays $\arg z=\pm\vartheta_{\mathrm{lip}}$ | $e^{-2\pi n\sin(\vartheta_{\mathrm{lip}}/p)\lvert z\rvert^{-1/p}}$; default angles: $e^{-2\pi n\sin(\frac{\pi}{3p})\lvert z\rvert^{-1/p}}$, $e^{-\sqrt3\pi n/\lvert z\rvert}$, $e^{-\sqrt2\pi n\lvert z\rvert^{-1/p}}$ | | |
| strip $\lvert\Im\xi\rvert\le\vartheta$ for $\ell_1$ | any $\vartheta>0$; default angles if $\vartheta\ge\frac{7\pi}{16p}$ | any $\vartheta>0$; default angles if $\vartheta\ge\frac{7\pi}{16}$ | any $\vartheta>0$; default angles if $\vartheta\ge\frac{3\pi}{8}$ |
| Hölder exponent $\nu_p$ (§5.9) | $\min(1/p,\frac12)$ | $\frac12$ | $\frac12$ |
| classical quotient parametrix at 0 | works for pure powers (not used) | fails (Remark 6.2) | not used |

TABLE 1. The ingredients of the Riemann–Hilbert analysis that depend on the geometry. Everything else in Sections 5–8 is the same for all $p>0$. The angles $\vartheta_{\mathrm{lip}}$ and $\vartheta_{\mathrm w}$ are the default values ($\mathsf m=1$ in (33)); for a thinner strip in ($\mathrm A_\vartheta$) both are multiplied by $\mathsf m=p\vartheta/\vartheta_{\mathrm w}^0<1$ (Remark 5.8). In the row for $H_n$, only the weaker bound $C\lvert z\rvert^{\nu_p}$ of (O3) is used.

*(O3), (O4).* On $\mathcal Z_n$, $\lvert\arg t\rvert=\lvert\arg z\rvert<\vartheta_{\mathrm w}<\pi/2$. For $\lvert z\rvert\le1$, $\lvert t\rvert\ge n$ and (58) gives $\lvert\log H_n\rvert=2\lvert j(t)\rvert\le C\lvert z\rvert/n$ and $\lvert(\log H_n)'\rvert\le C/n$; since $\nu_1=\frac12$, this is (O3). For $\lvert z\rvert\ge1$, $\lvert t\rvert\ge e^{-c_0}$ and $j(t)$ is bounded, which is (O4).

*(O5).* Fix $n$. Near 0 outside the lens, $u=-n/z$ has $\lvert u\rvert\to\infty$ and $\lvert\arg u\rvert=\pi-\lvert\arg z\rvert\le\pi-\frac34\vartheta_{\mathrm{lip}}$, so $j(u)\to0$ by (58) and $D\to1$; $K$ is bounded. In the lens $\tilde F=\tilde DK$ with $\tilde D\to1$. For $\lvert z\rvert\ge n+\delta'$ we have $\lvert u\rvert\le n/(n+\delta')<1$ and $\lvert1+u\rvert\ge\delta'/(2n)$, and $D=u^{-u-1/2}e^u\Gamma(1+u)/\sqrt{2\pi}$ is bounded above and below by $C_n\lvert z\rvert^{\pm1}$. Along $z=iy$, $D(iy)=(-iy/n)^{1/2}(2\pi)^{-1/2}(1+o(1))$ by (59), and $K(iy)=(-iy)^{-1/2}(1+o(1))$, so $F(iy)\to(2\pi n)^{-1/2}$. □

At $p=1$ the cut of $D$ disappears, so $p=1$ is a simplification of the case $p>1$. It does, however, break the classical construction of the hard-edge parametrix.

*Remark* 6.2 (The classical hard-edge parametrix at $p=1$). For pure powers and $p\ne1$ one can also build the hard-edge parametrix in the classical way (T2), from a square root $\tilde W$ of the band weight continued to a full neighbourhood of 0; for $p>1$ its jump on $(-\delta,0)$ is super-exponentially close to $I$, since $\lvert e^{\mp2i\theta}\rvert=\exp(-2\pi n\sin(\pi/p)\lvert x\rvert^{-1/p})$ there. At $p=1$ this construction fails, already for $\ell_1\equiv1$: on $(-\delta,0)$, $\theta=\pi n/x$ is real, the continuations of the

lens scalar from the two half-planes differ by the unimodular factor $\tilde F_+/\tilde F_- = -e^{-2i\theta}$, which is not close to 1, and the continuation of $H_n = e^{2j(n/z)}$ has double poles at $z = -n/m$, $m \in \mathbb{N}$, accumulating at 0. Our route never continues anything across $(-\delta, 0)$: neither $V$ nor $Q$ has a jump there, and the comparison parametrix of Section 7 uses only $D_W$. (The obstruction concerns this classical construction, not every quotient construction.)

6.2. **Common steps: Szegő functions, normalisation, parametrices.** From now on $F$ is the outer scalar of Lemma 6.1. Everything in this subsection uses only (O1)–(O5).

*Szegő functions.* Let $\mathcal{R}(z) := (z(z-4c))^{1/2}$, analytic off $[0, 4c]$ with $\mathcal{R}(z) \sim z$ at $\infty$, so that $\mathcal{R}_\pm(x) = \pm i\sqrt{x(4c-x)}$ on $(0, 4c)$. Let $q$ be a function on $(0, 4c)$ that is a non-zero constant multiple of a positive function $q_0$ with $\log q_0 \in L^1(\mathrm{d}x/\sqrt{x(4c-x)})$, and fix a value of the logarithm of the constant. Put

$$D_q(z) := \exp\Big(\frac{\mathcal{R}(z)}{2\pi}\int_0^{4c} \frac{\log q(x)}{\sqrt{x(4c-x)}}\frac{\mathrm{d}x}{z-x}\Big), \qquad D_q(\infty) = \exp\Big(\frac{1}{2\pi}\int_0^{4c}\frac{\log q(x)\,\mathrm{d}x}{\sqrt{x(4c-x)}}\Big). \tag{61}$$

$D_q$ is analytic and zero-free off $[0, 4c]$, $D_{q_1q_2} = D_{q_1}D_{q_2}$, and $D_q = q^{1/2}$ for a constant $q$. Where $\log q_0$ is Hölder continuous, the Sokhotski–Plemelj formula and $\mathcal{R}_+ = -\mathcal{R}_-$ give $D_{q,+}D_{q,-} = q$.

We shall estimate $D_q^2/q$ near the band by means of the following identity. Since

$$\frac{1}{\pi}\int_0^{4c}\frac{\mathrm{d}x}{\sqrt{x(4c-x)}\,(z-x)} = \frac{1}{\mathcal{R}(z)}$$

(the Stieltjes transform of the arcsine law, as in the proof of Lemma 5.4), we have for every $z \notin [0, 4c]$ and every number $\mathfrak{c}$

$$\log D_q(z)^2 - \mathfrak{c} = \frac{\mathcal{R}(z)}{\pi}\int_0^{4c}\frac{(\log q(x) - \mathfrak{c})\,\mathrm{d}x}{\sqrt{x(4c-x)}\,(z-x)}. \tag{62}$$

We use it with $\mathfrak{c} = \log q(z)$, the value of an analytic continuation of $\log q$. Put $\eta_* := \min(1, 2c)$ and $Z(r) := \{z : \ 0 < |z| < r, \ |\arg z| < \vartheta_{\mathrm{w}}\}$.

*The band weight and its Szegő function.* Let $W_0$ be the band weight (55), $D_W := D_{W_0}$ and $D_\infty := D_W(\infty)$. Writing $H_n^\circ := H_n/H_n(0^+)$,

$$D_W = \big(\mathsf{w}_n H_n(0^+)\big)^{1/2} D_{x^\alpha} D_{4c-x} D_{H_n^\circ} D_{r_n}. \tag{63}$$

All factors are defined: by (O3)–(O4), $\log H_n^\circ$ is bounded on $(0, 4c)$, and by Lemma 5.9(iii), $|\log r_n(x)| \le C\varepsilon_\Lambda(T)(1 + |\log x|)$. The constant $\mathsf{w}_n H_n(0^+)$ cancels in the ratio

$$\omega(z) := \frac{D_W(z)^2}{W_0(z)} = \frac{D_{x^\alpha}^2}{z^\alpha}\cdot\frac{D_{4c-x}^2}{4c-z}\cdot\frac{D_{H_n^\circ}^2}{H_n^\circ}\cdot\frac{D_{r_n}^2}{r_n}, \tag{64}$$

defined wherever $W_0$ is, i.e. on $\mathcal{Z}_n \setminus [0, 4c]$. In particular no bound on $H_n(0^+)$ is ever needed.

**Lemma 6.3** (Szegő function estimates)**.** *The constants below depend only on $c$, the stated $\mathfrak{M}$, and $\varepsilon_1$ in (iv).*

(i) *Let $\Upsilon(z) := (z^{1/2} + (z-4c)^{1/2})^2 = 2z - 4c + 2\mathcal{R}(z)$, with principal roots. Then, with principal powers, $D_{x^\alpha}(z)^2 = (4cz/\Upsilon(z))^\alpha$ and $D_{4c-x}(z)^2 = 4c(z-4c)/\Upsilon(z)$. $\Upsilon$ is analytic and zero-free off $[0, 4c]$, with continuous non-zero boundary values, $\Upsilon(z) \sim 4z$ at $\infty$, $\Upsilon(0) = -4c$, $\Upsilon(4c) = 4c$, and $\Upsilon(z) = -4c(1 + O(|z|^{1/2}))$ as $z \to 0$. In particular $|D_{x^\alpha}(z)^2| \asymp |z|^\alpha$ and $|D_{4c-x}(z)^2| \asymp |z - 4c|$ for $|z| \le 8c$, $z \notin [0, 4c]$.*

(ii) *(Hölder type at 0.) Let* $\nu \in (0,1]$, $\mathfrak{M} \geq 0$, $\mathfrak{c}_0 \in \mathbb{C}$. *Suppose* $\log q$ *continues analytically to* $Z(\eta_*)$ *and continuously to* $(0,\eta_*)$, *with* $|\log q(z) - \mathfrak{c}_0| \leq \mathfrak{M}|z|^\nu$ *and* $|(\log q)'(z)| \leq \mathfrak{M}|z|^{\nu-1}$ *there, and* $|\log q(x) - \mathfrak{c}_0| \leq \mathfrak{M}(1 + |\log(4c-x)|)$ *for* $\frac{1}{2}\eta_* \leq x < 4c$. *Then for* $z \in Z(\frac{1}{4}\eta_*) \setminus (0,\infty)$,

$$|\log D_q(z)^2 - \log q(z)| \leq C\mathfrak{M}|z|^{\min(\nu,1/2)} \log(2/|z|).$$

(iii) *(Logarithmic type at 0.) Suppose* $\log q$ *continues analytically to* $Z(\eta_*)$ *and continuously to* $(0,\eta_*)$, *with* $|(\log q)'(z)| \leq \mathfrak{M}/|z|$ *there and* $|\log q(x) - \log q(z)| \leq \mathfrak{M}(1 + |\log(x/|z|)|)$ *for* $z \in Z(\eta_*)$, $x \in (0,4c)$. *Then* $|\log D_q(z)^2 - \log q(z)| \leq C\mathfrak{M}$ *for* $z \in Z(\frac{1}{4}\eta_*) \setminus (0,\infty)$.

(iv) *(Near 4c.) Suppose* $\log q$ *continues analytically to* $U(4c, 2\varepsilon_1)$, *with* $|\log q - \mathfrak{c}_1| \leq \mathfrak{M}$ *there and* $|\log q(x) - \mathfrak{c}_1| \leq \mathfrak{M}(1 + |\log x|)$ *on* $(0,4c)$. *Then* $|\log D_q(z)^2 - \log q(z)| \leq C\mathfrak{M} \min(1, |z-4c|^{1/2})$ *for* $z \in U(4c,\varepsilon_1) \setminus [0,4c]$.

(v) *(Away from the band.) If* $|\log q(x) - \mathfrak{c}_1| \leq \mathfrak{M}(1 + |\log x| + |\log(4c-x)|)$ *on* $(0,4c)$, *then* $|\log D_q(z)^2 - \mathfrak{c}_1| \leq C_d\mathfrak{M}$ *for* $\operatorname{dist}(z,[0,4c]) \geq d$.

(vi) *(The band weight.) For* $n \geq n_0$, *with* $C$ *independent of* $n$: *(a)* $|\omega(z)|^{\pm 1} \leq C$ *for* $z \in Z(\frac{1}{4}\eta_*) \setminus (0,\infty)$, *in particular on the lips in* $U(0,\delta)$; *(b) in* $U(4c,2\varepsilon)$,

$$-\omega = \frac{4c}{\Upsilon}\, e^{L_1}, \qquad L_1 := \log \frac{D_{q_1}^2}{q_1}, \quad q_1 := x^\alpha H_n^\circ r_n, \tag{65}$$

*where* $L_1$ *is given by* (62) *with* $\mathfrak{c} = \log q_1(z)$; $L_1$ *is analytic in* $U(4c,2\varepsilon) \setminus [0,4c]$, *real on* $(4c, 4c+2\varepsilon)$, *satisfies* $L_{1,+} + L_{1,-} = 0$ *on* $(4c-2\varepsilon, 4c)$ *and* $|L_1(z)| \leq C\min(1, |z-4c|^{1/2})$; *in particular* $|\omega(z)|^{\pm 1} \leq C$ *there; (c)* $|\log \omega(z)| \leq C_d\big(1 + \log(1+|z|)\big)^2$ *for* $z \in \mathcal{Z}_n$ *with* $\operatorname{dist}(z,[0,4c]) \geq d$. *Moreover, for each fixed* $n$, $|D_W(z)|^{\pm 1} \leq C_{n,\eta}|z|^{\pm\alpha/2-\eta}$ *for every* $\eta > 0$ *and* $0 < |z| < \frac{1}{4}\eta_*$, $z \notin [0,\infty)$.

*Proof.* Part (i) identifies two explicit functions with the right jumps. Parts (ii)–(v) use the representation (62) and a splitting of $(0,4c)$ into the near zone $|x-z| < \frac{1}{2}|z|$ (or a neighbourhood of $4c$) and the rest. Part (vi) applies (i)–(v) to the four factors of (64).

(i) Write $\Upsilon = z(1+\mathfrak{r})^2$ with $\mathfrak{r} := \mathcal{R}(z)/z = \sqrt{1-4c/z}$, which has $\operatorname{Re}\mathfrak{r} > 0$ off $[0,4c]$. Then $4cz/\Upsilon = 4c/(1+\mathfrak{r})^2$ and $4c(z-4c)/\Upsilon = 4c\,(\mathfrak{r}/(1+\mathfrak{r}))^2$; both $1/(1+\mathfrak{r})$ and $\mathfrak{r}/(1+\mathfrak{r})$ lie in the disc $|\cdot - \frac{1}{2}| < \frac{1}{2}$, so their squares avoid $(-\infty,0]$ and the principal powers are analytic off $[0,4c]$. On the band, $\Upsilon_\pm = (\sqrt{x} \pm i\sqrt{4c-x})^2$, so $\Upsilon_+\Upsilon_- = 16c^2$, the two boundary values of each expression are complex conjugates, and the product of their principal powers is $x^{2\alpha}$, respectively $(4c-x)^2$. The expressions tend to $c$ at $\infty$. Hence they are the Szegő functions squared (the quotient of two functions with the same jump, bounded at $\infty$ and of at most logarithmic growth at 0 and $4c$, is constant). The values and the expansion of $\Upsilon$ at 0 and $4c$ are read off from $\Upsilon = (z^{1/2} + (z-4c)^{1/2})^2$.

(ii) Fix $z \in Z(\frac{1}{4}\eta_*) \setminus (0,\infty)$ and use (62) with $\mathfrak{c} = \log q(z)$. Note $|\mathcal{R}(z)| \leq C|z|^{1/2}$. Split $(0,4c)$ into $\mathcal{U}_z := \{x : |x-z| < \frac{1}{2}|z|\}$ and the rest. For $x \in \mathcal{U}_z$ the segment $[x,z]$ lies in $Z(\eta_*)$ (the sector is convex since $\vartheta_{\mathrm{w}} < \pi/2$) and at distance $\geq \frac{1}{2}|z|$ from 0, so $|\log q(x) - \log q(z)| \leq \mathfrak{M}(|z|/2)^{\nu-1}|x-z|$, and the contribution is at most $C|z|^{1/2}\mathfrak{M}|z|^{\nu-1}\int_{\mathcal{U}_z} x^{-1/2}\,\mathrm{d}x \leq C\mathfrak{M}|z|^\nu$. Off $\mathcal{U}_z$, $|x-z| \geq \frac{1}{8}(x+|z|)$ (if $x \leq 3|z|$ then $|x-z| \geq \frac{1}{2}|z|$; otherwise $|x-z| \geq x - |z| \geq \frac{1}{2}x$). On $(0, \frac{1}{2}\eta_*)$ we have $|\log q(x) - \log q(z)| \leq \mathfrak{M}(x^\nu + |z|^\nu)$, and

$$|z|^{1/2} \int_0^{\eta_*/2} \frac{\mathfrak{M}(x^\nu + |z|^\nu)\,\mathrm{d}x}{x^{1/2}(x+|z|)} \leq C\mathfrak{M}|z|^{\min(\nu,1/2)} \log(2/|z|),$$

by the substitution $x = |z|u$: the term $|z|^\nu$ gives $C\mathfrak{M}|z|^\nu$; the term $x^\nu$ gives $C\mathfrak{M}|z|^\nu$ if $\nu < \frac12$, $C\mathfrak{M}|z|^{1/2}\log(2/|z|)$ if $\nu = \frac12$, and $C\mathfrak{M}|z|^{1/2}$ if $\nu > \frac12$. On $[\frac12\eta_*, 4c)$, $|x-z| \ge \frac14\eta_*$ and $\int(1+|\log(4c-x)|)(4c-x)^{-1/2}\,\mathrm{d}x < \infty$ give $C\mathfrak{M}|z|^{1/2}$.

(iii) The same splitting. On $\mathcal{U}_z$ the derivative bound gives $|\log q(x) - \log q(z)| \le 2\mathfrak{M}|x-z|/|z|$ and a contribution $C\mathfrak{M}$. Off $\mathcal{U}_z$ the contribution is at most $C\mathfrak{M}|z|^{1/2}\int_0^{4c} \frac{(1+|\log(x/|z|)|)\,\mathrm{d}x}{\sqrt{x(4c-x)}\,(x+|z|)}$, and the substitution $x = |z|u$ bounds the part $x \le 2c$ by $C\mathfrak{M}\int_0^\infty (1+|\log u|)u^{-1/2}(1+u)^{-1}\,\mathrm{d}u$; the part $x \ge 2c$ is $O(\mathfrak{M}|z|^{1/2}\log(2/|z|))$.

(iv) By Cauchy's estimate $|(\log q)'| \le 2\mathfrak{M}/\varepsilon_1$ on $U(4c, \frac32\varepsilon_1)$. For $x \in (4c - \frac32\varepsilon_1, 4c)$ and $z \in U(4c,\varepsilon_1)$, $|\log q(x) - \log q(z)| \le C\mathfrak{M}|x-z|$, which cancels the kernel's singularity, and $\int(4c-x)^{-1/2}\,\mathrm{d}x < \infty$. For $x \le 4c - \frac32\varepsilon_1$, $|z-x| \ge \frac12\varepsilon_1$ and the integrand is $\le C\mathfrak{M}(1+|\log x|)x^{-1/2}$. So the integral in (62) is at most $C\mathfrak{M}$, and the prefactor satisfies $|\mathcal{R}(z)| \le C\min(1, |z-4c|^{1/2})$.

(v) In (62) with $\mathfrak{c} = \mathfrak{c}_1$, $|\mathcal{R}(z)/(z-x)| \le C_d$ for all $x \in [0,4c]$ when $\operatorname{dist}(z,[0,4c]) \ge d$, and the remaining integral is finite.

(vi)(a) Use the four factors of (64). The first two are bounded above and below by (i), since $z^\alpha$ and $(4cz/\Upsilon)^\alpha$ differ by a bounded factor $(4c/\Upsilon)^\alpha e^{2\pi i k\alpha}$. The third is bounded by (ii) with $q = H_n^\circ$, $\mathfrak{c}_0 = 0$, $\nu = \nu_p$ and $\mathfrak{M} = C$ from (O3)–(O4). The fourth is bounded by (iii) with $q = r_n$: by Lemma 5.9(iii) with $R = \max(4c,1)$, the hypotheses hold with $\mathfrak{M} = C\varepsilon_\Lambda(\log n - \frac1p\log\max(4c,1)) \le C$, and $|(\log r_n)'(z)| = |\Lambda'(\xi)|/(p|z|)$. (b) By (i), $D^2_{4c-x}/(4c-z) = -4c/\Upsilon$, and the three other factors of (64) combine, by multiplicativity, into $D^2_{q_1}/q_1$; this is (65). Near $4c$, $H_n^\circ$ and $r_n$ are analytic and positive on the real axis, with $|\log H_n^\circ| \le C$ and $|\log r_n| \le C$ (O4, Lemma 5.9(iii)), so (iv) applies to $q_1$ with $\mathfrak{M} = C$. The boundary values of the exponent in (61) add up to $\log q_1$ on the band (this is $D_{q_1,+}D_{q_1,-} = q_1$), so $L_{1,+} + L_{1,-} = 2\log q_1 - 2\log q_1 = 0$; and $L_1$ is real on $(4c, 4c+2\varepsilon)$ because $D_{q_1}$ and $q_1$ are positive there. (c) (v) with $q = W_0/(\mathsf{w}_n H_n(0^+))$ bounds $\log D_W^2 - \log(\mathsf{w}_n H_n(0^+))$, and $|\log(W_0(z)/(\mathsf{w}_n H_n(0^+)))| \le C(1+\log(1+|z|))^2$ on $\mathcal{Z}_n$ by (O3), (O4) and Lemma 5.9(iii) with $R = e^{pc_0}n^p$ (here $|\log r_n(z)| \le \frac1p\varepsilon_\Lambda(-c_0)(\vartheta_{\mathrm{w}} + |\log|z||)$). For the last claim, fix $n$ and $\eta > 0$. By Lemma 5.9(i), $|\log r_n(x)| \le C_{n,\eta} + \eta|\log x|$ for $x \in (0,4c)$, and $|r_n(z)|^{\pm1} \le C_{n,\eta}|z|^{-\eta}$ for $z \in \mathcal{Z}_n$. For $|\arg z| < \vartheta_{\mathrm{w}}$ the claim follows from (a) and $|W_0| \asymp |z|^\alpha |r_n|$. For $|\arg z| \ge \frac12\vartheta_{\mathrm{lip}}$ we have $|x-z| \ge (x+|z|)/C$ for $x > 0$, and (62) with $\mathfrak{c} = 0$ gives $|\log D_{H_n^\circ}(z)| \le C$ and

$$|\log D_{r_n}(z)^2| \le C|z|^{1/2}\int_0^{4c} \frac{C_{n,\eta} + \eta|\log x|}{\sqrt{x(4c-x)}(x+|z|)}\,\mathrm{d}x \le C_{n,\eta} + C\eta\log\frac{1}{|z|};$$

together with (i) for $D_{x^\alpha}$ and $D_{4c-x}$ this gives the claim (after renaming $\eta$). □

*The global parametrix.* Let $a(z) := ((z-4c)/z)^{1/4}$, analytic off $[0,4c]$ with $a(\infty) = 1$, and

$$P_0^\infty(z) := \begin{pmatrix} \frac{a+a^{-1}}{2} & \frac{a-a^{-1}}{2i} \\ -\frac{a-a^{-1}}{2i} & \frac{a+a^{-1}}{2} \end{pmatrix}, \qquad P^\infty := D_\infty^{\sigma_3} P_0^\infty D_W^{-\sigma_3}. \tag{66}$$

Then $\det P_0^\infty = 1$, $P_{0,+}^\infty = P_{0,-}^\infty J$ on $(0,4c)$ with $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ (because $a_+ = ia_-$ there), and $P_0^\infty(z) = I + O(1/z)$. Since $|a(z)|^{\pm1} = |z-4c|^{\pm1/4}|z|^{\mp1/4}$,

$$|P_0^\infty(z)| \le C|z|^{-1/4}|z-4c|^{-1/4}(1+|z|)^{1/2}. \tag{67}$$

As $D_{W,+}D_{W,-} = W_0$, $P^\infty$ has the band jump of $V$ (Lemma 5.10(a)). We put

$$Q := D_\infty^{-\sigma_3} V D_W^{\sigma_3}; \tag{68}$$

in the frame of $Q$ the global parametrix is $P_0^\infty$, which does not involve the weight. A matrix $X$ in the frame of $V$ corresponds to $D_\infty^{-\sigma_3} X D_W^{\sigma_3}$ in the frame of $Q$.

**Lemma 6.4** (Normalisation at infinity)**.** *Let* $\varkappa_\infty(z) := F(z)^{-1}e^{n(g(z)-\operatorname{Log} z)}$ *for* $|z| > n^p + \delta'$, $z \notin \mathbb{R}$. *Then* $\varkappa_\infty$ *extends to an analytic function on* $\{|z| > n^p + \delta'\} \cup \{\infty\}$ *with* $\varkappa_\infty(\infty) \neq 0$. *With* $C_\infty := \varkappa_\infty(\infty)^{\sigma_3}$ *in* (51), $V(z) = I + O(1/z)$ *and* $Q(z) = I + O(1/z)$ *as* $z \to \infty$, *and for* $|z| > n^p + \delta' + h_0$

$$Q(z) = k_\infty^{-\sigma_3}\big(U(z)z^{-n\sigma_3}\big)k(z)^{\sigma_3}, \qquad k := \frac{e^{n\gamma_p}D_W}{\varkappa_\infty}, \tag{69}$$

*where* $k$ *is analytic at* $\infty$ *with* $k_\infty := k(\infty) = e^{n\gamma_p}D_\infty/\varkappa_\infty(\infty) \neq 0$.

*Proof.* On $(-\infty, -n^p - \delta')$, $g$ and $\operatorname{Log} z$ both jump by $2\pi i$ and $F$ has no jump (O1). On $(n^p + \delta', \infty)$, $e^{n(g-\operatorname{Log} z)}$ jumps by $e^{2\pi i n\mu^*((x,\infty))} = e^{2i\theta}$ (Lemma 5.4(i)) and so does $F$ (O1). So $\varkappa_\infty$ has continuous boundary values that agree, and is analytic in $\{|z| > n^p + \delta'\}$ by Morera's theorem. By Lemma 5.4(ii) and (O5), $|\varkappa_\infty(z)| \le C_n|z|^{N_n}$, so $\infty$ is at most a pole; since $\varkappa_\infty(iy)$ is bounded and bounded away from 0 as $y \to \pm\infty$, it is a removable singularity and $\varkappa_\infty(\infty) \neq 0$.

In $\Omega_\infty$, by (52), $V = C_\infty e^{-n\gamma_p\sigma_3}(Uz^{-n\sigma_3})(z^n e^{n\varphi}F)^{\sigma_3}$, and $z^n e^{n\varphi}F = e^{n\gamma_p}e^{-n(g-\operatorname{Log} z)}F = e^{n\gamma_p}/\varkappa_\infty$. Hence $V = \varkappa_\infty(\infty)^{\sigma_3}e^{-n\gamma_p\sigma_3}(Uz^{-n\sigma_3})e^{n\gamma_p\sigma_3}\varkappa_\infty^{-\sigma_3} = I + O(1/z)$, by Lemma 5.1. Multiplying by $D_\infty^{-\sigma_3}$ on the left and $D_W^{\sigma_3}$ on the right gives (69), and $D_W(z)/D_\infty = 1 + O(1/z)$ gives $Q = I + O(1/z)$; $D_W$ is analytic at $\infty$ because the exponent in (61) is. □

**Lemma 6.5** (Lip exponentials and super-small entries)**.** *Let* $s_p := \sin(\frac{3}{4}\vartheta_{\mathrm{lip}}/p) > 0$. *For* $n \ge n_0$ *and* $z \in \Sigma_{\mathrm{lip}}^{\pm} \cap U(0,\delta)$*:*

(a) $|\kappa_\pm(z) - 1| \le 2\exp(-2\pi s_p n|z|^{-1/p})$;
(b) $|e^{-2n\tilde\varphi(z)}| \le \exp(-\pi s_p n|z|^{-1/p})$;
(c) $|e^{2n\varphi(z)}| \le \exp\big(-2\sin(\vartheta_{\mathrm{lip}}/2)\, n|f(z)|^{1/2}\big) \le e^{-n|z|^{1/2}/C}$.

*Proof.* On these lips $\frac{3}{4}\vartheta_{\mathrm{lip}} \le |\arg z| \le \frac{5}{4}\vartheta_{\mathrm{lip}}$ (Lemma 5.6), and $\frac{5}{4}\vartheta_{\mathrm{lip}}/p < \pi/2$ in all cases, so $\sin(|\arg z|/p) \ge s_p$. (a) follows from (41) and $|\kappa_\pm - 1| = |e^{\mp 2i\theta}|/|1 - e^{\mp 2i\theta}|$. (b) $e^{-2n\tilde\varphi} = e^{-2n\varphi}e^{\mp 2i\theta}$ and $|\operatorname{Re}\varphi| \le C|z|^{1/2}$ (Lemma 5.5(iii)), so $|e^{-2n\tilde\varphi}| \le \exp(2Cn|z|^{1/2} - 2\pi s_p n|z|^{-1/p})$, and $2C|z|^{1/2} \le \pi s_p|z|^{-1/p}$ for $|z| \le \delta$ small. (c) By Lemma 5.5(iii), $\operatorname{Re}\varphi = -2\operatorname{Re} f^{1/2}$, and on the lip $\arg f = \mp(\pi - \vartheta_{\mathrm{lip}})$, so $\operatorname{Re} f^{1/2} = |f|^{1/2}\sin(\vartheta_{\mathrm{lip}}/2)$; finally $|f(z)| \ge |z|/C$ on $U(0,\delta)$. □

*The soft edge.* In $U(4c,\varepsilon)$ the jumps of $Q$ are, by Lemma 5.10(a), (c), (d) and the conjugation by $D_W^{\sigma_3}$,

$$J \text{ on } (4c-\varepsilon, 4c), \qquad \begin{pmatrix} 1 & 0 \\ \omega e^{2n\tilde\varphi} & 1 \end{pmatrix} \text{ on } (4c, 4c+\varepsilon), \qquad \begin{pmatrix} 1 & -\omega^{-1}e^{-2n\tilde\varphi} \\ 0 & 1 \end{pmatrix} \text{ on } \Sigma_v^{\pm} \cap U(4c,\varepsilon), \tag{70}$$

the first two oriented from left to right and the last away from the real axis. This is the jump structure of [19, (6.4)]. We construct the local solution from Airy functions, in our variable and with our orientations, following the pattern of [19, (6.10)–(6.12)] (T1(c)); the facts about Ai that we use are those listed in T4.

Let $\mathsf{q} := e^{2\pi i/3}$ and define the vector functions

$$y_0(\zeta) := \begin{pmatrix} \mathrm{Ai}(\zeta) \\ \mathrm{Ai}'(\zeta) \end{pmatrix}, \qquad y_\pm(\zeta) := \begin{pmatrix} \mathrm{Ai}(\mathsf{q}^{\mp 1}\zeta) \\ \mathsf{q}^{\mp 1}\mathrm{Ai}'(\mathsf{q}^{\mp 1}\zeta) \end{pmatrix}.$$

By (9.2.12) of T4 and its derivative, $y_0 + \mathsf{q}^{-1}y_+ + \mathsf{q}\,y_- = 0$. With $k_{\mathrm{Ai}} := \sqrt{2\pi}\,e^{i\pi/12}$ and $M = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & i \\ i & 1 \end{smallmatrix}\right)$ (the matrix of T2(c)), put

$$\begin{aligned} \Psi^{\mathrm{I}} &:= k_{\mathrm{Ai}} e^{-i\pi\sigma_3/4}\big(y_+,\ \mathsf{q}\,y_0\big), & \Psi^{\mathrm{II}} &:= k_{\mathrm{Ai}} e^{-i\pi\sigma_3/4}\big(y_+,\ -\mathsf{q}^{-1}y_-\big), \\ \Psi^{\mathrm{III}} &:= k_{\mathrm{Ai}} e^{-i\pi\sigma_3/4}\big(-\mathsf{q}^{-1}y_-,\ -y_+\big), & \Psi^{\mathrm{IV}} &:= k_{\mathrm{Ai}} e^{-i\pi\sigma_3/4}\big(-\mathsf{q}^{-1}y_-,\ \mathsf{q}\,y_0\big). \end{aligned} \tag{71}$$

These are entire matrix functions of $\zeta$, independent of $n$. The relation $y_0 + \mathsf{q}^{-1}y_+ + \mathsf{q}y_- = 0$ gives

$$\Psi^{\mathrm{II}} = \Psi^{\mathrm{I}}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \Psi^{\mathrm{II}} = \Psi^{\mathrm{III}} J, \quad \Psi^{\mathrm{IV}} = \Psi^{\mathrm{III}}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \Psi^{\mathrm{I}} = \Psi^{\mathrm{IV}}\begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}, \tag{72}$$

and $\det\Psi^{\mathrm{I}} = -k_{\mathrm{Ai}}^2\mathsf{q}\,\mathcal{W}\{\mathrm{Ai}(\zeta), \mathrm{Ai}(\mathsf{q}^{-1}\zeta)\} = -2\pi e^{i\pi/6}\mathsf{q}\,e^{i\pi/6}/(2\pi) = 1$ by (9.2.8) of T4, hence $\det\Psi^j \equiv 1$ for all four. Let $\mathcal{S}_{\mathrm{I}} := \{0 \le \arg\zeta \le \frac{7\pi}{12}\}$, $\mathcal{S}_{\mathrm{II}} := \{\frac{5\pi}{12} \le \arg\zeta \le \pi\}$ and $\mathcal{S}_{\mathrm{III}}, \mathcal{S}_{\mathrm{IV}}$ their mirror images (with $\arg\zeta = \pm\pi$ meaning the boundary value from $\mathbb{C}_\pm$). Then, uniformly for $\zeta \in \mathcal{S}_j$, $|\zeta| \ge 1$,

$$\Psi^j(\zeta) = \zeta^{-\sigma_3/4}\big(M + O(|\zeta|^{-3/2})\big)e^{\frac{2}{3}\zeta^{3/2}\sigma_3}. \tag{73}$$

Indeed, in each $\mathcal{S}_j$ the three arguments $\zeta$, $\mathsf{q}^{\pm1}\zeta$ that occur have $|\mathrm{ph}| \le \frac{11\pi}{12}$, so the Airy asymptotics of T4 apply with relative errors $O(|\zeta|^{-3/2})$. For instance, for $\zeta$ in $\mathcal{S}_{\mathrm{I}} \cup \mathcal{S}_{\mathrm{II}}$ we have $(\mathsf{q}^{-1}\zeta)^{1/4} = e^{-i\pi/6}\zeta^{1/4}$ and $\frac{2}{3}(\mathsf{q}^{-1}\zeta)^{3/2} = -\frac{2}{3}\zeta^{3/2}$, so, entrywise with relative errors $O(|\zeta|^{-3/2})$, $y_+ = \frac{1}{2\sqrt{\pi}}e^{\frac{2}{3}\zeta^{3/2}}\zeta^{-\sigma_3/4}\left(\begin{smallmatrix} e^{i\pi/6} \\ e^{i\pi/6} \end{smallmatrix}\right)$ and $y_0 = \frac{1}{2\sqrt{\pi}}e^{-\frac{2}{3}\zeta^{3/2}}\zeta^{-\sigma_3/4}\left(\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right)$; and $k_{\mathrm{Ai}}e^{-i\pi\sigma_3/4}\frac{1}{2\sqrt{\pi}}\left(\begin{smallmatrix} e^{i\pi/6} & \mathsf{q} \\ e^{i\pi/6} & -\mathsf{q} \end{smallmatrix}\right) = M$. The other columns and sectors are the same computation with $(\mathsf{q}\zeta)^{1/4} = e^{-i\pi/3}\zeta^{1/4}$, $(\mathsf{q}\zeta)^{3/2} = \zeta^{3/2}$ in $\mathcal{S}_{\mathrm{II}}$, and the mirror relations in $\mathcal{S}_{\mathrm{III}} \cup \mathcal{S}_{\mathrm{IV}}$.

The disc $U(4c,\varepsilon)$ minus the contour has four components: $\Omega_+^{\triangle} \cap U(4c,\varepsilon)$, $\Omega_{\mathrm{b}}^+ \cap U(4c,\varepsilon)$, $\Omega_{\mathrm{b}}^- \cap U(4c,\varepsilon)$, $\Omega_-^{\triangle} \cap U(4c,\varepsilon)$; we label them I, II, III, IV. With $f_r$ of Lemma 5.5(iv) put $\zeta := n^{2/3}f_r(z)$ and $\Psi_r(\zeta) := \Psi^j(\zeta)$ for $z$ in component $j$. By Lemma 5.5(iv), $|\arg f_r(z) - \arg(z-4c)| \le \frac{\pi}{12}$, so $\zeta \in \mathcal{S}_j$ when $z$ is in component $j$. By Lemma 6.3(vi)(b) let

$$(-\omega)^{1/2} := \frac{2\sqrt{c}}{z^{1/2} + (z-4c)^{1/2}}\,e^{L_1/2} \qquad (z \in U(4c,2\varepsilon) \setminus [0,4c]), \tag{74}$$

a square root of $-\omega$ (as $4c/\Upsilon = (2\sqrt{c}/(z^{1/2} + (z-4c)^{1/2}))^2$). It is analytic in $U(4c,2\varepsilon) \setminus (4c - 2\varepsilon, 4c]$, positive on $(4c, 4c+2\varepsilon)$, bounded above and below uniformly in $n$, and its boundary values on the band satisfy

$$(-\omega)_+^{1/2}(-\omega)_-^{1/2} = \frac{4c}{(\sqrt{x} + i\sqrt{4c-x})(\sqrt{x} - i\sqrt{4c-x})}\,e^{(L_{1,+}+L_{1,-})/2} = 1. \tag{75}$$

(The square root of $\omega$ itself would not do: $\omega(z) \to -1$ as $z \to 4c$, so a branch of $\omega^{1/2}$ that is continuous across $(4c, 4c+\varepsilon)$ has $\omega_+^{1/2}\omega_-^{1/2} = -1$ on the band.) Define

$$\begin{aligned} \hat{P}_r(z) &:= A_r(z)\,n^{\sigma_3/6}M_{r,n}(z), \qquad A_r := P_0^{\infty}\,(-\omega)^{-\sigma_3/2}M^{-1}f_r^{\sigma_3/4}, \\ M_{r,n}(z) &:= \Psi_r\big(n^{2/3}f_r(z)\big)\big((-\omega)^{1/2}e^{n\tilde{\varphi}}\big)^{\sigma_3}, \end{aligned} \tag{76}$$

with the principal root $f_r^{1/4}$ (cut along the band).

**Lemma 6.6** (Airy parametrix)**.** *For $n \ge n_0$, with constants independent of $n$:*

(i) *$A_r$ extends analytically to $U(4c,\varepsilon)$, $\det A_r = 1$ and $|A_r^{\pm1}| \le C$ there.*

(ii) *$\hat{P}_r$ is analytic in $U(4c,\varepsilon) \setminus \Sigma$, $\det\hat{P}_r = 1$, and $\hat{P}_{r,+} = \hat{P}_{r,-}v$ on $\Sigma \cap U(4c,\varepsilon)$ with the jumps (70).*

(iii) $\hat P_r(z)P_0^\infty(z)^{-1} = I + O(1/n)$ *uniformly for* $\frac12\varepsilon \le |z-4c| < \varepsilon$, $z \notin \Sigma$.

(iv) *(Two scales.) With* $\zeta = n^{2/3} f_r(z)$: *if* $|\zeta| \ge 2$, *then* $M_{r,n}(z) = \zeta^{-\sigma_3/4}\Theta_\infty(z)$ *with* $|\Theta_\infty^{\pm1}| \le C$; *if* $|\zeta| \le 3$, *then* $M_{r,n}(z) = \Psi^{\mathrm I}(\zeta)\Theta_0(z)$ *with* $|\Theta_0^{\pm1}| \le C$, *where* $\Psi^{\mathrm I}$ *is entire, independent of* $n$, *with* $\det \Psi^{\mathrm I} = 1$. *The same holds for the boundary values on* $\Sigma$.

*Proof.* Parts (ii) and (iv) follow from the relations (72) and (73) of the model; part (i) is a cancellation of jumps on the band; part (iii) combines (i) and (73). Put $h := (-\omega)^{1/2} e^{n\tilde\varphi}$. By (74), $h^{-2} = -\omega^{-1}e^{-2n\tilde\varphi}$. On the band, $\tilde\varphi_+ + \tilde\varphi_- = \varphi_+ + \varphi_- = 0$ (Lemma 5.4(i); $\theta$ is analytic across the band), so $h_+h_- = 1$ by (75); on $(4c, 4c+\varepsilon)$ and on $\Sigma_v^\pm$, $h$ has no jump.

(ii) Analyticity off $\Sigma$ and $\det \hat P_r = 1$ are clear ($\det A_r = 1$ is shown in (i)). Since $A_r$ is analytic, the jumps of $\hat P_r$ are those of $M_{r,n}$, and these are $h_-^{-\sigma_3}(\Psi^{j-})^{-1}\Psi^{j+}h_+^{\sigma_3}$, where $j_\pm$ are the components on the two sides. On $\Sigma_v^+$ (oriented upwards, + side II, − side I) and on $\Sigma_v^-$ (oriented downwards, + side IV, − side III), (72) gives $h^{-\sigma_3}\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)h^{\sigma_3} = \left(\begin{smallmatrix}1&h^{-2}\\0&1\end{smallmatrix}\right)$, which is the third jump in (70). On $(4c, 4c+\varepsilon)$ (+ side I, − side IV) we get $h^{-\sigma_3}\left(\begin{smallmatrix}1&0\\-1&1\end{smallmatrix}\right)h^{\sigma_3} = \left(\begin{smallmatrix}1&0\\-h^2&1\end{smallmatrix}\right)$ and $-h^2 = \omega e^{2n\tilde\varphi}$. On the band (+ side II, − side III) we get $h_-^{-\sigma_3}Jh_+^{\sigma_3} = h_+^{\sigma_3}Jh_+^{\sigma_3} = J$, since $h_- = h_+^{-1}$ and $DJD = J$ for every diagonal $D$ with $\det D = 1$.

(i) On $(4c, 4c+\varepsilon)$ all factors of $A_r$ are analytic. On the band, $P_{0,+}^\infty = P_{0,-}^\infty J$; since $f_r$ is negative there and maps $\mathbb C_\pm$ into $\mathbb C_\pm$, $f_{r,+}^{\sigma_3/4} = f_{r,-}^{\sigma_3/4}\mathrm{diag}(i,-i)$; and $(-\omega)_+^{-\sigma_3/2} = (-\omega)_-^{\sigma_3/2}$ by (75). Using $JD = D^{-1}J$ for diagonal $D$ with $\det D = 1$, and the identity $MJM^{-1} = \mathrm{diag}(-i,i)$ (a direct computation),

$$A_{r,+} = P_{0,-}^\infty(-\omega)_-^{-\sigma_3/2}\, JM^{-1}\mathrm{diag}(i,-i)\, f_{r,-}^{\sigma_3/4} = P_{0,-}^\infty(-\omega)_-^{-\sigma_3/2}M^{-1}f_{r,-}^{\sigma_3/4} = A_{r,-}.$$

So $A_r$ is analytic in $U(4c,2\varepsilon)\backslash\{4c\}$. Near $4c$, $|P_0^\infty| \le C|z-4c|^{-1/4}$ by (67), $(-\omega)^{\pm1/2}$ is bounded and $|f_r^{\pm1/4}| \le C|z-4c|^{\pm1/4}$, so $|A_r(z)| \le C|z-4c|^{-1/2}$, and an isolated singularity with this growth is removable (as in T2(c)). Each factor has determinant 1, and $A_r^{-1}$ is the adjugate of $A_r$. On the circle $|z-4c| = \frac32\varepsilon$ all factors are bounded uniformly in $n$ (Lemma 6.3(vi)(b)), so $|A_r^{\pm1}| \le C$ on $U(4c,\varepsilon)$ by the maximum principle.

(iii) For $\frac12\varepsilon \le |z-4c| < \varepsilon$ we have $|\zeta| \ge n^{2/3}\varepsilon/C \ge 2$, and $\frac23\zeta^{3/2} = \frac23 n f_r^{3/2} = -n\tilde\varphi$ (principal powers; $n^{2/3} > 0$). By (73) and $\zeta^{-\sigma_3/4} = n^{-\sigma_3/6} f_r^{-\sigma_3/4}$,

$$\hat P_r(P_0^\infty)^{-1} = P_0^\infty(-\omega)^{-\sigma_3/2}M^{-1}\big(M + O(n^{-1})\big)(-\omega)^{\sigma_3/2}(P_0^\infty)^{-1} = I + O(1/n),$$

since $P_0^\infty$, $(P_0^\infty)^{-1}$ and $(-\omega)^{\pm1/2}$ are bounded there uniformly in $n$.

(iv) For $|\zeta| \ge 2$ put

$$\Theta_\infty := \zeta^{\sigma_3/4}\Psi_r(\zeta)e^{-\frac23\zeta^{3/2}\sigma_3}(-\omega)^{\sigma_3/2}.$$

By (73) it is $(M + O(|\zeta|^{-3/2}))(-\omega)^{\sigma_3/2}$, bounded with bounded inverse for $|\zeta| \ge R_0$; for $2 \le |\zeta| \le R_0$ it is bounded because the $\Psi^j$ are continuous, and so is its inverse, as $\det\Theta_\infty = 1$. For $|\zeta| \le 3$, $\Psi_r = \Psi^{\mathrm I}S_j$ with $S_j$ one of the constant matrices $I$, $\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$, $\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)J^{-1}$, $\left(\begin{smallmatrix}1&0\\1&1\end{smallmatrix}\right)$ by (72); put $\Theta_0 := S_j h^{\sigma_3}$. Here $|e^{\pm n\tilde\varphi}| = |e^{\mp\frac23\zeta^{3/2}}| \le e^{2\sqrt3}$ and $(-\omega)^{\pm1/2}$ is bounded. □

We now collect what Sections 7–8 use.

**Proposition 6.7** (Frame Proposition)**.** *Let* $p > 0$, $1 < s < 1 + 2p$, $\alpha = (s-1-p)/p$, *let* $\ell_1$ *satisfy* $(\mathrm A_\vartheta)$ *for some* $\vartheta > 0$, *let* $\vartheta_{\mathrm w}, \vartheta_{\mathrm{lip}}$ *be the angles* (33), *and let* $F$ *be the outer scalar of Lemma 6.1. The radii* $\delta, \varepsilon$ *and the contour of §5.6 may be taken as small as we wish (but fixed, independent of* $n$*). Let* $Q$ *be the matrix* (68). *There is* $n_0$ *such that for* $n \ge n_0$:

(F1) *(Contour and jumps.)* $Q$ *is analytic in* $\mathbb{C}\setminus\Sigma$, $Q(z)=I+O(1/z)$ *at* $\infty$, $Q$ *has continuous boundary values on* $\Sigma$ *minus its vertices, and* $Q_+=Q_-v$ *with* $\det v=1$, *where* $v=J=\left(\begin{smallmatrix}0&1\\-1&0\end{smallmatrix}\right)$ *on the band* $(0,4c)$ *(oriented from left to right), and on every other arc* $v$ *is the jump* $v_V$ *of Lemma 5.10 (same orientation) with* $W_0$ *replaced by* $\omega^{-1}$, *where* $\omega:=D_W^2/W_0$ *and* $W_0$ *on a lip is the analytic continuation of* $W_0$ *from the band. In particular, on the lips* $\Sigma_{\mathrm{lip}}^{\pm}$ *(oriented away from 0),* $v=\left(\begin{smallmatrix}1&0\\ \omega e^{2n\varphi}&1\end{smallmatrix}\right)+E$ *with* $E=\left(\begin{smallmatrix}\iota_\pm(\kappa_\pm-1)&\omega^{-1}e^{-2n\tilde\varphi}\\ (\kappa_\pm-1)\omega e^{2n\varphi}&\iota_\mp(\kappa_\pm-1)\end{smallmatrix}\right)$ *($\iota_+=1$, $\iota_-=0$); there is no jump on* $(-\infty,0)$, *on* $(n^p+\delta',\infty)$, *or (for* $p<1$*) on the seams.* $|v^{\pm1}|\le C$ *on* $\Sigma$. $\Sigma$ *is a finite union of* $C^2$ *arcs;* $\Sigma\cap(U(0,2\delta)\cup U(4c,2\varepsilon))$ *does not depend on* $n$; *the length of* $\Sigma$ *is* $O(n^p)$; $|\Sigma\cap U(z,\rho)|\le C\rho$ *for all* $z,\rho$; *and* $\|\mathcal{C}_\pm\|_{L^2(\Sigma)\to L^2(\Sigma)}\le C$.

(F2) *(Band weight.)* $W_0=\mathsf{w}_n x^\alpha(4c-x)H_n(x)r_n(x)$ *with* $\mathsf{w}_n\neq0$ *constant,* $r_n(z)=\exp(\Lambda(\log n-\frac1p\log z)-\Lambda(\log n))$, *and* $H_n>0$ *on* $(0,4c+\varepsilon]$. $H_n$ *and* $r_n$ *are analytic on* $\mathcal{Z}_n$, *which contains the closed sectors* $\{0<|z|\le\delta,\ |\arg z|\le\frac54\vartheta_{\mathrm{lip}}\}$ *and a neighbourhood of* $(0,4c+\varepsilon]$, *and, uniformly in* $n$,

$$|\log H_n(z)-\log H_n(0^+)|\le C|z|^{\nu_p},\quad |(\log H_n)'(z)|\le C|z|^{\nu_p-1}\qquad (z\in\mathcal{Z}_n,\ |z|\le1),$$
$$|\log H_n(x)-\log H_n(0^+)|\le C\qquad (0<x\le4c+\varepsilon).$$

*The constant* $\mathsf{w}_nH_n(0^+)$ *cancels in* $\omega=D_W^2/W_0$; $D_W$ *is given by (61) with* $q=W_0$.

(F3) *(Lips.) In* $U(0,\delta)$ *the lips are the preimages under* $\zeta=n^2f(z)$ *of the rays* $\arg\zeta=\mp(\pi-\vartheta_{\mathrm{lip}})$ *(the upper lip is mapped to the lower ray); they leave 0 at the angles* $\pm\vartheta_{\mathrm{lip}}$ *and satisfy* $\frac34\vartheta_{\mathrm{lip}}\le|\arg z|\le\frac54\vartheta_{\mathrm{lip}}$; *the Bessel rays are* $\arg\zeta=\pm(\pi-\vartheta_{\mathrm{lip}})$.

(F4) *(Super-small entries.)* $|E(z)|\le C\exp(-n|z|^{-1/p}/C)$ *on the lips in* $U(0,\delta)$.

(F5) *(Parametrices away from 0.)* $P_0^\infty$ *is given by (66):* $\det P_0^\infty=1$, $P_{0,+}^\infty=P_{0,-}^\infty J$ *on* $(0,4c)$, *and (67) holds. On* $\Sigma\setminus(U(0,\delta)\cup U(4c,\varepsilon)\cup[0,4c])$, $|v-I|\le Ce^{-n/C}$. *In* $U(4c,\varepsilon)$ *the Airy parametrix* $\hat P_r$ *of (76) has exactly the jumps* $v$, $\det\hat P_r=1$, *and* $\hat P_r(P_0^\infty)^{-1}=I+O(1/n)$ *uniformly on* $\frac12\varepsilon\le|z-4c|<\varepsilon$. *It has the* two-scale structure $\hat P_r=A_rn^{\sigma_3/6}M_{r,n}$, *where* $A_r^{\pm1}$ *are analytic on* $U(4c,\varepsilon)$ *and bounded uniformly in* $n$, $f_r$ *is conformal on a neighbourhood of* $\overline{U(4c,\varepsilon)}$ *and independent of* $n$, $f_r(4c)=0$, *and, with* $\zeta=n^{2/3}f_r(z)$: *for* $|\zeta|\ge2$, $M_{r,n}=\zeta^{-\sigma_3/4}\Theta_\infty$ *with* $|\Theta_\infty^{\pm1}|\le C$; *for* $|\zeta|\le3$, $M_{r,n}=\Psi^{\mathrm{I}}(\zeta)\Theta_0$ *with* $\Psi^{\mathrm{I}}$ *entire, independent of* $n$, $\det\Psi^{\mathrm{I}}=1$, *and* $|\Theta_0^{\pm1}|\le C$ *(also for the boundary values).*

(F6) *(Hard-edge map.)* $f$ *is conformal on a neighbourhood of* $\overline{U(0,\delta)}$, *independent of* $n$ *and of the weight, real on* $(-\delta,\delta)$, $f(0)=0$, $f'(0)<0$, $f=(\varphi\pm i\pi)^2/4$ *in* $U(0,\delta)\cap\mathbb{C}_\pm$; $\operatorname{Re}\varphi<0$ *on the lips and* $|e^{2n\varphi}|\le e^{-n|z|^{1/2}/C}$ *on the lips in* $U(0,\delta)$.

(F7) *(A priori bound.)* $Q_\pm$ *and* $(Q^{-1})_\pm$ *belong to* $I+\partial\mathcal{C}(L^2(\Sigma))$. *For each fixed* $n$ *and every* $\eta>0$, $|Q(z)|\le C_{n,\eta}(1+|z|^{-|\alpha|/2-\eta})$ *for* $z\notin\Sigma$.

(F8) *(Extraction.) For* $|z|>n^p+\delta'+h_0$, $Q=k_\infty^{-\sigma_3}(Uz^{-n\sigma_3})k^{\sigma_3}$ *with a scalar* $k$ *analytic at* $\infty$, $k(\infty)=k_\infty\neq0$ *(69). Consequently, with* $Q=I+Q_1/z+Q_2/z^2+\dots$, $\mathsf{b}(Q)=(Q_1)_{12}(Q_1)_{21}=n^{2p}b_n^2$ *and* $\mathsf{a}(Q)=(Q_2)_{12}/(Q_1)_{12}-(Q_1)_{22}=n^pa_n$.

*Proof.* Items (F1)–(F6) collect the lemmas of Sections 5 and 6; the only new arguments are the uniform Cauchy bound in (F1) and the arc-by-arc bound in (F5).

(F1). The jumps are those of Lemma 5.10 conjugated by $D_W^{\sigma_3}$, which has no jump off $[0,4c]$; on the band $D_{W,-}^{-\sigma_3}\left(\begin{smallmatrix}0&W_0\\-1/W_0&0\end{smallmatrix}\right)D_{W,+}^{\sigma_3}=J$ because $D_{W,+}D_{W,-}=W_0$; on the seams there is no jump by Lemma A.2(iv). The behaviour at $\infty$ is Lemma 6.4. Boundedness of $v^{\pm1}$ (note $\det v=1$): on the band $v=J$; on the lips in $U(0,\delta)$ by Lemma 6.3(vi)(a) and Lemma 6.5;

on the remaining arcs it follows from the bounds in (F5) below, from Lemma 6.3(vi)(b) in $U(4c,\varepsilon)$, and from $\operatorname{Re}\tilde\varphi<0$ on $(4c,4c+\varepsilon)$, $\operatorname{Re}\tilde\varphi>0$ on $\Sigma_v$ near $4c$ (Lemma 5.5(iv)). The arcs inside $\{\operatorname{Re} z\le 4c+1\}$ do not depend on $n$; the rest of $\Sigma$ consists of pieces of the three lines $\operatorname{Im} z\in\{0,\pm h_0\}$ and the top segment. Hence $|\Sigma\cap U(z,\rho)|\le C\rho$. For the Cauchy operator, split $\Sigma=\Sigma_1\cup\Sigma_2\cup\Sigma_3$ according to $\operatorname{Re} z\le 4c+1$, $4c+1<\operatorname{Re} z<4c+3$, $\operatorname{Re} z\ge 4c+3$ ($n\ge n_0$). The blocks between $\Sigma_1\cup\Sigma_2$ and itself are bounded because this curve is fixed and Carleson (T3(a)). $\Sigma_2\cup\Sigma_3$ is contained in a translate of the fixed infinite curve $\Gamma_\infty$ consisting of three horizontal half-lines ending on a vertical segment of length $2h_0$. $\Gamma_\infty$ is Carleson (a finite union of lines and segments), so its Cauchy operators are bounded by T3(a); the Cauchy operator is translation invariant, and restricting a bounded operator $\mathcal{C}_\pm$ to functions supported on, and evaluated on, a sub-curve is a compression, which does not increase the norm. So these blocks are bounded uniformly in $n$. The blocks between $\Sigma_1$ and $\Sigma_3$ have kernels bounded by $|s-z|^{-1}$ with $|s-z|\ge 2$, and $\int_{\Sigma_1}\int_{\Sigma_3}|s-z|^{-2}|\,\mathrm{d}s||\,\mathrm{d}z|\le C$, so they are Hilbert–Schmidt with a uniform bound.

(F2). This is (55) and (O2)–(O4); the sectors lie in $\mathcal{Z}_n$ because $\frac54\vartheta_{\mathrm{lip}}<\vartheta_{\mathrm{w}}$, and $|\log H_n-\log H_n(0^+)|\le C$ on $[1,4c+\varepsilon]$ by (O4). The cancellation is (64).

(F3). This is Lemma 5.6(a).

(F4). By Lemma 6.5(a), (b), Lemma 6.3(vi)(a) and $|e^{2n\varphi}|\le 1$, every entry of $E$ is at most $C\exp(-\pi s_p n|z|^{-1/p})$.

(F5). The properties of $P_0^\infty$ were shown above, and those of $\hat P_r$ are Lemma 6.6. We bound $|v-I|$ arc by arc, using Lemma 6.3(vi)(c) for $\omega$ (all these arcs have distance $\ge d>0$ from $[0,4c]$, except the parts near $4c$, where (vi)(b) applies). *Lips outside $U(0,\delta)$:* this part of the lips is compact and does not meet $[0,4c]$, so $\operatorname{Re}\varphi\le -1/C$ there (Lemma 5.5(i)), $|\operatorname{Im}\theta|\ge n/C$ (by (41), since $|\arg z|$ is bounded below and $|z|\le 4c+h_0$), and $\operatorname{Re}\tilde\varphi\ge 1/C$ by Lemma 5.6(b) and compactness; so all entries of $v-I$ are $O(e^{-n/C})$. *$\Sigma_v$ outside $U(4c,\varepsilon)$:* $\operatorname{Re}\tilde\varphi\ge 1/C$ by Lemma 5.5(iv) and compactness. *Saturated segment beyond $4c+\varepsilon$:* $|e^{2n\tilde\varphi}|=e^{2n(\gamma_p-U^*(x))}$ is at most $e^{-n/C}$ for $x\le x_0$ and at most $x^{-n}$ for $x\ge x_0$ (Lemma 5.4(iii)), while $|\omega|\le\exp(C(1+\log(1+x))^2)$ and $\log x\le C\log n$; so $|v-I|\le Ce^{-n/C}$. *$\Sigma^{\triangle\pm}$ and the top segment:* for $z=x\pm iy$ with $x>4c$, $\operatorname{Re} g(z)\ge U^*(x)$, so $\operatorname{Re}\varphi(z)\le\gamma_p-U^*(x)$, which is $\le -1/C$ for $x\ge 4c+\varepsilon$ and $\le -\frac12\log x$ for $x\ge x_0$; near $x=4c$ the bound $\operatorname{Re}\varphi\le -1/C$ on $\Sigma^{\triangle\pm}$ follows from Lemma 5.5(i) by compactness. Moreover $|\operatorname{Im}\theta(x\pm ih_0)|\ge h_0 n\,x^{-1-1/p}/C$, so $|\kappa_\pm|\le Cn^p$ on these arcs (and on the top segment, where $t$ is close to 1, $|1-e^{\mp 2i\theta}|\ge n^{-p}/C$). With $|\omega|\le\exp(C\log^2 n)$ this gives $|v-I|\le Cn^pe^{C\log^2 n}e^{-n/C}\le Ce^{-n/(2C)}$, and $\le Cn^{-pn/4}$ on the top segment.

(F6). Lemma 5.5(i), (iii) and Lemma 6.5(c); $f$ is built from $\mu^*$ alone.

(F7). This is Proposition B.4 and Corollary B.5 of Appendix B.

(F8). The first statement is Lemma 6.4; the second follows from (38) and Lemma 5.2. □

## 7. The hard edge: comparison with a Bessel model

After Sections 5 and 6 the orthogonal polynomials of $\sigma_{p,s,\ell_1}$ are encoded in a matrix $Q$, normalised at infinity, whose jumps are listed in the Frame Proposition (Proposition 6.7). Away from the two endpoints 0 and $4c$ of the band these jumps are those of an explicit global parametrix, up to exponentially small errors. At the soft edge $4c$ the Airy parametrix of Lemma 6.6 applies. This section treats the hard edge 0. We compare $Q$ with an explicitly solvable *comparison problem* and show that the two have the same first two moments at infinity up to $O(1/n)$; Section 8 then reads off $a_n$ and $b_n$.

*Why the usual local parametrix fails.* The slowly varying factor enters the band weight $W_0 = \mathsf{w}_n x^\alpha (4c-x) H_n(x) r_n(x)$ through $r_n(z) = \ell_1(nz^{-1/p})/\ell_1(n)$. In the variable $\xi = \log n - (\log z)/p$ the point $z = 0$ corresponds to $\xi = +\infty$, where $\Lambda$ has no expansion: $r_n$ is neither analytic at 0 nor of Fisher–Hartwig type. The Bessel parametrix of T2 needs a square root $\tilde{W}$ of $W_0$ continued around 0 across $(-\delta, 0)$. Continuing $r_n^{1/2}$ from the two half-planes produces two values on $(-\delta, 0)$ whose ratio is $\exp\big(\frac{1}{2}(\Lambda(\xi - i\pi/p) - \Lambda(\xi + i\pi/p))\big) = 1 + O(\varepsilon_\Lambda)$ (this even presupposes that $\Lambda$ is analytic on the strip $|\Im\xi| \le \pi/p$, which $(\mathrm{A}_\vartheta)$ does not provide). This tends to 0 with no rate in $n$, and after conjugation by the Bessel model at the scale $|z| \asymp n^{-2}$ it becomes $O(n\varepsilon_\Lambda)$, so a quotient argument does not close. A Bessel order that drifts with $n$ removes only the power-like part $z^{-\Lambda'(\log n)/p}$ of $r_n$. This is the situation of [2, Remark 4.10].

*The remedy, due to Conway and Deift.* We build the Bessel parametrix with $\tilde{W} := D_W$, the Szegő function of the whole band weight. It has exactly the band jump of $Q$ and no jump on $(-\delta, 0)$, and it differs from $Q$ only on the lips of the lens, where the true jump contains $\omega = D_W^2/W_0$ and the model jump contains the constant $\varrho_\pm = e^{\mp i\pi\alpha}$. The difference $\omega - \varrho_\pm$ is $o(1)$ uniformly on the lips, without a rate (§7.4). That suffices for a Neumann series once the comparison resolvent is bounded uniformly in $n$ (§7.5), and the difference formula of Conway and Deift turns the mismatch into a change of the moments (§7.6). The factor $1/n$ comes from *Bessel localisation*: the mismatch carries the factor $e^{2n\varphi}$, which confines it to $|z| \lesssim n^{-2}$ up to a tail, and there the comparison solution has size $n^{1/2}|n^2 z|^{-|\alpha|/2}$; so a mismatch $\varepsilon_n' = o(1)$ produces an error $O(\varepsilon_n'/n)$.

Unlike [2], our comparison problem is an explicit matrix (the global parametrix with the two local parametrices), so the matching errors on the two circles are part of the comparison and give the $O(1/n)$ terms as explicit integrals. The restriction $|\alpha| < 1$ enters only through square integrability and power-weighted Cauchy bounds at the Bessel scale (Remark 7.8).

7.1. **Conventions.** The standing assumptions of §5.1 hold. All matrices are $2 \times 2$, $|\cdot|$ is a fixed matrix norm, $e_1, e_2$ are the standard basis vectors and

$$J := \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad M = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}, \qquad \varrho_\pm := e^{\mp i\pi\alpha}.$$

Constants $C$ may depend on $p$, $s$, $\ell_1$ and on the radii $\delta$ (at 0) and $\varepsilon$ (at $4c$), but never on $n$ or on the point $z$; they may change from line to line. Exponentially small factors are written $e^{-n/C}$. Powers and logarithms are principal unless said otherwise. The Cauchy transform $\mathcal{C}$, its boundary values $\mathcal{C}_\pm$, the operator $\mathcal{C}_v f = \mathcal{C}_-(f(v - I))$ and the class $I + \partial\mathcal{C}(L^2)$ are as in T3.

We work in the frame of $Q$ (see (68)) and use only the items (F1)–(F8) of the Frame Proposition, for fixed radii $\delta, \varepsilon$ that may be taken as small as we wish. In $U(0, \delta)$ the contour consists of the band and two lips, with $v = J$ on $(0, \delta)$ and $v = \left(\begin{smallmatrix} 1 & 0 \\ \omega e^{2n\varphi} & 1 \end{smallmatrix}\right) + E$ on the lips, where $\omega = D_W^2/W_0$ (F1); there is no jump on $(-\delta, 0)$. Since $f'(0) < 0$ and $f$ is real on $(-\delta, \delta)$ (F6), $f$ maps $U(0, \delta) \cap \mathbb{C}_\pm$ into $\mathbb{C}_\mp$, the band $(0, \delta)$ into $(-\infty, 0)$ and $(-\delta, 0)$ into $(0, \infty)$; and $|f(z)| \ge |z|/C$, $|f'| + |f'|^{-1} \le C$ on $U(0, \delta)$. On the lips in $U(0, \delta)$

$$\tfrac{3}{4}\vartheta_{\mathrm{lip}} \le |\arg z| \le \tfrac{5}{4}\vartheta_{\mathrm{lip}} \tag{77}$$

by (F3). Finally, with the Hölder exponent $\nu_p = \min(1/p, \frac{1}{2})$ of §5.9, put $T_n := \log n - \frac{1}{p}\log\max(4c, 1)$ and

$$\varepsilon_n' := n^{-2\nu_p}\log n + \varepsilon_\Lambda(T_n) \;\longrightarrow\; 0. \tag{78}$$

7.2. **The Bessel model and its local form.** Let $\Psi_\alpha$ be the Bessel model of T2, built on the three rays $\arg\zeta = \pm(\pi - \vartheta_{\mathrm{lip}})$ and $\arg\zeta = \pi$ (angle freedom, T2(b); for $\vartheta_{\mathrm{lip}} = \pi/3$ these are the rays of [10, §6]). We orient them as in T2(a): all three rays point towards the origin, with the + side on the left. Thus $\Psi_\alpha$ is given by (6.23) of T2 in the sector $|\arg\zeta| < \pi - \vartheta_{\mathrm{lip}}$ and by the Hankel formulas in the two thin sectors, it has the jumps (6.16)–(6.18) on the three rays, and it satisfies the large-$\zeta$ expansion (6.28). We have $\det\Psi_\alpha \equiv 1$: in the central sector this is the Wronskian identity of T4, and the jump matrices have determinant 1.

In the central sector $|\arg 2\zeta^{1/2}| < \pi/2$, so all Bessel functions there are principal branches (T4, §10.25(ii)); and (6.28) remains valid for the rays at $\pm(\pi - \vartheta_{\mathrm{lip}})$, since the Bessel and Hankel functions in (6.23)–(6.25) are then evaluated in closed subsectors of the sectors of validity of the large-argument expansions T4: (10.40.1)–(10.40.5) for $I_\alpha$, $K_\alpha$ in the central sector, where $|\arg 2\zeta^{1/2}| \le \frac12(\pi - \vartheta_{\mathrm{lip}}) < \frac12\pi$, and (10.17.5)–(10.17.6) for the Hankel functions $H_\alpha^{(1,2)}(2(-\zeta)^{1/2})$ in the thin sectors, where $\arg 2(-\zeta)^{1/2}$ lies in $(-\frac12\vartheta_{\mathrm{lip}}, 0)$, respectively $(0, \frac12\vartheta_{\mathrm{lip}})$, deep inside both sectors $[-\pi+\delta, 2\pi-\delta]$ and $[-2\pi+\delta, \pi-\delta]$.

**Lemma 7.1** (local form of the Bessel model)**.** *Let $\alpha \in (-1,1)$ and let $\mathcal{S}_\pm$ denote the thin sectors $\pi - \vartheta_{\mathrm{lip}} < \pm\arg\zeta < \pi$.*

(i) *If $\alpha \neq 0$, there is an entire matrix function $\hat\Psi_\alpha$ with $\det\hat\Psi_\alpha = -2i\sin\pi\alpha$ such that*

$$\Psi_\alpha(\zeta) = \hat\Psi_\alpha(\zeta)\,\zeta^{\alpha\sigma_3/2}\,\mathsf{S}, \qquad \mathsf{S} := \begin{pmatrix} 1 & -\frac{i}{2\sin\pi\alpha} \\ 0 & \frac{i}{2\sin\pi\alpha} \end{pmatrix},$$

*in $|\arg\zeta| < \pi - \vartheta_{\mathrm{lip}}$, while in $\mathcal{S}_\pm$ the constant $\mathsf{S}$ is replaced by $\mathsf{S}$ times a constant jump matrix of $\Psi_\alpha$.*

(ii) *If $\alpha = 0$, there is an entire $\hat\Psi_0$ with $\det\hat\Psi_0 = 1$ such that*

$$\Psi_0(\zeta) = \hat\Psi_0(\zeta)\begin{pmatrix} 1 & \mathsf{k}_0\log\zeta \\ 0 & 1 \end{pmatrix}, \qquad \mathsf{k}_0 := -\frac{i}{2\pi},$$

*in $|\arg\zeta| < \pi - \vartheta_{\mathrm{lip}}$, while in $\mathcal{S}_\pm$ the right-hand side is multiplied on the right by a constant jump matrix.*

(iii) *Consequently, for $0 < |\zeta| \le 3$ off the rays,*

$$|\Psi_\alpha(\zeta)^{\pm1}| \le C|\zeta|^{-|\alpha|/2}, \qquad |\Psi_\alpha(\zeta)e_2e_1^{\mathsf{T}}\Psi_\alpha(\zeta)^{-1}| \le C|\zeta|^{-|\alpha|} \qquad (\alpha \neq 0),$$

*and the same with $|\zeta|^{-|\alpha|/2}$, $|\zeta|^{-|\alpha|}$ replaced by $1 + |\log|\zeta||$ and $(1+|\log|\zeta||)^2$ if $\alpha = 0$. The same bounds hold for the boundary values on the rays.*

*Proof.* We expand the entries of (6.23) by the power series T4(10.25.2) and read off the singular factors.

(i) Let $\nu = \pm\alpha$. By (10.25.2) with $x = 2\zeta^{1/2}$ we have $(\frac12 x)^{\nu+2k} = \zeta^{\nu/2+k}$, hence

$$I_\nu(2\zeta^{1/2}) = \zeta^{\nu/2}\mathfrak{p}_\nu(\zeta), \qquad 2\pi i\zeta^{1/2}I_\nu'(2\zeta^{1/2}) = \pi i\zeta^{\nu/2}\sum_{k\ge0}\frac{(\nu+2k)\zeta^k}{k!\,\Gamma(\nu+k+1)} =: \zeta^{\nu/2}\mathfrak{q}_\nu(\zeta),$$

with $\mathfrak{p}_\nu(\zeta) = \sum_k \zeta^k/(k!\Gamma(\nu+k+1))$; $\mathfrak{p}_\nu$ and $\mathfrak{q}_\nu$ are entire because $1/\Gamma$ is entire. Write $\mathbf{u}_\nu := (\mathfrak{p}_\nu, \mathfrak{q}_\nu)^{\mathsf{T}}$, so that the first column of (6.23) is $\zeta^{\alpha/2}\mathbf{u}_\alpha$. By (10.27.4), $\frac{i}{\pi}K_\alpha = \frac{i}{2\sin\pi\alpha}(I_{-\alpha} - I_\alpha)$, and differentiating, $-2\zeta^{1/2}K_\alpha'(2\zeta^{1/2}) = \frac{i}{2\sin\pi\alpha}\cdot 2\pi i\zeta^{1/2}(I_{-\alpha}' - I_\alpha')(2\zeta^{1/2})$ (both sides equal $-\frac{\pi}{\sin\pi\alpha}\zeta^{1/2}(I_{-\alpha}' - I_\alpha')$). So the second column of (6.23) is $\frac{i}{2\sin\pi\alpha}(\zeta^{-\alpha/2}\mathbf{u}_{-\alpha} - \zeta^{\alpha/2}\mathbf{u}_\alpha)$, and

$$\Psi_\alpha = \begin{pmatrix}\mathbf{u}_\alpha & \mathbf{u}_{-\alpha}\end{pmatrix}\begin{pmatrix}\zeta^{\alpha/2} & -\frac{i}{2\sin\pi\alpha}\zeta^{\alpha/2} \\ 0 & \frac{i}{2\sin\pi\alpha}\zeta^{-\alpha/2}\end{pmatrix} = \hat\Psi_\alpha\,\zeta^{\alpha\sigma_3/2}\mathsf{S}, \qquad \hat\Psi_\alpha := \begin{pmatrix}\mathbf{u}_\alpha & \mathbf{u}_{-\alpha}\end{pmatrix}.$$

Since $\det\Psi_\alpha = 1$ and $\det \mathsf{S} = i/(2\sin\pi\alpha)$, $\det\hat\Psi_\alpha = -2i\sin\pi\alpha \neq 0$. In a thin sector, $\Psi_\alpha$ is the continuation of the central formula multiplied on the right by a constant jump matrix, because the jumps of $\Psi_\alpha$ are constant and $\hat\Psi_\alpha \zeta^{\alpha\sigma_3/2}\mathsf{S}$ is analytic in $\mathbb{C}\setminus(-\infty,0]$.

(ii) By (10.31.2), $K_0(x) = -(\ln\frac{x}{2}+\gamma_{\mathrm{E}})I_0(x) + S_0(\frac14 x^2)$ with $S_0$ entire. At $x = 2\zeta^{1/2}$, $\ln\frac{x}{2} = \frac12\log\zeta$, so $\frac{i}{\pi}K_0(2\zeta^{1/2}) = -\frac{i}{2\pi}\log\zeta\cdot\mathfrak{p}_0(\zeta) + (\text{entire})$. Differentiating the same identity, $-2\zeta^{1/2}K_0'(2\zeta^{1/2}) = I_0(2\zeta^{1/2}) + (\frac12\log\zeta+\gamma_{\mathrm{E}})\,2\zeta^{1/2}I_0'(2\zeta^{1/2}) - 2\zeta S_0'(\zeta)$, and $2\zeta^{1/2}I_0'(2\zeta^{1/2}) = \mathfrak{q}_0(\zeta)/(\pi i)$. Hence the second column equals $\mathsf{k}_0\log\zeta\cdot\mathbf{u}_0(\zeta) + \mathbf{h}(\zeta)$ with $\mathsf{k}_0 = -i/(2\pi)$ and $\mathbf{h}$ entire, and $\Psi_0 = (\mathbf{u}_0\ \ \mathbf{h})\left(\begin{smallmatrix}1 & \mathsf{k}_0\log\zeta\\ 0 & 1\end{smallmatrix}\right)$; $\det(\mathbf{u}_0\ \mathbf{h}) = \det\Psi_0 = 1$. The thin sectors are treated as in (i).

(iii) On $|\zeta|\le 3$ the entire factors and their inverses are bounded, and the constant matrices are bounded. The middle factors satisfy $|\zeta^{\pm\alpha\sigma_3/2}| \le C|\zeta|^{-|\alpha|/2}$, respectively $|\left(\begin{smallmatrix}1 & \mathsf{k}_0\log\zeta\\ 0 & 1\end{smallmatrix}\right)^{\pm1}| \le C(1+|\log|\zeta||)$. For the rank-one product write

$$\Psi_\alpha e_2 e_1^{\mathsf{T}}\Psi_\alpha^{-1} = \hat\Psi_\alpha\,\zeta^{\alpha\sigma_3/2}(\mathsf{S}_j e_2)(e_1^{\mathsf{T}}\mathsf{S}_j^{-1})\zeta^{-\alpha\sigma_3/2}\,\hat\Psi_\alpha^{-1};$$

the entries of the middle matrix are constants times $1,\zeta^{\alpha},\zeta^{-\alpha}$, whence the bound $C|\zeta|^{-|\alpha|}$; for $\alpha=0$ the same computation gives entries of size $(1+|\log|\zeta||)^2$. □

*The model in the normalisation used below.* Put

$$\Phi_\alpha(\zeta) := \sigma_3\Psi_\alpha(\zeta)\sigma_3\, e^{-2\zeta^{1/2}\sigma_3}. \tag{79}$$

This is the left-endpoint model of T2(c) with the exponential factor attached; it does not depend on $n$. Orient the three rays $\Gamma_\alpha := (-\infty,0]\cup\{\arg\zeta = \pm(\pi-\vartheta_{\mathrm{lip}})\}$ *away* from the origin. Reversing the orientation of a ray interchanges its $\pm$ sides, and $\sigma_3 J^{-1}\sigma_3 = J$, $\sigma_3\left(\begin{smallmatrix}1&0\\x&1\end{smallmatrix}\right)^{-1}\sigma_3 = \left(\begin{smallmatrix}1&0\\x&1\end{smallmatrix}\right)$, and $e^{2w\sigma_3}\left(\begin{smallmatrix}1&0\\x&1\end{smallmatrix}\right)e^{-2w\sigma_3} = \left(\begin{smallmatrix}1 & 0\\ xe^{-4w} & 1\end{smallmatrix}\right)$. With (6.16)–(6.18) this gives:

$$\Phi_{\alpha,+} = \Phi_{\alpha,-}J \text{ on } (-\infty,0), \qquad \Phi_{\alpha,+} = \Phi_{\alpha,-}\begin{pmatrix}1 & 0\\ e^{\mp i\pi\alpha}e^{-4\zeta^{1/2}} & 1\end{pmatrix} \text{ on } \arg\zeta = \mp(\pi-\vartheta_{\mathrm{lip}}). \tag{80}$$

On $(-\infty,0)$ we used that the two boundary values $w_\pm$ of $\zeta^{1/2}$ satisfy $w_+ = -w_-$, so that $e^{2w_-\sigma_3}Je^{-2w_+\sigma_3} = J$. Moreover $\det\Phi_\alpha = 1$, and $\Phi_\alpha$ has the two-scale structure

$$\Phi_\alpha(\zeta) = \zeta^{-\sigma_3/4}\Theta_\infty(\zeta), \quad \Theta_\infty = (2\pi)^{-\sigma_3/2}\big(M^{-1}+O(\zeta^{-1/2})\big), \quad |\Theta_\infty^{\pm1}|\le C \quad (|\zeta|\ge 2), \tag{81}$$

$$\Phi_\alpha(\zeta) = \hat A(\zeta)\,\mathsf{P}(\zeta)\,\Theta_0(\zeta), \quad |\Theta_0^{\pm1}|\le C \qquad (|\zeta|\le 3), \tag{82}$$

where $\hat A = \sigma_3\hat\Psi_\alpha\sigma_3$ is entire with entire inverse, $\Theta_0 = \sigma_3\mathsf{S}_j\sigma_3 e^{-2\zeta^{1/2}\sigma_3}$ in the $j$-th sector, and $\mathsf{P}(\zeta) = \zeta^{\alpha\sigma_3/2}$ for $\alpha\neq 0$, $\mathsf{P}(\zeta) = I - \mathsf{k}_0(\log\zeta)e_1e_2^{\mathsf{T}}$ for $\alpha=0$. Indeed (81) follows from (6.28), since $\sigma_3M\sigma_3 = M^{-1}$ and the exponential factors cancel; the bound on $\Theta_\infty^{-1}$ holds for large $|\zeta|$ by the expansion and for $2\le|\zeta|\le R$ because $\det\Theta_\infty = 1$ and $\Theta_\infty$ is bounded. (82) is Lemma 7.1 conjugated by $\sigma_3$.

7.3. **The comparison parametrix and the comparison problem.** We first relate $e^{n\varphi}$ to the Bessel variable $\zeta = n^2 f(z)$. In $U(0,\delta)\cap\mathbb{C}_+$ both $(\varphi+i\pi)/2$ and the principal root $f^{1/2}$ are analytic (the latter because $f$ avoids $(-\infty,0]$ there), and their squares agree by (F6). So $(\varphi+i\pi)/2 = \pm f^{1/2}$ with a fixed sign. On the upper lip $\Re\varphi<0$ while $\Re f^{1/2}>0$, so the sign is $-$. The lower half-disc is treated in the same way, and we obtain, with $\zeta^{1/2} = nf^{1/2}$,

$$e^{n\varphi(z)} = (-1)^n e^{-2\zeta^{1/2}}, \qquad e^{2n\varphi(z)} = e^{-4\zeta^{1/2}} \qquad (z\in U(0,\delta)\setminus[0,\delta)). \tag{83}$$

On the lips $\arg\zeta = \mp(\pi-\vartheta_{\mathrm{lip}})$, so $\Re\zeta^{1/2} = |\zeta|^{1/2}\sin(\vartheta_{\mathrm{lip}}/2)$ and

$$|e^{2n\varphi(z)}| = e^{-4\sin(\vartheta_{\mathrm{lip}}/2)|\zeta|^{1/2}} \le e^{-n|z|^{1/2}/C} \qquad \text{on the lips in } U(0,\delta). \tag{84}$$

Following T2(c) with $\tilde{W}$ replaced by $D_W$, define in $U(0,\delta)$

$$\tilde{P}_0 := (-1)^n E_n\,\sigma_3\Psi_\alpha(n^2 f)\sigma_3\,D_W^{-\sigma_3}e^{n\varphi\sigma_3}, \qquad E_n := D_\infty^{\sigma_3}P_0^\infty M f^{\sigma_3/4}(2\pi n)^{\sigma_3/2}. \tag{85}$$

In the frame of $Q$ this becomes, by (83),

$$\hat{P}_0 := D_\infty^{-\sigma_3}\tilde{P}_0 D_W^{\sigma_3} = \hat{E}_n\,\Phi_\alpha(n^2 f), \qquad \hat{E}_n := P_0^\infty M f^{\sigma_3/4}(2\pi n)^{\sigma_3/2} = A_0\,(2\pi n)^{\sigma_3/2}, \tag{86}$$

where $A_0 := P_0^\infty M f^{\sigma_3/4}$. Note that $\hat{P}_0$ contains neither $D_W$ nor any other trace of the weight.

**Lemma 7.2** (the comparison parametrix)**.** (i) *The matrix $A_0$ has a removable singularity at 0, so it is analytic on $U(0,\delta)$; and $\det A_0 = 1$. It depends neither on $n$ nor on the weight, and $|A_0^{\pm1}| \le C$ there. In particular $E_n$ is analytic in $U(0,\delta)$, $\det E_n = 1$, and $D_\infty^{-\sigma_3}E_n(2\pi n)^{-\sigma_3/2} = A_0$.*

(ii) *$\hat{P}_0$ is analytic in $U(0,\delta)\setminus\Sigma$, $\det\hat{P}_0 = 1$, and*

$$\hat{P}_{0,+} = \hat{P}_{0,-}J \text{ on } (0,\delta), \qquad \hat{P}_{0,+} = \hat{P}_{0,-}\begin{pmatrix} 1 & 0 \\ \varrho_\pm e^{2n\varphi} & 1\end{pmatrix} \text{ on the lips in } U(0,\delta)\cap\mathbb{C}_\pm;$$

*it has no jump on $(-\delta,0)$. Equivalently, $\tilde{P}_0$ has the band jump of $V$, no jump on $(-\delta,0)$ (nor on the seams if $p<1$), and the lip jump $\left(\begin{smallmatrix} 1 & 0 \\ \varrho_\pm e^{2n\varphi}/D_W^2 & 1\end{smallmatrix}\right)$.*

(iii) *Uniformly for $\frac12\delta \le |z| \le \delta$ off $\Sigma$, $\hat{P}_0(z)P_0^\infty(z)^{-1} = I + O(1/n)$, and $|\hat{P}_0^{\pm1}(z)| \le C$.*

*Proof.* (i) On $(0,\delta)$, $P_{0,+}^\infty = P_{0,-}^\infty J$, and since $f$ is negative there and maps $\mathbb{C}_+$ to $\mathbb{C}_-$, $f_+^{\sigma_3/4} = f_-^{\sigma_3/4}\mathrm{diag}(-i,i)$. A direct computation gives $M^{-1}JM = \mathrm{diag}(i,-i)$, so $A_{0,+} = P_{0,-}^\infty JMf_-^{\sigma_3/4}\mathrm{diag}(-i,i) = P_{0,-}^\infty M\,\mathrm{diag}(i,-i)\mathrm{diag}(-i,i)f_-^{\sigma_3/4} = A_{0,-}$. On $(-\delta,0)$ both $P_0^\infty$ and $f^{1/4}$ are analytic. So $A_0$ is analytic in $U(0,\delta)\setminus\{0\}$. At 0, $|P_0^\infty| \le C|z|^{-1/4}$ by (F5) and $|f^{\pm1/4}| \le C|z|^{\pm1/4}$, so $|A_0(z)| \le C|z|^{-1/2}$; an isolated singularity with this growth is removable. This is the removability argument of T2(c), applied to a matrix that involves only $P_0^\infty$, $M$ and $f$, and therefore neither $n$ nor the weight; in particular the hypothesis of [10] that the weight be analytic at the endpoint plays no role. $\det A_0 = 1$ since each factor has determinant 1, and $A_0^{-1}$ is the adjugate of $A_0$. Shrinking $\delta$ slightly, $A_0^{\pm1}$ are bounded on $U(0,\delta)$.

(ii) $\hat{E}_n$ is analytic, so the jumps of $\hat{P}_0$ are those of $\zeta \mapsto \Phi_\alpha(\zeta)$ at $\zeta = n^2 f(z)$. The map $z \mapsto n^2 f(z)$ is conformal, hence preserves the sides of an oriented curve; it maps the band (left to right) onto part of $(-\infty,0)$ traversed away from 0, the lips (away from 0) onto the rays at $\mp(\pi - \vartheta_{\mathrm{lip}})$ traversed away from 0, and $(-\delta,0)$ into $(0,\infty)$, where $\Phi_\alpha$ is analytic. So (80) and (83) give the stated jumps. For $\tilde{P}_0 = D_\infty^{\sigma_3}\hat{P}_0 D_W^{-\sigma_3}$, conjugation by the diagonal matrix $D_W^{-\sigma_3}$ turns $J$ into the band jump of $V$ because $D_{W,+}D_{W,-} = W_0$, turns the lip jump into the stated one, and creates no jump on $(-\delta,0)$ because $D_W$ is analytic off $[0,4c]$.

(iii) For $\frac12\delta \le |z| \le \delta$ we have $|\zeta| = n^2|f(z)| \ge n^2\delta/C \ge 2$. By (81) and $(n^2 f)^{-\sigma_3/4} = n^{-\sigma_3/2}f^{-\sigma_3/4}$,

$$\begin{aligned}\hat{P}_0 &= A_0(2\pi n)^{\sigma_3/2}(n^2 f)^{-\sigma_3/4}(2\pi)^{-\sigma_3/2}\big(M^{-1} + O(n^{-1})\big)\\ &= P_0^\infty M f^{\sigma_3/4}f^{-\sigma_3/4}\big(M^{-1} + O(n^{-1})\big) = P_0^\infty\big(I + O(n^{-1})\big),\end{aligned}$$

and $P_0^\infty$, $(P_0^\infty)^{-1}$ are bounded on the annulus. Both claims follow. □

*The comparison problem.* Let $\Sigma_\circ := \partial U(0,\delta)\cup\partial U(4c,\varepsilon)$, with both circles oriented clockwise (so that their + side is the outside), and $\hat\Sigma := \Sigma\cup\Sigma_\circ$. Define

$$\tilde Q(z) := \begin{cases} \hat P_0(z), & z\in U(0,\delta),\\ \hat P_r(z), & z\in U(4c,\varepsilon),\\ P_0^\infty(z), & \text{otherwise,}\end{cases} \qquad z\notin\hat\Sigma. \tag{87}$$

Then $\tilde Q$ is analytic off $\hat\Sigma$, $\det\tilde Q=1$, $\tilde Q\to I$ at infinity, and $\tilde Q_+ = \tilde Q_-\tilde v$ on $\hat\Sigma$ with

$$\tilde v = \begin{cases} J & \text{on } (0,4c),\\ \left(\begin{smallmatrix} 1 & 0\\ \varrho_\pm e^{2n\varphi} & 1\end{smallmatrix}\right) & \text{on the lips in } U(0,\delta)\cap\mathbb{C}_\pm,\\ v & \text{on } \Sigma\cap U(4c,\varepsilon),\\ I & \text{on } \Sigma\setminus\big(U(0,\delta)\cup U(4c,\varepsilon)\cup[0,4c]\big),\\ \tilde Q_{\text{in}}^{-1}\tilde Q_{\text{out}} & \text{on } \Sigma_\circ,\end{cases} \tag{88}$$

by Lemma 7.2(ii), (F5), and the fact that $P_0^\infty$ has only the band jump $J$. Here $\tilde Q_{\text{in}}$, $\tilde Q_{\text{out}}$ are the values from inside and outside the circle. We extend $v$ by $I$ to $\Sigma_\circ$; $Q$ has no jump there. We call the problem "$\tilde Q_+=\tilde Q_-\tilde v$ on $\hat\Sigma$, $\tilde Q\to I$" the *comparison problem*; it is solved explicitly by (87).

The two problems differ in three places. On the circles, $v-\tilde v = I-\tilde Q_{\text{in}}^{-1}\tilde Q_{\text{out}} = O(1/n)$, by Lemma 7.2(iii) and (F5). On $\Sigma\setminus(U(0,\delta)\cup U(4c,\varepsilon)\cup[0,4c])$, $|v-\tilde v|\le Ce^{-n/C}$ by (F5). On the lips in $U(0,\delta)$, $v-\tilde v = d_n e_2 e_1^{\mathsf T}+E$ with

$$d_n(z) := \Big(\frac{D_W(z)^2}{W_0(z)}-\varrho_\pm\Big)e^{2n\varphi(z)}. \tag{89}$$

This last difference is the subject of the next subsection.

## 7.4. **The lip mismatch.**

**Lemma 7.3** (lip mismatch)**.** *For $n\ge n_0$ and $z$ on the lips in $U(0,\delta)\cap\mathbb{C}_\pm$,*

$$\big|\omega(z)-\varrho_\pm\big| \le C\big(|z|^{\nu_p}\log(2/|z|)+\varepsilon_\Lambda(T_n)\big), \qquad \omega=\frac{D_W^2}{W_0}. \tag{90}$$

*Consequently* $\sup_{\text{lips}}|d_n|\le C\varepsilon_n'$, *and*

$$\|v-\tilde v\|_{L^\infty(\hat\Sigma)} \le C\big(\varepsilon_n'+n^{-1}\big)\longrightarrow 0. \tag{91}$$

*The convergence in (91) has no rate beyond that of $\varepsilon_\Lambda(T_n)$, about which nothing is assumed.*

The proof treats the four factors of $\omega$ in (64) separately, by Lemma 6.3. The first two are explicit; the third is Hölder at 0; the fourth, the slowly varying factor, is handled by the scale-invariant path bound of Lemma 5.9(iii), which is the point of the lemma: its right side depends on $x$ and $z$ only through $\arg z$ and $x/|z|$.

*Proof.* Let $z$ be on a lip in $U(0,\delta)\cap\mathbb{C}_\pm$. Since $\delta\le\frac14\eta_*$ and (77) holds, $z\in Z(\frac14\eta_*)$ (notation of Lemma 6.3). We write $\log\omega$ as the sum of the logarithms of the four factors in (64).

*The factor $x^\alpha$.* By Lemma 6.3(i), $D_{x^\alpha}^2=(4cz/\Upsilon)^\alpha$ and $4cz/\Upsilon=(-z)(1+O(|z|^{1/2}))$. On the lips $|\arg(-z)|\le\pi-\frac34\vartheta_{\text{lip}}$, so for $\delta$ small the principal powers multiply, $(4cz/\Upsilon)^\alpha=(-z)^\alpha(1+O(|z|^{1/2}))$, and $(-z)^\alpha=e^{\mp i\pi\alpha}z^\alpha$ in $\mathbb{C}_\pm$. Hence $D_{x^\alpha}(z)^2/z^\alpha=\varrho_\pm(1+O(|z|^{1/2}))$.

*The factor $4c-x$.* By Lemma 6.3(i), $D_{4c-x}^2/(4c-z)=-4c/\Upsilon=1+O(|z|^{1/2})$.

*The factor $H_n^\circ$.* By (O3)–(O4), Lemma 6.3(ii) applies to $q=H_n^\circ$ with $\mathfrak{c}_0=0$, $\nu=\nu_p$ and $\mathfrak{M}=C$, and gives $|\log(D_{H_n^\circ}^2/H_n^\circ)|\le C|z|^{\nu_p}\log(2/|z|)$.

*The slowly varying factor.* By Lemma 5.9(iii) with $R = \max(4c, 1)$, for $z' \in Z(\eta_*)$ and $x \in (0, 4c)$,

$$|\log r_n(x) - \log r_n(z')| \le \frac{\varepsilon_\Lambda(T_n)}{p}\big(\vartheta_{\mathrm{w}} + |\log(x/|z'|)|\big), \qquad |(\log r_n)'(z')| = \frac{|\Lambda'(\xi(z'))|}{p|z'|} \le \frac{\varepsilon_\Lambda(T_n)}{p|z'|},$$

since $\operatorname{Re}\xi \ge T_n$ and $|\operatorname{Im}\xi| \le \vartheta_{\mathrm{w}}/p \le \vartheta$ on $Z(\eta_*)$. So Lemma 6.3(iii) applies to $q = r_n$ with $\mathfrak{M} = C\varepsilon_\Lambda(T_n)$ and gives $|\log(D_{r_n}^2/r_n)| \le C\varepsilon_\Lambda(T_n)$.

*Conclusion.* Hence $\omega = \varrho_\pm e^w$ with $|w| \le C(|z|^{1/2} + |z|^{\nu_p}\log(2/|z|) + \varepsilon_\Lambda(T_n)) \le C_0$; since $|e^w - 1| \le e^{C_0}|w|$, $|\varrho_\pm| = 1$ and $|z|^{1/2} \le |z|^{\nu_p}$, (90) follows. By (84),

$$|d_n(z)| \le C\big(|z|^{\nu_p}\log(2/|z|) + \varepsilon_\Lambda(T_n)\big)e^{-n|z|^{1/2}/C} \le C\varepsilon_n',$$

because with $t = n|z|^{1/2}$ the first term is $t^{2\nu_p}n^{-2\nu_p}(\log 2 + 2\log n - 2\log t)e^{-t/C} \le Cn^{-2\nu_p}\log n$. On the lips $|E| \le C\exp(-n|z|^{-1/p}/C) \le C\exp(-n\delta^{-1/p}/C)$ by (F4). Together with the bounds on the circles and on the rest of $\Sigma$ stated at the end of §7.3 this gives (91). □

7.5. **Uniform invertibility.** The comparison problem is explicit, but its solution is unbounded near 0 at the Bessel scale: on $|\zeta| \le 2$ it has size $n^{1/2}|\zeta|^{-|\alpha|/2}$. The following proposition says that, nevertheless, the singular integral operator attached to it is boundedly invertible on $L^2$, uniformly in $n$.

**Proposition 7.4** (uniform resolvent bound for the comparison problem)**.** *There are $n_0$ and $C$ such that for $n \ge n_0$:*

(i) $\tilde{Q}_\pm \in I + \partial\mathcal{C}(L^2(\hat{\Sigma}))$ *and* $\tilde{Q}_\pm^{-1} \in I + \partial\mathcal{C}(L^2(\hat{\Sigma}))$*;*

(ii) $\|\mathcal{C}_\pm\|_{L^2(\hat{\Sigma})\to L^2(\hat{\Sigma})} \le C$*, and $1 - \mathcal{C}_{\tilde{v}}$ is invertible on $L^2(\hat{\Sigma})$ with $\|(1-\mathcal{C}_{\tilde{v}})^{-1}\| \le C$.*

The proof is in Appendix D; we sketch it (the uses of $|\alpha| < 1$ are listed in Remark 7.8). By the easy half of T3(e) it suffices to solve $m_+ = m_-\tilde{v} + \mathfrak{g}$, $m_\pm \in \partial\mathcal{C}(L^2)$, with $\|m_\pm\| \le C\|\mathfrak{g}\|$ for $\mathfrak{g}$ in a dense subspace, and to show uniqueness, which follows from T3(b) once (i) is known. The explicit candidate is $m := \mathcal{C}(\mathfrak{g}\tilde{Q}_+^{-1})\tilde{Q}$. Away from the two discs $\tilde{Q}^{\pm 1}$ is bounded and the bound reduces to that of $\mathcal{C}_\pm$. Near 0 we write $\tilde{Q} = A_0(2\pi)^{\sigma_3/2}n^{\sigma_3/2}\Phi_\alpha(n^2 f)$ and pass to the variable $\zeta$: the analytic prefactor contributes Hilbert–Schmidt commutators, and what remains is, on a star of three rays, the Cauchy operator conjugated by $|\zeta|^{\pm 1/4}$ for $|\zeta| \ge 2$ and by $|\zeta|^{\pm\alpha/2}$ for $|\zeta| \le 3$ (Lemma C.1 with $\beta = \pm\alpha/2$; at $\alpha = 0$ a logarithmic commutator, Lemma C.2). These operators are dilation invariant, so their norms do not depend on $n$. The soft edge is treated in the same way with the Airy model; no $\alpha$ is involved there. Membership of $m_\pm$ in $\partial\mathcal{C}(L^2)$ is checked for smooth data vanishing near the vertices by Cauchy's integral formula, and the uniform bound extends the inverse to all of $L^2$.

**Theorem 7.5** (uniform resolvent bound for the true problem)**.** *Extend $v$ by $I$ to $\Sigma_\circ$. There is $n_1$ such that for $n \ge n_1$ the operator $1 - \mathcal{C}_v$ is invertible on $L^2(\hat{\Sigma})$ and $\|(1-\mathcal{C}_v)^{-1}\| \le 2C$, with $C$ from Proposition 7.4. Moreover $Q_\pm \in I + \partial\mathcal{C}(L^2(\hat{\Sigma}))$ and $Q_- = (1-\mathcal{C}_v)^{-1}I$.*

*Proof.* Since $\mathcal{C}_v f - \mathcal{C}_{\tilde{v}} f = \mathcal{C}_-(f(v-\tilde{v}))$, Proposition 7.4(ii) and (91) give $\|\mathcal{C}_v - \mathcal{C}_{\tilde{v}}\| \le C(\varepsilon_n' + n^{-1})$, which is at most $1/(2C)$ for large $n$. The Neumann-series step T3(d) of Conway and Deift then gives the bound. Here only the convergence $\varepsilon_n' \to 0$ is used, not its rate.

By (F7), $Q - I = \mathcal{C}_\Sigma(Q_+ - Q_-)$. On $\Sigma_\circ$ the function $Q$ has no jump, and it is bounded there by (F7) (the circles stay away from 0), so its values on $\Sigma_\circ$ are the boundary values of the same Cauchy integral, now regarded on $\hat{\Sigma}$ with density 0 on $\Sigma_\circ$. Hence $Q_\pm \in I + \partial\mathcal{C}(L^2(\hat{\Sigma}))$. Then $Q_- - I = \mathcal{C}_-(Q_+ - Q_-) = \mathcal{C}_-(Q_-(v - I))$, i.e. $(1 - \mathcal{C}_v)Q_- = I$ in $L^2(\hat{\Sigma})$ ($\hat{\Sigma}$ has finite length, so $I \in L^2(\hat{\Sigma})$), and $Q_- = (1-\mathcal{C}_v)^{-1}I$. □

7.6. **The effect of the mismatch on the moments.** Write $Q = I + Q_1/z + Q_2/z^2 + O(z^{-3})$ and $P_0^\infty = I + P_1/z + P_2/z^2 + O(z^{-3})$ at infinity; outside the discs $\tilde{Q} = P_0^\infty$, so $\tilde{Q}_k = P_k$. The circle terms

$$\Xi_k := -\frac{1}{2\pi i}\oint_{\Sigma_\circ} s^{k-1}\big(\tilde{Q}_{\mathrm{in}}(s)\tilde{Q}_{\mathrm{out}}(s)^{-1} - I\big)\,\mathrm{d}s \qquad (k = 1, 2) \tag{92}$$

are $O(1/n)$ by Lemma 7.2(iii) and (F5); they are the first-order corrections of the comparison problem itself.

**Proposition 7.6** (the mismatch moves the moments by $O(\varepsilon_n'/n)$)**.** *For $n \geq n_1$,*

$$Q_1 = P_1 + \Xi_1 + O\big(\varepsilon_n'/n + n^{-2}\big),$$
$$Q_2 = P_2 + \Xi_2 + (Q_1 - P_1)P_1 + O\big(\varepsilon_n'/n + n^{-2}\big).$$

*The lip mismatch $d_n$ contributes $O(\varepsilon_n'/n)$; the remaining errors come from the interaction of the circle terms with themselves and are $O(n^{-2})$. In particular $Q_k = P_k + O(1/n)$ for $k = 1, 2$.*

*Proof.* We split $v - \tilde{v}$ into the lip mismatch, the circle terms and an exponentially small rest; the lip mismatch is estimated pointwise and then integrated using Bessel localisation, and the correction term is bounded by Cauchy–Schwarz. We apply T3(b) and its consequence (18) with $A = Q$, $B = \tilde{Q}$, $v_A = v$, $v_B = \tilde{v}$ on $\hat{\Sigma}$, with exponents $r = r' = 2$. The hypotheses hold: $v, \tilde{v}^{\pm 1} \in L^\infty$, $Q_\pm \in I + \partial\mathcal{C}(L^2)$ by Theorem 7.5, and $\tilde{Q}_\pm^{\pm 1} \in I + \partial\mathcal{C}(L^2)$ by Proposition 7.4(i). With $\mathcal{E} := Q_-(v - \tilde{v})\tilde{v}^{-1}\tilde{Q}_-^{-1}$, (18) reads

$$Q_1 - P_1 = -\frac{1}{2\pi i}\int_{\hat{\Sigma}} \mathcal{E}\,\mathrm{d}s, \qquad Q_2 - P_2 = -\frac{1}{2\pi i}\int_{\hat{\Sigma}} s\,\mathcal{E}\,\mathrm{d}s + (Q_1 - P_1)P_1.$$

Split $(v - \tilde{v})\tilde{v}^{-1} = \Delta_1 + \Delta_2 + \Delta_3$, where $\Delta_1 := d_n e_2 e_1^{\mathsf{T}}$ on the lips in $U(0, \delta)$ (we used $e_1^{\mathsf{T}}\tilde{v}^{-1} = e_1^{\mathsf{T}}$ there), $\Delta_2 := \tilde{v}^{-1} - I$ on $\Sigma_\circ$, and $\Delta_3$ is the rest: $E\tilde{v}^{-1}$ on the lips and $v - I$ on $\Sigma \setminus (U(0, \delta) \cup U(4c, \varepsilon) \cup [0, 4c])$. Then $\|\Delta_3\|_{L^\infty} \leq Ce^{-n/C}$, on a set of length $O(n^p)$. Write $Q_- = \tilde{Q}_- + (Q_- - \tilde{Q}_-)$.

*Pointwise bounds on the lips.* On the lips in $U(0, \delta)$ we have, by (86) and (83),

$$\tilde{Q}_-\Delta_1\tilde{Q}_-^{-1} = \big(\omega - \varrho_\pm\big)e^{-4\zeta^{1/2}}\,A_0(2\pi)^{\sigma_3/2}\Big[n^{\sigma_3/2}\Phi_{\alpha,-}e_2e_1^{\mathsf{T}}\Phi_{\alpha,-}^{-1}n^{-\sigma_3/2}\Big](2\pi)^{-\sigma_3/2}A_0^{-1}.$$

If $|\zeta| \geq 2$, then $n^{\sigma_3/2}\Phi_\alpha = f^{-\sigma_3/4}\Theta_\infty$ by (81), so the bracket is at most $C|f|^{-1/2} \leq C|z|^{-1/2}$, and $|e^{-4\zeta^{1/2}}| \leq e^{-n|z|^{1/2}/C}$ by (84). If $|\zeta| \leq 2$, the identity

$$\Phi_\alpha e_2 e_1^{\mathsf{T}}\Phi_\alpha^{-1} = -e^{4\zeta^{1/2}}\sigma_3\Psi_\alpha e_2 e_1^{\mathsf{T}}\Psi_\alpha^{-1}\sigma_3$$

and Lemma 7.1(iii) bound the bracket times $e^{-4\zeta^{1/2}}$ by $Cn|\zeta|^{-|\alpha|}$. With the right side of (90),

$$\mathfrak{m}_n(z) := |z|^{\nu_p}\log(2/|z|) + \varepsilon_\Lambda(T_n),$$

we obtain

$$|\tilde{Q}_-\Delta_1\tilde{Q}_-^{-1}| \leq C\mathfrak{m}_n(z)\cdot\begin{cases} n|\zeta|^{-|\alpha|}, & |\zeta| \leq 2, \\ |z|^{-1/2}e^{-n|z|^{1/2}/C}, & |\zeta| \geq 2. \end{cases} \tag{93}$$

In the same way, $\Phi_\alpha e_2 = \zeta^{-\sigma_3/4}\Theta_\infty e_2$ and $e_1^{\mathsf{T}}\Phi_\alpha^{-1} = e_1^{\mathsf{T}}\Theta_\infty^{-1}\zeta^{\sigma_3/4}$ for $|\zeta| \geq 2$, and $|\Phi_\alpha^{\pm 1}| \leq C|\zeta|^{-|\alpha|/2}$ for $|\zeta| \leq 2$, give

$$|d_n\tilde{Q}_-e_2| + |d_ne_1^{\mathsf{T}}\tilde{Q}_-^{-1}| \leq C\mathfrak{m}_n(z)\cdot\begin{cases} n^{1/2}|\zeta|^{-|\alpha|/2}, & |\zeta| \leq 2, \\ |z|^{-1/4}e^{-n|z|^{1/2}/C}, & |\zeta| \geq 2. \end{cases} \tag{94}$$

For $\alpha = 0$, replace $|\zeta|^{-|\alpha|}$ by $(1 + |\log|\zeta||)^2$ and $|\zeta|^{-|\alpha|/2}$ by $1 + |\log|\zeta||$.

*Bessel localisation.* We integrate (93) along the lips, on which $|\mathrm{d}z| \le C\,\mathrm{d}|z|$ and $|\zeta| \ge n^2|z|/C$. The region $|\zeta| \le 2$ lies in $|z| \le Cn^{-2}$, where $\mathfrak{m}_n \le C\varepsilon_n'$; its contribution is at most

$$C\varepsilon_n' \int_0^{Cn^{-2}} n\,(n^2 r/C)^{-|\alpha|}\,\mathrm{d}r = \frac{C\varepsilon_n'}{n}\int_0^{C} u^{-|\alpha|}\,\mathrm{d}u = O(\varepsilon_n'/n),$$

because $|\alpha| < 1$. In the region $|\zeta| \ge 2$ substitute $t = nr^{1/2}$:

$$\begin{aligned}\int_0^{\delta} \mathfrak{m}_n(r) r^{-1/2} e^{-nr^{1/2}/C}\,\mathrm{d}r &= \frac{2}{n}\int_0^{n\delta^{1/2}} \mathfrak{m}_n(t^2/n^2) e^{-t/C}\,\mathrm{d}t \\ &\le \frac{C}{n}\big(\varepsilon_\Lambda(T_n) + n^{-2\nu_p}\log n\big) = O(\varepsilon_n'/n).\end{aligned}$$

So $\int |\tilde Q_- \Delta_1 \tilde Q_-^{-1}|\,|\mathrm{d}s| = O(\varepsilon_n'/n)$; the same computation applied to the square of (94) gives

$$\|d_n \tilde Q_- e_2\|_{L^2}^2 + \|d_n e_1^{\mathsf T} \tilde Q_-^{-1}\|_{L^2}^2 = O(\varepsilon_n'^2/n). \tag{95}$$

(For $\alpha = 0$ the factor $\int_0^C u^{-|\alpha|}\,\mathrm{d}u$ becomes $\int_0^C (1 + |\log u|)^2\,\mathrm{d}u < \infty$.)

*The other main terms.* On $\Sigma_\circ$, $\tilde Q_- \Delta_2 \tilde Q_-^{-1} = \tilde Q_{\mathrm{in}} \tilde Q_{\mathrm{out}}^{-1} - I$, whose integrals are exactly the circle terms $\Xi_k$. The term $\Delta_3$ contributes $O(e^{-n/C})$: on $\Sigma \setminus (U(0,\delta) \cup U(4c,\varepsilon))$, $\tilde Q^{\pm 1} = (P_0^\infty)^{\pm 1}$ is bounded and $|s| \le Cn^p$, and on the lips $|\tilde Q_-^{\pm 1}| \le Cn^{1/2}|z|^{-1/2}$ (by (81), (82) and Lemma 7.1(iii)), so $|E||\tilde Q_-||\tilde Q_-^{-1}| \le C\exp(-n|z|^{-1/p}/C)\,n|z|^{-1}$ by (F4), whose integral is $O(e^{-n/C})$.

*The correction term.* By Theorem 7.5 and T3(c) applied to both problems, $Q_- = I + \mathcal{C}_v Q_-$ and $\tilde Q_- = I + \mathcal{C}_{\tilde v}\tilde Q_-$, so $(1 - \mathcal{C}_v)(Q_- - \tilde Q_-) = \mathcal{C}_-(\tilde Q_-(v - \tilde v))$ and

$$\|Q_- - \tilde Q_-\|_{L^2} \le 2C\,\|\mathcal{C}_-\|\,\|\tilde Q_-(v - \tilde v)\|_{L^2} \le C\big(\|d_n \tilde Q_- e_2\|_{L^2} + n^{-1} + e^{-n/C}\big) \le C\big(\varepsilon_n' n^{-1/2} + n^{-1}\big),$$

using (95), the bound $O(1/n)$ for $v - \tilde v$ on the circles (where $\tilde Q_-$ is bounded), and $\Delta_3$. Similarly $\|(v - \tilde v)\tilde v^{-1}\tilde Q_-^{-1}\|_{L^2} \le C(\varepsilon_n' n^{-1/2} + n^{-1})$. By the Cauchy–Schwarz inequality,

$$\left|\int (Q_- - \tilde Q_-)(v - \tilde v)\tilde v^{-1}\tilde Q_-^{-1}\,\mathrm{d}s\right| \le C\big(\varepsilon_n' n^{-1/2} + n^{-1}\big)^2 \le C\big(\varepsilon_n'^2/n + n^{-2}\big).$$

The factor $s$ in the second moment is bounded on the lips and the circles, and costs at most $n^p$ on the rest, where everything is exponentially small. Collecting terms gives the proposition. □

*Remark* 7.7 (what is not proved). Relative to the comparison problem, the lip mismatch moves $Q_1$ and $Q_2$ by $O(\varepsilon_n'/n)$. The comparison problem itself depends on $\ell_1$ at order $1/n$, but only through the soft edge: on $\partial U(0,\delta)$, $\tilde Q_{\mathrm{in}}\tilde Q_{\mathrm{out}}^{-1} = \hat P_0 (P_0^\infty)^{-1}$ involves only $\alpha$, $f$ and $c$, whereas the Airy parametrix contains $\omega = D_W^2/W_0$ (76), so the circle terms $\Xi_k$ may depend on $\ell_1$. We do not analyse them; this is why the $O(1/n)$ in Theorem 3 is absolute, not relative (Remark 1.5).

*Remark* 7.8 (Where $|\alpha| < 1$ is used; the case $\alpha \ge 1$). The restriction $|\alpha| < 1$ is used in an essential way in the following three places, and nowhere else. (The vertex conditions in Appendix D need only $|\alpha| < 2$, and the bound on $\Delta_3$ in the proof of Proposition 7.6 uses $|\alpha| \le 1$ only to bound a power of $|z|$ that is multiplied by an exponentially small factor.) (a) Square integrability at the Bessel scale: $|z|^{-|\alpha|/2-\eta}$ is square integrable on the lips near 0 for small $\eta$, which gives $Q_\pm \in I + \partial\mathcal{C}(L^2)$ (F7) (Appendix B) and $\tilde Q_\pm^{\pm 1} \in I + \partial\mathcal{C}(L^2)$ (Proposition 7.4(i)); these are the hypotheses of T3(b) with $r = r' = 2$, used in Theorem 7.5 and Proposition 7.6. (b) The power-weighted Cauchy bound, Lemma C.1 with $\beta = \pm\alpha/2$, $|\beta| < \frac12$, in the uniform

bound near 0 (Lemma D.1). (c) The integrability of $|\zeta|^{-|\alpha|}$ on a ray near 0: the Hilbert–Schmidt bounds of Lemma D.1 and the Bessel localisation in Proposition 7.6. For $\alpha \geq 1$ all three fail: $\tilde{Q}$ is of size $n^{1/2}|\zeta|^{-\alpha/2}$ at the Bessel scale and is not square integrable on the lips near 0; Lemma C.1 needs $|\beta| = \alpha/2 < \frac{1}{2}$; and $\int_0^2 u^{-\alpha}\,\mathrm{d}u$ diverges. (Since $s > 1$, always $\alpha > -1$.) This is why the base window $|\alpha| < 1$ is combined with the Christoffel shift (Theorem 2). It is consistent with the remark of Deift and Piorkowski quoted in §1.4, that for a weight vanishing to first order at an endpoint the crucial step of [2] requires a different proof. An analysis of the weight itself for $\alpha \geq 1$ remains open (Section 10).

## 8. Extraction: proof of Theorem 3

The recurrence coefficients are read off from the first two coefficients at infinity, through the functionals $\mathsf{a}$ and $\mathsf{b}$ of (35): by (F8), $\mathsf{b}(Q) = n^{2p}b_n^2$ and $\mathsf{a}(Q) = n^p a_n$. Both come from the *same* matrix of degree $n$, so the ratio $a_n/(2b_n)$ in Theorem 3 pairs the diagonal entry at $P_n$ with the off-diagonal entry $b_n$ between $P_{n-1}$ and $P_n$, as in (10); this matters for any statement about the $1/n$ terms (§10.1).

*Proof of Theorem 3.* Let $p > 0$, $1 < s < 1 + 2p$ and let $\ell_1$ satisfy $(\mathrm{A}_\vartheta)$ for some $\vartheta > 0$. For $n \geq n_1$, Proposition 7.6 gives $Q_1 = P_1 + O(1/n)$ and $Q_2 = P_2 + O(1/n)$, where $P_1, P_2$ are the coefficients of the global parametrix (66). We compute them. With $a(z) = (1 - 4c/z)^{1/4}$,

$$a = 1 - \frac{c}{z} - \frac{3c^2}{2z^2} + O(z^{-3}), \qquad a^{-1} = 1 + \frac{c}{z} + \frac{5c^2}{2z^2} + O(z^{-3}),$$

so $\frac{1}{2}(a + a^{-1}) = 1 + \frac{c^2}{2z^2} + O(z^{-3})$ and $\frac{1}{2i}(a - a^{-1}) = \frac{ic}{z} + \frac{2ic^2}{z^2} + O(z^{-3})$. Hence

$$(P_1)_{12} = ic, \quad (P_1)_{21} = -ic, \quad (P_1)_{22} = 0, \quad (P_2)_{12} = 2ic^2,$$

and $\mathsf{b}(P_0^\infty) = c^2$, $\mathsf{a}(P_0^\infty) = 2c$. By (F8),

$$n^{2p}b_n^2 = (Q_1)_{12}(Q_1)_{21} = \big(ic + O(\tfrac{1}{n})\big)\big(-ic + O(\tfrac{1}{n})\big) = c^2\big(1 + O(\tfrac{1}{n})\big),$$

so $n^p b_n = c(1 + O(1/n))$ since $b_n > 0$. Since $(Q_1)_{12} = ic + O(1/n) \neq 0$ for large $n$,

$$n^p a_n = \mathsf{a}(Q) = \frac{2ic^2 + O(1/n)}{ic + O(1/n)} - O(\tfrac{1}{n}) = 2c\big(1 + O(\tfrac{1}{n})\big),$$

and therefore $a_n/(2b_n) = 1 + O(1/n)$. The constants depend on $p$, $s$ and $\ell_1$. □

The limits come entirely from the global parametrix: $\mathsf{b}(P_0^\infty) = c^2$ and $\mathsf{a}(P_0^\infty) = 2c$ are the squared off-diagonal and the diagonal entry of the constant Jacobi matrix with diagonal $2c$ and off-diagonal $c$, whose spectrum is the band $[0, 4c]$; the length $4c(p)$ of the band is fixed by the saturation of the constrained equilibrium problem (Proposition 2.4). The weight, including the slowly varying factor, affects only the $O(1/n)$ terms.

## 9. Numerical illustrations

The computations in this section are floating-point illustrations; they play no role in the proofs. The scripts will be made available as supplementary material (repository URL and commit hash: to be supplied before submission).

*What is shown.* Table 2 compares the factor $\ell(k) = \exp\sqrt{\log(k + e)}$, which satisfies $(\mathrm{A}_\vartheta)$ for every $\vartheta < \pi$, with $\ell \equiv 1$, for $p = \frac{1}{2}$, 1, 2 and $s = 2$. By Theorem 1(ii) and Corollary 1.6 the normalised deviations $n(n^p b_n/c(p) - 1)$ and $n(a_n/(2b_n) - 1)$ stay bounded. The Bessel orders are $\alpha = 1$ (outside the window of Theorem 3, reached through the shift from $s = \frac{3}{2}$), $\alpha = 0$ (the logarithmic case of the Bessel model) and $\alpha = -\frac{1}{2}$.

*How the numbers were computed.* The atoms $k \le K$ alone do not suffice: the spectral weight of $P_n$ near the hard edge sits at indices $k \asymp n^{1+2/p}$, about $10^{16}$ for $p = \frac{1}{2}$ and $n = 1600$. A truncated measure has a soft edge at its smallest atom and obeys a different law; for $p = \frac{1}{2}$, $s = \frac{3}{2}$, the same $\ell$ and the atoms $k \le 2 \cdot 10^5$ alone, $n(n^p b_n/c(p) - 1)$ equals $-1.81$, $-13.3$ and $-105.9$ at $n = 100$, 400 and 1600. We therefore used the atoms $k \le K$ together with a Gauss–Legendre quadrature of the continuous tail $\int_{K+1/2}^{T} t^{-s}\ell(t)\,\delta_{t^{-p}}\,\mathrm{d}t$ on logarithmic panels (ratio 1.02, eight nodes per panel), with $T = 10^{48}$ for $p = \frac{1}{2}$ and $T = 10^{16}$ for $p = 1, 2$. The Jacobi matrix of this discrete measure was computed in extended (long double) precision by the updating algorithm that adds one atom at a time and restores tridiagonal form by plane rotations, for $K = 10^5$ and $K = 2 \cdot 10^5$, and for $K = 2 \cdot 10^5$ with $T$ reduced by a factor $10^4$. The largest change of any tabulated entry under these variations is $10^{-7}$ ($p = \frac{1}{2}$), $2 \cdot 10^{-8}$ ($p = 1$) and $2 \cdot 10^{-11}$ ($p = 2$). The column $\ell \equiv 1$ at $p = 2$ agrees in every digit shown with the closed form of Remark 1.7. That closed form was in turn verified in interval (ball) arithmetic, with rigorous enclosures, for all $n < 300$: the coefficients were computed by the Chebyshev algorithm from the moments $\zeta(2j + 2)$, and the enclosures agree with the closed form to a relative accuracy better than $10^{-2000}$ (supplementary material, certified suite).

| | $n\big(n^p b_n/c(p) - 1\big)$ | | $n\big(a_n/(2b_n) - 1\big)$ | |
|---|---|---|---|---|
| $n$ | $\ell$ | $\ell \equiv 1$ | $\ell$ | $\ell \equiv 1$ |
| $p = \frac{1}{2}$, $s = 2$, $\alpha = 1$ (reached through the shift) | | | | |
| 100 | −0.4039 | −0.5069 | −0.2442 | −0.2408 |
| 200 | −0.4069 | −0.5052 | −0.2470 | −0.2453 |
| 400 | −0.4101 | −0.5039 | −0.2484 | −0.2477 |
| 800 | −0.4133 | −0.5028 | −0.2492 | −0.2488 |
| 1600 | −0.4163 | −0.5020 | −0.2496 | −0.2494 |
| $p = 1$, $s = 2$, $\alpha = 0$ | | | | |
| 50 | −0.3821 | −0.4934 | −0.4954 | −0.4950 |
| 100 | −0.3893 | −0.4967 | −0.4978 | −0.4975 |
| 200 | −0.3958 | −0.4983 | −0.4989 | −0.4987 |
| 400 | −0.4015 | −0.4992 | −0.4994 | −0.4994 |
| 800 | −0.4065 | −0.4996 | −0.4997 | −0.4997 |
| $p = 2$, $s = 2$, $\alpha = -\frac{1}{2}$ | | | | |
| 50 | −0.3794 | −0.4938 | −0.9756 | −0.9780 |
| 100 | −0.3868 | −0.4969 | −0.9878 | −0.9889 |
| 200 | −0.3936 | −0.4984 | −0.9939 | −0.9944 |
| 400 | −0.3995 | −0.4992 | −0.9970 | −0.9972 |
| 800 | −0.4046 | −0.4996 | −0.9985 | −0.9986 |

TABLE 2. Normalised deviations from the pointwise law for $\sigma_{p,2,\ell}$, with $\ell(k) = e^{\sqrt{\log(k+e)}}$ and for $\ell \equiv 1$, computed in extended precision from the atoms $k \le K$ and a quadrature of the continuous tail; all entries are stable under doubling of $K$ and under a change of the tail cut-off (see the text). The limits $-(s-1)/2$ and $-p/2$ in the columns $\ell \equiv 1$ are observed, not proved. The slow drift of the first column is the effect of $\Lambda'(\log n)$ (§10.1). Source: supplementary script `table2`.

*What the table shows.* All entries stay bounded, as the theorems require. For the pure powers the two normalised deviations approach $-(s-1)/2 = -\frac{1}{2}$ and $-p/2$ in all three geometries.

For $\ell$ the deviation of $n^p b_n$ is smaller in absolute value, and the difference from the pure power drifts slowly towards 0, roughly like $\frac{1}{2}\Lambda'(\log n)$ with $\Lambda(\xi) = \sqrt{\log(e^\xi + e)}$; see §10.1. The factor $\ell$ has almost no effect on $a_n/(2b_n)$ at order $1/n$.

*A factor analytic only in a thin strip.* Theorem 1(ii) needs $(\mathrm{A}_\vartheta)$ for some $\vartheta > 0$ only. As a test we took $p = 2$ and $\Lambda(\xi) = \sqrt{\xi+1} + (\xi+2)^{-1/2}\log(1 - q\cos 2\xi)$ with $q = 1/\cosh(0.6)$, the example of §1.2. Here $1 - q\cos 2\xi$ vanishes at $\xi = \pi m \pm 0.3i$ for every integer $m$, so $\Lambda$ satisfies $(\mathrm{A}_\vartheta)$ exactly for $\vartheta < 0.3$, and the singularities are not confined to a bounded part of the strip (singularities near $\Re\xi = 0$ alone would be a weaker test, since they concern only the head of the measure, whose influence is exponentially small by Proposition 4.5). The factor $\ell(k) = e^{\Lambda(\log k)}$ grows like $e^{\sqrt{\log k}}$ and carries a log-periodic oscillation of relative size $(\log k)^{-1/2}$. With the same numerical scheme, at $n = 50 \cdot 2^{j/4}$, $0 \le j \le 24$ (so $50 \le n \le 3200$), the deviation $n(n^2 b_n/c(2) - 1)$ oscillates log-periodically in $[-0.603, -0.029]$ for $s = 2$ ($\alpha = -\frac{1}{2}$) and in $[-1.101, -0.549]$ for $s = 3$ ($\alpha = 0$); its maximum modulus is 0.598 on $50 \le n < 400$ and 0.589 on $400 < n \le 3200$ for $s = 2$, and 1.097, respectively 1.088, for $s = 3$, while $n(a_n/(2b_n) - 1)$ stays in $[-1.009, -0.951]$. The range of $n$ covers only about 1.3 periods of the oscillation (period $\pi$ in $\log n$), so this is weak evidence against growth; the deviations are bounded, consistent with Theorem 1(ii); the oscillation of the $1/n$ term reflects that of $\Lambda'$, which here has no sign (compare §10.1).

## 10. Further remarks and open problems

10.1. **The $1/n$ term.** Theorem 1(ii) and Corollary 1.6 give the rate $O(1/n)$ but not the coefficient of $1/n$. The computations of Section 9, and similar ones for other $s$ (for $p = \frac{1}{2}$, $s \in \{1.5, 1.75, 2, 2.5\}$; for $p = 1$, $s \in \{2, 2.5, 3\}$), suggest the following. Throughout, $a_n/(2b_n)$ pairs the diagonal entry at $P_n$ with the off-diagonal entry $b_n$ between $P_{n-1}$ and $P_n$, as in (2); in the other pairing $a_n/(2b_{n+1})$ the coefficient $-p/2$ below becomes $+p/2$, since $b_{n+1}/b_n = 1 - p/n + O(n^{-2})$.

**Conjecture 10.1.** (a) *For every $p > 0$ and $s > 1$, the pure powers satisfy*

$$\frac{n^p b_n(\sigma_{p,s})}{c(p)} = 1 - \frac{s-1}{2n} + o\Big(\frac{1}{n}\Big), \qquad \frac{a_n(\sigma_{p,s})}{2b_n(\sigma_{p,s})} = 1 - \frac{p}{2n} + o\Big(\frac{1}{n}\Big).$$

(b) *If $\ell$ satisfies $(\mathrm{A}_\vartheta)$ for some $\vartheta > 0$, with $\ell(k) = \exp\Lambda(\log k)$, then*

$$\frac{n^p b_n(\sigma_{p,s,\ell})}{c(p)} = 1 - \frac{s - 1 - \Lambda'(\log n)}{2n} + o\Big(\frac{\Lambda'(\log n)}{n}\Big) \qquad \textit{whenever} \quad n\Lambda'(\log n) \to \infty.$$

Part (a) holds for $(p, s) = (2, 2)$, by the closed form of Remark 1.7. Part (b) says that, at order $1/n$, a slowly varying factor acts like a change of the exponent $s$ by the local logarithmic derivative $\Lambda'(\log n)$, i.e. like a hard-edge Bessel order that drifts with $n$. In Table 2 the difference between the first two columns is 0.91 to 0.99 times $\frac{1}{2}\Lambda'(\log n)$, and this ratio increases with $n$ in all three geometries. The $1/n$ coefficient behaves well under the shift: by the identities of Section 3, (a) at $s$ implies (a) at $s+p$ provided the chain-sequence parameters of Section 3 have increments $n^p u_n - (n-1)^p u_{n-1} = o(1/n)$, a condition we can only check numerically. Within our method, (b) amounts to computing the main term of the comparison estimate of Section 7, as Conway and Deift do for their weight [2, (5.5)]; this would need a Bessel parametrix with a drifting order.

10.2. **Rough weights.** The announcement in §1.4 shows that the pointwise law can fail for weights $k^{-s}e^{\epsilon_k}$ with $|\epsilon_k| \le \varepsilon$, although the averaged law holds for them. Two questions remain. First, between slow variation and arbitrary bounded oscillation there is a range of regularity

conditions for which we do not know the answer. A simple test case is $\epsilon_k = \varepsilon \sin\log k$: its increments $\epsilon_{k+1} - \epsilon_k$ tend to 0, but $e^{\epsilon_k}$ is not slowly varying, so Theorem 1 does not apply. Second, for bounded rough $\epsilon_k$ one may ask whether the law holds along a subsequence of density one; we do not know.

10.3. **Other nodes, other frames, larger Bessel orders.** Our nodes are exact powers $k^{-p}$. For nodes $x_k$ that are regularly varying in $k$ of index $-p$ the averaged law is known [15], but the pointwise law is open. Heuristically one expects $b_n/x_n \to c(p)$ under smooth regular variation of the nodes, and failure under rough displacements of the nodes, in analogy with the weights.

For Bessel orders $\alpha \geq 1$ we pass through the shift (Theorem 2). An RH analysis of the weight itself for $\alpha \geq 1$ would require a different comparison frame (Remark 7.8); it would be needed to compute the $1/n$ term directly in that range.

At $p = 1$ one can also work in the symmetric variable $y = \pm x^{1/2}$ of [19], where 0 is an interior Fisher–Hartwig point of the band; the comparison method then covers only $1 < s < 2$ directly, and $1 < s < 3$ after folding. We do not pursue this. For pure powers and $p \neq 1$ a classical quotient parametrix also works (Remark 6.2).

## Appendix A. Construction of the outer scalar for $p \neq 1$

This appendix proves Lemma 6.1 for $p > 1$ (§A.1) and for $0 < p < 1$ (§A.2). We use the notation of Section 6: Binet's function $j$, $\tilde{D} = e^{-j(t)}$, $D = \kappa_\pm \tilde{D}$, $K = (4c - z)^{-1/2}$, and the bounds (58), (59). In both cases the difficulty is the cut of $t(z)$ on $(-\infty, 0)$, where $D$ has a jump that is not of the form required by (O1). For $p > 1$ the jump is removed exactly by a Cauchy integral; for $p < 1$, $D$ is used only in an inner sector and glued to an elementary function outside.

A.1. **The case $p > 1$: the Binet-cut scalar.** For $p > 1$, $|\arg t| < \pi/p < \pi$ on the slit plane, so $\Gamma(t)$ has no poles there, and the jump of $D$ on $(-\infty, 0)$ is a bounded phase $e^{i\mathfrak{b}}$.

**Lemma A.1** (Binet-cut scalar)**.** *Let $p > 1$.*

(i) *$\tilde{D}$ is analytic and zero-free on $\mathbb{C} \setminus (-\infty, 0]$, and $D$ is analytic and zero-free in $\mathbb{C}_\pm$, with continuous boundary values on $\mathbb{R} \setminus (\{X_k\} \cup \{0\})$. On $(-\infty, 0)$, $D_- = \overline{D_+}$.*

(ii) *Let $\mathfrak{b}(y) := 2\,\mathrm{Im}\big(-j(t_+(y)) - \mathrm{Log}(1 - e^{-2i\theta_+(y)})\big)$ for $y < 0$, where $t_+, \theta_+$ are boundary values from $\mathbb{C}_+$. Then $D_+/D_- = e^{i\mathfrak{b}}$ on $(-\infty, 0)$; $\mathfrak{b}$ is real-analytic, $|\mathfrak{b}(y)| \leq C \min(1, |y|^{1/p})$ with $C$ independent of $n$, $|\mathfrak{b}'(y)| \leq C_n |y|^{1/p-1}$ for $-1 \leq y < 0$, $\mathfrak{b}'$ is bounded on $(-\infty, -\frac{1}{2}]$, and $\mathfrak{b}(y) \to \pi/p - \pi$ as $y \to -\infty$, with $\int_{-\infty}^{-1} |\mathfrak{b}(y) - \pi/p + \pi|\,|y|^{-1}\,\mathrm{d}y < \infty$.*

(iii) *The function*

$$B(z) := \exp\Big(-\frac{1}{2\pi}\int_{-\infty}^{0} \mathfrak{b}(y)\Big[\frac{1}{y - z} - \frac{1}{y - 1}\Big]\,\mathrm{d}y\Big) \tag{96}$$

*is analytic and zero-free on $\mathbb{C} \setminus (-\infty, 0]$, positive on $(0, \infty)$, has continuous boundary values on $(-\infty, 0)$ with $B_+ = B_- e^{-i\mathfrak{b}}$, and $DB$ extends analytically across $(-\infty, 0)$. For $z \in \mathcal{Z}_n$, with $C$ independent of $n$,*

$$\begin{gathered} |\log B(z) - \log B(0)| \leq C|z|^{1/p}, \quad |(\log B)'(z)| \leq C|z|^{1/p-1} \quad (|z| \leq 1), \\ |\log B(z)| \leq C(1 + \log(1 + |z|)). \end{gathered} \tag{97}$$

(iv) *$F := DBK$ satisfies (O1)–(O5), with $\tilde{F} = \tilde{D}BK$ and $H_n = (\tilde{D}B)^{-2}$, $H_n(0^+) = B(0)^{-2}$.*

*Proof.* Part (i) is the absence of poles of $\Gamma$ in the sector; (ii) estimates the jump $\mathfrak{b}$ by Stirling's formula for large $|t|$ and by (59) for small $|t|$; (iii) is the Sokhotski–Plemelj formula together with a splitting of the Cauchy integral at $|y| = |z|$; (iv) checks (O1)–(O5) one by one.

(i) On the slit plane $|\arg t| < \pi/p < \pi$, so $\Gamma(t)$ has neither zeros nor poles and $\tilde{D}$ is analytic and zero-free. By (41), $|e^{\mp 2i\theta}| < 1$ in $\mathbb{C}_\pm$ (here $\pi p > \pi$), so $D$ is analytic and zero-free in $\mathbb{C}_\pm$. On $(-\infty, 0)$, $t_- = \overline{t_+}$ and $\Gamma$, $j$ are real on the real axis, so $D_- = \overline{D_+}$.

(ii) Since $|e^{-2i\theta_+}| < 1$ on $(-\infty, 0)$, $\log D_+ := -j(t_+) - \operatorname{Log}(1 - e^{-2i\theta_+})$ is a continuous logarithm, real-analytic in $y$, and $D_+/D_- = e^{2i \operatorname{Im} \log D_+} = e^{i\mathfrak{b}}$. With $|t_+| = n|y|^{-1/p}$ and $\arg t_+ = -\pi/p$: for $|t_+| \geq 1$, (58) and $|\operatorname{Log}(1 - e^{-2i\theta_+})| \leq 2\exp(-2\pi \sin(\pi/p)|t_+|)$ give $|\mathfrak{b}| \leq C/|t_+| = C|y|^{1/p}/n$. For $|t_+| \leq 1$ the expansion (59) and $\operatorname{Log}(1 - e^{-2\pi i t}) = \log(2\pi|t|) + i(\frac{\pi}{2} + \arg t) + O(t)$ (note $\frac{\pi}{2} + \arg t_+ \in (-\frac{\pi}{2}, \frac{\pi}{2})$) show that $\mathfrak{b}$ is bounded and

$$\mathfrak{b}(y) = 2\big(\tfrac{1}{2}\arg t_+ - \tfrac{\pi}{2} - \arg t_+\big) + O(|t_+|\log(2/|t_+|)) = \frac{\pi}{p} - \pi + O\big(n|y|^{-1/p}\log|y|\big).$$

This gives the bounds on $\mathfrak{b}$ and the integrability. For the derivative, $\mathrm{d}t_+/\,\mathrm{d}y = -t_+/(py)$, and $\frac{\mathrm{d}}{\mathrm{d}t}\big(j(t) + \operatorname{Log}(1 - e^{-2\pi i t})\big)$ is $O(|t|^{-2})$ for $|t| \geq 1$ and $O(|t|^{-1})$ for $|t| \leq 1$; multiplied by $|t|/|y|$ this gives $|\mathfrak{b}'(y)| \leq C|y|^{1/p-1}/n$ for $|y| \leq n^p$ and $|\mathfrak{b}'(y)| \leq C/|y|$ otherwise.

(iii) The integral converges absolutely, since $\mathfrak{b}$ is bounded and the bracket is $(z-1)/((y-z)(y-1))$. By the Sokhotski–Plemelj formula (the density is real-analytic on $(-\infty, 0)$), $\log B$ has continuous boundary values and $(\log B)_+ - (\log B)_- = -i\mathfrak{b}$. So $(DB)_+ = (DB)_-$ on $(-\infty, 0)$, and $DB$, being continuous across, is analytic there by Morera's theorem. $B$ is positive on $(0, \infty)$ because the integrand is real there.

For (97), let $z \in \mathcal{Z}_n$. Then $|y - z|^2 = y^2 + |z|^2 + 2|y||z|\cos(\arg z) \geq \frac{1}{2}(|y| + |z|)^2$ for $y < 0$, since $|\arg z| < \vartheta_{\mathrm{w}} < \pi/2$. We have $\log B(z) - \log B(0) = -\frac{z}{2\pi}\int_{-\infty}^{0} \frac{\mathfrak{b}(y)\,\mathrm{d}y}{y(y-z)}$, so by (ii), for $|z| \leq 1$,

$$|\log B(z) - \log B(0)| \leq C|z| \int_0^\infty \frac{\min(1, u^{1/p})}{u(u + |z|)}\,\mathrm{d}u \leq C|z|\Big(\int_0^1 \frac{u^{1/p-1}\,\mathrm{d}u}{u + |z|} + 1\Big) \leq C|z|^{1/p},$$

because $1/p - 1 \in (-1, 0)$. The same splitting applied to $(\log B)'(z) = -\frac{1}{2\pi}\int \mathfrak{b}(y)(y-z)^{-2}\,\mathrm{d}y$ gives $C|z|^{1/p-1}$, and applied to the original integral gives $C(1 + \log(1 + |z|))$.

(iv) *(O1).* $F$ is analytic and zero-free off $[0, \infty)$ by (i), (iii) and the choice of $K$. On $(4c, \infty)$ off the nodes, $F_+/F_- = (D_+/D_-)(K_+/K_-) = (-e^{2i\theta})(-1)$, since $B$ is analytic there.

*(O2).* $\tilde{F} = F/\kappa_\pm = \tilde{D}BK$ is analytic and zero-free on $\mathbb{C}\setminus\big((-\infty, 0]\cup[4c, \infty)\big)$, and $\tilde{F}_+ = -\tilde{F}_-$ on $(4c, \infty)$ because of $K$. Thus $H_n = 1/((4c - z)\tilde{F}^2) = (\tilde{D}B)^{-2}$, which is positive on $(0, \infty)$, and $H_n(0^+) = B(0)^{-2}$ because $\tilde{D}(z) \to 1$ as $z \to 0$ in $\mathcal{Z}_n$ by (58).

*(O3), (O4).* $\log H_n - \log H_n(0^+) = 2j(t) - 2(\log B - \log B(0))$. For $z \in \mathcal{Z}_n$, $|z| \leq 1$, we have $|t| \geq n$ and $|\arg t| \leq \vartheta_{\mathrm{w}}/p$, so $|j(t)| \leq C|z|^{1/p}/n$ and $|\frac{\mathrm{d}}{\mathrm{d}z} j(t)| = |j'(t)||t|/(p|z|) \leq C|z|^{1/p-1}/n$. Together with (97) and $\nu_p \leq 1/p$, this is (O3). For $|z| \geq 1$ in $\mathcal{Z}_n$ we have $|t| \geq e^{-c_0}$, so $j(t)$ is bounded (by (58) for $|t| \geq 1$ and by continuity on the compact set $\{e^{-c_0} \leq |t| \leq 1,\ |\arg t| \leq \vartheta_{\mathrm{w}}/p\}$), and (97) gives (O4).

*(O5).* Fix $n$. Near 0 and outside the lens, $|\arg z| \geq \frac{3}{4}\vartheta_{\mathrm{lip}}$, so $\kappa_\pm \to 1$ by (41) and $\tilde{D} \to 1$; $K$ is bounded; and $\log B$ is bounded on $U(0, \delta) \setminus (-\delta, 0]$ uniformly up to the cut. For the last point split $\int \mathfrak{b}(y)/(y - z)\,\mathrm{d}y$ into the window $|y - \operatorname{Re} z| \leq |\operatorname{Re} z|/2$ and the rest. On the window write $\mathfrak{b}(y) = \mathfrak{b}(\operatorname{Re} z) + O(|y - \operatorname{Re} z|\,|\operatorname{Re} z|^{1/p-1})$ by (ii); the constant part integrates to $\mathfrak{b}(\operatorname{Re} z)\operatorname{Log}(\cdot)$ with a bounded logarithm (the end points of the window are at comparable distances from $z$), and the rest is bounded; outside the window $|y - z| \geq (|y| + |z|)/C$ and $|\mathfrak{b}(y)| \leq C|y|^{1/p}$ give a bounded contribution. In the lens, $\tilde{F} = \tilde{D}BK$ is bounded above and below by the same facts.

Near $\infty$, the same window argument on $(-\infty, -\frac{1}{2}]$, where $\mathfrak{b}'$ is bounded, gives $|\log B(z)| \le C_n(1+\log|z|)$ for $|z| \ge 2$ off the cut. For $|z| \ge n^p + \delta'$ we have $|t| < 1$, and $t$ stays away from 1, so $D$ is bounded above and below by a power of $|z|$; hence $|F|^{\pm 1} \le C_n|z|^{N_n}$. Finally, along $z = iy$, $y \to \pm\infty$: by (59), $|D(iy)| = |t|^{-1/2}(2\pi)^{-1/2}(1+o(1)) \asymp |y|^{1/(2p)}$; $|K(iy)| \asymp |y|^{-1/2}$; and splitting $\mathfrak{b} = (\pi/p - \pi)\mathbf{1}_{(-\infty,-1)} + \mathfrak{e}$, the first part contributes $-\frac{\pi/p-\pi}{2\pi}\operatorname{Log}\frac{z+1}{2}$ to $\log B$ (an explicit integral), while the second contributes a quantity with a finite limit, by dominated convergence and the integrability in (ii). So $|B(iy)| \asymp |y|^{\frac{1}{2}-\frac{1}{2p}}$, and $|F(iy)| \asymp 1$. □

A.2. **The case $0 < p < 1$: seams.** For $p < 1$ the poles $t = -1, -2, \dots$ of $\Gamma$ lie on the rays $\arg z = \pm\pi p$, inside the slit plane, and $D$ has no useful continuation to the negative axis. We use $D$ only in the inner sector, where $|\arg t| < \pi/2$, and an elementary function outside; a Cauchy integral $\chi$ over the two separating rays glues them together exactly. Put

$$\vartheta_{\text{seam}} := \tfrac{\pi p}{2}, \qquad S_{\text{in}} := \{z \ne 0 : \ |\arg z| < \vartheta_{\text{seam}}\}, \qquad S_{\text{out}} := \mathbb{C} \setminus (\overline{S_{\text{in}}} \cup \{0\}),$$
$$\Sigma^{\text{s}}_{\pm} := \{re^{\pm i\vartheta_{\text{seam}}} : \ r > 0\},$$

both seams oriented away from 0, and

$$\varpi := \tfrac{1}{2} - \tfrac{1}{2p} < 0, \qquad B_{\text{in}}(z) := (1+z)^{\varpi}, \qquad B_{\text{out}}(z) := \Big(1 - \frac{z}{2\pi n}\Big)^{1/2},$$

with principal powers; the cut $(-\infty, -1]$ of $B_{\text{in}}$ lies in $S_{\text{out}}$, and the cut $[2\pi n, \infty)$ of $B_{\text{out}}$ lies in $S_{\text{in}}$. The lens, the saturated regions and the top segment lie in $\mathcal{Z}_n \subset S_{\text{in}}$, since $\vartheta_{\text{w}} < \vartheta_{\text{seam}}$ by (34).

**Lemma A.2** (Seam scalar)**.** *Let $0 < p < 1$.*

(i) *On $S_{\text{in}}$, $|\arg t| < \pi/2$; $\tilde{D} = e^{-j(t)}$ is analytic and zero-free there, and $D = \kappa_{\pm}\tilde{D}$ is analytic and zero-free in $S_{\text{in}} \cap \mathbb{C}_{\pm}$.*

(ii) *On $\Sigma^{\text{s}}_{\pm}$ put, with principal logarithms,*

$$\chi_{\pm}(\zeta) := -j(t(\zeta)) - \operatorname{Log}\big(1 - e^{\mp 2\pi i t(\zeta)}\big) + \varpi \operatorname{Log}(1+\zeta) - \tfrac{1}{2}\operatorname{Log}\Big(1 - \frac{\zeta}{2\pi n}\Big). \tag{98}$$

*Then $\chi_{\pm}$ is real-analytic on the open ray, $e^{\chi_{\pm}} = DB_{\text{in}}/B_{\text{out}}$ there, $\chi_{\pm}(\zeta) \to 0$ as $\zeta \to 0$, and $\chi_{\pm}(\zeta) = O\big(n|\zeta|^{-1/p}\log|\zeta| + n|\zeta|^{-1}\big)$ as $\zeta \to \infty$. In particular $\chi_{\pm}$ has no winding: it is the continuous logarithm of $DB_{\text{in}}/B_{\text{out}}$ that vanishes at both ends. Moreover $\chi_-(\bar{\zeta}) = \overline{\chi_+(\zeta)}$.*

(iii) *The* seam scalar

$$\chi(z) := \frac{1}{2\pi i}\Big[\int_{\Sigma^{\text{s}}_+} \frac{\chi_+(\zeta)\,\mathrm{d}\zeta}{\zeta - z} - \int_{\Sigma^{\text{s}}_-} \frac{\chi_-(\zeta)\,\mathrm{d}\zeta}{\zeta - z}\Big] \tag{99}$$

*is analytic off $\Sigma^{\text{s}}_+ \cup \Sigma^{\text{s}}_-$, real on $\mathbb{R} \setminus \{0\}$, tends to 0 at $\infty$, has continuous boundary values on the open seams, and $\chi_{\text{out}} - \chi_{\text{in}} = \chi_{\pm}$ on $\Sigma^{\text{s}}_{\pm}$, where "out" and "in" refer to the boundary values from $S_{\text{out}}$ and $S_{\text{in}}$.*

(iv) *The function*

$$F := DB_{\text{in}}Ke^{\chi} \text{ in } S_{\text{in}}, \qquad F := B_{\text{out}}Ke^{\chi} \text{ in } S_{\text{out}}, \tag{100}$$

*extends analytically across $\Sigma^{\text{s}}_{\pm}$ and satisfies (O1)–(O5), with $\tilde{F} = \tilde{D}B_{\text{in}}Ke^{\chi}$ and $H_n = (\tilde{D}B_{\text{in}}e^{\chi})^{-2}$, $H_n(0^+) = e^{-2\chi(0)}$.*

*Proof.* (i) $|\arg t| = |\arg z|/p < \pi/2$ on $S_{\text{in}}$, so $\Gamma(t)$ has no poles or zeros there; and $\pm\operatorname{Im}\theta < 0$ in $S_{\text{in}} \cap \mathbb{C}_{\pm}$ by (41).

(ii) Each term of (98) is continuous on the open ray $\Sigma_+^{\mathrm{s}}$: $j$ is analytic off $(-\infty,0]$; $\arg t = -\pi/2$, so $e^{-2\pi i t} = e^{-2\pi|t|} \in (0,1)$; $1+\zeta \notin (-\infty,0]$ since $\arg\zeta \in (0,\pi)$; and $1-\zeta/(2\pi n)$ has non-zero imaginary part. The exponential of the sum is $\tilde{D}\kappa_+ B_{\mathrm{in}}/B_{\mathrm{out}}$. As $\zeta\to 0$, $|t|\to\infty$ and each term tends to 0 (by (58) for $j$). As $\zeta\to\infty$, $t\to 0$ and, by (59), using $\arg t = -\pi/2$, $\arg\zeta = \vartheta_{\mathrm{seam}}$ and $\operatorname{Log} t = \log n - \frac{1}{p}\operatorname{Log}\zeta$:

$$\begin{aligned} -j(t) &= \tfrac{1}{2}\operatorname{Log} t + \tfrac{1}{2}\log(2\pi) + O(|t|\log(2/|t|)),\\ -\operatorname{Log}(1-e^{-2\pi i t}) &= -\log(2\pi) - \operatorname{Log} t - \tfrac{i\pi}{2} + O(|t|),\\ \varpi\operatorname{Log}(1+\zeta) &= \varpi\operatorname{Log}\zeta + O(|\zeta|^{-1}),\\ -\tfrac{1}{2}\operatorname{Log}\big(1-\tfrac{\zeta}{2\pi n}\big) &= -\tfrac{1}{2}\operatorname{Log}\zeta + \tfrac{1}{2}\log(2\pi n) + \tfrac{i\pi}{2} + O(n|\zeta|^{-1}); \end{aligned}$$

the second line uses $1-e^{-2\pi i t} = 2\pi i t(1+O(t))$ and $\arg(it)=0$, the last uses $\arg(-\zeta/(2\pi n)) = \vartheta_{\mathrm{seam}} - \pi \in (-\pi,0)$. The sum of the main terms is $-\frac{1}{2}\operatorname{Log} t + \frac{1}{2}\log n + (\varpi - \frac{1}{2})\operatorname{Log}\zeta = \big(\frac{1}{2p} + \varpi - \frac{1}{2}\big)\operatorname{Log}\zeta = 0$. The symmetry follows from $\overline{t(\bar\zeta)} = t(\zeta)$ and the reality of $j$.

(iii) By (ii), $\chi_\pm(\zeta)/\zeta$ is integrable on $\Sigma_\pm^{\mathrm{s}}$ near 0 and at $\infty$ (since $1/p>1$), so the integrals converge and $\chi\to 0$ at $\infty$. The Sokhotski–Plemelj formula gives the jumps: $\Sigma_+^{\mathrm{s}}$ is oriented outwards with $S_{\mathrm{out}}$ on its left, so $\chi_{\mathrm{out}} - \chi_{\mathrm{in}} = \chi_+$; on $\Sigma_-^{\mathrm{s}}$ the left side is $S_{\mathrm{in}}$ and the density carries a minus sign, so again $\chi_{\mathrm{out}} - \chi_{\mathrm{in}} = \chi_-$. Reality on $\mathbb{R}$ follows from the symmetry in (ii).

(iv) On $\Sigma_\pm^{\mathrm{s}}$, $F_{\mathrm{in}}/F_{\mathrm{out}} = DB_{\mathrm{in}}e^{\chi_{\mathrm{in}}}/(B_{\mathrm{out}}e^{\chi_{\mathrm{out}}}) = e^{\chi_\pm}e^{-\chi_\pm} = 1$, so $F$ is continuous, hence analytic, across the seams.

*(O1).* In $S_{\mathrm{out}}$ every factor is analytic and zero-free, and $(-\infty,0)\subset S_{\mathrm{out}}$. In $S_{\mathrm{in}}\setminus[0,\infty)$ the same holds by (i). On $(4c,\infty)$, $B_{\mathrm{in}}$ and $\chi$ are analytic, so $F_+/F_- = (-e^{2i\theta})(-1)$.

*(O2).* $\tilde F = \tilde D B_{\mathrm{in}} K e^{\chi}$ on $S_{\mathrm{in}}\supset\mathcal{Z}_n$, analytic and zero-free off $[4c,\infty)$, with $\tilde F_+ = -\tilde F_-$ there. $H_n = (\tilde D B_{\mathrm{in}} e^{\chi})^{-2} > 0$ on $(0,\infty)$ by (iii), and $H_n(0^+) = e^{-2\chi(0)}$, where $\chi(0) := \lim_{z\to 0}\chi(z)$ exists by Lemma A.3.

*(O3), (O4).* Points of $\mathcal{Z}_n$ satisfy $|\arg z| < \vartheta_{\mathrm{w}}$, hence $\operatorname{dist}(z,\Sigma_+^{\mathrm{s}}\cup\Sigma_-^{\mathrm{s}}) \ge \sin(\vartheta_{\mathrm{seam}}-\vartheta_{\mathrm{w}})|z|$ (note $0<\vartheta_{\mathrm{seam}}-\vartheta_{\mathrm{w}}<\pi/2$). So Lemma A.3(ii) applies with $\tau_0 = \sin(\vartheta_{\mathrm{seam}}-\vartheta_{\mathrm{w}})$. For $|z|\le 1$,

$$\begin{aligned} |\log H_n - \log H_n(0^+)| &\le 2\big(|j(t)| + |\varpi||\operatorname{Log}(1+z)| + |\chi-\chi(0)|\big)\\ &\le C\big(|z|^{1/p}/n + |z| + |z|(1+\log\tfrac{1}{|z|})\big) \le C|z|^{1/2}, \end{aligned}$$

and similarly $|(\log H_n)'| \le C(|z|^{1/p-1}/n + 1 + \log\frac{1}{|z|}) \le C|z|^{-1/2}$. Since $\nu_p = \frac{1}{2}$, this is (O3). For $|z|\ge 1$ in $\mathcal{Z}_n$, $j(t)$ is bounded, $|\operatorname{Log}(1+z)| \le C(1+\log|z|)$, and $|\chi-\chi(0)| \le C(1+\log|z|)^2$ by Lemma A.3(ii); this is (O4).

*(O5).* Fix $n$. Near 0: in $S_{\mathrm{in}}$ outside the lens, $\kappa_\pm\to 1$ (as $\frac{3}{4}\vartheta_{\mathrm{lip}}/p \le |\arg t| < \frac{\pi}{2}$ there, so that $\sin|\arg t| \ge \sin(\frac{3}{4}\vartheta_{\mathrm{lip}}/p) > 0$ in (41)) and $\tilde D, B_{\mathrm{in}}\to 1$; in $S_{\mathrm{out}}$, $B_{\mathrm{out}}\to 1$; $K$ is bounded; and $\chi$ is bounded near 0 uniformly up to the seams by Lemma A.3(iii). In the lens, $\tilde F$ is bounded above and below by the same facts. Near $\infty$: in $S_{\mathrm{out}}$ (which contains the imaginary axis, as $\vartheta_{\mathrm{seam}}<\pi/2$), $F = B_{\mathrm{out}}Ke^{\chi}\to(2\pi n)^{-1/2}$, because $B_{\mathrm{out}}(z)K(z) = \big(\frac{-z}{2\pi n}\big)^{1/2}(-z)^{-1/2}(1+o(1))$ with consistent principal branches in $S_{\mathrm{out}}$. In $S_{\mathrm{in}}\cap\{|z|\ge n^p+\delta'\}$, $|t|<1$ and $t$ stays away from 1, so $D$, $B_{\mathrm{in}}$, $K$ and $e^{\chi}$ are bounded above and below by powers of $|z|$. □

The seams carry no weight: $\chi_\pm$ and $\chi$ are built from $j$, $e^{\mp 2\pi i t}$, $B_{\mathrm{in}}$ and $B_{\mathrm{out}}$ only, and $F$ has no jump across them. Neither $V$ nor $Q$ has a jump on the seams, and the comparison parametrix of Section 7 has none either, since it is built from $D_W$. The only trace of the seams in the hard-edge analysis is the factor $e^{-2\chi}$ in $H_n$, which is controlled by the following lemma. The constant $\chi(0)$ depends on $n$, but no bound on it is needed: it enters only through

$H_n(0^+) = e^{-2\chi(0)}$, hence only through the constant $\mathsf{w}_n H_n(0^+)$, which cancels in $\omega = D_W^2/W_0$ (see (63)).

**Lemma A.3** (Corner modulus of the seam scalar)**.** *Let $0 < p < 1$ and $\tau_0 \in (0,1)$. There are $C = C(p,\tau_0)$ and $n_0$ such that for $n \ge n_0$:*

(i) *$|\chi_\pm(\zeta)| \le C|\zeta|$ for $\zeta \in \Sigma_\pm^{\mathrm{s}}$, $|\zeta| \le 1$; $|\chi_\pm(\zeta)| \le C(1+\log|\zeta|)$ for $|\zeta| \ge 1$; and $|\chi_\pm'(\zeta)| \le C$ for $|\zeta| \le 2$.*

(ii) *For $z$ with $\operatorname{dist}(z, \Sigma_+^{\mathrm{s}} \cup \Sigma_-^{\mathrm{s}}) \ge \tau_0|z|$: $|\chi(z) - \chi(0)| \le C|z|(1 + \log\frac{1}{|z|})$ and $|\chi'(z)| \le C(1 + \log\frac{1}{|z|})$ if $|z| \le 1$, and $|\chi(z) - \chi(0)| \le C(1+\log|z|)^2$ if $|z| \ge 1$. Here $\chi(0) := \frac{1}{2\pi i}\big[\int_{\Sigma_+^{\mathrm{s}}} \chi_+(\zeta)\frac{\mathrm{d}\zeta}{\zeta} - \int_{\Sigma_-^{\mathrm{s}}} \chi_-(\zeta)\frac{\mathrm{d}\zeta}{\zeta}\big]$.*

(iii) *For each fixed $n$, $\chi$ is bounded on $U(0,1) \setminus (\Sigma_+^{\mathrm{s}} \cup \Sigma_-^{\mathrm{s}})$, uniformly up to the seams.*

*Proof.* (i) On $\Sigma_+^{\mathrm{s}}$, $t = -i|t|$ with $|t| = n|\zeta|^{-1/p}$. If $|\zeta| \le 1$, then $|t| \ge n$; by (58) with $\phi = \pi/2$, $|j(t)| \le \frac{1}{6|t|} \le |\zeta|^{1/p}/n \le |\zeta|$; $|\operatorname{Log}(1 - e^{-2\pi|t|})| \le 2e^{-2\pi n|\zeta|^{-1/p}} \le C|\zeta|$; $|\varpi \operatorname{Log}(1+\zeta)| \le C|\zeta|$; $|\operatorname{Log}(1 - \zeta/(2\pi n))| \le C|\zeta|/n$. If $|\zeta| \ge 1$ and $|t| \ge 1$, then $|j(t)| \le \frac{1}{6}$; if $|t| < 1$, then (59) (with a bounded remainder on $\{|t| \le 1, \operatorname{Re} t \ge 0\}$) and $|\log|t|| \le \frac{1}{p}\log|\zeta|$ give $|j(t)| \le C(1+\log|\zeta|)$. Next $|\operatorname{Log}(1 - e^{-2\pi|t|})| \le C(1 + |\log|t||) \le C(1+\log|\zeta|)$, since $1 - e^{-y} \ge (1 - e^{-2\pi})y/(2\pi)$ for $0 < y \le 2\pi$. Finally $|1 - \zeta/(2\pi n)|$ lies between $\sin\vartheta_{\mathrm{seam}}$ and $1 + |\zeta|$, because $\zeta/(2\pi n)$ lies on the ray of angle $\vartheta_{\mathrm{seam}} \in (0, \pi/2)$. For the derivative on $|\zeta| \le 2$ (where $|t| \ge n2^{-1/p} \ge 1$): $\frac{\mathrm{d}}{\mathrm{d}\zeta} j(t) = -j'(t)t/(p\zeta) = O(|t|^{-1}|\zeta|^{-1}) = O(|\zeta|^{1/p-1}/n)$, which is bounded since $1/p > 1$; the derivative of $\operatorname{Log}(1 - e^{-2\pi i t})$ is $O(|t||\zeta|^{-1}e^{-2\pi|t|})$, also bounded; the other two terms have bounded derivatives. $\Sigma_-^{\mathrm{s}}$ follows by conjugation.

(ii) For $z \notin \Sigma_\pm^{\mathrm{s}}$, $\chi(z) - \chi(0) = \frac{z}{2\pi i}\sum_\pm (\pm 1)\int_{\Sigma_\pm^{\mathrm{s}}} \frac{\chi_\pm(\zeta)\,\mathrm{d}\zeta}{\zeta(\zeta - z)}$ and $\chi'(z) = \frac{1}{2\pi i}\sum_\pm(\pm 1)\int_{\Sigma_\pm^{\mathrm{s}}} \frac{\chi_\pm(\zeta)\,\mathrm{d}\zeta}{(\zeta - z)^2}$. Under the hypothesis, $|\zeta - z| \ge \tau_0|z|$, and $|\zeta - z| \ge |\zeta|/2$ if $|\zeta| \ge 2|z|$; hence $|\zeta - z| \ge \frac{\tau_0}{3}(|\zeta| + |z|)$. With (i), for $|z| \le 1$,

$$|\chi(z) - \chi(0)| \le C|z|\Big[\int_0^1 \frac{\mathrm{d}\rho}{\rho + |z|} + \int_1^\infty \frac{1 + \log\rho}{\rho^2}\,\mathrm{d}\rho\Big] \le C|z|\big(1 + \log\tfrac{1}{|z|}\big),$$

and $|\chi'(z)| \le C[\int_0^1 \rho(\rho + |z|)^{-2}\,\mathrm{d}\rho + \int_1^\infty (1+\log\rho)\rho^{-2}\,\mathrm{d}\rho] \le C(1 + \log\frac{1}{|z|})$. For $|z| \ge 1$ the same splitting gives $|z|\int_0^1 \frac{\mathrm{d}\rho}{|z|} + |z|\int_1^\infty \frac{(1+\log\rho)\,\mathrm{d}\rho}{\rho(\rho+|z|)} \le C(1+\log|z|)^2$.

(iii) Let $z \in U(0,1)$ be close to $\Sigma_+^{\mathrm{s}}$, with nearest point $z^* \in \Sigma_+^{\mathrm{s}}$, $r := |z^*|$ (other $z$ are covered by (ii)). Split $\int_{\Sigma_+^{\mathrm{s}}} \chi_+(\zeta)\frac{\mathrm{d}\zeta}{\zeta - z}$ into the part $|\zeta - z^*| \le r/2$ and the rest. On the first part write $\chi_+(\zeta) = \chi_+(z^*) + O(|\zeta - z^*|)$ by (i): the constant contributes $\chi_+(z^*)$ times a logarithm whose modulus is at most $\pi + \log C$, because the two end points $z^* \pm \frac{r}{2}e^{i\vartheta_{\mathrm{seam}}}$ are at comparable distances from $z$; since $|\chi_+(z^*)| \le Cr$, this is bounded, and so is the Lipschitz part. On the rest, $|\zeta - z| \ge (|\zeta| + r)/C$, and the contribution is at most $C\int_{\Sigma_+^{\mathrm{s}}} |\chi_+(\zeta)|(|\zeta| + r)^{-1}|\,\mathrm{d}\zeta|$, which is finite for fixed $n$ because $\chi_+(\zeta) = O(|\zeta|)$ at 0 and $\chi_+(\zeta)/\zeta$ is integrable at $\infty$ (Lemma A.2(ii)). The integral over $\Sigma_-^{\mathrm{s}}$ is bounded in the same way, since $|\zeta - z| \ge (|\zeta| + |z|)/C$ for $\zeta \in \Sigma_-^{\mathrm{s}}$. □

## Appendix B. The a priori bound at the hard edge

The comparison argument of Section 7 applies Theorem 4.4 of Conway and Deift (T3) to $Q$. Its hypothesis is $Q_\pm \in I + \partial\mathcal{C}(L^2(\Sigma))$: the boundary values of $Q - I$ must be Cauchy integrals of $L^2$ functions. We never solve a Riemann–Hilbert problem for $Q$; $Q$ is built from the orthogonal polynomials, and we only have to check that it is not too singular at the vertices of $\Sigma$. The only delicate vertex is the hard edge 0, where infinitely many nodes accumulate.

There we bound the Cauchy sums in $Y$ directly (Lemma B.2), except in thin sectors around the band, where the poles are removed by $R$ and a contour argument is needed (Lemma B.3). Throughout this appendix $n \geq n_0$ is *fixed*, and constants may depend on $n$.

B.1. **Three facts about Cauchy integrals.** The following facts are used here and in Appendix D. In this subsection $\Gamma$ is a bounded, finite union of oriented $C^2$ arcs with $|\Gamma \cap U(z,\rho)| \leq C_\Sigma \rho$ for all $z \in \mathbb{C}$, $\rho > 0$, and $\mathcal{V}$ is the finite set of endpoints and intersection points of the arcs. We say that a function $\mathfrak{F}$ analytic in $\mathbb{C} \setminus \Gamma$ is *continuous up to* $\Gamma$ if near every point of $\Gamma \setminus \mathcal{V}$ its restriction to each side of $\Gamma$ extends continuously to the arc; $\mathfrak{F}_\pm$ then denote these extensions. We use only Cauchy's integral formula, Green's formula and the facts of T3(a).

**Lemma B.1.** (a) *For $\mathfrak{h} \in L^2(\Gamma)$ and $z \notin \Gamma$, $|\mathcal{C}\mathfrak{h}(z)| \leq \pi^{-1} C_\Sigma^{1/2} \|\mathfrak{h}\|_{L^2} \operatorname{dist}(z,\Gamma)^{-1/2}$.*

(b) *Let $\mathfrak{k}$ be $C^1$ (in arc length) on each arc and vanish near $\mathcal{V}$. Then $\mathcal{C}\mathfrak{k}$ is analytic near each point of $\mathcal{V}$ and continuous up to $\Gamma$, and $(\mathcal{C}\mathfrak{k})_+ - (\mathcal{C}\mathfrak{k})_- = \mathfrak{k}$ on $\Gamma \setminus \mathcal{V}$. In particular $(\mathcal{C}\mathfrak{k})_\pm = \mathcal{C}_\pm \mathfrak{k}$.*

(c) *Let $\mathfrak{F}$ be analytic in $\mathbb{C} \setminus \Gamma$ and continuous up to $\Gamma$, with $\mathfrak{F}_\pm \in L^2(\Gamma)$, $\mathfrak{F}(z) = O(1/z)$ as $z \to \infty$, and $\rho \sup\{|\mathfrak{F}(z)| : |z - z_0| = \rho,\ z \notin \Gamma\} \to 0$ as $\rho \to 0$ for every $z_0 \in \mathcal{V}$. Then $\mathfrak{F} = \mathcal{C}(\mathfrak{F}_+ - \mathfrak{F}_-)$ in $\mathbb{C} \setminus \Gamma$; hence $\mathfrak{F}_\pm = \mathcal{C}_\pm(\mathfrak{F}_+ - \mathfrak{F}_-)$, i.e. $\mathfrak{F}_\pm \in \partial\mathcal{C}(L^2(\Gamma))$.*

*Proof.* (a) By Cauchy–Schwarz it suffices to show $\int_\Gamma |s-z|^{-2}|\,ds| \leq 4C_\Sigma/\mathfrak{d}$ with $\mathfrak{d} = \operatorname{dist}(z,\Gamma)$. Split $\Gamma$ into the shells $\{2^k\mathfrak{d} \leq |s-z| < 2^{k+1}\mathfrak{d}\}$, $k \geq 0$; the $k$-th contributes at most $C_\Sigma 2^{k+1}\mathfrak{d}\,(2^k\mathfrak{d})^{-2}$, and the sum is $4C_\Sigma/\mathfrak{d}$.

(b) Analyticity near $\mathcal{V}$ is clear, since $\mathfrak{k}$ vanishes there. By a partition of unity we may assume that $\mathfrak{k}$ is supported in a compact sub-arc of the interior of one arc $\gamma$. Choose an open disc $D$ which meets $\Gamma$ only in an open sub-arc of $\gamma$ containing that support and which $\gamma$ cuts into two parts, $D_+$ on the left and $D_-$ on the right. Using normal coordinates along $\gamma$ (which are $C^1$ because $\gamma$ is $C^2$), extend $\mathfrak{k}$ to a $C^1$ function $K$ on $\mathbb{C}$ with compact support in $D$ and $K|_\gamma = \mathfrak{k}$. Green's formula on $D_+$ (the Cauchy–Pompeiu formula) gives, since the positively oriented boundary of $D_+$ runs along $\gamma$ in its own direction and $K = 0$ on $\partial D$,

$$\mathcal{C}\mathfrak{k}(z) = K(z)\mathbf{1}_{D_+}(z) + \mathfrak{P}(z), \qquad \mathfrak{P}(z) := \frac{1}{\pi}\iint_{D_+} \frac{\bar\partial K(w)}{w - z}\,dA(w), \qquad z \notin \Gamma.$$

Since $\bar\partial K$ is bounded with compact support and $|w-z|^{-1}$ is locally integrable uniformly in $z$, $\mathfrak{P}$ is continuous on $\mathbb{C}$. So $\mathcal{C}\mathfrak{k}$ is continuous up to $\gamma$ from both sides, with $(\mathcal{C}\mathfrak{k})_+ - (\mathcal{C}\mathfrak{k})_- = K = \mathfrak{k}$. As continuous extensions, these are the non-tangential limits $\mathcal{C}_\pm\mathfrak{k}$ of T3.

(c) Fix $z \notin \Gamma$. For small $\rho$ and large $R$ apply Cauchy's integral formula to $\mathfrak{F}$ on the open set $U(0,R) \setminus \big(\Gamma \cup \bigcup_{z_0 \in \mathcal{V}} \overline{U(z_0,\rho)}\big)$, whose boundary consists of the two sides of the arcs of $\Gamma$ outside the small discs, the circles $|s - z_0| = \rho$ and the circle $|s| = R$; $\mathfrak{F}$ is continuous on the closure (taken separately on the two sides of each arc). The two sides of an arc contribute $\frac{1}{2\pi i}\int (\mathfrak{F}_+ - \mathfrak{F}_-)(s)\,ds/(s-z)$, the left side being traversed in the direction of the arc. As $\rho \to 0$ the small circles contribute at most $\sum_{z_0} 2\rho \sup_{|s-z_0|=\rho} |\mathfrak{F}(s)|/|z - z_0| \to 0$ (for $\rho \leq |z-z_0|/2$), and the arc integrals converge to $\mathcal{C}(\mathfrak{F}_+ - \mathfrak{F}_-)(z)$ because $\mathfrak{F}_+ - \mathfrak{F}_- \in L^2(\Gamma) \subset L^1(\Gamma)$. As $R \to \infty$ the large circle contributes $O(1/R)$. Hence $\mathfrak{F}(z) = \mathcal{C}(\mathfrak{F}_+ - \mathfrak{F}_-)(z)$. Taking boundary values from either side, $\mathfrak{F}_\pm = \mathcal{C}_\pm(\mathfrak{F}_+ - \mathfrak{F}_-)$ wherever the non-tangential limits $\mathcal{C}_\pm$ exist, i.e. almost everywhere. □

B.2. **The bound at the hard edge.** Recall from Lemma 5.1 that

$$U_{j2}(z) = \sum_{k \geq 1} \frac{w(X_k)\,U_{j1}(X_k)}{z - X_k} \qquad (j = 1, 2), \qquad w(X_k) = n^p w_k, \tag{101}$$

and that $U_{11}, U_{21}$ are polynomials. Put $\varphi_1 := \frac{1}{2}\vartheta_{\mathrm{lip}}$.

**Lemma B.2** (Sum bound)**.** *For every $\eta > 0$ there is $C$ (depending on $n$ and $\eta$) such that $|U_{j2}(z)| \le C|z|^{-\eta}(1+|z|^{\alpha})$ for $0 < |z| \le 1$, $\varphi_1 \le |\arg z| \le \pi$.*

*Proof.* For such $z$ and $X > 0$, $|z - X| \ge \sin\varphi_1 \max(|z|, X)$. By Lemma 2.1 (or Lemma 5.9(i)), $w_k \le C_\eta k^{-s+\eta}$, and $U_{j1}$ is bounded on $[0, n^p]$. Let $t_z := |t(z)| = n|z|^{-1/p}$, so that $X_k \ge |z|$ if and only if $k \le t_z$. Then

$$|U_{j2}(z)| \le C \sum_{k \le t_z} \frac{k^{-s+\eta}}{(n/k)^p} + \frac{C}{|z|} \sum_{k > t_z} k^{-s+\eta} \le C\big(1 + t_z^{1+p-s+\eta} + \log t_z\big) + C|z|^{-1} t_z^{1-s+\eta}.$$

Since $t_z^{1+p-s} = n^{1+p-s}|z|^{\alpha}$ and $|z|^{-1}t_z^{1-s} = n^{1-s}|z|^{\alpha}$ (recall $(s-1)/p - 1 = \alpha$), and $t_z^{\eta} = n^{\eta}|z|^{-\eta/p}$, the claim follows after renaming $\eta$. □

**Lemma B.3** (Lens sectors)**.** *For every $\eta > 0$ there are $\delta_{\mathrm{s}} \in (0, \delta)$ and $C$ such that $|R_{j2}(z)| \le C|z|^{-\eta}(1+|z|^{\alpha})$ for $0 < |z| \le \delta_{\mathrm{s}}/2$, $0 < |\arg z| < \varphi_1$, and also for $z \in \Omega_{\mathrm{b}} \cap U(0, \delta_{\mathrm{s}}/2)$.*

*Proof.* Inside the lens the Cauchy sum (101) has poles that $R$ removes; we move the sum onto the boundary of a sector by the residue theorem and bound the boundary integral. Take $z$ with $0 < \arg z < \varphi_1$; the case $\arg z < 0$ is the mirror image. In $\Omega_{\mathrm{b}}^{+}$, $R_{j2} = U_{j2} - U_{j1} W e^{-2i\theta}\kappa_{+}$ by (50). Put

$$\mathcal{K}_j(s) := \frac{U_{j1}(s) W(s)}{e^{2i\theta(s)} - 1}, \qquad s \in \mathcal{G} := \{s : \ r < |s| < \delta_{\mathrm{s}}, \ |\arg s| < \vartheta_{\mathrm{lip}}\},$$

where $\delta_{\mathrm{s}}$ and $r$ are chosen with $t(\delta_{\mathrm{s}}), t(r) \in \mathbb{N} + \frac{1}{2}$, and $r < \frac{1}{2}|z|$, $|z| < \frac{1}{2}\delta_{\mathrm{s}}$. Note $e^{-2i\theta}\kappa_{+} = 1/(e^{2i\theta} - 1)$, so $R_{j2}(z) = U_{j2}(z) - \mathcal{K}_j(z)$. The function $\mathcal{K}_j$ is meromorphic on a neighbourhood of $\overline{\mathcal{G}}$: $W$ is analytic there (Lemma 5.7), and $e^{2i\theta} = 1$ only where $t \in \mathbb{Z}$, i.e. at the nodes (the points with $t \in -\mathbb{N}$ have $|\arg s| = \pi p > \vartheta_{\mathrm{lip}}$). At a node, $e^{2i\theta(s)} - 1 = 2i\theta'(X_k)(s - X_k)(1 + O(s - X_k))$, so $\operatorname{Res}_{X_k} \mathcal{K}_j = U_{j1}(X_k) W(X_k)/(2i\theta'(X_k)) = w(X_k) U_{j1}(X_k)$. The residue theorem on $\partial\mathcal{G}$ (positively oriented) and (101) give

$$R_{j2}(z) = \sum_{X_k \notin \mathcal{G}} \frac{w(X_k) U_{j1}(X_k)}{z - X_k} - \frac{1}{2\pi i} \oint_{\partial\mathcal{G}} \frac{\mathcal{K}_j(s)\,\mathrm{d}s}{s - z}. \tag{102}$$

We bound the terms. (1) The nodes with $X_k \ge \delta_{\mathrm{s}}$ are finitely many and at distance $\ge \delta_{\mathrm{s}}/2$ from $z$; those with $X_k \le r$ contribute at most $\frac{2}{|z|}\sum_{X_k \le r} w(X_k)|U_{j1}(X_k)|$, which tends to 0 as $r \to 0$. (2) On the ray $\arg s = \vartheta_{\mathrm{lip}}$, $|e^{2i\theta} - 1|^{-1} = |e^{-2i\theta}|/|1 - e^{-2i\theta}| \le 2\exp(-2\pi n|s|^{-1/p}\sin(\vartheta_{\mathrm{lip}}/p))$ by (41), which beats $|W(s)| \le C|s|^{\alpha-\eta}$ (Lemmas 5.3, 5.9(ii)); and $|s - z| \ge |s|\sin(\vartheta_{\mathrm{lip}} - \varphi_1)$. So this part is bounded. (3) On the ray $\arg s = -\vartheta_{\mathrm{lip}}$ we write $1/(e^{2i\theta} - 1) = -1 - e^{2i\theta}/(1 - e^{2i\theta})$, where now $|e^{2i\theta}| < 1$ is super-small, so $\mathcal{K}_j = -U_{j1}W + (\text{bounded})$. Since $|s - z| \ge (|s| + |z|)/C$,

$$\int_0^{\delta_{\mathrm{s}}} \frac{|W(\rho e^{-i\vartheta_{\mathrm{lip}}})|\,\mathrm{d}\rho}{\rho + |z|} \le C \int_0^{\delta_{\mathrm{s}}} \frac{\rho^{\alpha-\eta}\,\mathrm{d}\rho}{\rho + |z|} \le C|z|^{-\eta}(1 + |z|^{\alpha}),$$

because $\alpha - \eta > -1$. (4) On the arcs $|s| = r$ and $|s| = \delta_{\mathrm{s}}$, $|t| \in \mathbb{N} + \frac{1}{2}$. For such $t = u + iv$ we have $|e^{2\pi i t} - 1| \ge 1/C > 0$: if $|v| \ge \frac{1}{4}$ this is clear, and if $|v| < \frac{1}{4}$ then $u$ is at distance at least $\frac{3}{8}$ from $\mathbb{Z}$, so $|e^{2\pi i t} - 1| = 2e^{-\pi v}|\sin \pi t| \ge 2e^{-\pi/4}\sin(3\pi/8)$. Hence $|\mathcal{K}_j| \le C|W| \le C|s|^{\alpha-\eta}$ there. The arc $|s| = r$ has length $\le 2\vartheta_{\mathrm{lip}} r$ and $|s - z| \ge \frac{1}{2}|z|$, so its contribution is $\le Cr^{1+\alpha-\eta}/|z| \to 0$ as $r \to 0$; the arc $|s| = \delta_{\mathrm{s}}$ contributes a bounded amount. Letting $r \to 0$ in (102) proves the bound for $0 < \arg z < \varphi_1$. For $z \in \Omega_{\mathrm{b}} \cap U(0, \delta_{\mathrm{s}}/2)$ with $\varphi_1 \le |\arg z| < \frac{5}{4}\vartheta_{\mathrm{lip}}$, $R_{j2} = U_{j2} - U_{j1} W e^{\mp 2i\theta}\kappa_{\pm}$ is bounded by Lemma B.2 plus a super-small term. □

**Proposition B.4** (A priori bound)**.** *Fix $n \geq n_0$. Then $\det Q \equiv 1$, and:*

(i) *for every $\eta > 0$ there is $C_{n,\eta}$ with $|Q(z)| \leq C_{n,\eta}|z|^{-|\alpha|/2-\eta}$ for $z \in U(0,\delta) \setminus \Sigma$;*
(ii) *$Q$ is bounded on every bounded set at positive distance from 0, and $Q(z) \to I$ as $z \to \infty$;*
(iii) *$Q$ has continuous boundary values on $\Sigma$ minus its vertices.*

*The same holds for $Q^{-1}$.*

*Proof.* $\det U = \det Y = 1$ (Lemma 5.1), and every other factor in (52) and in $Q = D_\infty^{-\sigma_3} V D_W^{\sigma_3}$ has determinant 1; so $\det Q = 1$ and $Q^{-1}$ has the same entries up to sign and order.

(i) Let $\eta > 0$ and $|z| < \delta_s/2$ (the annulus $\delta_s/2 \leq |z| < \delta$ is covered by (ii)). By (52) and (68), $Q$ is a constant diagonal matrix times $U\,M\,D_W^{\sigma_3}$. We use: $|e^{\pm n\varphi}| \leq e^{Cn|z|^{1/2}} \leq C$ (Lemma 5.5(iii)); $|F|^{\pm 1} \leq C$ outside the lens and $|\tilde F|^{\pm 1} \leq C$ in the lens near 0 (O5); $|D_W(z)|^{\pm 1} \leq C|z|^{\pm\alpha/2-\eta}$ (Lemma 6.3(vi)); $|1/W(z)| \leq C|z|^{-\alpha-\eta}$ in the lens (Lemmas 5.3, 5.9(ii)); and $U_{j1}$ bounded.

*Outside the lens* ($z \in \Omega_\infty$), $|\arg z| \geq \frac{3}{4}\vartheta_{\text{lip}} > \varphi_1$ and $M = e^{n\varphi\sigma_3} F^{\sigma_3}$. The first column of $Q$ is a constant times $U_{\cdot 1} e^{n\varphi} F D_W$, of size $\leq C|z|^{\alpha/2-\eta}$; the second is a constant times $U_{\cdot 2} e^{-n\varphi} F^{-1} D_W^{-1}$, of size $\leq C|z|^{-\eta}(1+|z|^\alpha)|z|^{-\alpha/2-\eta}$ by Lemma B.2. Both are $\leq C|z|^{-|\alpha|/2-2\eta}$.

*In the lens* $\Omega_b^\pm$, $UM = R\, e^{n\varphi\sigma_3} A_\pm \tilde F^{\sigma_3}$, and $e^{n\varphi\sigma_3} A_\pm = \left(\begin{smallmatrix} e^{n\varphi} & 0 \\ \mp e^{n\varphi}/W & e^{-n\varphi} \end{smallmatrix}\right)$. The first column of $R$ is that of $U$, and the second is bounded by Lemma B.3. Hence the first column of $Q$ is at most $C(1+|z|^{-\alpha-2\eta}(1+|z|^\alpha))|z|^{\alpha/2-\eta} \leq C|z|^{-|\alpha|/2-3\eta}$, and the second is at most $C|z|^{-\eta}(1+|z|^\alpha)|z|^{-\alpha/2-\eta} \leq C|z|^{-|\alpha|/2-2\eta}$. Renaming $\eta$ gives (i).

(ii) Away from 0 each of $U$ (meromorphic, with finitely many nodes in any compact set not containing 0), the triangular and diagonal factors in (50)–(51), $F$, $\tilde F$ and $D_W$ is analytic in each region up to its boundary, except for the poles of $U$ and of $\kappa_\pm$ at the nodes, which cancel in $R$ (Step 1 of §5.7). Near $4c$ the only singular factors are $\tilde F \sim (4c-z)^{-1/2}$ and $D_W \sim (4c-z)^{1/2}$, and they occur only in the product $\tilde F D_W$, whose square $(\tilde F D_W)^2 = \mathsf{w}_n z^\alpha r_n \omega$ is bounded above and below on $U(4c,\varepsilon)$ (Lemma 6.3(vi)(b)). At the remaining vertices ($4c \pm ih_0$, $n^p + \delta' \pm ih_0$, $n^p + \delta'$) all factors are analytic and bounded. At $\infty$, $Q \to I$ by Lemma 6.4. (iii) follows from the same description and Lemma 5.10. □

**Corollary B.5** ($Q$ is an admissible solution)**.** *For each fixed $n \geq n_0$, $Q_\pm - I \in L^2(\Sigma)$ and $Q = I + \mathcal{C}(Q_+ - Q_-)$ off $\Sigma$. Thus $Q_\pm \in I + \partial\mathcal{C}(L^2(\Sigma))$ in the sense of T3, and the same holds for $Q^{-1}$.*

*Proof.* We apply Lemma B.1(c) to $\mathfrak{F} := Q - I$ on $\Gamma = \Sigma$, which for fixed $n$ is bounded and satisfies $|\Sigma \cap U(z,\rho)| \leq C\rho$ (F1). By Proposition B.4(iii), $\mathfrak{F}$ is continuous up to $\Sigma$. By Proposition B.4(i) with $\eta = (1-|\alpha|)/4$, $|Q(z)| \leq C|z|^{-(1+|\alpha|)/4}$ near 0, and $(1+|\alpha|)/4 < \frac{1}{2}$; with (ii) this gives $\mathfrak{F}_\pm \in L^2(\Sigma)$ and $\rho \sup_{|z|=\rho} |\mathfrak{F}(z)| = O(\rho^{1-(1+|\alpha|)/4}) \to 0$ at the vertex 0, while at every other vertex the corresponding quantity is $O(\rho)$. Finally $\mathfrak{F} = O(1/z)$ at $\infty$ (Lemma 6.4). The argument for $Q^{-1}$ is identical. □

This is the content of (F7). Any $\eta \in (0, \frac{1-|\alpha|}{2})$ in (F7) makes the boundary values of $Q$ square integrable at 0; this is the use (a) of $|\alpha| < 1$ in Remark 7.8.

## Appendix C. Two $L^2$ bounds on a star of arcs

The uniform resolvent bound of Appendix D reduces, at the two edges, to two operators on a star of arcs through a point: a Cauchy operator conjugated by a power weight, and a Cauchy operator with a logarithmic commutator. Both are dilation invariant, which is what makes the bounds uniform in $n$. We prove them here.

*Setting.* A *star* is a set $\Gamma = \bigcup_{j=1}^{m} \gamma_j([0,R_j])$ of finitely many $C^2$ arcs $\gamma_j$, parametrised by arc length, with $\gamma_j(0)=0$, pairwise disjoint except at 0, and with distinct tangent directions $\gamma_j'(0) = e^{i\theta_j}$. Each arc carries an orientation, and $\mathcal{C}_\pm$ are the boundary values of $\mathcal{C}$ on $\Gamma$ as in T3. By compactness and transversality there is $C_\Gamma \ge 1$ with

(103)
$$|\gamma_j'(\rho) - e^{i\theta_j}| \le C_\Gamma \rho, \qquad |\gamma_j(\rho) - \gamma_i(r)| \ge (\rho + r)/C_\Gamma \ (i \ne j), \qquad |\gamma_j(\rho) - \gamma_j(r)| \ge |\rho - r|/C_\Gamma.$$

A finite union of $C^2$ arcs is a Carleson curve, so $\mathcal{C}_\pm$ are bounded on $L^2(\Gamma)$ by T3(a). For $\lambda > 0$ the dilation $(U_\lambda \mathfrak{h})(z) := \lambda^{-1/2}\mathfrak{h}(z/\lambda)$ is unitary from $L^2(\Gamma)$ onto $L^2(\lambda\Gamma)$, and since $\mathrm{d}s/(s-z)$ is invariant under $(s,z) \mapsto (\lambda s, \lambda z)$, $U_\lambda$ intertwines the Cauchy operators of $\Gamma$ and $\lambda\Gamma$. The same holds for the operators below, whose kernels are $\mathrm{d}s/(s-z)$ times a function of $(s/|s|,\, z/|z|,\, |z|/|s|)$.

We use one elementary bound repeatedly. Let $\mathfrak{k}$ be measurable on $(0,\infty)$ with

$$\|\mathfrak{k}\|_* := \int_0^\infty |\mathfrak{k}(u)| u^{-1/2}\,\mathrm{d}u < \infty, \qquad (\mathcal{M}_\mathfrak{k}\mathfrak{h})(r) := \int_0^\infty \mathfrak{k}(\rho/r)\mathfrak{h}(\rho)\frac{\mathrm{d}\rho}{r}.$$

Then

$$\|\mathcal{M}_\mathfrak{k}\mathfrak{h}\|_{L^2(0,\infty)} \le \|\mathfrak{k}\|_* \, \|\mathfrak{h}\|_{L^2(0,\infty)}. \tag{104}$$

Indeed $(\mathcal{M}_\mathfrak{k}\mathfrak{h})(r) = \int_0^\infty \mathfrak{k}(u)\mathfrak{h}(ru)\,\mathrm{d}u$, and $\|\mathfrak{h}(\cdot\, u)\|_{L^2} = u^{-1/2}\|\mathfrak{h}\|_{L^2}$, so (104) is Minkowski's integral inequality. Operators with kernels that are bounded on a bounded set are Hilbert–Schmidt, and we use that without comment.

**Lemma C.1** (power-weighted Cauchy operator on a star)**.** *Let $\Gamma$ be a star and $|\beta| < \frac12$. Then the operators*

$$\mathfrak{h} \ \longmapsto \ |z|^\beta \, \mathcal{C}_\pm\big(|s|^{-\beta}\mathfrak{h}\big)(z)$$

*are bounded on $L^2(\Gamma)$, with a norm $N_\beta$ that depends only on $\beta$ and $\Gamma$. The same bound holds on $\lambda\Gamma$ for every $\lambda > 0$, on every subset $\Gamma' \subset \lambda\Gamma$ (functions supported on, and evaluated on, $\Gamma'$), and for the weights $z^\beta$, $s^{-\beta}$ (any fixed continuous branches of the argument on each arc) in place of $|z|^\beta$, $|s|^{-\beta}$, with $N_\beta$ multiplied by $e^{4\pi|\beta|}$.*

*Proof.* The plan is to split off the plain Cauchy operator, which is bounded, and to compare the remainder, which has a kernel regular on the diagonal, with its analogue on the star of tangent rays.

Write $T_\pm\mathfrak{h} := |z|^\beta \mathcal{C}_\pm(|s|^{-\beta}\mathfrak{h}) = \mathcal{C}_\pm\mathfrak{h} + R\mathfrak{h}$, where

$$(R\mathfrak{h})(z) = \frac{1}{2\pi i}\int_\Gamma \Big(\frac{|z|^\beta}{|s|^\beta} - 1\Big)\frac{\mathfrak{h}(s)\,\mathrm{d}s}{s-z}.$$

The kernel of $R$ is bounded near the diagonal of each arc (see below), so $R\mathfrak{h}$ has no boundary-value ambiguity, and it suffices to bound $R$. In arc-length coordinates, $z = \gamma_i(r)$, $s = \gamma_j(\rho)$, the kernel of $R$ is

$$\mathfrak{r}_{ij}(\rho, r) = \frac{1}{2\pi i}\Big(\frac{|\gamma_i(r)|^\beta}{|\gamma_j(\rho)|^\beta} - 1\Big)\frac{\gamma_j'(\rho)}{\gamma_j(\rho) - \gamma_i(r)}.$$

Its flat analogue, on the star of rays $\{\rho e^{i\theta_j}\}$, is $\mathfrak{k}_{ij}(\rho/r)/r$ with

$$\mathfrak{k}_{ij}(u) := \frac{1}{2\pi i}\,\frac{(u^{-\beta} - 1)\, e^{i\theta_j}}{u e^{i\theta_j} - e^{i\theta_i}}.$$

For $i \ne j$ we have $|ue^{i\theta_j} - e^{i\theta_i}| \ge \sin(|\theta_i - \theta_j|/2)(1+u)$, and for $i = j$ the quotient $(u^{-\beta}-1)/(u-1)$ is continuous at $u = 1$. So $|\mathfrak{k}_{ij}(u)| \le C(1+u^{-\beta})$ for $u \le 2$ and $|\mathfrak{k}_{ij}(u)| \le C(u^{-\beta}+1)u^{-1}$ for $u \ge 2$, and $\|\mathfrak{k}_{ij}\|_* < \infty$ precisely because $-\frac12 < \beta < \frac12$: near 0 the integrand is $O(u^{-1/2-\max(\beta,0)})$,

near infinity $O(u^{-3/2-\min(\beta,0)})$. By (104) the flat operator is bounded on $L^2$ of the rays, hence on $L^2$ of the segments $[0,R_j]$.

The difference $\mathfrak{r}_{ij}(\rho,r)-\mathfrak{k}_{ij}(\rho/r)/r$ is a sum of two terms. Put $w_j(\rho):=(|\gamma_j(\rho)|/\rho)^\beta$; by (103) and $\gamma_j\in C^2$, $w_j$ is $C^1$ on $[0,R_j]$ and bounded above and below. The first term is

$$\frac{1}{2\pi i}\Big(\frac{r}{\rho}\Big)^\beta\Big(\frac{w_i(r)}{w_j(\rho)}-1\Big)\frac{\gamma_j'(\rho)}{\gamma_j(\rho)-\gamma_i(r)};$$

since $|w_i(r)-w_j(\rho)|\le C|\rho-r|$ for $i=j$ and $\le C(\rho+r)$ for $i\ne j$, (103) bounds it by $C(r/\rho)^\beta$. The second term is $\frac{1}{2\pi i}((r/\rho)^\beta-1)\,\mathfrak{e}_{ij}(\rho,r)$ with

$$\mathfrak{e}_{ij}(\rho,r):=\frac{\gamma_j'(\rho)}{\gamma_j(\rho)-\gamma_i(r)}-\frac{e^{i\theta_j}}{\rho e^{i\theta_j}-re^{i\theta_i}}.$$

We claim $|\mathfrak{e}_{ij}|\le C$. For $i\ne j$ both denominators are at least $(\rho+r)/C$, and the numerator of the difference, $\gamma_j'(\rho)(\rho e^{i\theta_j}-re^{i\theta_i})-e^{i\theta_j}(\gamma_j(\rho)-\gamma_i(r))$, is $O(\rho(\rho+r))+O(\rho^2+r^2)$ by (103). For $i=j$ the flat term is $1/(\rho-r)$, and $\gamma_j(\rho)-\gamma_j(r)=(\rho-r)(\gamma_j'(\rho)+O(\rho-r))$, so $\gamma_j'(\rho)/(\gamma_j(\rho)-\gamma_j(r))=(\rho-r)^{-1}(1+O(\rho-r))$, which gives $\mathfrak{e}_{jj}=O(1)$. Here the factor $\gamma_j'(\rho)$, i.e. the arc-length element $\mathrm{d}s=\gamma_j'(\rho)\,\mathrm{d}\rho$, must be kept: without it the difference is only $O(r/|\rho-r|)$ on a single arc. Hence the difference kernel is at most $C(1+(r/\rho)^\beta)$, and $\iint_{[0,R]^2}(1+(r/\rho)^{2\beta})\,\mathrm{d}\rho\,\mathrm{d}r<\infty$ for $|\beta|<\frac12$: it is Hilbert–Schmidt.

So $R$ is bounded on $L^2(\Gamma)$ (arc length makes the parametrisation unitary), and so is $T_\pm$. Dilation invariance gives the bound on $\lambda\Gamma$. Restriction to $\Gamma'$ is a compression. Finally $z^\beta=|z|^\beta e^{i\beta\arg z}$ with $|\arg z|\le 2\pi$, so the complex weights differ from the absolute ones by multiplication operators of norm at most $e^{2\pi|\beta|}$ on either side. □

**Lemma C.2** (logarithmic commutator on a star)**.** *Let $\Gamma$ be a star and fix on each arc a continuous branch $\mathrm{l}(z)=\log|z|+i\arg z$ of the logarithm. Then the operator*

$$(L\mathfrak{h})(z):=\frac{1}{2\pi i}\int_\Gamma\big(\mathrm{l}(z)-\mathrm{l}(s)\big)\frac{\mathfrak{h}(s)\,\mathrm{d}s}{s-z}$$

*is bounded on $L^2(\Gamma)$, with the same norm on $\lambda\Gamma$ for every $\lambda>0$ and on subsets.*

*Proof.* The kernel is regular on the diagonal, since l is Lipschitz in arc length away from 0 and the factor $\mathrm{l}(z)-\mathrm{l}(s)$ vanishes at $s=z$; we split it into the part with $\log|z|-\log|s|$ and the part with $\arg z-\arg s$.

*The modulus part.* On the star of rays its kernel is $\mathfrak{k}_{ij}(\rho/r)/r$ with

$$\mathfrak{k}_{ij}(u)=-\frac{1}{2\pi i}\,\frac{\log u\;e^{i\theta_j}}{ue^{i\theta_j}-e^{i\theta_i}}.$$

This has no singularity at $u=1$ even for $i=j$ ($\log u/(u-1)\to1$), and $|\mathfrak{k}_{ij}(u)|u^{-1/2}$ is $O(|\log u|u^{-1/2})$ at 0 and $O(|\log u|u^{-3/2})$ at infinity, so $\|\mathfrak{k}_{ij}\|_*<\infty$ and (104) applies. On the curved star the difference from the flat kernel is

$$\frac{1}{2\pi i}\Big[(\lambda_i(r)-\lambda_j(\rho))\frac{\gamma_j'(\rho)}{\gamma_j(\rho)-\gamma_i(r)}+\log\frac{r}{\rho}\;\mathfrak{e}_{ij}(\rho,r)\Big],$$

with $\lambda_j(\rho):=\log(|\gamma_j(\rho)|/\rho)$, a $C^1$ function on $[0,R_j]$. As in the proof of Lemma C.1 this is $O(1+|\log(r/\rho)|)$, and $\iint_{[0,R]^2}(1+\log^2(r/\rho))\,\mathrm{d}\rho\,\mathrm{d}r<\infty$.

*The argument part.* Its kernel is $\frac{i}{2\pi i}(\arg\gamma_i(r)-\arg\gamma_j(\rho))\gamma_j'(\rho)/(\gamma_j(\rho)-\gamma_i(r))$. For $i=j$, $\arg\gamma_j(\rho)=\theta_j+O(\rho)$ is $C^1$ on $[0,R_j]$, so by (103) the kernel is bounded. For $i\ne j$ the

numerator is bounded and the denominator is at least $(\rho+r)/C_\Gamma$, so the kernel is at most $C/(\rho+r) = C\,\mathfrak{k}(\rho/r)/r$ with $\mathfrak{k}(u) = (1+u)^{-1}$, $\|\mathfrak{k}\|_* = \pi$.

Both parts are bounded on $L^2(\Gamma)$. The kernel depends on $(s,z)$ only through $s/|s|$, $z/|z|$ and $|z|/|s|$, so the operator commutes with $U_\lambda$ and the bound holds on $\lambda\Gamma$. □

*Remark* C.3 (The condition $|\beta| < \frac{1}{2}$). For $\Gamma$ a union of rays the curvature terms vanish, and Lemma C.1 reduces to the boundedness of the Mellin multiplier $\int_0^\infty u^{s-1}\,\mathrm{d}u/(u-e^{i\phi_{ij}})$ on $\Re s = \frac{1}{2}-\beta$; on a single ray its principal value is $-\pi\cot(\pi s)$ (by (5.12.3) and (5.5.3) of T4), which is bounded on that line exactly when $0 < \frac{1}{2}-\beta < 1$. This explains the role of the condition $|\beta| < \frac{1}{2}$.

## Appendix D. Uniform invertibility of the comparison problem

This appendix proves Proposition 7.4.

We use the notation of Section 7. In particular $\hat\Sigma = \Sigma \cup \Sigma_\circ$, $\tilde Q$ is the explicit solution (87) of the comparison problem with jump $\tilde v$, and $n \ge n_0$. We use the three facts about Cauchy integrals of §B.1, prove a uniform bound for the singular integral operator near an edge where $\tilde Q$ has a two-scale structure (§D.1), and assemble the proof (§D.2).

D.1. **The model block.** The following lemma covers both edges. At the hard edge it is applied with $\mathtt{t}_n = n^2$, at the soft edge with $\mathtt{t}_n = n^{2/3}$.

**Lemma D.1** (uniform bound near an edge)**.** *Let $z_0 \in \mathbb{C}$, $0 < r_1 < r_0$, $\mathtt{t}_n \ge 1$, and let $f_0$ be conformal on a neighbourhood of $\overline{U(z_0,r_0)}$, independent of $n$, with $f_0(z_0) = 0$. Assume that $\hat\Sigma \cap U(z_0,r_0)$ consists of finitely many $C^2$ arcs from $z_0$ with distinct tangents and is independent of $n$, and put $\Sigma_1 := \hat\Sigma \cap U(z_0,r_1)$, $\Gamma_1 := f_0(\Sigma_1)$ (a star with vertex 0, in the sense of Appendix C). Assume that for $z \in U(z_0,r_0)\setminus\hat\Sigma$, with $\zeta := \mathtt{t}_n f_0(z)$,*

$$\tilde Q(z) = \mathcal{A}(z)\,\mathtt{t}_n^{\sigma_3/4} M_n(z),$$

*where, with a constant $C_1$ independent of $n$, and with the same statements for the boundary values on $\hat\Sigma$:*

- (B1) *$\mathcal{A}^{\pm1}$ are analytic on $U(z_0,r_0)$ and $|\mathcal{A}^{\pm1}| \le C_1$;*
- (B2) *if $|\zeta| \ge 2$, then $M_n(z) = \zeta^{-\sigma_3/4}\Theta_\infty(z)$ with $|\Theta_\infty^{\pm1}(z)| \le C_1$;*
- (B3) *if $|\zeta| \le 3$, then $M_n(z) = \hat A(\zeta)\mathsf{P}(\zeta)\Theta_0(z)$, where $\hat A$ does not depend on $n$, $\hat A^{\pm1}$ are analytic on $|\zeta| < 4$ with $|\hat A^{\pm1}| \le C_1$, $|\Theta_0^{\pm1}(z)| \le C_1$, and either $\mathsf{P}(\zeta) = \zeta^{\beta\sigma_3}$ with $|\beta| < \frac{1}{2}$, or $\mathsf{P}(\zeta) = I + \mathsf{k}(\log\zeta)e_1e_2^{\mathsf{T}}$ with a constant $\mathsf{k}$.*

*Then the operators $\mathfrak{g} \mapsto 1_{\Sigma_1}\mathcal{C}_\pm(\mathfrak{g}1_{\Sigma_1}\tilde Q_+^{-1})\tilde Q_\pm$ are bounded on $L^2(\Sigma_1)$ uniformly in $n$, and $\int_{\Sigma_1}(|\tilde Q_\pm|^2 + |\tilde Q_\pm^{-1}|^2)|\,\mathrm{d}s| \le C$.*

*Proof.* We remove the analytic prefactor $\mathcal{A}$ at the cost of a Hilbert–Schmidt error, pass to the variable $\zeta$, and split the resulting operator according to $|\zeta| \le 3$ and $|\zeta| \ge 2$. Throughout, $C$ depends on $C_1$, $f_0$, $r_0$, $r_1$, $\beta$ and the geometry, but not on $n$.

*Weights.* Put $\mathfrak{W}(z) := |f_0(z)|^{-1/4}$ if $|\zeta| \ge 2$ and $\mathfrak{W}(z) := \mathtt{t}_n^{1/4}\mathfrak{w}(\zeta)$ if $|\zeta| < 2$, where $\mathfrak{w}(\zeta) = |\zeta|^{-|\beta|}$, respectively $\mathfrak{w}(\zeta) = 1 + |\log|\zeta||$ in the logarithmic case. Since $\mathtt{t}_n^{\sigma_3/4}\zeta^{-\sigma_3/4} = f_0^{-\sigma_3/4}$, (B2)–(B3) give

$$|\mathtt{t}_n^{\sigma_3/4}M_n(z)| + |M_n(z)^{-1}\mathtt{t}_n^{-\sigma_3/4}| \le C\,\mathfrak{W}(z), \qquad |M_n(z)^{\pm1}| \le C\mathfrak{w}(\zeta)\ (|\zeta| \le 3). \tag{105}$$

On $\Sigma_1$ we have $|\,\mathrm{d}z| \le C|\,\mathrm{d}\zeta|/\mathfrak{t}_n$, $|f_0(z)| \ge |z-z_0|/C$, and $\Gamma_1$ satisfies an Ahlfors bound $|\Gamma_1 \cap U(\zeta,t)| \le Ct$, as does $\mathfrak{t}_n\Gamma_1$ with the same constant. Hence

$$\int_{\Sigma_1} \mathfrak{W}^2 |\,\mathrm{d}z| \le C\int_{\Sigma_1} |z-z_0|^{-1/2}|\,\mathrm{d}z| + C\mathfrak{t}_n^{1/2}\mathfrak{t}_n^{-1}\int_{\mathfrak{t}_n\Gamma_1\cap\{|\zeta|\le 2\}} \mathfrak{w}(\zeta)^2|\,\mathrm{d}\zeta| \le C, \tag{106}$$

since $2|\beta|<1$. With (105) and (B1) this gives the bound on $\int_{\Sigma_1}|\tilde{Q}_\pm^{\pm1}|^2$.

*Step 1: the prefactor.* For $s,z \in U(z_0,r_1)$ write $\mathcal{A}(s)^{-1}\mathcal{A}(z) = I + (s-z)\mathfrak{k}_A(s,z)$; by Cauchy's estimates on the convex set $U(z_0,r_1)$, $|\mathfrak{k}_A| \le C$. Then

$$\mathcal{C}(\mathfrak{g}\tilde{Q}_+^{-1})(z)\tilde{Q}(z) = \mathcal{C}(\mathfrak{g}M_{n,+}^{-1})(z)M_n(z) + \frac{1}{2\pi i}\int_{\Sigma_1} \mathfrak{g}(s)\big[M_{n,+}(s)^{-1}\mathfrak{t}_n^{-\sigma_3/4}\big]\mathfrak{k}_A(s,z)\big[\mathfrak{t}_n^{\sigma_3/4}M_n(z)\big]\,\mathrm{d}s.$$

By (105) the kernel of the second term is at most $C\,\mathfrak{W}(s)\mathfrak{W}(z)$, so by (106) it is Hilbert–Schmidt with norm at most $C$.

*Step 2: the variable $\zeta$.* Let $\mathfrak{k}_0(s,z) := f_0'(s)/(f_0(s)-f_0(z)) - 1/(s-z)$, which is analytic and bounded on $U(z_0,r_1)^2$. Then $\mathrm{d}s/(s-z) = \mathrm{d}\zeta_s/(\zeta_s-\zeta_z) - \mathfrak{k}_0(s,z)\,\mathrm{d}s$. The term with $\mathfrak{k}_0$ is a scalar kernel times $M_{n,+}(s)^{-1}M_n(z) = [M_{n,+}(s)^{-1}\mathfrak{t}_n^{-\sigma_3/4}][\mathfrak{t}_n^{\sigma_3/4}M_n(z)]$, hence Hilbert–Schmidt with norm at most $C$ as in Step 1. The main term is $\mathcal{S}\mathfrak{h} := \mathcal{C}^{\mathfrak{t}_n\Gamma_1}(\mathfrak{h}\mathbb{M}_+^{-1})\mathbb{M}$ with $\mathbb{M}(\zeta) := M_n(z)$ and $\mathfrak{h}(\zeta) = \mathfrak{g}(z)$; as $|f_0'|^{\pm1} \le C$, the map $\mathfrak{g}\mapsto\mathfrak{h}$ changes $L^2$ norms by the factor $\mathfrak{t}_n^{1/2}$ up to constants, the same on input and output, so it suffices to bound $\mathcal{S}$ on $L^2(\mathfrak{t}_n\Gamma_1)$ uniformly. The conformal map preserves the sides of the contour, so $\pm$ boundary values correspond.

*Step 3: splitting in $\zeta$.* Let $\mathcal{B}_{\le3} := \{|\zeta|\le 3\}$, $\mathcal{B}_{\ge2} := \{|\zeta|\ge 2\}$, $\mathcal{B}_{2,3} := \{2\le|\zeta|\le 3\}$. The pairs $(\zeta',\zeta)$ (source, target) are partitioned into $\mathcal{B}_{\le3}\times\mathcal{B}_{\le3}$; $(\mathcal{B}_{\ge2}\times\mathcal{B}_{\ge2})\setminus(\mathcal{B}_{2,3}\times\mathcal{B}_{2,3})$; $\{|\zeta'|<2, |\zeta|>3\}$; and $\{|\zeta'|>3, |\zeta|<2\}$. Accordingly $\mathcal{S} = 1_{\mathcal{B}_{\le3}}\mathcal{S}1_{\mathcal{B}_{\le3}} + \big(1_{\mathcal{B}_{\ge2}}\mathcal{S}1_{\mathcal{B}_{\ge2}} - 1_{\mathcal{B}_{2,3}}\mathcal{S}1_{\mathcal{B}_{2,3}}\big) + 1_{\{|\zeta|>3\}}\mathcal{S}1_{\{|\zeta|<2\}} + 1_{\{|\zeta|<2\}}\mathcal{S}1_{\{|\zeta|>3\}}$, and we bound each part.

$\mathcal{B}_{\le3}\times\mathcal{B}_{\le3}$. Here $\mathbb{M} = \hat{A}\mathsf{P}\Theta_0$. Write $\hat{A}(\zeta')^{-1}\hat{A}(\zeta) = I + (\zeta'-\zeta)\hat{\mathfrak{k}}(\zeta',\zeta)$ with $|\hat{\mathfrak{k}}| \le C$ on $\{|\zeta|,|\zeta'|\le 3\}$. The resulting commutator kernel is $\Theta_{0,+}(\zeta')^{-1}\mathsf{P}_+(\zeta')^{-1}\hat{\mathfrak{k}}\,\mathsf{P}(\zeta)\Theta_0(\zeta)$, of size at most $C\mathfrak{w}(\zeta')\mathfrak{w}(\zeta)$, and $\int_{\mathfrak{t}_n\Gamma_1\cap\mathcal{B}_{\le3}}\mathfrak{w}^2|\,\mathrm{d}\zeta| \le C$: Hilbert–Schmidt with norm $\le C$. The remaining part is $\mathfrak{h}\mapsto\big[\mathcal{C}(\mathfrak{h}'\mathsf{P}_+^{-1})\mathsf{P}\big]\Theta_0$ with $\mathfrak{h}' := \mathfrak{h}\Theta_{0,+}^{-1}$. If $\mathsf{P} = \zeta^{\beta\sigma_3}$, its first column is $\zeta^\beta\mathcal{C}(\zeta'^{-\beta}\mathfrak{h}'e_1)$ and its second column is $\zeta^{-\beta}\mathcal{C}(\zeta'^{\beta}\mathfrak{h}'e_2)$; both are bounded on $L^2(\mathfrak{t}_n\Gamma_1\cap\mathcal{B}_{\le3})$ by Lemma C.1 with exponents $\pm\beta$, uniformly in $n$ by dilation invariance. If $\mathsf{P}$ is logarithmic, $\mathsf{P}_+(\zeta')^{-1}\mathsf{P}(\zeta) = I + \mathsf{k}(\log\zeta - \log_+\zeta')e_1e_2^{\mathsf{T}}$, and the second term is the logarithmic commutator of Lemma C.2 applied to $\mathfrak{h}'e_1$ (the branches $\log_\pm$ on the arc $(-\infty,0)$ differ by the constant $2\pi i$, which adds a multiple of the Cauchy operator).

$(\mathcal{B}_{\ge2}\times\mathcal{B}_{\ge2})\setminus(\mathcal{B}_{2,3}\times\mathcal{B}_{2,3})$. On $\mathcal{B}_{\ge2}$, $\mathbb{M} = \zeta^{-\sigma_3/4}\Theta_\infty$, so $1_{\mathcal{B}_{\ge2}}\mathcal{S}1_{\mathcal{B}_{\ge2}}$ is $\Theta_\infty$ times the operator $\mathfrak{h}''\mapsto\mathcal{C}(\mathfrak{h}''\zeta'^{\sigma_3/4})\zeta^{-\sigma_3/4}$, with $\mathfrak{h}'' = \mathfrak{h}\Theta_{\infty,+}^{-1}$; this is Lemma C.1 with $\beta = \mp\frac14$. On $\mathcal{B}_{2,3}$ the matrices $\mathbb{M}^{\pm1}$ are bounded, so $1_{\mathcal{B}_{2,3}}\mathcal{S}1_{\mathcal{B}_{2,3}}$ is bounded by $\|\mathcal{C}_\pm\|$ on $\mathfrak{t}_n\Gamma_1$ (Lemma C.1 with $\beta=0$).

*The separated parts.* If $|\zeta'|<2$ and $|\zeta|>3$, then $|\zeta-\zeta'|\ge|\zeta|/3$, $|\mathbb{M}(\zeta')^{-1}| \le C\mathfrak{w}(\zeta')$ and $|\mathbb{M}(\zeta)| \le C|\zeta|^{1/4}$. The kernel is at most $C\mathfrak{w}(\zeta')|\zeta|^{-3/4}$, and $\int_{|\zeta'|<2}\mathfrak{w}^2\int_{|\zeta|>3}|\zeta|^{-3/2}|\,\mathrm{d}\zeta||\,\mathrm{d}\zeta'| \le C$ by the Ahlfors bound. The other part is symmetric. □

*The two instances.* At the hard edge, $z_0=0$, $r_0=\delta$, $r_1=\frac34\delta$, $f_0=f$, $\mathfrak{t}_n=n^2$, $\mathcal{A} = A_0(2\pi)^{\sigma_3/2}$ (Lemma 7.2(i)) and $M_n(z) = \Phi_\alpha(n^2f(z))$; (B2) and (B3) are (81) and (82), with $\beta=\alpha/2$, or the logarithmic case if $\alpha=0$. Here $|\beta|<\frac12$ is exactly $|\alpha|<1$. At the soft edge, $z_0=4c$, $r_0=\varepsilon$, $r_1=\frac34\varepsilon$, $f_0=f_r$, $\mathfrak{t}_n=n^{2/3}$, $\mathcal{A}=A_r$ and $M_n=M_{r,n}$: by the two-scale

structure in (F5) (Lemma 6.6(i), (iv)), (B1)–(B3) hold with $\mathsf{P} = I$ and $\hat{A} = \Psi^{\mathrm{I}}$, the entire Airy model matrix. No $\alpha$ is involved.

D.2. **Proof of Proposition 7.4.** *Step 0: the contour.* $\hat{\Sigma}$ is a finite union of $C^2$ arcs with $|\hat{\Sigma} \cap U(z,\rho)| \le C_\Sigma \rho$, $C_\Sigma$ independent of $n$: this holds for $\Sigma$ by (F1) and for the two fixed circles. We show $\|\mathcal{C}_\pm\|_{L^2(\hat{\Sigma})} \le C$. The operator splits into blocks according to source and target in $\Sigma$ or $\Sigma_\circ$. The block $\Sigma \to \Sigma$ is bounded by (F1), and $\Sigma_\circ \to \Sigma_\circ$ is the Cauchy operator of two fixed circles. For the mixed blocks let $\Sigma_{\mathrm{loc}} := \Sigma \cap (U(0,2\delta) \cup U(4c,2\varepsilon))$, which is independent of $n$; on the fixed curve $\Sigma_{\mathrm{loc}} \cup \Sigma_\circ$ the Cauchy operator is bounded by T3(a). The rest of $\Sigma$ has distance at least $\min(\delta,\varepsilon)$ from $\Sigma_\circ$, and there the kernel is Hilbert–Schmidt: $\int_{\Sigma_\circ}\int_{\Sigma\setminus\Sigma_{\mathrm{loc}}} |s-z|^{-2}|\,\mathrm{d}s||\,\mathrm{d}z| \le |\Sigma_\circ|\,4C_\Sigma/\min(\delta,\varepsilon)$ by the proof of Lemma B.1(a).

*Step 1: part (i).* Fix $n$. By (82), $|\tilde{Q}^{\pm1}(z)| \le C_n|z|^{-|\alpha|/2}$ near 0 (with $C_n(1+|\log|z||)$ if $\alpha = 0$). By (F5) and (B3) with $\mathsf{P} = I$, $\tilde{Q}^{\pm1}$ is bounded near $4c$. Elsewhere $\tilde{Q}^{\pm1}$ is bounded, and it is continuous up to $\hat{\Sigma}$ in the sense of §B.1, being built from explicit functions that are continuous up to the arcs from each side. Since $|\alpha| < 1$, the boundary values of $\tilde{Q}^{\pm1} - I$ lie in $L^2(\hat{\Sigma})$; and $\rho \sup_{|z|=\rho} |\tilde{Q}^{\pm1}(z) - I| = O(\rho^{1-|\alpha|/2}) \to 0$, while at the other vertices the corresponding quantity is $O(\rho)$. Finally $\tilde{Q}^{\pm1} - I = O(1/z)$ at infinity. Lemma B.1(c), applied for fixed $n$, gives $\tilde{Q}_\pm^{\pm1} \in I + \partial\mathcal{C}(L^2(\hat{\Sigma}))$. This is part (i).

*Step 2: reduction.* Let $\mathcal{D}$ be the space of functions on $\hat{\Sigma}$ that are $C^1$ on each arc and vanish near the vertices; it is dense in $L^2(\hat{\Sigma})$, and $\mathfrak{f}(\tilde{v} - I) \in \mathcal{D}$ for $\mathfrak{f} \in \mathcal{D}$, because $\tilde{v}$ is smooth on the arcs away from the vertices. Suppose that

(R1) for every $\mathfrak{g} \in \mathcal{D}$ there is $m$ with $m_\pm \in \partial\mathcal{C}(L^2)$, $m_+ = m_-\tilde{v} + \mathfrak{g}$ and $\|m_\pm\| \le C_2\|\mathfrak{g}\|$, with $C_2$ independent of $n$ and $\mathfrak{g}$;

(R2) if $m_\pm \in \partial\mathcal{C}(L^2)$ and $m_+ = m_-\tilde{v}$, then $m = 0$.

Then $1 - \mathcal{C}_{\tilde{v}}$ is invertible on $L^2(\hat{\Sigma})$ with $\|(1-\mathcal{C}_{\tilde{v}})^{-1}\| \le 1 + C_2\|\tilde{v} - I\|_\infty$. This is the easy half of T3(e). Indeed, for $\mathfrak{f} \in \mathcal{D}$ solve (R1) with $\mathfrak{g} := \mathfrak{f}(\tilde{v} - I)$ and put $\mu := m_- + \mathfrak{f}$. By (R2) (proved in Step 3, which uses only Step 1) the solution $m$ of (R1) is unique, so $\mathfrak{f} \mapsto \mu$ is linear. Then $\mu(\tilde{v} - I) = m_-\tilde{v} - m_- + \mathfrak{f}(\tilde{v} - I) = m_+ - m_-$; as $m_\pm \in \partial\mathcal{C}(L^2)$, $m_- = \mathcal{C}_-(m_+ - m_-) = \mathcal{C}_-(\mu(\tilde{v} - I))$, so $\mu - \mathcal{C}_{\tilde{v}}\mu = \mathfrak{f}$ and $\|\mu\| \le (1 + C_2\|\tilde{v} - I\|_\infty)\|\mathfrak{f}\|$. The map $\mathfrak{f} \mapsto \mu$ extends by continuity to a bounded operator $\mathcal{R}$ on $L^2(\hat{\Sigma})$ with $(1 - \mathcal{C}_{\tilde{v}})\mathcal{R} = 1$, since $1 - \mathcal{C}_{\tilde{v}}$ is bounded. Conversely, if $\mu = \mathcal{C}_{\tilde{v}}\mu$, then $m := \mathcal{C}(\mu(\tilde{v} - I))$ satisfies $m_\pm \in \partial\mathcal{C}(L^2)$, $m_- = \mu$ and $m_+ = \mu\tilde{v} = m_-\tilde{v}$, so $m = 0$ by (R2) and $\mu = 0$. Hence $1 - \mathcal{C}_{\tilde{v}}$ is injective with right inverse $\mathcal{R}$, and $(1 - \mathcal{C}_{\tilde{v}})^{-1} = \mathcal{R}$.

*Step 3: uniqueness.* Let $m_\pm \in \partial\mathcal{C}(L^2)$ with $m_+ = m_-\tilde{v}$. Put $X := \tilde{Q} + m$. Then $X_+ = X_-\tilde{v}$, $X_\pm \in I + \partial\mathcal{C}(L^2)$, $\tilde{Q}_\pm^{-1} \in I + \partial\mathcal{C}(L^2)$ by Step 1, and $\tilde{v}^{\pm1} \in L^\infty$. By T3(b) with $A = X$, $B = \tilde{Q}$, $v_A = v_B = \tilde{v}$ and exponents $r = r' = 2$, $X\tilde{Q}^{-1} = I$, i.e. $m = 0$. This is (R2). Step 3 uses only Step 1.

*Step 4: the explicit solution and the uniform bound.* Let $\mathfrak{g} \in \mathcal{D}$. Then $\mathfrak{k} := \mathfrak{g}\tilde{Q}_+^{-1} \in \mathcal{D}$, since $\tilde{Q}_+^{-1}$ is smooth on the arcs away from the vertices. Put $m := \mathcal{C}\mathfrak{k} \cdot \tilde{Q}$. By Lemma B.1(b), $m$ is continuous up to $\hat{\Sigma}$ and $m_+ - m_-\tilde{v} = \big((\mathcal{C}\mathfrak{k})_+ - (\mathcal{C}\mathfrak{k})_-\big)\tilde{Q}_+ = \mathfrak{g}$.

To bound $\|m_\pm\|_{L^2}$ uniformly, let $\Sigma_A := \hat{\Sigma} \cap U(0, \frac{3}{4}\delta)$, $\Sigma_B := \hat{\Sigma} \cap U(4c, \frac{3}{4}\varepsilon)$ and $\Sigma_R := \hat{\Sigma} \setminus (U(0, \frac{1}{2}\delta) \cup U(4c, \frac{1}{2}\varepsilon))$. On $\Sigma_R$ the matrices $\tilde{Q}^{\pm1}$ are bounded uniformly in $n$: outside the discs $\tilde{Q} = P_0^\infty$ by (66); in $\frac{1}{2}\delta \le |z| \le \delta$ by Lemma 7.2(iii); in $\frac{1}{2}\varepsilon \le |z - 4c| \le \varepsilon$ by (B2) for the Airy instance, since $|\zeta| \ge 2$ there. Partition the source–target pairs into $\Sigma_A \times \Sigma_A$, $\Sigma_B \times \Sigma_B$, $(\Sigma_R \times \Sigma_R) \setminus (\Sigma_A \times \Sigma_A \cup \Sigma_B \times \Sigma_B)$ and the rest. The first two blocks are bounded by Lemma D.1. The third is $1_{\Sigma_R} T 1_{\Sigma_R} - 1_{\Sigma_R \cap \Sigma_A} T 1_{\Sigma_R \cap \Sigma_A} - 1_{\Sigma_R \cap \Sigma_B} T 1_{\Sigma_R \cap \Sigma_B}$ with $T\mathfrak{g} = \mathcal{C}_\pm(\mathfrak{g}\tilde{Q}_+^{-1})\tilde{Q}_\pm$, and each

term is bounded by Step 0 since $\tilde{Q}^{\pm1}$ is bounded on $\Sigma_R$. In the remaining pairs one point lies in $U(0,\frac{1}{2}\delta)$ (or $U(4c,\frac{1}{2}\varepsilon)$) and the other outside $U(0,\frac{3}{4}\delta)$ (resp. $U(4c,\frac{3}{4}\varepsilon)$), so $|s-z|\geq \frac{1}{4}\min(\delta,\varepsilon)$. The kernel is at most $C|\tilde{Q}_+^{-1}(s)||\tilde{Q}(z)||s-z|^{-1}$, and its Hilbert–Schmidt norm is bounded by the $L^2$ bounds of Lemma D.1 on the discs, the boundedness of $\tilde{Q}^{\pm1}$ on $\Sigma_R$, and $\int_{|s-z|\geq t}|s-z|^{-2}|\,\mathrm{d}s|\leq 4C_\Sigma/t$. Hence $\|m_\pm\|\leq C_2\|\mathfrak{g}\|$ with $C_2$ independent of $n$ and of $\mathfrak{g}\in\mathcal{D}$.

*Step 5: $m_\pm\in\partial\mathcal{C}(L^2)$.* We apply Lemma B.1(c) to $\mathfrak{F}:=m$, for fixed $n$. Since $\mathfrak{k}$ vanishes near the vertices, $\mathcal{C}\mathfrak{k}$ is analytic and bounded near each vertex, and $\mathcal{C}\mathfrak{k}=O(1/z)$ at infinity; hence $|m|\leq C_n|\tilde{Q}|$ near each vertex, so that $\rho\sup_{|z-z_0|=\rho}|m|=O(\rho^{1-|\alpha|/2})\to 0$ (by Step 1), $m=O(1/z)$ at infinity, and $m_\pm\in L^2(\hat{\Sigma})$ by Step 4. Lemma B.1(c) gives $m_\pm\in\partial\mathcal{C}(L^2)$. This proves (R1).

Working with data in $\mathcal{D}$ avoids any analysis of $m$ at the vertex 0, where $\tilde{Q}$ is unbounded: for such data the singular factor $\tilde{Q}$ only multiplies a function that is analytic near 0, and the uniform estimate of Step 4, which is where the structure at 0 enters, extends the inverse to all of $L^2$.

Steps 2–5 (with (R2) from Step 3) prove part (ii) of Proposition 7.4, with $\|(1-\mathcal{C}_{\tilde{v}})^{-1}\|\leq 1+C_2\sup_n\|\tilde{v}-I\|_\infty$, which is finite because $\tilde{v}^{\pm1}$ is bounded uniformly in $n$ (on the lips $|\varrho_\pm e^{2n\varphi}|\leq 1$, on the circles $\tilde{v}=I+O(1/n)$, elsewhere $\tilde{v}\in\{I,J\}$ or $\tilde{v}=v$). □

## References


1. J. Baik, T. Kriecherbauer, K. T.-R. McLaughlin, and P. D. Miller, *Uniform asymptotics for polynomials orthogonal with respect to a general class of discrete weights and universality results for associated ensembles: announcement of results*, arXiv:math/0212149v1.
2. T. O. Conway and P. Deift, *Asymptotics of polynomials orthogonal with respect to a logarithmic weight*, SIGMA Symmetry Integrability Geom. Methods Appl. **14** (2018), Paper 056, arXiv:1711.01590v4.
3. D. Dai and L. Yao, *Uniform asymptotics for the discrete Laguerre polynomials*, arXiv:2104.03563v1 (2021).
4. Z. Dai, D. Piao, and J. Qiao, *Construction and characterization of oscillatory chain sequences*, arXiv:2508.07653v1 (2025).
5. P. Deift, A. Its, and I. Krasovsky, *Asymptotics of Toeplitz, Hankel, and Toeplitz+Hankel determinants with Fisher–Hartwig singularities*, Ann. of Math. (2) **174** (2011), 1243–1299, arXiv:0905.0443v3.
6. P. Deift and M. Piorkowski, *Recurrence coefficients for orthogonal polynomials with a logarithmic weight function*, SIGMA Symmetry Integrability Geom. Methods Appl. **20** (2024), Paper 004, arXiv:2307.09277v2.
7. M. Derevyagin and V. Derkach, *Darboux transformations of Jacobi matrices and Padé approximation*, Linear Algebra Appl. (2011), doi:10.1016/j.laa.2011.05.035; arXiv:1012.3712v2.
8. *NIST digital library of mathematical functions*, https://dlmf.nist.gov; release used: to verify.
9. A. Haynes and W. Zudilin, *Hankel determinants of zeta values*, SIGMA Symmetry Integrability Geom. Methods Appl. **11** (2015), Paper 101, arXiv:1510.01901.
10. A. B. J. Kuijlaars, K. T.-R. McLaughlin, W. Van Assche, and M. Vanlessen, *The Riemann–Hilbert approach to strong asymptotics for orthogonal polynomials on* $[-1,1]$, Adv. Math. **188** (2004), 337–398, arXiv:math/0111252v2 (the version read).
11. E. Levin and D. S. Lubinsky, *Asymptotics associated with exponential weights*, Announcement (1999), PDF on the second author's homepage; publication data to verify.
12. K. T.-R. McLaughlin and P. D. Miller, *The $\overline{\partial}$ steepest descent method for orthogonal polynomials on the real line with varying weights*, arXiv:0805.1980v1.
13. H. Monien, *Hankel determinants of Dirichlet series*, arXiv:0901.1883v1. Publication status: to verify.
14. M. Piorkowski, *Riemann–Hilbert theory without local parametrix problems: applications to orthogonal polynomials*, arXiv:1910.00564v2.
15. Gordon [surname], *Hankel determinants of discrete measures with power-law atoms, and of zeta values*, 2026, Companion paper, draft of 14 October 2026.
16. ———, *Rough weights and the pointwise asymptotics of recurrence coefficients*, 2026, In preparation; title provisional.
17. T. Trogdon, *On the asymptotics of orthogonal polynomials on multiple intervals with non-analytic weights*, SIGMA Symmetry Integrability Geom. Methods Appl. **22** (2026), Paper 006, arXiv:2412.18656v5.

18. W. Van Assche, *Compact Jacobi matrices: from Stieltjes to Krein and $M(a, b)$*, Ann. Fac. Sci. Toulouse Math. (6) **numéro spécial Stieltjes** (1996), 195–215, arXiv:math/9510214.
19. X.-B. Wu, Y. Lin, S.-X. Xu, and Y.-Q. Zhao, *Uniform asymptotics for discrete orthogonal polynomials on infinite nodes with an accumulation point*, Anal. Appl. (Singap.) **14** (2016), 705–737, arXiv:1410.3948v1 (the version read).